Deutsche Bank
Research

Europe
France

Aerospace & Defence
Aerospace & Defence

Industry
Defense

Date
22 September 2026

Industry Update

German Defense: An Ecosystem in Transformation

Michael Kuhn
Research Analyst
+49-69-910-46002

Tim Rokossa
Head of European Equity Research
+49-69-910-31998

Christoph Laskawi
Research Analyst
+49-69-910-31924

Christophe Menard
Research Analyst
+33-1-4495-9315

Sriram Krishnan, CFA
Research Analyst
+44-20-7541-0983

Launching Private Company Coverage with the German Defense Sector

Traditional analysis that focused on public companies overlooked a critical market segment: the many important competitors that are privately owned, typically by families or private equity firms. To address this, we are launching dedicated coverage of Europe's private-market giants, starting with Germany, where a traditionally dominant private sector means large firms outnumber public ones almost two to one. We begin with the German defense sector, where understanding the private ecosystem is essential: it is the source of critical innovation, the solution to supply-chain issues, and the venue for a future wave of M&A, consolidation, and IPOs.

From Incremental Modernization to Industrial Scaling

Germany's defense landscape is undergoing its most significant transformation in decades, shifting from a paradigm of 'incremental modernization' to 'industrial scaling'. Triggered by the *Zeitenwende* in 2022, the new era is underpinned by substantial financial commitments, including a €100bn special fund and constitutional reforms enabling sustained long-term spending. The objective is no longer merely to fill discrete capability gaps but to execute a full 'ecosystem build-out' that addresses structural choke points and enhances industrial capacity. This transformation is creating a more resilient and scalable German Defense Industrial Base (DIB), prepared to meet surging domestic and European demand.

Technology, Integration, and the Rise of New Players

This strategic pivot is also reshaping the industrial and technological environment. EU initiatives (EDF, EDIRPA, ASAP) are fostering joint funding and integrated supply chains, while the market is experiencing a 'next wave' of technological change, with a clear shift in value towards drones, software-defined systems, electronic warfare, and AI-driven capabilities. These dynamics force industry consolidation but also create openings for agile start-ups, which are vital for innovation, as speed and software become paramount. This report provides an in-depth primer on this evolving sector. We identify Germany's industrial strengths and technology gaps, analyze the key private players poised to benefit, and examine the structural themes shaping the market - from funding dynamics and procurement reforms to technology roadmaps. Companies analyzed in this report: **Diehl Group, Rohde & Schwarz, Helsing, STARK Defence, Quantum Systems, Alpine Eagle, Dynamit Nobel Defence, ARX Robotics, Reflex Aerospace, AUTOFLUG, and Anschütz.**

Deutsche Bank AG

IMPORTANT RESEARCH DISCLOSURES AND ANALYST CERTIFICATIONS LOCATED IN APPENDIX 1. Deutsche Bank does and seeks to do business with companies covered in its research reports. Thus, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.

Distributed on: 22/09/2026 16:16:19 GMT

Table Of Contents

Defense Companies Featured in this Report 4

Half the Industry is Invisible 5
We are Changing that 5
Executive Summary - Eight Points Carry this Report 5

The Argument in Five Parts 8
1. The Strategic Landscape 9
2. New Wars Change the Old Battlefield 13
3. How To Gain Exposure: Investment Opportunities in Private Gems 17
4. Trend: Europe Buys European and Europe Buys Between Middle Pow 20
5. The Regulatory Landscape & Investment Constraints 22

The Supply Chain: Who Delivers What to Whom 24
Energetics: the binding choke point 26

Spotlights 28

What This Coverage Is - and What It Is Not 35

Diehl Group 37
Europe's Air-Defense Anchor 37

Rohde & Schwarz 51
From 5G Networks to Battlefield Electronics 51

Helsing 65
Germany's Most Valuable Startup 65

STARK Defence 79
Fusing AI-Enabled UAVs / USVs with a Unified Integration Platform 79

Quantum Systems 93
Fusing UAS with AI Software for Superior Aerial Intelligence 93

Alpine Eagle 108
Coordinated Air-To-Air Defence 108

Dynamit Nobel Defence 125
From the Creator of Dynamite to Modern Anti-Tank Weaponry 125

ARX Robotics 137
Dual-Use Autonomous Ground Systems 137

Reflex Aerospace 152
Custom-Built, Dual-Use Satellite Technology 152



Table Of Contents

AUTOFLUG GmbH.....166
Custom Seating & Cabin Solutions.....166

Anschütz GmbH.....180
Navigation Systems Maritime Excellence.....180

Glossary.....193

Defense Companies Featured in this Report

Figure 1: Defense Companies - Overview

Company	HQ (City)	Founded	Ownership	Revenue (last rep.)	EBIT (last rep.)	Employees (approx.)	Land	Air	Sea	Ammunition	Space	Cyber / EW / Software	Defense % of revenue	Sales by geography (where disclosed)
Diehl Group	Nuremberg	1902	Family / foundation	€5,420m	€703m	~19,000	○	●	○	●		○	43%	~68% outside Germany (2024); Defence €1,258m intl vs €569m Germany
Rohde & Schwarz	Munich	1933	Family (100%)	€3,160m	€412m	15,000+	○	○	○			●	n/d	Global (~70 countries); split n/d
Quantum Systems	Gilching	2015	VC / strategic-backed	~€300m	"Double-digit profit."	~1,000	●	●				○	n/d	DE, US, UK, Ukraine, Romania, Australia; split n/d
Helsing	Munich	2021	VC / strategic-backed	€27m	-€8m	~900	●	●	○		○	●	~100%	Pan-European (DE/UK/FR) + US; split n/d
STARK Defence	Berlin	2024	VC-backed	n/d	n/d	~400	○	●	○			○	~100%	DE, Ukraine, UK, Greece, Sweden; split n/d
Dynamit Nobel Defence	Burbach	2004	Rafael group (private)	€146m	€8m	>500	●			●			~100%	Germany 28%, other EU 16%, non-EU 56% (2023)
ARX Robotics	Oberding	2022	VC-backed	€15m	-€14m	~150	●					○	~100%	DE, UK, Ukraine; split n/d
Alpine Eagle	Munich	2023	VC-backed	7-digit (self-rep.)	n/d	~50	○	●				○	~100%	DE, UK, Netherlands, Ukraine; split n/d
Reflex Aerospace	Berlin / Munich	2021	VC-backed	n/d	n/d	>100					●	○	n/d (dual-use)	Germany-centered; split n/d
AUTOFLUG	Rellingen	1919	Family (Sedlmayr)	~€53m	~€0-1m	~220	○	●					n/d (dual-use)	Heavily Germany-concentrated; split n/d
Anschütz	Kiel	1905	DMB (Murmann family)	€104m	-€3m	>700			●			○	n/d (dual-use)	DE core + intl subsidiaries; split n/d

Legend: ● = primary / core defense exposure ○ = secondary / adjacent exposure (blank) = no material activity

Source : Company disclosures, annual reports, press releases, Bundestag Lobby Register, Deutsche Bank. 1. Rohde & Schwarz and AUTOFLUG report FY2024/25; Quantum Systems figures refer to unaudited, company-stated 2025 data. All others report on a calendar-year basis. 2. Diehl figures are reported at group level; defense share is derived from the Defence division. 3. Rohde & Schwarz does not separately disclose EBIT or defense revenue share. 4. Quantum Systems revenue, profitability and employee data are company-stated. 5. Helsing valuation reflects the July 2026 funding round. 6. STARK Defence (SKD) and Reflex Aerospace financials for one financial year only and with limited explanation power.. ARX Robotics and Anschütz figures are based on Bundestag Lobby Register filings. 7. Alpine Eagle revenue is based on investor disclosures; no public financial statements are available. 8. Dynamit Nobel Defence is privately held within the Rafael group. 9. Defense % of revenue: "~100%" denotes predominantly defense-focused business models; "n/d (dual-use)" indicates no disclosed defense/civil revenue split. 10. Only Diehl and Dynamit Nobel Defence disclose meaningful geographic revenue splits; all other entries reflect operational footprint rather than revenue allocation. 11. Sector tags are Deutsche Bank analytical classifications. "Cyber / EW / Software" includes EW, SIGINT, autonomy, mission and C2 software.

Half the Industry is Invisible

We are Changing that.

Equity research has always looked at the listed half of the economy. That half is shrinking. Companies stay private for longer, the largest capital pools are now private, and in Germany the listed universe was never the industry to begin with. **There are roughly 120 listed German companies with sales above €1bn - and roughly 200 unlisted ones.** In our home market, large private companies outnumber public ones almost two to one.

That gap is not a curiosity. It is an analytical problem.

If a sector's innovation, its supply-chain frictions, its pricing behavior and its eventual consolidation sit largely outside the listed universe, then covering only the listed names means covering only part of the argument. With this report we start closing that gap. We are launching dedicated, ongoing coverage of Europe's private-market giants, beginning in Germany.

Why we are starting with German defense

Defense is the sharpest version of the problem we are trying to solve. Germany is in the middle of its largest rearmament cycle since reunification, but much of the capability being bought is not primarily produced by the listed defense groups. It is produced by an ecosystem: family-owned system houses that have supplied the Bundeswehr for decades, venture-capital-backed companies that did not exist three years ago, and the tier-2 and tier-3 suppliers who sit on the actual constraints - energetics, seekers, qualified electronics, test capacity, cleared labor. That ecosystem is where the innovation happens, where the supply chain either breaks or scales, and where the next wave of M&A, consolidation and IPOs will take place. Understanding it is not a side-project to covering Rheinmetall, Hensoldt or Renk. It is a prerequisite.

Companies covered: Diehl Group, Rohde & Schwarz, Helsing, STARK Defence, Quantum Systems, Alpine Eagle, Dynamit Nobel Defence, ARX Robotics, Reflex Aerospace, AUTOFLUG and Anschütz.

Executive Summary - Eight Points Carry this Report

1. **The spending shift is structural, not cyclical.** The €100bn special fund was the headline; the 2025 constitutional reform exempting defense spending above 1% of GDP from the debt brake was the substance. Government budgets now foresee more than €150bn of annual defense spending by 2029, against roughly €52bn of core budget in 2024. NATO's 2025 pledge of 3.5% of GDP on core defense by 2035 puts a floor under the back-end.
2. **Budgets are not revenue.** Between an approved budget and a delivered capability sit parliamentary approval for anything above €25m, tender design, contracting, qualification, test capacity and acceptance. The February 2026 procurement acceleration reform helps, but the conversion chain - not demand - sets the delivery slope.
3. **Value is moving to software, autonomy and effects.** Drones, software-defined systems, electronic warfare and AI-enabled decision layers are taking share from platform-centric spend. That favors companies whose iteration cycle is measured in weeks, and it is the clearest reason private



and venture-capital-backed firms now sit inside mainstream force design.

4. **Production is deterrence - and production is where Germany is short.** Scaling breaks at specific points: energetics permits, seekers and guidance electronics, qualification burden, specialized labor, and limited test windows. Almost every one of those nodes is occupied, at least in part, by a private company.
5. **The private ecosystem is now procurement-relevant, not just innovation-relevant.** The February 2026 Bundeswehr loitering-munitions framework, awarded to Helsing and STARK from an initial package of roughly €540m, is the clearest proof point. Sovereign buyers are contracting with companies two to five years old.
6. **Scale still sits with the incumbents.** Diehl generated €5.4bn of sales and €703m of EBIT in FY2025, with Defence at roughly €2.3bn; Rohde & Schwarz passed €3bn of revenue in FY2024/25 with no bank debt. These are not start-ups with a defense angle - they are among Europe's more significant defense industrial assets, and they happen to be family-owned.
7. **A consolidation and financing wave is coming.** Capital intensity is rising precisely as venture-stage companies move from software into hardware and factories. That points to further strategic rounds, industrial partnerships, JV structures, selective M&A and, in time, IPOs. For a bank, this is the most commercially relevant conclusion in the report.
8. **Regulation is the binding constraint, in both directions.** Export licensing, foreign investment screening, NATO interoperability and compliance intensity limit scaling and exits. But policy also accelerates - fast-track procurement, EU joint programs, state-backed vehicles for defense tech.

Why defense, in numbers

The money. Germany's core defense budget has moved from roughly €47bn in 2021 to a projected path above €150bn by 2029, with the €100bn special fund bridging the middle of that ramp. The 2025 debt-brake reform matters more than the fund itself: by exempting defense spending above 1% of GDP, it converts a one-off program into a sustained baseline. Procurement approvals have already responded: 97 major defense projects approved in 2024 against 55 in 2023, with over €58bn allocated.

The conversion chain. What is in doubt is throughput. Contracts above €25m require binding Bundestag budget-committee approval, and tendering, supplier orders, integration, testing and acceptance each add time. Europe's fragmentation compounds it: European states operate around 179 different major weapon systems against 33 in the US, and 19 different main battle tanks against one. Between 2020 and 2024, European arms imports rose 155%, and European NATO members sourced 64% of those imports from the US, up from 52% in 2015-19. Whether 'buy European' becomes policy rather than rhetoric is the single largest swing factor for the domestic industrial base.

Where the private sector sits. The constrained nodes are consistent across almost every program: energetics capacity, defense-qualified electronics, test and certification, and cleared labor. Private companies are disproportionately present in exactly those nodes. Diehl's acquisition of Dynamit Nobel was, in substance, the acquisition of explosives-related operating authorizations it describes as unique in Germany. STARK's dependence on TDW for German-manufactured warheads is the same story one tier down. Rohde & Schwarz occupies the secure-communications and SIGINT/EW layer no platform program can route around.



Where the new money is going. Germany attracted close to €1bn of defense and security venture-capital funding in 2025, second only to the US. The deal mix since 2021: unmanned systems (34 deals), surveillance and space (24), other defense tech (24), electronic warfare (10) and cyber (8). Helsing raised $1.8bn at an $18bn valuation in July 2026; STARK Defence raised €500m in June 2026 above €3.5bn. Capital of that scale changes the industrial map.

The five debates - and where we go next

1. **Can Germany convert budget into capability fast enough?** Demand is settled; throughput is not. Watch acceptance capacity, qualification timelines and framework design rather than headline budgets (*Executive Summary, Sections 1 & 5*).
2. **Do the newcomers industrialize, or stall at the factory door?** Helsing, STARK Defence, ARX Robotics and Quantum Systems are all making the same transition from software and prototypes into serial hardware. That is where defense start-ups historically fail. The next 18 months are the test.
3. **Do incumbents get disrupted, or absorb?** The likelier outcome is a denser partner network - Diehl with Hensoldt and Airbus, Helsing with Saab and Kongsberg, ARX Robotics with RENK and Daimler Truck. Partnership maps are becoming a better guide to competitive position than product catalogs.
4. **Does 'buy European' become real?** A conscious substitution of even part of the 64% US import share would be a structural repricing of the European industrial base. Fragmentation and interoperability standards cut against it *(Executive Summary, Section 4*).
5. **Who is financed, and how?** Rising capital intensity, thin private disclosure, constrained exit routes under foreign-investment screening, and uneven ESG treatment will all influence the financing and consolidation wave.

How we selected the companies. Coverage is selective by design; the eleven names are not a ranking. We selected based on three criteria: position in a constrained or strategically important node of the value chain; scale or capability sufficient to influence sector dynamics; and visibility - enough publicly available information to support analysis worth publishing. The result is deliberately mixed: two established family-owned groups of genuine scale (Diehl, Rohde & Schwarz); two venture-capital-backed companies that have crossed into sovereign procurement (Helsing, STARK Defence); four specialists in unmanned, autonomous and counter-air systems (Quantum Systems, Alpine Eagle, ARX Robotics, Reflex Aerospace); and three Mittelstand suppliers in narrow but hard-to-replace positions (Dynamit Nobel Defence, AUTOFLUG, and Anschütz).

What comes next. We will maintain and update these profiles, add companies where the ecosystem justifies it, and extend the same framework to further German sectors where the unlisted universe carries the argument - industrial technology and automation, energy transition, payments, healthcare services, and the wider Mittelstand 'hidden champion' base. ***Over time, we want Deutsche Bank Research to be the reference point for how Europe's private industrial base connects to its listed one. This report is the first installment.***

How this report is structured. The report builds these conclusions across five parts, each opening with a short summary before its deep dive. The eleven company profiles follow, preceded by a note on how we work.

The Argument in Five Parts

1. The Strategic Landscape: Russia's 2022 invasion of Ukraine, combined with the distancing of the US as Europe's protector, represents a turning point for the safety of Europe and its defense industry. The shift in policy is creating immense demand across Europe for arms in the decades to come, leading to significant M&A and investing opportunities. German companies are uniquely suited to serve the rising demand due to their immense technical know-how and their already-existing industrial sector. Combined with targeted funding programs, these developments both stimulate existing companies and incentivize start-ups that push the technical frontier forward.

2. New Wars Change the Old Battlefield: Recent wars in Ukraine and the Middle East have reshaped modern warfare, shifting capital from traditional contractors to start-ups that can keep pace with rapid innovation cycles. Key trends defining this new era include: unmanned systems creating a transparent battlefield that demands adaptability; huge munitions consumption proving that production itself is deterrence; the battlespace expanding into commercial space and cyberspace; and societal resilience against disinformation becoming a vital security element. This all occurs amid a new great power competition, heightening the risk of multi-theater wars and the use of nuclear threats to back conventional aggression.

3. How to Gain Exposure – Investment Opportunities in Private Gems: Rapid innovation in modern warfare is creating significant investment opportunities in Germany's defense sector, as start-ups outperform slow, traditional procurement. This operational shift is fueled by a long-term surge in European defense spending targeted at R&D and equipment. Investors can gain exposure through key 'private gems', from established firms like Diehl Group and Rohde & Schwarz to high-growth start-ups like Helsing (AI) and Quantum Systems (drones). These start-ups leverage a mix of venture capital and government grants with innovative business models, making the German private defense market a compelling investment area.

4. Trend – Europe Buys European & Europe Buys between Middle Powers: Recent energy and security shocks have shattered Europe's post-Cold War assumptions, creating an urgent push for strategic self-reliance amid uncertainty over the long-term US security guarantee. This pivot is evident in new security partnerships, such as Germany deepening defense ties in the Indo-Pacific (Japan) and North Atlantic (Canada). Paradoxically, Europe's rearmament has increased its reliance on US hardware (64% of imports '20-'24) and its defense sector remains highly fragmented. The core challenge is whether Europe's quest for autonomy will lead to a 'buy European' policy; doing so would unlock a monumental internal market, driving consolidation and creating a truly independent defense capability.

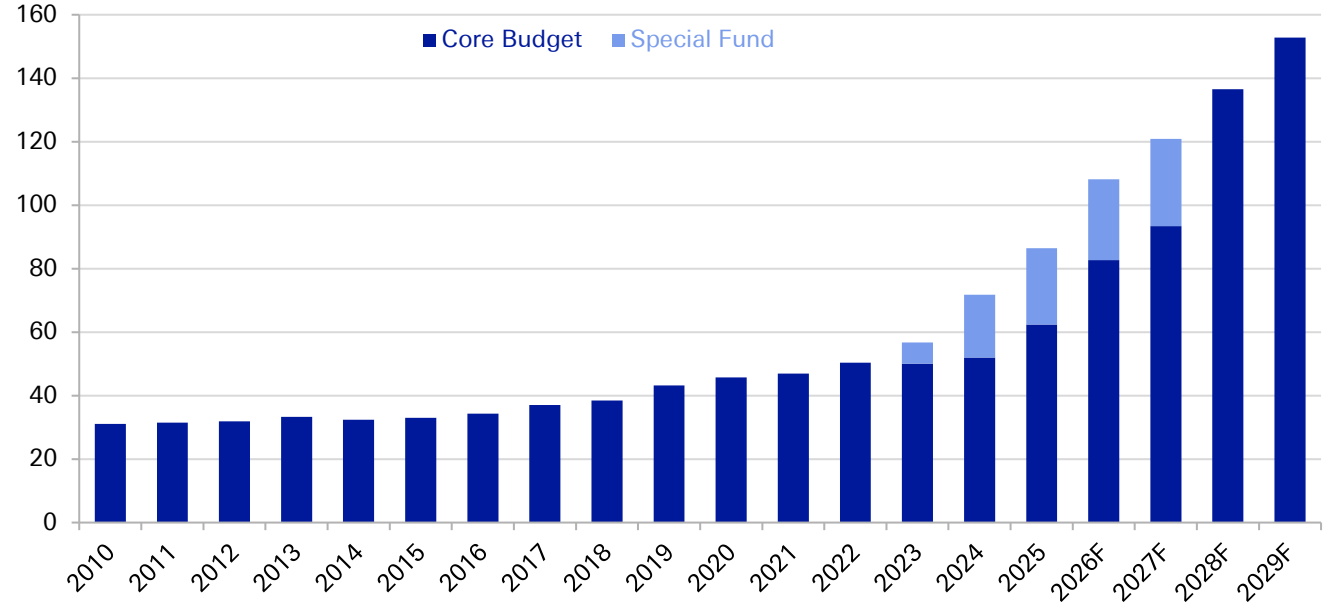
5. The Regulatory Landscape & Investment Constraints: The defense industry is uniquely constrained and regulated, and investors must understand its landscape. Growth is not driven simply by demand but is filtered through government budgets, procurement laws, and export controls. This creates risks: budgets don't translate to immediate revenue, market access can be denied politically, and production cannot be scaled freely. High compliance costs and national security reviews that limit exit strategies add further complexity. Successful investment, therefore, requires mastering these political and regulatory realities, not just identifying demand.

1. The Strategic Landscape

Russia's 2022 invasion of Ukraine, combined with the distancing of the US as Europe's protector, represents a turning point for the safety of Europe. The effect: a multi-decade rearmament cycle has started. For investors, this creates immense investment opportunities in the defense industry that will unfold for decades to come. This report is structured to identify precisely where to understand and find these opportunities.

In 2022, Germany acknowledged the 'Zeitenwende' (watershed) with the German Bundestag marking a major turning point for the German Defense Industry. A €100bn special fund was established for the Bundeswehr, substantially raising the baseline defense budget. A further decisive development was the 2025 debt brake reform package, which effectively exempts defense spending above 1% of GDP from previous debt limits. This reform, along with the creation of a €500bn, 12-year infrastructure fund, fundamentally alters the financial constraints, allowing Germany to pursue higher defense spending beyond the one-off special fund and providing the multi-year visibility crucial for industrial planning.

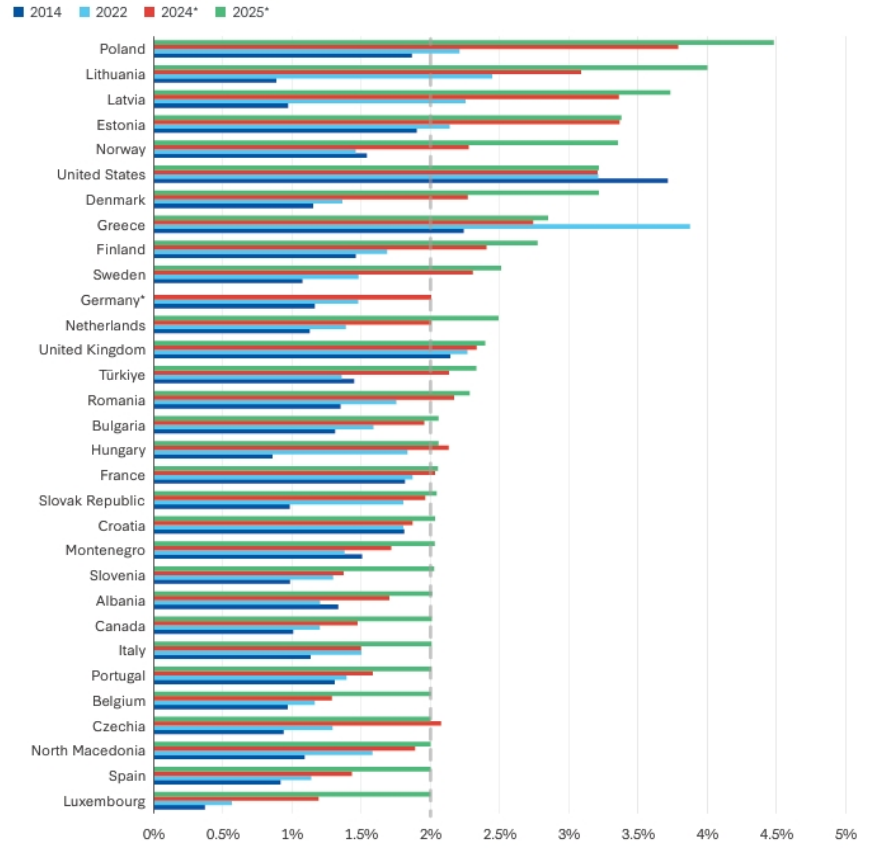
Figure 2: German defense budget (€bn) - government budgets foresee >€150bn annual spending by the end of the decade



Source : German government, Deutsche Bank research

The landmark pledge from **NATO's 2025 summit, requiring members to allocate at least 3.5% of GDP to core defense by 2035**, presents a significant opportunity for **Germany**'s industrial sector. The country's existing strength is undeniable; it already ranks among the world's top-five arms exporters, accounting for **5.6% of global trade between 2020 and 2024**. This foundation is reinforced by a network of highly specialized 'hidden champion' firms in key areas like optics, vehicles, and electronics, ensuring a strong domestic capability to scale production.

Figure 3: NATO Members' Percentage of GDP Spent on Defence

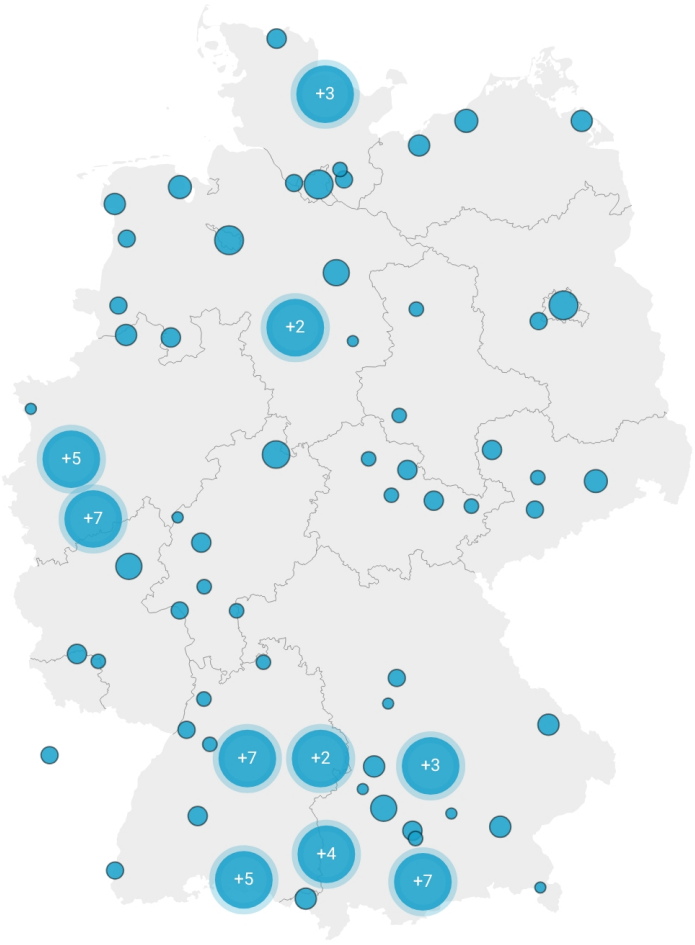


Source : CSIS, Deutsche Bank research
Note: 2024 & 2025 Data based on estimates.
**No 2025 Data reported for Germany*

In parallel, EU-level initiatives such as the EDF (European Defence Fund), EDIRPA (European Defence Industry Reinforcement through common Procurement Act), and ASAP (Act in Support of Ammunition Production) are mobilizing joint funding for R&D and procurement, shaping how supply chains are structured and how 'European content' is defined. These programs incentivize collaborative R&D, joint procurement favoring EU-based firms, and a ramp-up in ammunition and missile production.

Private sector companies are poised to benefit from this development and will likely play an increasingly crucial role in future conflicts. Germany's unique defense industrial base, spanning both global 'hidden champions' and public defense groups, presents significant M&A and IPO opportunities. Notably, about 70% more large German defense firms exist outside the stock market—such as family-owned enterprises like Diehl and Rohde & Schwarz, or conglomerates with defense exposure like Bosch and ZF Friedrichshafen—than on it, meaning a major portion of national capability can be opaque. This industry is concentrated in key regional hubs, including Bavaria (home to MTU Aero Engines, Airbus Helicopters, and major land system providers), Baden-Württemberg (with the ZEISS defense-relevant optics ecosystem, a dense base of precision engineering suppliers, and small-arms manufacturers like Heckler & Koch), North Rhine-Westphalia (hosting Rheinmetall's corporate HQ and a broad aerospace and security ecosystem), and Northern Germany, which remains central for naval defense with TKMS shipyards and also hosts a number of air & space companies.

Figure 4: Germany Defence Locations (Companies)

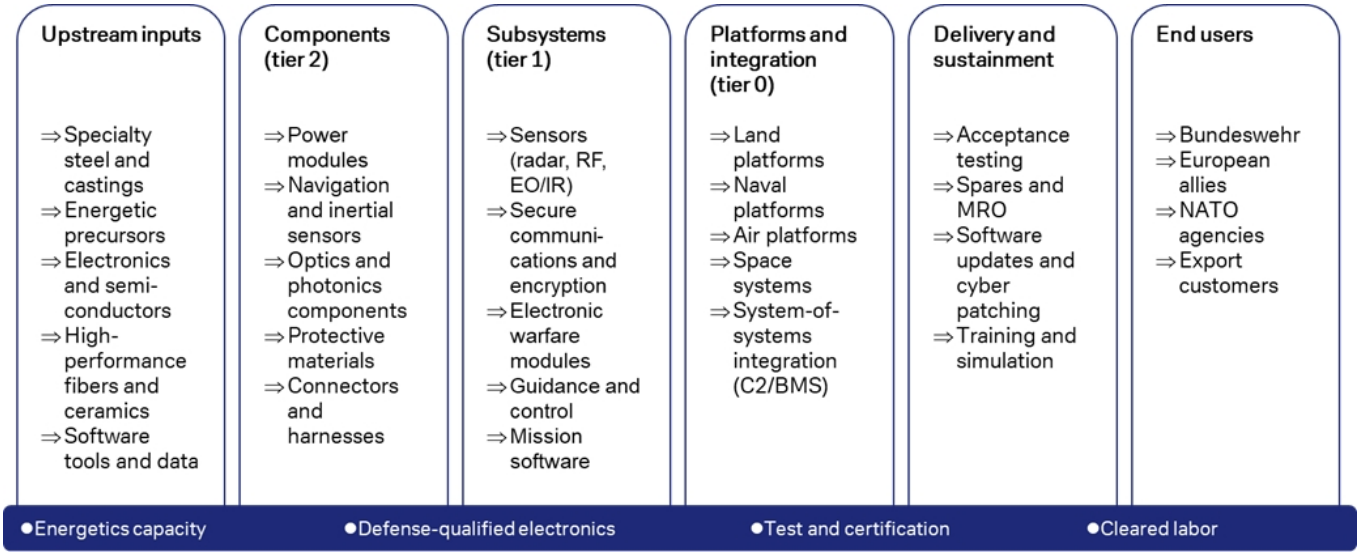


Source : Company Data, created with Datawrapper

However, the Zeitenwende-driven surge in demand for weapons and ammunitionstarkly contrasts with the industry's historically artisanal production models and lean inventories, creating significant choke points. Critical shortages exist for materials like nitrocellulose—a key propellant where 'no nitrocellulose means no shells'—prompting firms like Rheinmetall to acquire domestic production plants. Other choke points include semiconductor chips, high-grade steel and titanium, GPS components, precision optics, and specialized engines, while looming labor constraints are exacerbated by the Bundeswehr's own plan to significantly increase its number of troops. In response, the national strategy now explicitly calls for simplifying production processes and increasing scalability. Key players navigating these challenges include listed primes like Rheinmetall AG, Airbus/MBDA, and ThyssenKrupp Marine Systems, alongside influential private or security-focused firms such as the Diehl Group and Hensoldt AG.

A 2024 German government strategy (Nationale Sicherheits- und Verteidigungsindustriestrategie) acknowledges a Zeitenwende of surging demand versus decades of decline, and stresses the need to scale up production beyond artisanal methods.

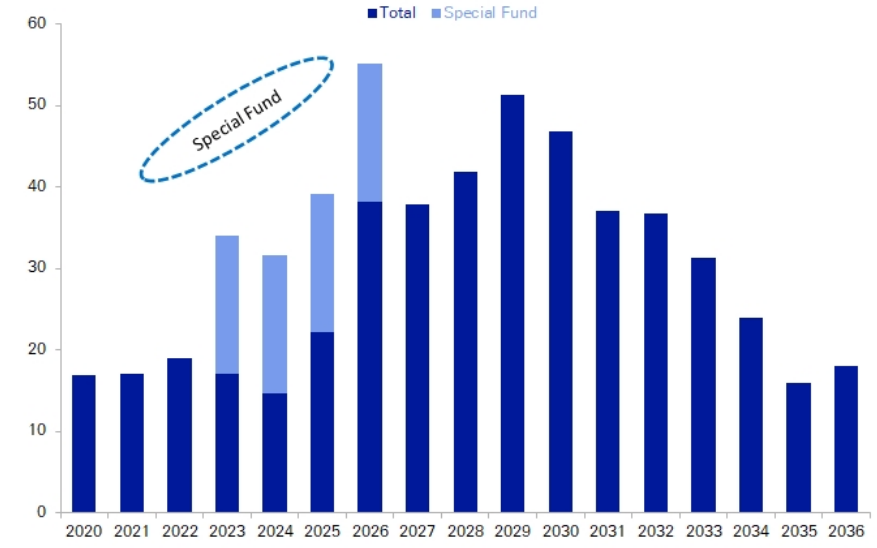
Figure 5: German Defense Industrial Value Chain



Source : Deutsche Bank research

In summary, prompted by major geopolitical shifts, Germany's 'Zeitenwende' policy has launched a multi-decade rearmament cycle, fueled by a €100bn special fund and structural budget reforms. This creates significant long-term investment opportunities within its defense industry, which, despite a strong industrial base of both public primes and private 'hidden champions', must now navigate critical supply-chain and labor bottlenecks to meet the historic surge in demand.

Figure 6: Germany Procurement Spending - future years based on commitment appropriations for multi-year procurement programs



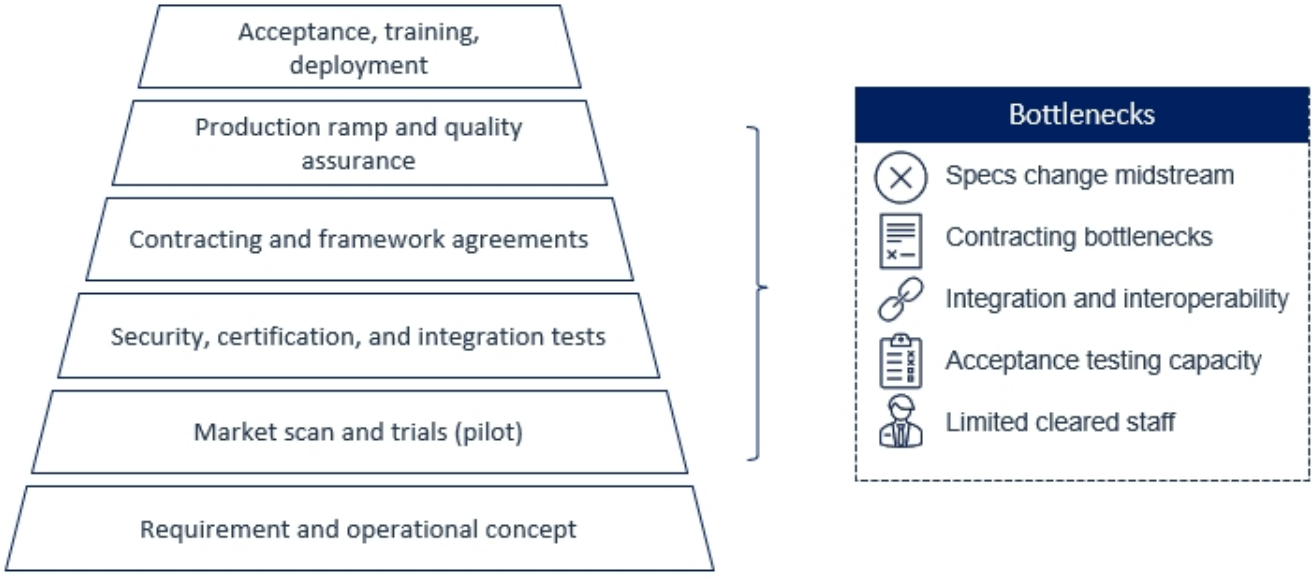
Source : BMVg

2. New Wars Change the Old Battlefield

The wars in Ukraine and the Middle East are acting as powerful catalysts, fundamentally altering the modern battlefield and forcing a significant shift in capital allocation away from rigid, traditional defense structures toward agile start-ups that can match the rapid innovation cycles seen in current conflicts.

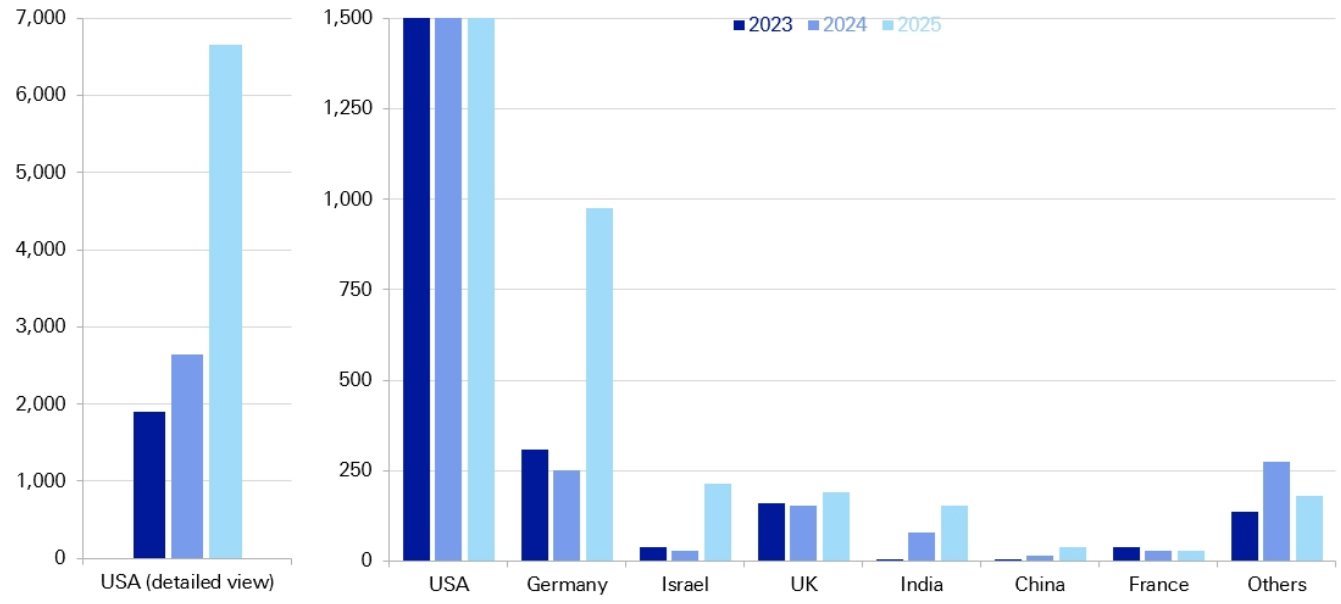
1. New Battlefield Innovation Cycles Driven by Start-ups: The battlefield now operates on an extremely rapid six-week learning cycle of technological measures and countermeasures. This pace is primarily driven by agile start-ups and commercial firms, creating a demand for battlefield adaptation that traditional, slow-moving state processes cannot meet on their own.

Figure 7: Procurement Funnel: From Need to Fielded Capability



Source : Deutsche Bank research

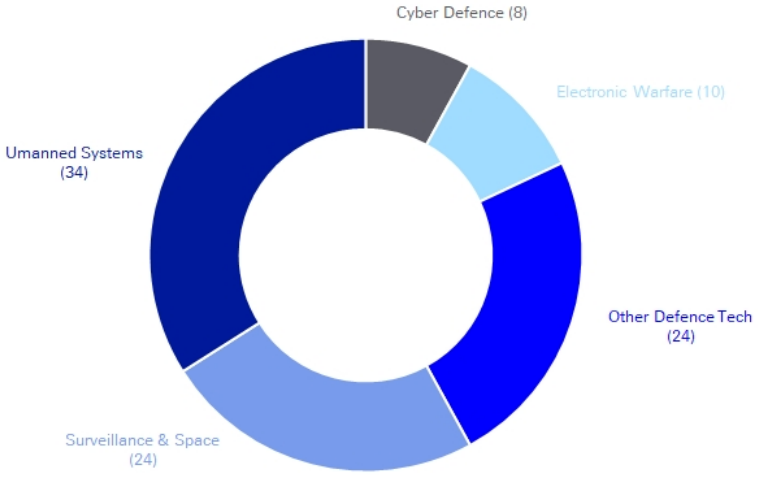
Figure 8: Venture Capital Investments in Defense and Security Start-Ups per Country (USDm)



Source : Handelsblatt Research Institute

2. The Rise of Unmanned Systems and a Transparent Battlefield: The widespread use of unmanned systems (UASs, USVs, etc.) has transformed the battlefield. These systems have evolved from a few hundred to millions being manufactured annually, leading to a 'transparent battlefield' where movement is easily detected. This has made large-scale maneuvers highly vulnerable and forced troops to operate in smaller, more cautious units. The integration of these drones with sensor networks has created dynamic 'kill webs', making speed and adaptability critical.

Figure 9: VC-Deals 2021-2025 by Category in Germany



Source : WirtschaftsWoche, FCF

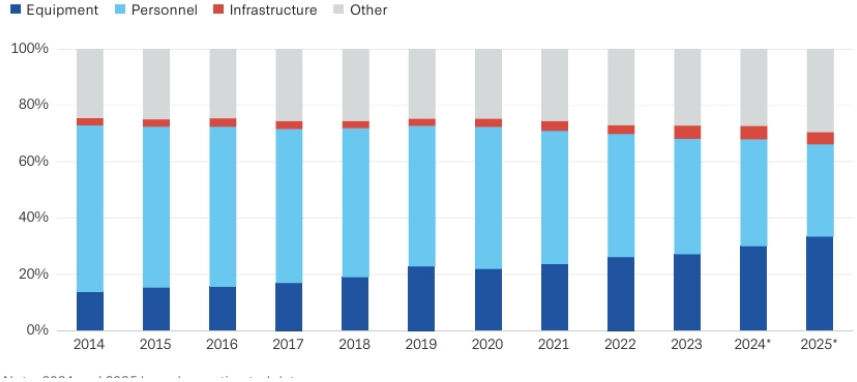
3. The Re-emergence of Industrial-Scale Warfare: The conflicts have highlighted the critical importance of a robust industrial base capable of producing munitions and equipment at scale for a protracted war. The consumption rates of artillery shells and other munitions have been staggering, far exceeding peacetime forecasts and straining the production capacity of the US and its allies. This has led to a renewed understanding that 'production is deterrence' and that industrial readiness is essential for sustaining a high-intensity conflict.

Figure 10: Munitions and Missiles: Where Scaling Breaks - main obstacles to overcome

Energetics inputs (propellants, explosives)	*Permits and hazardous-process capacity*
Critical components (seekers, guidance electronics)	*Long lead times and scarce suppliers*
Sub-assembly and safety certification	*Qualification burden and safety constraints*
Final assembly and integration	*Specialized labor and throughput balancing*
Testing and qualification	*Limited test capacity and scheduling windows*
Storage and replenishment logistics	*Handling rules and inventory constraints*

Source : Deutsche Bank

Figure 11: Percentage spending of NATO by cost category



Source : CSIS



4. The Centrality of Societal Resilience and Irregular Warfare: The wars have shown that societal resilience—the will and ability of a nation to resist and recover from pressure—is a vital element of national security and deterrence. Adversaries increasingly use irregular warfare tactics like sabotage, disinformation, and cyberattacks as a prelude to or alternative to conventional war, aiming to undermine a nation's cohesion and will to fight.

5. The Battlespace Has Expanded to Space and Cyberspace: Recent conflicts have marked a turning point for the role of space, which is now central to the daily conduct of armed conflict. The 'democratization' of space technology through commercial providers like Starlink and satellite imagery companies has become an equalizer, allowing less powerful nations to gain significant battlefield advantages. Consequently, space-based systems (for communications, navigation, and surveillance) and cyberspace are now primary targets in the opening stages of a conflict.

6. A New Era of Great Power Competition and Nuclear Risk: The conflicts are occurring in a renewed era of global competition, marked by a coalition of hostile powers (China, Russia, Iran, and North Korea) collaborating across military, economic, and political domains. This has raised the prospect of simultaneous, multi-theater wars. This competition is further complicated by the resurgence of nuclear risk, with adversaries leveraging their nuclear arsenals to support expansionist goals, thereby increasing the potential for miscalculation and nuclear escalation in regional conflicts.

Private Companies

Argument: The Primacy of Speed and Software

The very nature of modern warfare is now defined by capabilities where the private sector possesses a distinct advantage. Wars today demand faster iteration cycles than ever before; adversaries must constantly adapt to tactical and technological changes that unfold not over years or months, but in a matter of weeks. This accelerated pace renders traditional government procurement models obsolete, thereby creating a vital role for agile private companies, especially start-ups, that can innovate rapidly. Such speed is essential to master a battlefield increasingly dominated by technology, particularly Open-Source Intelligence (OSINT), software-defined networks, and AI-powered data fusion.

Example: The Private Sector in Action

The growing indispensability of the private sector is clearly demonstrated by recent events and evolving spending patterns. In Israel's recent conflicts, start-ups have become integral to the war effort, with analyses indicating they produced approximately 50% of the anti-drone systems deployed by the Israel Defense Forces. Similarly, in Ukraine, tech giants such as Microsoft, Amazon Web Services (AWS), and SpaceX (with Starlink) have provided the critical cloud and communications infrastructure that has sustained the country's defense. This reliance is further reflected in government spending. NATO allies are significantly increasing their defense budgets, with a notable shift in capital allocation towards equipment procurement, projected to reach an average of 33% of defense spending by 2025.

Relevance: A Structural Shift in Defense Markets

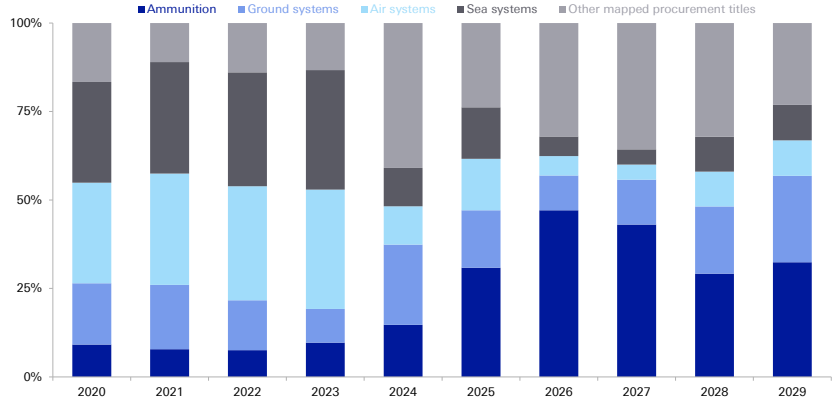
The key takeaway for the market is that the increased defense spending represents not a temporary surge but a structural, long-term shift.

- Increased Overall Investment: Following Russia's invasion, European nations have reversed years of underinvestment. Nearly all NATO members are now on a trajectory to dramatically increase defense spending as a percentage of their GDP, creating a massive and durable market for military hardware.
- Targeted Capital Allocation: This new capital is flowing directly to the industry. Allies are now allocating a significantly larger portion of their budgets towards the procurement of equipment and R&D. This trend directly benefits arms producers and technology companies, making nations with strong industrial bases, such as Germany, particularly well-positioned to capitalize on the heightened and sustained demand.

3. How To Gain Exposure: Investment Opportunities in Private Gems

The following provides a classification of key private companies in the German defense sector, broken down by their primary operational domain. These companies represent a significant portion of the innovation and industrial base in the private market.

Figure 12: Spending Percentage of Germany on Equipment by Category



Source : BMVG – Budget

Air Systems

- Diehl Group: A major systems supplier for air defense, producing guided missiles like the IRIS-T SLM, which is a key component of Germany's air defense capabilities.
- Alpine Eagle: Focuses on coordinated air-to-air defense systems, developing networked solutions to counter threats from drones and cruise missiles.
- AUTOFLUG GmbH: Provides custom seating and cabin solutions for aircraft and helicopters, including crash-safe seats for pilots and crew.
- Quantum Systems: Specializes in the fusion of Unmanned Aerial Systems (UAS) with AI software for superior aerial intelligence and reconnaissance missions.
- Reflex Aerospace: Designs and builds custom, dual-use satellites, positioning itself as a key new player in the aerospace and defense sector.



Sea Systems

- Anschütz GmbH: A leader in maritime navigation systems, offering integrated solutions for submarines, naval surface vessels, and commercial ships, including gyro compasses and steering gear systems.

Ground Systems

- Dynamit Nobel Defence: Develops advanced anti-tank weaponry, including shoulder-launched munitions like the RGW series, used by infantry for ground combat.
- AUTOFLUG GmbH: In addition to its aviation products, the company also supplies safety seats and spall liners for land-based military vehicles.
- ARX Robotics: Focuses on creating scalable, dual-use autonomous ground systems for both military and civilian applications.

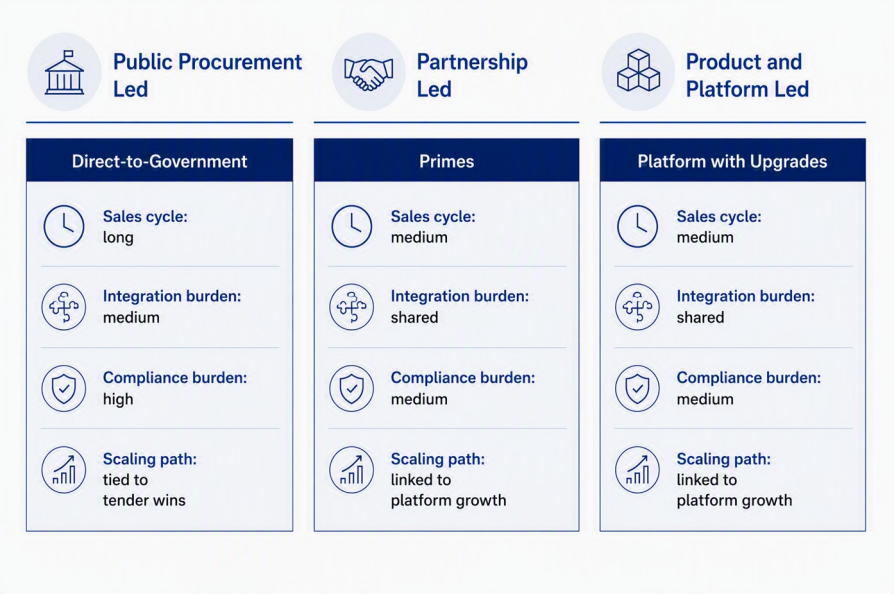
Multi-Domain / Electronics

- Rohde & Schwarz: A technology group that provides solutions across multiple domains. Its products range from 5G network components to secure communications, electronic warfare systems, and battlefield electronics for air, land, and sea platforms.
- Helsing: Described as "Germany's Most Valuable Startup," Helsing is a software and AI company that develops capabilities for defense and security systems across air, land, and sea domains.
- STARK Defence: Develops a unified integration platform for AI-enabled unmanned aerial, ground, and surface vehicles (UAVs/UGVs/USVs).

Introduction: The Start-Up Model

Start-Ups are characterized by their remarkably short prototyping cycles and a focus on developing highly tailored, software-enabled products. This technology-centric approach allows them to deliver specific, advanced solutions much faster than traditional industry players.

Figure 13: Defense Tech Commercialization Routes



Source : Deutsche Bank research

The operational and financial models of these start-ups are as innovative as their technologies. Their funding is secured through a sophisticated mix of venture capital, supplemented by strategic EU/NATO grants and corporate venture arms. Defense-specific accelerators, such as the Cyber Innovation Hub, alongside national government funds and NATO's DIANA (Defense Innovation Accelerator for the North Atlantic) initiative, provide the crucial early-stage support needed to navigate the high-risk, high-reward defense market. Their business strategies are equally diverse, moving beyond traditional Business-to-Government (B2G) contracts to include Business-to-Business (B2B) technology licensing and emerging Platform-as-a-Service (PaaS) offerings. This multi-faceted approach allows them to penetrate the market and scale effectively.

Example: Navigating the Path to Integration

The go-to-market strategy for these firms demonstrates a clear pathway from domestic validation to international expansion. Initial entry is often secured through federal procurement tenders (Federal Office of Bundeswehr Equipment, Information Technology and In-Service Support, BAAINBw) or direct trials with the Bundeswehr, establishing crucial credibility. Success in these domestic programs and at key defense exhibitions generates export leads, which are frequently pursued through strategic partnerships with established international defense firms. However, significant challenges remain. These innovators must navigate complex and lengthy procurement regulations, overcome a historical preference for established contractors, and manage the intricate demands of intellectual property protection and export controls. Despite these hurdles, the rise of these start-ups marks an irreversible trend towards greater agility and innovation within the German defense sector.

4. Trend: Europe Buys European and Europe Buys Between Middle Powers

A series of rapid and profound shocks has shattered Europe's post-Cold War security assumptions, acting as a powerful wake-up call and creating an urgent push for greater strategic independence. The 2022 gas crisis, which exposed the danger of relying on a single adversarial energy supplier, was compounded by a severe oil price spike in April 2026. This spike saw Brent crude prices surge towards ~$120 per barrel after military conflict in the Middle East led to the closure of the Strait of Hormuz, a chokepoint for nearly 20% of the world's oil supply. These events, coupled with persistent uncertainty over the long-term reliability of the American security guarantee, have exposed deep vulnerabilities and are forcing the continent to build its own capacity for energy and military security.

Figure 14: Share of Global Arms Imports Ranked by Country – America Ranks 1st, Germany 5th

Importer	Share of Global Arms Imports (%) 2020–24	1st Main Supplier (%)	2nd Main Supplier (%)	3rd Main Supplier (%)
Ukraine	8.8	USA (45%)	Germany (12%)	Poland (11%)
India	8.3	Russia (36%)	France (33%)	Israel (13%)
Qatar	6.8	USA (48%)	Italy (20%)	UK (15%)
Saudi Arabia	6.8	USA (74%)	Spain (10%)	France (6.2%)
Pakistan	4.6	China (81%)	Netherlands (5.5%)	Türkiye (3.8%)
Japan	3.9	USA (97%)	UK (2.0%)	Germany (0.5%)
Australia	3.5	USA (81%)	Spain (15%)	Norway (1.3%)
Egypt	3.3	Germany (32%)	Italy (27%)	France (19%)
United States	3.1	UK (18%)	France (15%)	Israel (13%)
Kuwait	2.9	USA (63%)	Italy (29%)	France (7.1%)
UAE	2.6	USA (42%)	France (17%)	Türkiye (11%)
South Korea	2.6	USA (86%)	Germany (9.7%)	UK (2.6%)
United Kingdom	2.5	USA (86%)	Israel (6.9%)	Spain (3.3%)
Poland	2.4	USA (45%)	South Korea (42%)	Italy (3.5%)
Israel	1.9	USA (66%)	Germany (33%)	Italy (1.0%)
China	1.8	Russia (72%)	France (13%)	Ukraine (12%)
Netherlands	1.8	USA (97%)	Italy (0.7%)	Finland (0.6%)
Norway	1.5	USA (91%)	South Korea (4.2%)	Italy (2.4%)
Greece	1.3	France (64%)	USA (19%)	UK (7.5%)
Singapore	1.2	Germany (40%)	USA (32%)	France (14%)
Algeria	1.2	Russia (48%)	China (19%)	Germany (14%)
Türkiye	1.1	Spain (34%)	Italy (24%)	Germany (19%)
Bahrain	1.1	USA (97%)	Türkiye (1.4%)	UK (0.9%)
Italy	1.1	USA (94%)	Germany (2.0%)	UK (1.5%)
Brazil	1.0	France (53%)	Sweden (20%)	Italy (11%)
Kazakhstan	0.9	Russia (88%)	Spain (6.4%)	Türkiye (1.5%)
Philippines	0.9	South Korea (33%)	Israel (27%)	USA (20%)
Indonesia	0.9	USA (33%)	France (15%)	South Korea (12%)
Denmark	0.9	USA (79%)	Switzerland (7.6%)	Germany (7.2%)
Romania	0.8	USA (61%)	Norway (15%)	Switzerland (8.4%)
Morocco	0.7	USA (64%)	France (15%)	Israel (11%)
NATO	0.7	France (74%)	UK (17%)	USA (9.9%)
Germany	0.7	USA (70%)	Israel (13%)	Sweden (5.1%)
Belgium	0.7	Spain (49%)	USA (43%)	Germany (8.1%)
Canada	0.7	Spain (30%)	USA (26%)	Australia (22%)
Serbia	0.7	China (57%)	Russia (20%)	France (7.4%)
Hungary	0.7	Germany (51%)	USA (17%)	Norway (11%)
New Zealand	0.7	USA (76%)	South Korea (10%)	UK (9.1%)
Taiwan	0.6	USA (98%)	Italy (1.2%)	Germany (1.0%)
Thailand	0.6	China (43%)	USA (14%)	South Korea (12%)
Total European Import Share	23,9			

Source : SIPRI

This strategic pivot towards self-reliance is clearly visible in a new network of security partnerships that extends beyond traditional alliances. A prime example is



Germany's effort to deepen defense ties with key democratic partners. In late March 2026, German Defense Minister Boris Pistorius proposed a "Reciprocal Access Agreement" (RAA) to Japan, a pact designed to significantly reduce bureaucratic hurdles for stationing troops in each other's territory for joint training and exercises. This move, mirroring pacts Japan has with Australia and the UK, signals a shift towards more structured, long-term military cooperation, explicitly linking the security of Europe with that of the Indo-Pacific.

Figure 15: Biggest Arms Importer Middle East vs Europe

Importer	Share of Global Arms Imports (%) 2020–24	1st Main Supplier (%)	2nd Main Supplier (%)	3rd Main Supplier (%)
Qatar	6.8	USA (48%)	Italy (20%)	UK (15%)
United Kingdom	2.5	USA (86%)	Israel (6.9%)	Spain (3.3%)

Source : SIPRI

Similarly, Germany and Norway have launched a major diplomatic and industrial effort to bring Canada into a trilateral submarine partnership. The offer is centered on Canada acquiring the advanced German-made Type 212CD submarine, a platform already being jointly procured by Germany and Norway. The pitch emphasizes the creation of a 'familiar family' and a long-term strategic alliance, offering full NATO interoperability, shared resources, and significant domestic production and knowledge transfer to Canada. This cooperative approach is part of a broader strategy to bolster NATO's capabilities in the North Atlantic and Arctic, regions of growing strategic importance.

Figure 16: Number of Different Weapon Systems in Use, 2023

Europe (No. of Systems)	Weapon System	USA (No. of Systems)
19	Main Battle Tanks	1
23	Armored Infantry Fighting Vehicles	3
28	152/155-mm Howitzers	2
20	Tactical Combat Aircraft	7
4	Attack Helicopters	2
12	Anti-Ship Missiles	5
13	Air-to-Air Missiles	3
27	Destroyers and Frigates	4
17	Torpedoes	2
10	Conventional Submarines	0
6	Nuclear-Powered Submarines	4
179	**Total Different Weapon Systems**	**33**

Source : Handelsblatt, McKinsey

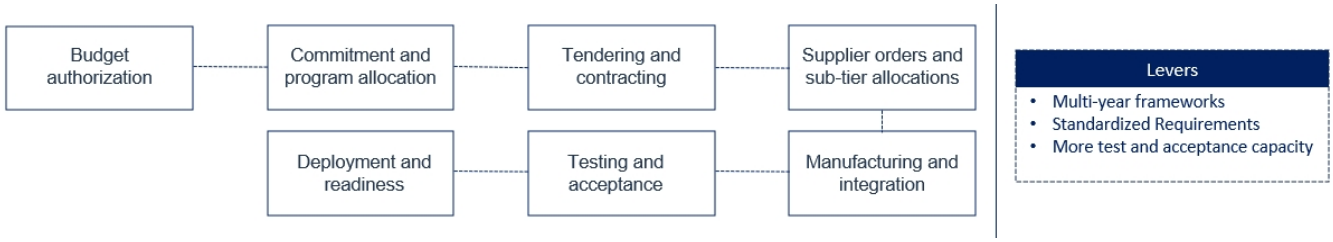
This drive for European self-reliance has profound implications for the continent's massive defense market. Between 2020 and 2024, arms imports by European states surged by 155% in response to regional threats. However, this rearmament has deepened reliance on foreign suppliers; European NATO members sourced 64% of their arms imports from the US during that period, an increase from 52% in 2015-2019. The European defense sector remains highly fragmented, with EU states operating 17 different types of tanks compared with just one main model in the US If Europe's quest for strategic autonomy leads to a conscious policy to 'buy European' and replace even a portion of these American imports with domestically developed and produced systems, it would unlock a vast internal market. Such a shift would represent a monumental commercial opportunity for the continent's own industrial base, fostering consolidation and creating a more resilient and independent defense capability.

5. The Regulatory Landscape & Investment Constraints

The financing backdrop has changed materially. Defense was long screened out by parts of the European investment community, both because ESG frameworks often excluded it and because the sector lacked obvious structural growth. That is no longer the case. Over recent years, private equity investors, asset managers, and even some ESG-focused pools have been re-evaluating defense exposure, while Euronext said it would revisit ESG index methodologies to limit blanket exclusions on defense companies, as reported by Reuters. The EIB has also expanded its support for security and defense-related investment.

However, defense remains a highly regulated market, and that still matters for investment analysis. In Germany, contracts above €25m require binding approval from the Bundestag's Budget Committee before they can be signed. The government has tried to ease that constraint through the procurement acceleration reform, which entered into force on 14 February 2026. At the same time, NATO interoperability rules continue to matter through STANAG standardization, and export control remains central: Germany's 2025 arms export report recorded around €13.3bn of military export licenses. This is not a normal industrial market in which companies can scale solely by commercial demand. Growth is filtered through budgets, procurement law, export licensing, and alliance standards.

Figure 17: Budget to Output: The Conversion Chain



Source : Deutsche Bank research

For investors, the implication is twofold. First, the sector is more financeable and commercially interesting than it was three years ago. Second, it remains politically mediated, compliance-heavy, and more path-dependent than most industrial verticals. The right quick overview for the intro is therefore not just "defense is now investable," but "defense is investable if one understands procurement cycles,



export controls, interoperability requirements, and the difference between contract visibility and true industrial throughput." The main takes:

1. **Procurement timing versus industrial reality:** Even when budgets are approved, money does not immediately convert into revenue. Parliamentary approvals, tender design, contract negotiation, milestone acceptance, and certification can all delay conversion. Investors may overestimate the speed at which higher budgets translate into sales profits and cash flows.
2. **Export controls and political approval risk:** Defense companies do not control their own addressable market in the same way normal industrial companies do. Export licenses can be delayed, restricted, or denied, and destination mix can change with elections or geopolitical events. This is especially relevant for firms depending on exports.
3. **Order visibility versus production permissions:** In many defense segments, companies cannot simply build inventory freely in anticipation of demand. Production often follows contract timing, budget allocation, and regulatory approval. That makes the sector less flexible than outsiders may assume, and it can cap upside in the short term even when demand is clearly there.
4. **Working capital and pre-financing burdens:** Even in a strong demand environment, defense production can be cash intensive. Companies may need to fund tooling, labor buildup, supplier reservation, lengthy qualification work, or capex before volumes fully ramp.
5. **Compliance intensity as a barrier and cost factor:** Security clearances, documentation requirements, product traceability, cybersecurity standards, anti-corruption rules, and procurement compliance all create friction. This can be a barrier to entry, which is positive for incumbents, but it also raises the cost base and makes scaling harder, especially for start-ups.
6. **NATO and interoperability standards:** Thinking needs to go beyond national procurement markets. NATO standards, qualification requirements, and interoperability rules shape product design and buying decisions. That favors firms with installed positions, certifications, and long-standing defense relationships, and it can slow newcomers despite strong technology.
7. **Foreign investment screening and ownership sensitivity:** In defense, capital is not fully fungible. Cross-border acquisitions, minority stakes, and control changes may face national security scrutiny. That matters for private market investors because exit routes, buyer universe, and deal structures can be more constrained than in normal industrial sectors.
8. **ESG has improved, but has not disappeared as a constraint:** The easy narrative is that defense is investable again. That is true, but only partly. Some pools of capital remain restricted, some mandates still exclude parts of the sector, and the exact treatment differs by weapon type, end-market exposure, and jurisdiction. This may affect valuation dispersion and funding access.
9. **Policy risk cuts both ways:** Investors tend to focus on regulatory burden, but policy can also accelerate the sector through fast-track procurement, subsidized financing, national stockpiling, EU joint procurement vehicles, or state-backed industrial initiatives.

The Supply Chain: Who Delivers What to Whom

Our value-chain analysis identified four structurally constrained nodes in the German defense industrial base - energetics capacity, defense-qualified electronics, test and certification, and cleared labor. This section maps the covered private companies onto that chain and documents the specific, publicly disclosed supply and integration links between them, their listed partners, and the wider ecosystem. The picture that emerges is not a neat linear chain but a dense web in which several of the covered companies are simultaneously suppliers, integrators, and increasingly acquirers of their own upstream inputs.

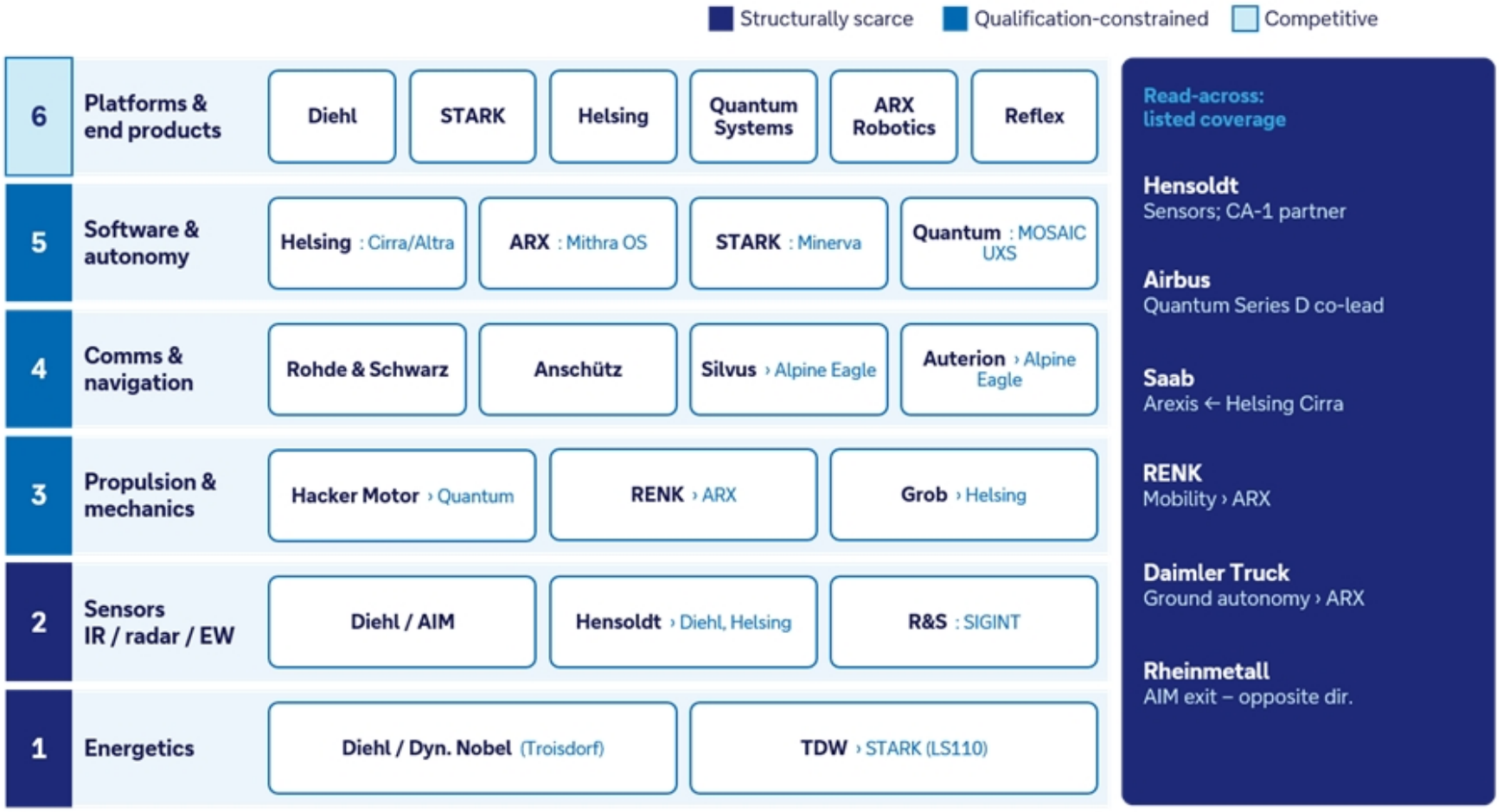
A tiered map

Read from the top down, the covered companies occupy every layer of the chain. At Tier 0 (platforms and integration) sit the end-product owners: Diehl (missiles, air defense), STARK Defence and Helsing (loitering munitions and strike drones), Quantum Systems and ARX Robotics (unmanned air and ground systems), and Reflex Aerospace (satellites). At Tier 1 (subsystems) the split is between mission software - Helsing's Cirra/Altra, ARX's Mithra OS, STARK's Minerva, Quantum's MOSAIC UXS - and secure communications and navigation, where Rohde & Schwarz (SOVERON, NAVICS) and Anschütz (SYNAPSIS) dominate, and where Alpine Eagle integrates third-party radios from Silvus and Auterion. At Tier 2 (components) the chain runs through propulsion and structure (Hacker Motor into Quantum; RENK into ARX; Grob airframes and Schaeffler components into Helsing; AeroMass into STARK) and sensors (Diehl's AIM for infrared; Hensoldt radar into both Diehl and Helsing). At Tier 3 (upstream energetics) the base narrows sharply to a handful of holders of explosives know-how and permits - the subject of the dedicated sub-section below.

Who supplies whom

The publicly documented links can be stated precisely. In energetics, TDW (an MBDA subsidiary) supplies the LS110 warhead to STARK Defence (first German hot strike, December 2025). In sensors, Hensoldt supplies radar for Diehl's IRIS-T SLM and sensors for Helsing's CA-1 Europa. In propulsion, Hacker Motor supplies Quantum Systems (before being acquired by it), RENK supplies drivetrain and mobility to ARX Robotics, and Schaeffler supplies components for Helsing's HX-2 scaling. In software, the direction of flow reverses: Helsing's Cirra EW software is embedded inside Saab's Arexis suite, and ARX's Mithra OS is the autonomy layer for Daimler Truck's fleets - here the private companies are the suppliers to larger, listed names. In counter-air, Origin Robotics supplies the BLAZE interceptor into Alpine Eagle's Sentinel (June 2026).

Figure 18: German Defense Supply Chain



Pattern 2025/26: Vertical integration downwards. Diehl (AIM, Dynamit Nobel), Quantum (Hacker Motor, Spleenlab), Helsing (Grob, Blue Ocean) and STARK (AeroMass) are securing bottleneck nodes — not end products.

Source : Deutsche Bank research, company data

The dominant pattern: vertical integration downward

The single clearest trend of 2025-2026 is that the platform owners are buying their way into the bottleneck tiers rather than acquiring more end-product capability. Diehl secured energetics and sensors (Dynamit Nobel, AIM, EST Energetics, plus the Tauber ordnance-services acquisition); Quantum Systems secured propulsion and AI perception (Hacker Motor, Spleenlab, AirRobot, FERNRIDE); Helsing secured airframes and maritime hardware (Grob Aircraft, Blue Ocean); STARK Defence secured its manufacturing supply chain (AeroMass); and Reflex Aerospace is building out its own Munich MicroFactory. In each case, the acquisition targets a scarce, qualification-heavy node - not a new product line. This is the supply chain telling you where the real constraints are: companies do not vertically integrate into abundant inputs.

Read-across to listed coverage

The map has direct implications for listed names our colleagues cover. Hensoldt sits upstream of both Diehl and Helsing as a sensor supplier, while also being a CA-1 partner and a Quantum investor - a multi-directional relationship. Airbus co-led Quantum's Series D and partners on IRIS-T command-and-control. Saab integrates Helsing's EW software. RENK supplies mobility and drivetrain to ARX Robotics, and Daimler Truck relies on ARX for ground autonomy. OHB is Helsing's partner in the KIRK space joint venture. The one relationship running the opposite way is Rheinmetall's exit from AIM (sold to Diehl), a rare case of an incumbent ceding a choke point node. For the listed desk, the actionable point is that these partnership maps are becoming a better guide to competitive position than product catalogs - a private-company development can be a leading indicator for a listed supplier's order book.

Energetics: the binding choke point

Energetics - propellants, explosives, and warhead fills - is the node where the German supply chain is most constrained and least substitutable, and it deserves separate treatment. Unlike electronics or software, energetics capacity cannot be scaled quickly: it depends on hazardous-process permits, site-level safety certification, and specialized labor that take years to replicate and are effectively impossible for a new entrant to stand up at speed.

Two positions dominate

Diehl, through its 2024 acquisition of Dynamit Nobel, controls the Troisdorf explosives-related operating authorizations that the company itself describes as "unique in Germany," and it has since deepened the position with the June 2026 acquisition of EST Energetics from General Atomics Europe and the addition of Diehl Energy Products (thermal batteries). One tier down, STARK Defence's dependence on TDW for German-manufactured warheads and fuzing is the same story: even a fast-moving, venture-capital-backed platform company remains structurally dependent on a scarce energetics supplier it does not control.

The strategic implication is that energetics is where scaling either happens or stalls. A drone airframe can be sourced or built in months; the warhead and its permits cannot. This makes the holders of energetics capacity - Diehl above all - occupants of one of the most durable moats in the entire ecosystem, and it makes energetics the single node most worth watching as the rearmament cycle moves from budgets into delivered rounds. It is also the clearest example of why the private half of this industry cannot be analyzed as an afterthought to the listed lead contractors: the



choke point that governs the whole chain sits inside privately held hands.

Defense Conference Calendar

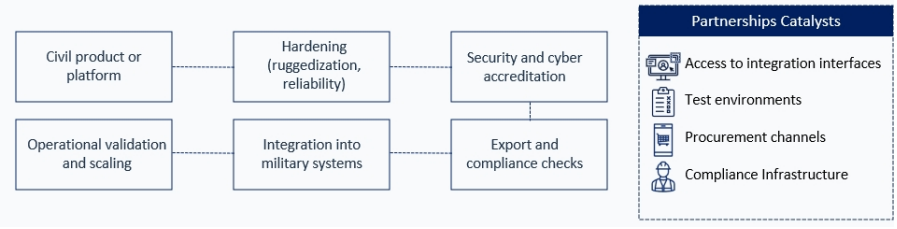
- AUSA Annual Meeting & Exposition | 12-14 October 2026 | Washington, D.C., USA
- Euronaval | 3-6 November 2026 | Paris Nord Villepinte, France
- IAV - International Armoured Vehicles | 19-21 January 2027 | Farnborough, UK
- Munich Security Conference | 12-14 February 2027 | Munich, Germany
- AI in Defense Summit | 1 March 2027 | Brussels, Belgium
- Sea-Air-Space | 4-7 April 2027 | National Harbor, USA
- World Defense Show | 16-20 January 2028 | Riyadh, Saudi Arabia
- DIMDEX | 27-30 March 2028 | Doha, Qatar
- Eurosatory | 19-23 June 2028 | Paris Nord Villepinte, France

Spotlights

Spotlight 1: Dual-Use & Civil-Military Convergence

Many emerging defense technologies originate in the commercial sector. Examples include drones (initially civilian UAVs), satellite communications (5G, small sats), cybersecurity tools, and artificial intelligence algorithms (image recognition, logistics). Germany's strong civil tech base (Siemens, SAP, Infineon, Deutsche Telekom) is now engaged via dual-use applications. For instance, civilian telecom networks (5G/6G) underpin military comms; AI labs develop defense analytics; robotics firms adapt to mine-clearing or unmanned vehicles.

Figure 19: Dual-Use Pathway: Civil Technology to Military Deployment



Source : Deutsche Bank research

Regulatory Context: The EU has strengthened its dual-use controls through Regulation (EU) 2021/821 and subsequent annual updates. These regulations restrict sensitive technologies such as cryptography, sensors, and navigation systems, necessitating export licenses. General authorizations are available for trusted partners. Companies must navigate German BAFA licenses and consider potential US ITAR constraints if their products incorporate US components. German firms entering the defense sector also face new compliance regimes, including military product marking and security clearances.

Private Sector: The private sector demonstrates a dynamic landscape with 'spin-ins' exemplified by start-ups like drone manufacturer Quantum Systems or AI firm Aignostics. Large Original Equipment Manufacturers (OEMs) such as Airbus' Aalto, which develops Zephyr drones, further blur traditional distinctions. While some traditional industrial giants like Bosch and ThyssenKrupp maintain separate dual divisions, others, such as Hensoldt (a sensor spin-off from Airbus in 2017), have explicitly branded themselves for defense.

Economics and Valuation: Dual-use companies, while potentially accessing broader markets, may also experience valuation impacts due to security concerns. Export limitations or ESG concerns, though arguably less pronounced now, can deter commercial investors. Conversely, stable government contracts (B2G sales) and partnerships with prime contractors can enhance valuation. The sector benefits from a diverse funding landscape, including EU grants (e.g., EDF EDIDP grants), the NATO Innovation Fund (supporting areas like AI and autonomy), venture capital, and strategic investments by prime contractors (e.g., Rheinmetall's venture arm).

Risks & Policy: The convergence of dual-use technologies introduces ethical and espionage risks, leading to increased controls on technology transfer, particularly for components like AI chips. Policy levers in this area include targeted R&D grants,



clearer definitions of dual-use applications, and the establishment of public-private innovation hubs, such as Germany's Cyber Innovation Hub and the EU DIANA initiative.

Spotlight 2: Workforce & Skills Shortages

Talent Shortages in Germany: Germany is currently experiencing chronic shortages in Science, Technology, Engineering, and Mathematics (STEM) talent within its defense sector. The Bundeswehr aims to expand its personnel from approximately 180,000 to 260,000 soldiers by the early 2030s, in addition to roughly 200,000 reservists, to fulfill NATO targets. Achieving this ambitious goal necessitates a significant surge in recruitment and training efforts. Consequently, major defense contractors are also compelled to undertake mass hiring initiatives. Key skills in high demand include aerospace engineers, software developers (particularly in AI and cybersecurity), mechatronics technicians, and munitions technicians.

Developments 2020–2026: In response to these challenges, the German government has reintroduced voluntary military service, vocational programs, and STEM outreach initiatives targeting young people. Simultaneously, private firms have established partnerships with technical schools (Fachhochschulen) and are offering internships and cooperative education programs. However, the hiring process is often slowed by lengthy security clearance procedures required for classified work. Furthermore, many skilled professionals are attracted to the high-tech civil sectors, such as Big Tech and the automotive industry.

Spotlight 3: New Threats in a New Environment

This new paradigm of conflict also introduces three primary strategic risks for the industry.

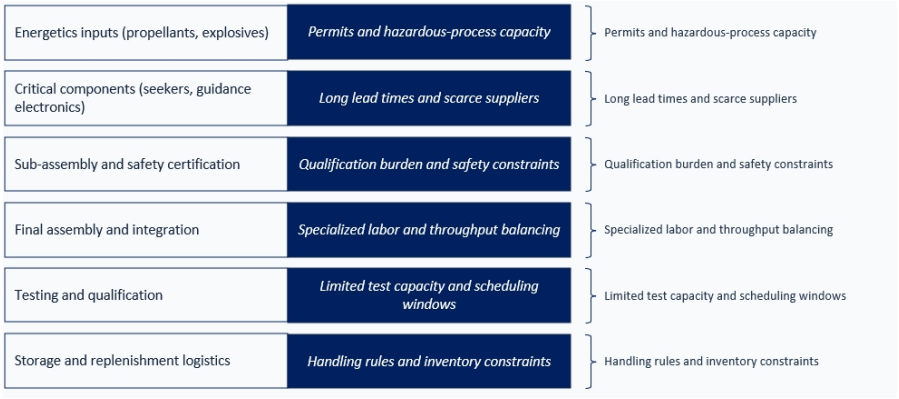
- **Demand Volatility:** The current surge in investment is primarily driven by active conflicts. A future de-escalation or a severe economic downturn could prompt governments to revert to 'peacetime fiscal restraint', leading to reduced military budgets and slower procurement. This, in turn, could undermine the long-term investment necessary for industrial readiness.
- **Supply-Chain Disruption:** Western defense production exhibits high vulnerability to supply-chain exploitation, particularly due to its reliance on potential adversaries like China for critical minerals and legacy electronics. This dependency could be weaponized during a crisis, severely crippling the West's capacity to produce and sustain military hardware.
- **Commercial Assets as Targets:** As private companies become increasingly indispensable to modern military operations, their infrastructure and personnel are more frequently being perceived as legitimate military targets. In future conflicts, adversaries are likely to conduct cyber or kinetic attacks against critical commercial assets, such as satellite networks and data centers, to gain a military advantage, thereby placing civilian assets and lives directly in harm's way.

Spotlight 4: Procurement Reform & Bureaucratic Friction

German defense procurement is characterized by slow and complex procedures. The Bundeswehr procurement agency (BAAINBw) operates under strict adherence to EU and German tender law; consequently, contracts exceeding €25 million necessitate a lengthy review by the Parliamentary Budget Committee. This often results in delays or fragmentation of acquisitions. However, in late 2024 and 2025, the government took steps to accelerate these processes. A

Bundeswehrbeschaffungsbeschleunigungsgesetz (Procurement Acceleration Act) was passed, simplifying defense-related rules and exempting certain deals from standard procurement ceilings.

Figure 20: Munitions and Missiles: Where Scaling Breaks



Source : Deutsche Bank research

Recent Developments: In 2025, Germany approved 103 major defense projects, vs 97 in 2024 and 55 in 2023, largely due to faster executive approvals. Over €83 billion was allocated that year, including from the special fund. Statements from the Federal Ministry of Defence (BMVg) boast "significantly accelerated" procurement. Nevertheless, critics point out that ammunition backlogs and maintenance deficits persist, and that rapid approvals do not always translate into delivered capabilities. Furthermore, the procurement office faces ongoing challenges related to IT modernization and capacity-building, including the need for new staff and digital tools.

Key Actors: The BAAINBw, operating under the BMVg, and the BWB (defense logistics agency) are primarily responsible for acquisitions. Ultimately, the Parliamentary Budget Committee provides final approval for large contracts. New legislative tools, such as fast-track procurement committees and emergency acquisition powers, have been utilized. Germany also coordinates with NATO and EU procurement efforts for common goods, including howitzers and Airbus A400M transport aircraft.

Opportunities: Simplified processes are opening doors for Small and Medium-sized Enterprises (SMEs) and start-ups, as discussed in Section 5. Germany is also increasingly utilizing advanced contracting methods, such as prize competitions and rapid prototyping, alongside end-to-end digital procurement systems. Defense lead contractors have, in turn, scaled up their in-house bid response teams.

Risks & Levers: Administrative inertia remains a significant risk that can still derail deals. While corruption is less prevalent in Germany, compliance standards are stringent. Policy levers for further improvement include legislative reforms, such as expanding exceptions to public tender rules, streamlining certification processes, and fostering closer ties with EU defense procurement initiatives, such as the European Defence Industrial Development Programme (EDIDP) framework.

Spotlight 5: Internationalization & Exports

German defense firms are increasingly globalizing their operations. According to SIPRI data from 2020 to 2024, Germany ranks fifth in arms exports, accounting for



5.6% of the world total, trailing only the US, France, Russia, and China. Major export customers include Ukraine and Egypt, each representing approximately 19% of recent German exports, and Israel, which accounts for around 11%. Non-European markets, particularly in the Middle East and Asia, are growing areas of focus. For instance, Diehl Defence is supplying IRIS-T SLM air-defense systems to Egypt, and Typhoon jets (produced in collaboration with the UK and Italy) are being sold to Qatar. German firms rigorously adhere to strict export vetting procedures under the Kriegswaffenkontrollgesetz, administered by BAFA.

Recent Trends: Arms export approvals reached record values in 2024, reflecting increased aid to Ukraine, including anti-aircraft missiles and small arms sent to Kyiv. Concurrently, many dual-use German goods, such as electronics and software, fall under US ITAR control if US-origin technology is involved. This complicates joint ventures, such as a German electronics firm partnering with a US prime contractor. Furthermore, the EU's new EDIRPA regulation funds joint procurement and also establishes shared export terms among EU buyers.

Companies: Traditional exporters include Rheinmetall (tanks, artillery), Hensoldt (sensors to NATO allies), TKMS (submarines, frigates), Diehl BGT (IRIS-T missiles), and Diehl's launcher vehicles. Private players like Heckler & Koch (small arms) and Rheinmetall's truck arm also engage in international sales. German SMEs often serve as subcontractors in larger export initiatives, providing specialized components and subsystems to major international prime contractors and multinational weapons programs.

Supply Chain: German exports frequently flow to allied laboratories and bases, illustrating major export routes. However, critical dependencies exist, as many exported systems still rely on US or third-party technology; for instance, a German-made jet might use GE engines subject to ITAR regulations.

Implications: Diversifying markets provides companies with protection against domestic economic downturns. Joint foreign projects, such as the Kongsberg-Rheinmetall collaboration for medium artillery, help share R&D costs. Nevertheless, political risk remains high; for example, sales to Gulf states face scrutiny over human rights concerns. Export controls continue to be a significant lever, with Germany historically limiting high-tech sales to certain regions. NATO interoperability requirements also influence export design, necessitating the adoption of standards like STANAG interfaces.

Spotlight 6: Industrial Clusters & Regional Footprints

Germany's defense industry exhibits a distinct regional distribution across the country:

- **Bavaria:** The Munich area hosts key players such as Airbus Helicopters, Diehl in Übersee, and Hensoldt in the Munich area.
- **Baden-Württemberg:** The Stuttgart/Heilbronn region is home to ThyssenKrupp Marine Systems (with significant operations also in Rostock/Wilhelmshaven) and Bosch Defense tech.
- **North Rhine-Westphalia:** Cologne features Airbus C295 assembly and antenna factories, while Rendsburg houses Rheinmetall Air Defence.
- **Schleswig-Holstein/Hamburg:** Kiel is known for its naval yards, including HDW, and Hamburg has a presence of Airbus and sensor technology companies.



- **Eastern States:** Dresden is a hub for laser technology for sensors, and Oberpfaffenhofen is home to the German Aerospace Center, focusing on space-related activities.

Local economic development offices frequently promote defense clusters, such as 'Verteidigungs-Cluster Bayern'. State governments in Bavaria and Baden-Württemberg have industrial strategies that are aligned with the Defense Industrial Base (DIB).

Clusters: Proximity among these industries fosters robust supply networks. For instance, southern Germany's clusters integrate automotive and turbine engine manufacturers (like MTU) with defense. Northern coastal clusters, on the other hand, specialize in shipbuilding and maritime systems. Cyber/C4ISR firms tend to gravitate towards tech hubs such as Berlin, Cologne, and Dortmund.

Investments: EU cohesion funds and German re-industrialization programs are likely to favor key regions, for example, Hamburg for shipyard modernization or Bavaria for electronics fabrication facilities. Funding for critical infrastructure, including ports, testing grounds, and ranges, is also increasing, evidenced by a new drone testing corridor in Brandenburg.

Risks & Policy: Regional disparities mean that some areas, particularly in former East Germany, are less integrated into these defense industrial clusters. Policy initiatives can incentivize relocation through measures such as tax credits for defense R&D and grants in states with lower GDPs. A German map highlighting leading defense companies by location would further illustrate these regional footprints.

Spotlight 7: Cyber & Electronic Warfare Capabilities

German industry demonstrates significant strength in cybersecurity and Electronic Warfare (EW). Rohde & Schwarz, based in Munich, stands out as a leading provider of battlefield radios, Signal Intelligence (SIGINT)/EW systems, and joint communication suites. Hensoldt, located in Taufkirchen/Munich, specializes in producing radar systems (for applications such as the MEADS air defense) and infrared sensors. In the realm of pure cyber, companies like Rohde & Schwarz Cybersecurity and Infineon (known for secure chips), and even SAP (contributing defense analytics), play important roles. The Bundeswehr's own Cyber- und Informationsraum (CIR) command actively collaborates with these companies to develop national cyber tactics.

Germany consistently adheres to NATO and EU standards, such as NATO STANAGs for cryptography and ETSI standards for communications. German firms prioritize interoperability; for instance, Airbus provides encryption for Tiger helicopters, and Rohde & Schwarz radios are compatible with NATO waveforms.

State-of-Play (2020–26): Data protection laws akin to GDPR and BSI guidelines have significantly commercialized the cybersecurity sector. Companies like Nexware in Munich are now building secure defense-grade networking equipment. German shipyards, including TKMS, have integrated advanced EW suites into frigates. The anti-drone systems market has also seen innovation from start-ups, such as Ordrum in Berlin.

Key Players: Beyond Rohde & Schwarz and Hensoldt, mid-tier firms like Rheinmetall Electronics, Diehl EW, and ESG Elektroniksystem- und Logistik GmbH



(Dortmund) are active contributors. These companies frequently collaborate with academic institutions, such as Fraunhofer FKIE, and engage with international partners.

Implications: Cyber and EW represent a significant growth niche for German exports to allied nations. Private cybersecurity firms particularly benefit from defense R&D spin-offs. Furthermore, government funding, channeled through national cybersecurity strategies, provides a boost to testing laboratories.

Risks: Cyber firms face the constant challenge of technological obsolescence and talent poaching, as they compete with the banking and technology sectors for skilled professionals. Export control presents another hurdle, given the highly regulated nature of cryptography. Policy responses to these challenges include funded training programs, such as the KI Masters at Ruhr-Universität, and participation in NATO cooperative exercises like Cyber Coalition.

Spotlight 8: Ownership Structures & Governance

German defense companies exhibit a variety of ownership forms:

- Family-owned/Private: Examples include Diehl Stiftung & Co. (where the family retains 100% ownership of Diehl Group), Bayern-Chemie, and Rheinmetall Air Defence GmbH (a howitzer manufacturer), which is partly family-owned. RUAG International, though Swiss-owned, has German operations in Munster and is fully state-owned by Switzerland.
- Publicly Listed: Rheinmetall AG is publicly traded, governed by an executive board and a supervisory board. TKMS - thyssenkrupp's former marine-systems arm - has itself been publicly listed since October 2025, with thyssenkrupp AG retaining a 51% majority stake and 49% in free float on the Frankfurt Stock Exchange. Hensoldt AG was majority-owned by private equity firm KKR; now being listed, the German State and Leonardo are the major shareholders. Infineon (chips) has partial state ownership, with a stake held by the German government.
- Joint Ventures: MBDA, formed by Airbus, BAE Systems, and Leonardo, is Europe's leading missile joint venture, operating with national divisions in Germany, France, the UK, and Italy under a shared governance structure.
- Strategic Investments: In 2023-2024, the German government considered acquiring 'golden shares' in key firms, such as MTU Aero Engines, though this particular initiative did not materialize. While there is no explicit 'national champion' law, some contractors have advisory boards comprising former military personnel and ex-politicians.

Governance: Defense firms are tasked with balancing profit motives with national security imperatives. Supervisory or advisory boards often include former generals or ex-government officials. Institutional investors frequently hold significant blocks of shares. Insider ownership by founders and management can ensure stability but also risks opaque control, particularly in family-owned firms.

Risks: Obscure ownership structures can potentially mask foreign influence, particularly concerning the takeover of mid-tier defense SMEs. Policy levers to mitigate these risks include rigorous vetting processes (to determine if takeover proposals require government sign-off under IT security law) and ensuring that boards possess adequate expertise in security matters.

Spotlight 9: Technology Roadmaps & Investment Needs



Over the coming decade, priority technologies for defense include: Artificial Intelligence (AI) and autonomy (encompassing unmanned drones/swarms and decision aids), hypersonic weapons, advanced cyber defense tools, next-generation air and space systems, and the replenishment of munitions stockpiles. Germany actively participates in EU/NATO programs, such as the Franco-German Future Combat Air System (FCAS) and the Main Ground Combat System (MGCS), which define joint roadmaps for systems expected between 2035 and 2050. The Bundeswehr's Integrated Plan outlines specific needs, including more Unmanned Aerial Vehicles (UAVs, particularly loitering munitions), Ground-Based Air Defense (GBAD) systems (such as IRIS-T SLM upgrades), and enhanced space capabilities (through initiatives like the German Space Situational Awareness and COMSATSat programs).

Funding Calls: The EU's European Defence Investment Programme (EDIP) for 2025–2027 has earmarked €1.5 billion for projects focusing on areas like drone autonomy, space sensors, and AI software. Similarly, NATO's defense planning, particularly its Capability Targets, identifies AI and cyber as key investment areas. Germany's own technology fund, ZUKuF (Zukunftsfonds), may be seeded by the defense special budget.

Industry Response: Prime contractors are strategically pivoting their Research and Development (R&D) efforts. Rheinmetall, for instance, operates laboratories dedicated to AI-driven vehicles, Hensoldt is developing advanced space radars, and Rohde & Schwarz is building quantum-safe cryptography solutions. Start-ups specializing in drone swarms, such as ARTIS and Project SIRIUS, are working on AI coordination. NATO's Defence Innovation Accelerator for the North Atlantic (DIANA) also selects German Small and Medium-sized Enterprises (SMEs) in its calls, exemplified by its latest Innovation Fund (IF) project for AI.

Capability Gaps: Analysts caution that Europe currently lacks the capacity to produce sufficient missiles, ammunition, or certain advanced platforms to effectively deter a near-peer conflict. Germany, in particular, previously had a deficit in domestic long-range artillery (now being addressed with PULS orders) and its hypersonics program remains nascent. To fully meet established targets, investment needs are likely to range from €20–30 billion per year over the baseline for R&D and modernization.

Implications for Investors: The massive planned spending signals lucrative long-term contracts and export opportunities, such as exportable components of the FCAS. VC opportunities exist in niche technologies, including sensors, satellite communications, and simulation. PPPs, exemplified by German Space Agency projects and various research consortia, also channel significant funds. However, risks persist concerning budgeting stability and intense technological competition, particularly regarding leadership in AI from countries like the US or China.

What This Coverage Is - and What It Is Not

Private company research is a different product from equity research, and we want to be precise about the difference before the first company page.

It is coverage, not a one-off

This is the first report of a continuing franchise. The companies profiled here will be maintained and updated, and the sector will be revisited as the ecosystem changes. We chose defense first because it is where the private half of the industry matters most today.

Sources and method

We work exclusively from publicly available information: audited filings and Bundesanzeiger (Federal Gazette) accounts, company disclosure, commercial and lobby registers, government and EU procurement material, partner and customer disclosure, and credible independent reporting. Specialist private-company databases are used to structure and cross-check that material. AI is used systematically for aggregation, structuring and cross-checking of fragmented public information; analysis, judgment and conclusions are the analysts'. Where a conclusion is an assumption rather than a disclosed fact, we say so explicitly - deliberately, and often. Any direct contact with the companies covered follows the bank's standard procedures; whether a company has been contacted, and on what basis, is determined by Compliance.

No ratings, no target prices, no recommendations

There is no traded instrument, so there is nothing to rate. We publish no ratings, no target prices and no investment recommendations on these companies. Where we discuss financial metrics, they are drawn from public accounts. Where we discuss valuation, it is high-level benchmarking against listed peers or publicly disclosed transactions, with the appropriate caveats attached.

How to read the company pages

Each company is presented in a consistent structure: what it is and why it matters now, its position in the value chain, products and capabilities, customers and route to market, footprint, supply chain and dependencies, ownership and governance, competitive positioning, risks and open questions, and what to watch next. Each profile opens with a summary table covering ownership, domain, value-chain position, key products, core capabilities and customers, with historical financials shown directly beneath where disclosure supports them.

The honest limitation

Private disclosure is thinner than listed disclosure, and in places materially so. For most of these companies, there is no public backlog, no divisional profitability, no customer-concentration table and no current cap table. We do not fill those gaps with false precision. The value of this work is not a forecast - it is a structurally accurate map of an industry currently analyzed as if half of it did not exist.

Figure 21: Diehl Group - overview

Diehl Group			
Ownership:	Independent family-owned group organized under Diehl Stiftung GmbH & Co. KG, combining family ownership with a foundation-based holding structure.		
HQ:	Nuremberg, 90478, Germany	**Founded:**	1902
Domain:	Multi-domain – diversified industrial group; defense relevance is centered in Diehl Defence's air-defense, missile and ammunition activities		
Value chain:	Hybrid – system integrator plus missile/ammunition supplier		
Why it matters	Diehl is not a pure-play defense name, but its defense arm has become one of Europe's key air-defense suppliers as IRIS-T and related programs scale		
Key products	IRIS-T SLM/SLS – ground-based air defense; IRIS-T air-to-air missiles; guided missiles and ammunition; training and protection systems; newer counter-drone effectors such as CICADA/KDV		
Core capabilities	Guided-missile integration, ground-based air-defense engineering, ammunition manufacturing and growing missile-production capacity		
Customers	Diehl Defence is a long-time partner to the Bundeswehr and international armed forces; the IRIS-T family now spans 21 user countries, with public recent orders/procurements including Germany, Denmark, Sweden, Slovenia and Bulgaria		

Key financials (31 Dec)	2021	2022	2023	2024	2025
P&L (EURm)					
Sales	3,167	3,506	3,883	4,696	5,420
Gross profit	-	-	-	-	-
EBITDA	117	265	368	584	842
EBIT	18	165	257	460	703
EBT	-23	134	269	501	754
Net income	-84	64	174	343	492
Order intake	-	-	-	-	-
Cash flow statement (EURm)					
Cash flow from operations	31	276	1,674	746	3,099
Cash flow from investing activities	-117	-122	-227	-346	-536
Cash flow from financing activities	34	-31	-57	-35	-188
Balance sheet (EURm)					
Cash & cash equivalents	121	255	1,646	2,014	4,430
Tangible fixed assets	584	608	724	954	1,176
Goodwill & intangible assets	21	21	21	26	29
Inventories	747	615	0	0	0
Accounts receivable	393	421	450	565	674
Total assets	2,200	2,331	3,303	4,041	6,976
Equity	480	527	668	993	1,346
Minorities	59	65	77	94	0
Borrowings	88	83	75	69	74
Accounts payable	268	299	381	513	509
Other ST liabilities	57	65	725	890	3,149
Net debt (+) / net cash (-)	-35	-175	-1,572	-1,947	-3,602
Key company metrics					
Book/bill (x)					
Sales growth (%)	6	11	11	21	15
Net income growth (%)	-51	-176	170	97	43
EBITDA margin (%)	4	8	9	12	16
EBIT margin (%)	1	5	7	10	13
ROE (%)	-18	12	26	35	37
Capex/sales (%)	4	3	6	7	10
Capex/depreciation (x)	1.2	1.2	2.1	2.8	3.9
Net debt/equity (%)	-7	-33	-235	-196	-268
Net interest cover (x)	0	5	-22	-11	-14

Source : Deutsche Bank research, company data, Bloomberg Finance LP, Gain.pro; 2023 & 2024 inventories fully netted against prepayments; sales excl. inventory changes

Diehl Group

Europe's Air-Defense Anchor

What type of company it is

Diehl Group is a privately held, family-owned German technology group headquartered in Nuremberg. It operates through five divisions and serves both civilian and defense markets. In plain language, Diehl is a portfolio of specialized industrial businesses, but the center of gravity is shifting toward defense and aviation as European demand for air defense, missiles, ammunition, and aerospace systems rises.

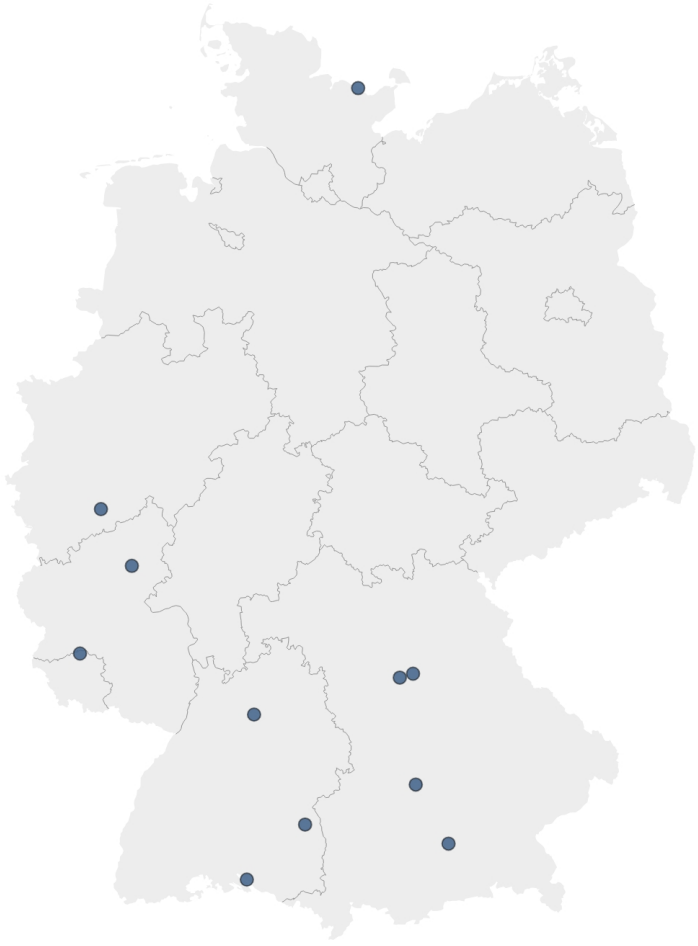
Investment-style or strategy-style summary

Diehl is best understood as a privately held, family-owned German industrial group whose strategic core operational anchor is shifting toward defense and aerospace, even though it remains a genuinely diversified portfolio. The five divisions, Defence, Aviation, Metering, Controls, and Metall, span guided missiles and air defense, aircraft systems, utility metering, appliance electronics, and metal processing, which makes Diehl a mixed model rather than a defense pure-play. That structure is both a strength and a complication: it diversifies cash flow and engineering depth, but it also dilutes the clarity of the defense case relative to a single-segment defense manufacturer.

The defense relevance is nonetheless rising fast. Diehl Defence sits in several of the capabilities Europe is trying to rebuild quickly - ground-based air defense, missiles, and ammunition - anchored by the IRIS-T family, which now spans 21 user countries. In 2025, group sales rose to €5.4bn, EBIT to ~€700m, and management guided to ~€6bn of sales for 2026, with Defence and Aviation driving the growth while Metall and Controls were pressured by weak industrial end markets. SIPRI's external estimate that arms revenue grew about 53% year on year underlines how much of the current momentum is defense-led.

Financially, the group looks robust and self-funding, with ~€4.4bn of cash and more than €2.5bn of free cash flow in 2025, no visible dependence on external equity, and a deliberate strategy of securing constrained subsystem and energetics capacity through M&A (Dynamit Nobel, AIM, Tauber). The main limitations are analytical rather than operational: as a private company, Diehl does not disclose divisional profitability, backlog, order intake, or customer concentration, so program-level economics - especially the degree of IRIS-T dependence - must be derived rather than read directly.

Figure 22: Diehl Defence Locations Germany



Source : Company Data, created with Datawrapper

1. Why this company is relevant now

Diehl has moved from being a broadly diversified private German industrial group to being, increasingly, a company whose strategic relevance is shaped by air defense, missiles, ammunition, and aerospace. In 2025, group sales rose 15.4% to €5.4bn, EBIT to €702.6m, and net income to €492.3m; management guides to around €6bn of sales for 2026. The biggest acceleration came from Defence and Aviation.

2. What type of company it is

Diehl is not a pure defense contractor. It is a strategic management holding with five divisions: Metall, Controls, Defence, Aviation, and Metering. That makes it a mixed model: in some areas it is a system house and integrator, in others a tier-1 supplier, component specialist, or industrial technology provider.

3. Three things that matter most

Diehl Defence is now large enough to matter at the European system level, especially in ground-based air defense and selected missile and ammunition programs.

The group's civilian businesses still matter a lot. Defence was 39% of 2024 sales by company disclosure, which means the overall economic profile is broader, more



diversified, and less 'pure-play defense' than some readers may assume.

Diehl appears to be selectively deepening control over choke point capabilities, including explosives-related licenses, infrared modules, and missile integration capacity. That is strategically important in the current European rearmament cycle.

4. At a glance

Diehl is relevant now because its defense arm sits in several of the areas that Europe is trying to rebuild quickly: ground-based air defense, missile production, ammunition supply, and related industrial capacity. The Aviation division adds a large aerospace footprint, while Metering, Controls, and Metall provide diversification and industrial depth. On the latest figures, the group generated €5.4bn of sales and €702.6m of EBIT in 2025, with Defence contributing roughly €2.3bn, and management guides to about €6bn of sales in 2026.

As a private company, Diehl discloses more than many peers but still materially less than a listed group. There is no public breakdown of backlog, order intake, customer concentration, or divisional profitability in the reviewed sources, so several conclusions about program mix and margin quality rest on analysis of the annual report, transactions, and product roles rather than on disclosed segment data.

Key takeaways

- Diehl is a mixed industrial group, not a single-line defense contractor.
- 2025 sales were €5.4bn and EBIT was €703m, up sharply from 2024 (when the numbers were €4.7bn and €460m, respectively).
- Defence was the largest division in 2024 at ~€1.8bn of sales, followed by Aviation at ~€1.3bn. Together they represented about two-thirds of group revenue. This last point is a calculation based on company figures.
- The group remains family-owned and describes itself as financially independent.
- Diehl Defence is publicly described by the company as a leading system integrator for guided missiles and ground-based air defense systems.
- Public evidence suggests Diehl is becoming more vertically integrated in key defense bottlenecks through acquisitions and capacity additions. This is a logical deduction based on the Dynamit Nobel (not Dynamit Nobel Defence GmbH) acquisition, the AIM transaction (buying the remaining 50% in AIM Infrarot-Module GmbH from Rheinmetall), and current site expansion.
- International exposure is material. In 2024, ~€3.2bn of group sales came from outside Germany.
- Public transparency is meaningful for a private company, but still limited. There is no full public disclosure of backlog, order intake, customer concentration, or divisional profitability in the sources reviewed. This is based on the annual report and key figures pages reviewed, but not explicitly stated by the company.

5. Corporate history and legal structure

Diehl has been an independent family business since 1902. The group is organized as a strategic management holding, with the operating divisions running their businesses close to the market. That structure matters because Diehl is best understood as a federation of specialized industrial businesses rather than a



centrally integrated pure-play. Its legal structure centers on Diehl Stiftung & Co. KG and Diehl Verwaltungs-Stiftung, combining family ownership with a foundation-based holding. No listed securities were identified, and the family ownership split, succession framework, and broader shareholder economics are not publicly disclosed.

A short timeline is useful:

- 1902: Diehl founded as a family business.
- 2010: Diehl Metering became an official group division.
- 2024: Defence and Aviation drove the strongest group growth; Dynamit Nobel was acquired to secure the Troisdorf explosives-related site and licenses; the TD Services JV was established with TAUBER.
- 2025: Diehl Defence announced the move to sole ownership of AIM and signed to acquire Tauber Group, subject to approval.
- 2026: Diehl opened a new missile integration center in Nonnweiler and announced a new air and missile defense cooperation with Indra.

6. Business overview

As just outlined, Diehl is best understood as a group of specialized industrial businesses rather than a centrally integrated pure-play.

The five operating divisions are:

- **Diehl Defence:** guided missiles, ground-based air defense, ammunition, training, protection, and related systems.
- **Diehl Aviation:** avionics, aircraft systems, cabin solutions, and the Diehl Aerospace joint venture with Thales.
- **Diehl Metering:** smart water and heat metering solutions, software, analytics, and services.
- **Diehl Controls:** electronic systems for home appliance and HVAC/R applications.
- **Diehl Metall:** metal processing, copper and copper-alloy products, and battery-related cell contact systems.

7. Position in the value chain

At group level, Diehl sits across multiple layers of the value chain. It is not just a lead contractor, just a component supplier, or just a services company. It is all of these, depending on the division.

Within defense, Diehl is strongest as a system house and integrator in ground-based air defense and guided missiles, while also supplying ammunition, training, protection, and components. The company explicitly says it is a leading system integrator for guided missiles and ground-based air defense systems and, in IRIS-T SLM, it is the system integrator and general contractor, while Hensoldt provides radar and Airbus provides the command post software.

That means Diehl touches:

- Air defense and missiles: clearly core.
- Land munitions and rocket artillery effectors: meaningful and growing.



- Naval weapons: relevant through RAM, RBS15 and naval ammunition.
- Training, protection, counter-UAS: present, though less central than missiles and air defense.
- Military avionics: relevant through Diehl Aviation and Diehl Aerospace.

Strategic criticality varies by niche. As a group, Diehl is important but not universally irreplaceable. In specific niches, especially IRIS-T-based air defense integration, some missile subsystems, thermal batteries, infrared modules, and explosives-related industrial permissions, replacement would likely be slow and difficult. This is an assumption based on public product roles, transactions, and capacity actions, but not a company statement.

8. Products, services and core capabilities

Defence

The disclosed defense portfolio includes guided missiles, air defense systems, ammunition, training, reconnaissance and protection systems, counter-UAS, customer support, and components. Major named products include IRIS-T, IRIS-T SL, IDAS, RAM, RBS15 Mk3, Spike, Sidewinder, and PARS 3 LR.

In practical terms, the operational anchor is clearly air defense and missile effects. IRIS-T SLM and SLS address short- and medium-range ground-based air defense. Publicly disclosed contracts show Diehl selling to government customers and export users, including Germany, Denmark, and Sweden, while Reuters reported broader demand across ESSI (European Sky Shield Initiative) members and export customers.

Diehl also has meaningful ammunition exposure, including 155mm artillery ammunition through the ARGE DiNa 155mm working group with Nammo, plus rocket artillery ammunition via cooperation with Elbit for PULS and EuroPULS.

This looks like a mix of:

- Mature and scaled products: IRIS-T family, RAM, conventional ammunition, training/support.
- Next-generation or development-stage work: IRIS-T Block II, IRIS-T SLX, HYDEF, BEAST, FCAS-linked activities, navalized IRIS-T variants, and newer counter-UAS offerings.

Aviation

Diehl Aviation develops products in cabin interiors, aircraft systems, and supply systems. Publicly disclosed products include side panels, overhead stowage, lavatories, galleys, ventilation systems, and military avionics solutions such as cockpit, flight and engine control, and integrated systems.

The key capability here is not just component making. It is systems integration at tier-1 level, reinforced by Diehl Aerospace, the JV with Thales, which develops and supports systems and components from cockpit to cabin for both commercial and military aircraft.

Civilian divisions

The non-defense divisions matter because they add scale, engineering breadth, and financial diversification:



- Metering combines meters with software, analytics, and implementation services.
- Controls ships millions of electronics used in heat pumps, washing machines, and similar appliances.
- Metall combines traditional copper/copper-alloy processing with battery cell contact systems and mobility applications.

9. Customers and routes-to-market

The customer base is multi-channel and differs sharply by division.

For **Defence**, the route to market is primarily direct to governments, procurement authorities, and armed forces, often with partner companies embedded in the solution. Publicly disclosed examples include Germany's BAAINBw, the German Air Force, Denmark's Ministry of Defence procurement, and Sweden's FMV. Some programs run through JVs such as EuroSpike, PARSYS, and RAMSYS.

For **Aviation**, customers are mainly B2B and include Airbus, Boeing, Bombardier, Embraer, Gulfstream, airlines, operators, and military partners.

For **Metering**, the customer logic is utilities, municipalities, and industrial infrastructure owners. For **Controls**, it is appliance and HVAC/R OEMs. For **Metall**, it is industrial and mobility customers.

Geographically, Diehl is clearly international. In 2024, ~68% of group sales came from outside Germany, and foreign sales were generated mainly in other European countries, North America, and China. Defence alone generated €1.258bn outside Germany versus €569m in Germany.

What is not publicly disclosed is the precise concentration by customer, program, or ministry. The reviewed sources do not provide a quantified backlog or a breakdown of revenue by major defense program.

10. Key financials and economic profile

Publicly disclosed 2025 group figures show a clear step-up in profitability: Sales €5,420.5m (up 15.4% from €4,695.8m); EBIT €702.6m (from €460.3m); Net income €492.3m (from €343.5m); headcount 19,449 (up 769). On simple calculation, the EBIT margin was about 13.0% in 2025, up from 9.8% in 2024. Management guides to around €6bn of sales in 2026 with broadly stable EBIT. The 2024 reference figures for cash (€1,988.7m), free cash flow (€420.6m), and R&D (€397.0m) remain the latest fully broken-out cash-flow disclosures pending the detailed 2025 annual report.

By division, Defence sales grew to roughly €2.3bn in 2025 (from €1.827bn in 2024), remaining the group's largest and fastest-growing segment, while Metall and Controls stayed pressured by weak end markets. Full 2025 divisional splits for Aviation, Metering, Metall and Controls await the detailed annual report.

SIPRI offers a useful external cross-check. In its 2024 (latest available) Top 100 arms company data, SIPRI ranked Diehl 67th globally and estimated arms revenue at $2.11bn, or 42% of total revenue, up 53% year over year (broadly in line with company data). That is not company guidance. It is an external estimate, but it is useful because it suggests Diehl is now large enough to matter in international defense market mapping. On 2025 company figures, Defence revenue of roughly



€2.3bn implies arms revenue is now well above the 2024 SIPRI estimate, consistent with the 15%+ group growth.

Economic profile assessment.

Revenue quality appears mixed. Defense and aviation should carry longer program durations, higher qualification barriers, and stronger aftermarket/service potential than controls or metal processing, but the company does not disclose recurring revenue share. This is an assumption based on business mix, not a disclosed metric.

Working capital is likely structurally heavy but partly buffered by customer advances. Inventories excluding advances were €1.996bn at year-end 2024, while advances received were €2.781bn, and cash flow commentary explicitly linked liquidity to advance payments. This points to milestone- and contract-funded economics in parts of the portfolio, especially defense and aerospace. This is an inference based on public balance sheet and cash flow disclosures.

Pricing power likely differs by division. It is probably strongest in qualified defense systems and aviation content, weaker in appliance electronics, metering, and metal products where competition and customer price pressure are more visible. Again, this is analysis, not a company disclosure.

11. Footprint and industrial setup

The group is headquartered in Nuremberg. Its legal structure centers on Diehl Stiftung & Co. KG and Diehl Verwaltungs-Stiftung.

The industrial footprint is broad, but several nodes stand out:

- Defence: Überlingen is the headquarters. Other major German sites include Röthenbach, Nonnweiler, Todendorf, Troisdorf, and Heilbronn, plus affiliated entities such as AIM, DynITEC, EuroSpike, PARSYS, and RAMSYS.
- Aviation: Laupheim is the headquarters. The division also has major German operations in Hamburg, Gilching, and Dresden, plus international sites in Hungary, Mexico, Romania, Singapore, the UAE, and elsewhere. The company states it has 26 sites.
- Metering: Ansbach is the headquarters, with important development and production roles in Nuremberg, Apolda, Saint-Louis, and Bażanowice, plus a wider international sales and manufacturing network.
- Controls: development in the US, production in Mexico for North America, plus broader European and Asian presence.

The footprint looks industrially credible and scalable, especially for a private company. Germany remains the core, but the manufacturing and sales network clearly leverages European and nearshore cost locations, North American proximity, and Asian presence. This is particularly visible in Aviation and Metering.

12. Supply chain and key dependencies

Several broad dependency categories are visible from public sources.

Energetics and explosives. This appears to be a genuine bottleneck area. Diehl Defence acquired Dynamit Nobel including the essential explosives-related operating authorizations at Troisdorf, which the company called "unique in Germany." That strongly suggests access to licensed energetics capacity is



strategically valuable and hard to replicate.

Infrared and seekers. AIM, now moving to full Diehl ownership, supplies infrared detectors, thermal imaging devices, and Stirling cooling machines for industrial, space, and military use. Given Diehl's missile portfolio, this looks like a strategically sensitive capability node. This is an assumption based on AIM's product profile and Diehl's own product mix.

Special batteries and power. Diehl Energy Products publicly lists customers and programs including NSPA, Raytheon, Lockheed, MBDA, Airbus, Leonardo, NAMMO, and programs such as IRIS-T, PATRIOT, RAM and ESSM. That indicates a meaningful position in a niche but important subsystem category.

Sensors, radars, and battle management software. In IRIS-T SLM, Hensoldt and Airbus are named partners. That implies Diehl does not fully internalize all key sensor and command layers in its main air defense offering.

Metal inputs. The group discloses long-term commitments including quantity contracts for metal and commodity futures. That points to raw-material exposure in Metall and likely some cross-divisional purchasing sensitivity.

Overall, Diehl looks vulnerable where regulation, qualification, and specialized energetics or electronics capacity are constrained, but it also seems to be deliberately reducing this vulnerability through acquisitions and capacity expansion. That is a logical deduction based on the AIM and Dynamit Nobel transactions, missile integration expansion, and current production commentary.

13. Management, board, ownership, and governance

Diehl is a family-owned private company. The annual report describes it as a financially independent, family-owned company, and the group website says it has been an independent family business since 1902.

The most recent public governance materials reviewed show:

- Supervisory Board: Markus Diehl (Chairman), Konstantin Diehl, Klaus Helmrich, Frank Felix Werdin.
- Executive Board / current website legal notice: Jürgen Reimer, Jens Böhlke, Dr. Christof Bosbach, Dieter Landgraf, Helmut Rauch, Dr. Jörg Schuler.

Governance features worth noting:

- The holding structure appears formalized and institutional for a family-owned group.
- Klaus Helmrich's 2024 appointment added a supervisory board member with major industrial and technology background.
- Compliance infrastructure is explicitly described, including a compliance committee, corporate compliance officer, and external ombudsman.

What is not publicly disclosed in detail is the family ownership split, succession framework, or broader shareholder economics.



14. Competitive positioning

There is no single clean peer group for Diehl. The most meaningful comparisons are by division, not at group level. That itself is important. It makes Diehl harder to benchmark and, strategically, more resilient than a narrow pure-play in some scenarios.

In **Defence**, likely reference peers include MBDA, Kongsberg, Saab, Rafael, Hensoldt in adjacent layers, Rheinmetall in ammunition and some air-defense adjacencies, and Nammo in ammunition. This is an analysis based on product overlap, partnerships, and public program context, not a company-issued peer list.

Diehl's apparent advantages in defense are:

- qualification and customer trust in specific missile and air defense programs,
- system integration know-how,
- partner ecosystem depth,
- growing installed base,
- and a deliberate move to secure constrained subsystem capacity.

Its likely vulnerabilities are:

- dependence on government procurement timing,
- export approvals,
- capacity scaling pressure,
- and concentration around a few flagship product families, especially IRIS-T. The exact concentration is not publicly disclosed, so this is an assumption.

15. Strategic relevance and market implications

Diehl matters because it sits at the intersection of several live European policy themes: rebuilding air defense, restoring ammunition depth, increasing sovereign industrial capacity, and reducing dependence on fragile or non-European supply chains.

From a market perspective, the most relevant listed-company connections are:

- **Hensoldt** and **Airbus** as named partners in IRIS-T SLM/SLX system architecture.
- **Thales** through Diehl Aerospace.
- **Rheinmetall** through the former AIM shareholding and broader overlap in German defense industrial capacity.
- **Elbit** in rocket artillery effectors for PULS/EuroPULS.

Strategically, Diehl looks more like a selective capability anchor than a broad-spectrum defense giant. It is especially relevant in bottleneck-rich niches, not because it covers the whole battlespace, but because it covers parts of it that are currently supply-constrained and politically important. This is an assumption based on the company's portfolio, transactions, and production actions.



16. Dual-use and civilian exposure

Civilian exposure is substantial and should not be treated as incidental. In disclosed 2024 group figures, Defence was 39% of sales, which means most group revenue still came from outside the defense division. SIPRI's external estimate of 42% arms revenue points in the same general direction: Diehl is a significant defense player, but not a defense-only company.

This civilian base has two opposite implications. It likely helps by diversifying cash flow, engineering capability, and industrial culture across sectors such as aerospace, utilities, appliances, and mobility. But it may also dilute the simplicity of the defense case relative to a pure-play. That is a logical deduction based on the portfolio structure.

17. Legal, regulatory, and compliance risks

The clearest public compliance issue identified is the European Commission's 2023 hand-grenades cartel decision. The Commission found market allocation and exchange of competitively sensitive information in relation to military hand grenades over 2007-2021, and the public decision text states that Diehl received a 50% leniency reduction before settlement reduction. Reuters reported the fine at €1.2m.

The company also publicly describes a compliance organization with executive-board involvement, a corporate compliance officer, and external reporting channels.

Beyond that, the main sensitive issue is export policy exposure, not a disclosed company-specific scandal. As with any defense exporter, contracts can be affected by political decisions in Germany and customer countries. That is a structural risk, but we did not identify other major public compliance events in the reviewed sources.

18. Funding, financing, and strategic optionality

There is no sign in the public materials reviewed that Diehl depends on external equity funding. The company describes itself as financially independent and generated strong liquidity in 2024, with nearly €2.0bn of cash and cash equivalents and €420.6m of free cash flow.

No public funding rounds were identified. Instead, strategic change is happening through internal investment and M&A: Dynamit Nobel, AIM, Tauber, and continued capacity additions.

Strategic Optionality: An IPO is possible in the abstract, but there is no public indication that this is under consideration. Based on the family-owned structure and financing profile, the more plausible default is that Diehl remains privately held and continues to compound through selective investment and bolt-on acquisitions. That is an assumption, not a public plan.

19. Moat analysis

Diehl's moat is an engineering, certification, and installed-base moat rather than a brand or network-effects one, concentrated where substitution is hardest: ground-based air defense and guided missiles. It is anchored by the multi-role IRIS-T family, now spanning 21 user countries across both air-to-air and surface-to-air layers; once a customer adopts IRIS-T-based air defense, switching is slow and costly because integration, qualification, training, and munition logistics become



embedded across the force.

The moat looks stable to modestly widening, with the clearest evidence being Diehl's move to secure bottleneck nodes rather than add end-product breadth - the Troisdorf energetics authorizations ("unique in Germany") via Dynamit Nobel, infrared detectors via AIM, and ammunition logistics via Tauber. Each targets a scarce, regulation-heavy input rivals cannot easily replicate. The sources are thus switching costs, qualification assets, and regulated capacity rather than cost leadership; the main limitation is concentration around IRIS-T, while the civilian divisions carry thinner, more cyclical moats that dilute group-level clarity. The widening reading is an inference from deal logic rather than a company statement.

20. SWOT analysis

Strengths

- Strong position in ground-based air defense and missile integration.
- Broad industrial base across defense, aviation, utilities, and industrial electronics.
- Private ownership and strong liquidity can support long-cycle investment.
- Increasing control over constrained subsystem and energetics nodes.

Weaknesses

- Limited public transparency on backlog, program mix, and divisional profitability.
- Group complexity reduces pure-play clarity.
- Some divisions remain exposed to cyclical civilian end markets.

Opportunities

- European air defense replenishment and ESSI-related demand.
- Ammunition stockpiling and local production mandates.
- Further vertical integration and support/sustainment expansion.
- Aviation recovery and defense-aircraft content growth.

Threats

- Export-policy constraints and procurement timing risk.
- Scaling bottlenecks in constrained inputs and qualified labor.
- Program concentration around flagship systems.
- Competition from larger and more specialized defense houses.

21. Key risks and open questions

- How concentrated is Diehl Defence economically? Public sources do not disclose revenue or margin by major program. IRIS-T clearly matters, but the degree of dependence is not publicly confirmed.
- How durable are current defense growth rates? 2024 & 2025 were unusually strong. The medium-term slope will depend on procurement timing, exports, and production ramp execution.



- Can production scale without new bottlenecks? The new Nonnweiler missile integration center and Reuters-reported unit-capacity expansion are positive signals, but upstream dependencies remain real.
- How much margin upside sits in Defence versus dilution elsewhere? The group does not publish divisional EBIT, so the internal mix quality is hard to assess publicly.
- How much strategic value sits in in-house subsystems? AIM, Diehl Energy Products, DynITEC, and related capabilities may be more important than public revenue labels suggest, but the company does not quantify that value. This is an assumption based on disclosed products and transactions.

22. What to watch next

Over the next 6 to 18 months, the most important markers are:

- **IRIS-T production ramp:** Reuters reported a plan to expand annual firing-unit capacity to as many as 16 within two years.
- **Nonnweiler utilization:** the new missile integration center is a concrete capacity milestone.
- **AIM integration:** whether full ownership translates into visible product, capacity, or sovereign-supply messaging.
- **Tauber transaction outcome and integration:** especially if Diehl uses it to broaden ammunition logistics, storage, and ordnance-clearance offerings.
- **Additional IRIS-T and artillery contracts:** Germany, ESSI members, and export customers remain the key demand watchpoints.
- **European partnership expansion:** Indra, Elbit, Hensoldt, Airbus, and other partners indicate whether Diehl is evolving into a denser European defense-network node.
- **Aviation footprint ramp:** the Mexico and Romania expansions are the clearest public markers.

23. News flow and recent developments

Diehl's recent news flow has been dense and overwhelmingly defense-led, reinforcing the group's shift toward air defense, missiles, and energetics. The headline event was the FY2025 result, reported on 7 July 2026: group sales rose 15.4% to €5.4bn, EBIT climbed to €703m, and net income reached €492m, with the Defence division contributing roughly €2.3bn; management guided to around €6bn of sales for 2026 with broadly stable EBIT.

Capacity expansion has been the most concrete theme. In January 2026, Diehl Defence opened a new 3,400 sqm missile-integration center at Nonnweiler - the first of more than 20 planned construction projects at that site - as part of a company-wide investment of around €1.5bn to scale production, and in July 2026 it opened a "Battery Tech Hub" at Röthenbach through subsidiary Diehl Energy Products. On the demand side, Diehl signed further IRIS-T contracts with BAAINBw at the turn of the year covering air-to-air and ground-based SLS and SLM variants, and in June 2026 delivered the first IRIS-T SLM fire unit to the Estonian Air Force.

M&A and vertical integration continued to deepen the group's control over constrained upstream capacity. Diehl signed to acquire the Tauber Group (around 700 employees and roughly €100m of sales) in October 2025 to add explosive-ordnance-disposal and ammunition-storage capability, subject to antitrust approval, and in June 2026 acquired EST Energetics from General Atomics Europe,

strengthening its energetics supply. Both fit the pattern of securing bottleneck nodes around missiles and ammunition.

Finally, June to August 2026 brought an unusually thick cluster of partnerships and program wins: the ERAP (Extended Range Artillery Projectile) award alongside Leonardo and General Dynamics Ordnance and Tactical Systems; a SkyStriker loitering-munitions partnership with Elbit Systems for the Bundeswehr; cooperation agreements with Terma and with SMS on HYDEF2; membership in Boeing's German MQ-28 Ghost Bat industry team alongside Rohde & Schwarz; a naval-defense cooperation with BAE Systems; and a ground-based air-defense MoU with AEL Sistemas in Brazil. Taken together, the flow shows a company converting elevated European demand into contracts, capacity, and a widening European partner network.

24. Source notes and transparency limits

Company sources used

Diehl Group Annual Reports incl. business outlook.
Diehl Group key business figures, corporate divisions, locations, and legal notice pages.
Diehl Defence, Aviation, Metering, Controls, Metall, and Diehl Energy Products product, company, and press pages.

Government and regulatory sources used

European Commission decision materials on the hand-grenades cartel.
Bundeskartellamt clearance reporting for AIM.

Independent external sources used

Reuters on IRIS-T production scaling and customer footprint.
SIPRI Top 100 2024 data and commentary.

Transparency limits

Diehl is private, so public disclosure is materially less granular than for listed peers. No detailed public backlog, order intake, divisional EBIT, or customer concentration breakdown was identified in the reviewed sources.
Some strategic conclusions above are reasonable assumptions based on public transactions, product roles, and capacity actions, and are labeled as such where important.

Figure 23: Rohde & Schwarz - overview

Rohde & Schwarz GmbH & Co KG

Ownership:	Privately held, family-owned and independent, with no public free float		
HQ:	Munich, 81617, Germany	**Founded:**	1933
Domain:	Enabling systems – secure communications, SIGINT/EW, spectrum monitoring and naval/air C4ISR		
Value chain:	Tier-1 supplier and turnkey systems integrator		
Why it matters	R&S is a sovereign RF and secure-communications house with deep roles in tactical radios, military communications, EW and spectrum dominance, plus long-standing government relationships		
Key products	SOVERON tactical radios and secure communications; SIGINT/EW systems; radiomonitoring and radiolocation systems; M4ACS and NAVICS mission communications; crypto devices for protected voice/data		
Core capabilities	Deep RF engineering, secure military networking, turnkey ISR/EW integration and active participation in NATO/European interoperability standards		
Customers	R&S positions itself as a partner to government customers worldwide; named public examples include the Bundeswehr, the Belgian Ministry of Defense / Air Component, the Royal Netherlands Navy and Belgian Navy, and the company says it has been a long-term partner to more than 40 navies		

Key financials (30 Jun)	2021	2022	2023	2024	2025
P&L (EURm)					
Sales	2,275	2,525	2,784	2,933	3,160
Gross profit	1,678	1,815	2,350	2,360	2,639
EBITDA	485	457	470	387	551
EBIT	352	315	316	243	412
EBT	246	220	300	281	483
Net income	179	156	208	343	362
Order intake	-	-	3,155	3,873	~5,000
Cash flow statement (EURm)					
Cash flow from operations	502	311	500	410	188
Cash flow from investing activities	-302	-304	-127	-194	-186
Cash flow from financing activities	-151	-148	-207	-106	-90
Balance sheet (EURm)					
Cash & cash equivalents	545	402	575	680	592
Tangible fixed assets	429	436	157	486	524
Goodwill & intangible assets	64	129	117	83	52
Inventories	776	833	1,301	1,303	1,322
Accounts receivable	409	469	476	563	777
Other assets					
Total assets	4,246	4,515	4,584	5,120	5,425
Equity	1,738	1,735	1,751	2,041	2,240
Minorities	6	7	9	9	5
Borrowings	0	0	0	0	0
Accounts payable	81	89	90	89	128
Other ST liabilities	-	-	-	-	-
Net debt (+) / net cash (-)*	789	1,091	951	850	998
Key company metrics					
Book/bill (x)	-	-	1.1	1.3	1.6
Sales growth (%)	-12	11	10	5	8
Net income growth (%)	-15	-13	33	65	6
EBITDA margin (%)	21	18	17	13	17
EBIT margin (%)	15	12	11	8	13
ROE (%)	10	9	12	17	16
Capex/sales (%)	13	12	5	7	6
Capex/depreciation (x)	2.3	2.1	0.8	1.3	1.3
Net debt/equity (%)	45	63	54	42	45
Net interest cover (x)	3	3	20	-6	-6

Source : Deutsche Bank research, company data, Bloomberg Finance LP, Gain.pro, incl. pension liabilities and liabilities to shareholders

Rohde & Schwarz

From 5G Networks to Battlefield Electronics

What type of company it is

Rohde & Schwarz is not a typical defense company. It is a private German technology company that operates across three main areas.

Its biggest business is making test-and-measurement equipment for electronics, which is sold all over the world. It also has a defense division that, for the battlefield, supplies secure, jam-proof radios for soldiers, vehicles, and aircraft. Lastly, it has a cybersecurity business that helps protect computer networks.

This mix is a key advantage. It means the company is more stable than one that sells only to the defense sector, and its products are more critical than those of a simple equipment-maker.

Figure 24: Modernizing Bundeswehr Communications under the D-LBO Project



Source : Rohde & Schwarz

Investment-style or strategy-style summary

Rohde & Schwarz is a privately-owned German technology group with a diverse business model spanning electronic test and measurement (T&M), defense systems, and cybersecurity. Strategically, the company is a critical 'national champion' for Germany, providing sovereign and secure communications technology for military and government clients. The company's financial profile is exceptionally strong and conservative, with over €3.16 billion in recent annual revenue, high profitability, a cash-rich balance sheet, and no bank debt. Its primary moat is built on world-leading expertise in radio frequency (RF) technology, a high degree of vertical integration, and the deep trust it has cultivated with government customers.



The investment case is anchored in this dual strength: a stable, profitable T&M business combined with rising demand in its strategically important defense and security segments. Its private, family-owned structure allows it to make long-term R&D investments without market pressure, positioning it well for future technologies like 6G and quantum computing.

However, the main challenge for an outside observer is the lack of public transparency compared with listed peers, particularly on the profitability of individual business lines. While the defense business is growing in relevance, its exact economic weight within the group is not publicly disclosed. The investment thesis therefore rests on the company's clear technological leadership, its indispensable role in European sovereign security, and its pristine financial health. The story is one of a deeply entrenched, highly profitable, and strategically vital company that benefits from both commercial and government spending cycles.

1. Why this company is relevant now

Rohde & Schwarz is relevant now because European security spending is moving from broad political commitment toward concrete procurement, industrial capacity, and sovereign technology choices. The group is exposed to several of the priority areas that matter in that shift: secure tactical communications, naval communications, SIGINT/EW, counter-drone, air-traffic and air-sovereignty systems, and trusted cyber/network infrastructure. At the same time, its T&M franchise keeps it tied to future technology cycles such as 6G, AI-RAN, automotive electrification, radar, and satellite test.

The timing also matters because FY2024/25 was a record revenue year, while 2025 added a series of strategically coherent moves: ZES ZIMMER in power electronics test, Munich Innovation Labs in OSINT/data analytics, an Orbint stake in space-based SIGINT, and a TRUMPF partnership in counter-drone. Those are not random adjacencies. They reinforce a broader move toward independent sensing, analytics, and secure system capability.

2. What type of company it is

Rohde & Schwarz is best described as a privately held high-technology group with a mixed industrial and sovereign-systems profile. It is simultaneously one of the world’s largest electronic test-and-measurement manufacturers, a provider of communications and intelligence systems for military and government users, and a European cybersecurity and secure-networking supplier.

In defense value-chain terms, it is not primarily a complete platform manufacturer. It looks more like a hybrid of subsystem specialist, system house, secure-communications provider, spectrum-intelligence supplier, and selected turnkey integrator, especially in naval communications, ATC communications, and electronic-support domains. That assessment is partly inference from the product and contract set, but it matches the company’s own positioning as a supplier of integrated communications systems, SIGINT/EW, and modular secure radio families.

3. Three things that matter most

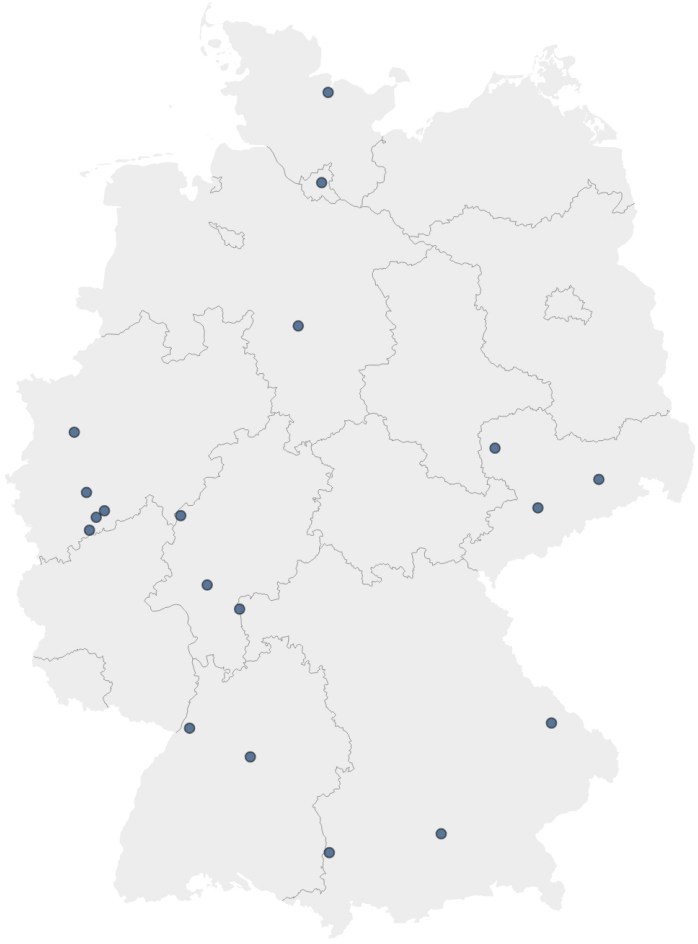
The first core takeaway is that Rohde & Schwarz has unusually strong sovereign relevance for a private mid-sized industrial technology group. Its portfolio maps into national and European priorities around spectrum dominance, trusted communications, cyber resilience, and secure infrastructure.

The second is that the company's business quality likely benefits from diversification. Test and measurement gives it exposure to civilian and dual-use R&D cycles, while defense and cybersecurity give it strategic demand tailwinds and policy relevance. That should make the earnings base less binary than that of a single-program contractor, although public segment profit disclosure is limited.

The third is that ownership and capital structure are part of the thesis. A 100%-family-owned, self-financed group with substantial cash, no bank debt, and high vertical integration can invest through cycles and pursue long-payback technology positions more easily than many listed peers can.

4. At a glance

Figure 25: Rhode & Schwarz Locations



Source: Company Data, created with Datawrapper

Rohde & Schwarz is headquartered in Munich and operates globally across more than 70 local subsidiaries.

The business mix spans three divisions. Test & Measurement covers electronic instrumentation, wireless, EMC, aerospace-and-defense test, and related solutions. Technology Systems covers secure communications, SIGINT/EW, air-



defense-adjacent and ATC systems, drone defense, and related security solutions. Networks & Cybersecurity combines LANCOM Systems and Rohde & Schwarz Cybersecurity around secure networking, trusted cloud-managed infrastructure, and high-assurance cyber products.

What makes the company matter is not just size. It is the combination of RF depth, long product history, sovereign customer relevance, vertical integration, and the ability to move across defense, security, industrial, and research use cases. That makes Rohde & Schwarz more strategically embedded than a simple electronics manufacturer and more diversified than a narrow defense electronics specialist.

Key takeaways

- Private, family-owned Munich technology group with FY2024/25 revenue of €3.16bn (first year above €3bn) and more than 15,000 employees.
- Three-division structure spanning Test & Measurement, Technology Systems, and Networks & Cybersecurity, providing exposure to T&M, defense/security systems, and cyber/networking markets.
- Strategic relevance is rising due to activities in sovereign communications, SIGINT/EW, cyber, trusted European infrastructure, and air-defense-adjacent capabilities.
- Test & Measurement remains the largest visible business, but defense and security relevance is increasing, although divisional revenue disclosure is limited.
- High vertical integration supports resilience, delivery control, technological independence, and sovereign positioning.
- Conservative financial profile with 100% founding-family ownership, self-financed growth, a cash-rich balance sheet, and no bank debt as of June 2025.
- Disclosure is relatively strong for a private company but remains below listed-peer standards, particularly on segment profitability, program economics, customer concentration, and revenue mix.
- The key analytical question is the future balance between the increasingly strategic defense-security businesses and the larger, less politically sensitive T&M franchise.

5. Corporate history and legal structure

The company was founded on November 17, 1933 in Munich by Dr. Lothar Rohde and Dr. Hermann Schwarz as the Physikalisch-Technische Entwicklungslabor. Its early focus was RF engineering and measurement technology. Over time, it expanded from measuring instruments into broadcasting, communications, radio-monitoring, and later secure communications, IT security, and broader sovereign systems.

The current top-level legal entity is Rohde & Schwarz GmbH & Co. KG, seated in Munich and registered at Amtsgericht München under HRA 16 270. The personally liable general partner is RUSEG Verwaltungs-GmbH, registered under HRB 7 534. The current management listed in the legal notice is Christian Leicher as chairman, alongside Andreas Pauly and Markus Fischer.

The group structure appears strategically centralized and operationally distributed. That is a logical deduction from a Munich-centered parent, global subsidiaries, and

major German industrial sites, alongside dozens of country companies and specialist subsidiaries such as LANCOM Systems, Rohde & Schwarz Cybersecurity, Rohde & Schwarz SIT, Zurich Instruments, ipoque, and R&S Marinesysteme. Public sources do not indicate listed group equity securities.

Key inflection points are visible in the company's official history. The company was founded in 1933, built Europe's first VHF transmitter in 1949, entered spectrum analysis in 1986, rode the GSM testing wave from 1992, acquired SIT and entered IT security in 1996, formalized a modern SDR family in 1999, entered quantum applications through Zurich Instruments in 2021, and added ZES ZIMMER and Munich Innovation Labs in 2025.

6. Business overview

The business today is organized into three divisions, which is a helpful way to understand the company economically and strategically. Test & Measurement is the broad instrumentation and solutions business. Technology Systems is the sovereign systems arm focused on government, military, and security use cases. Networks & Cybersecurity addresses secure networking and cyber requirements with a more software and IT-centric profile.

The current structure reflects a long shift from pure instrumentation toward a wider connectivity-and-security portfolio. The history pages show expansion from RF and measurement into mobile communications, encryption, radio-monitoring, software-defined radios, quantum measurement, and now OSINT and counter-drone. That evolution matters because it means the defense-security activities are not an opportunistic add-on. They are the result of decades of adjacent technical development.

In our judgment, T&M remains the central economic pillar by breadth, installed base, and global market visibility, while Technology Systems likely carries the greatest strategic and political relevance. Networks & Cybersecurity is smaller in terms of visibility but important because it anchors the group in European digital-sovereignty themes. That weighting is partly based on assumptions because public divisional revenue is not disclosed.

7. Position in the value chain

Rohde & Schwarz occupies multiple positions in the defense and industrial value chain. In T&M, it is clearly an OEM and solutions provider for research, design, validation, manufacturing, and regulatory testing. In secure communications and naval systems, it acts more like a subsystem provider and system integrator. In SIGINT/EW and selected defense programs, it appears to operate as a specialist systems supplier with integration capability rather than as a complete-platform manufacturer.

The naval communications business is the cleanest example of a higher-value-chain role. The company describes itself as a turnkey supplier for naval communications and presents NAVICS plus SOVERON as integrated onboard and external communication solutions. In air and multi-domain applications, the role looks more mixed: provider of radios, communications, SIGINT payloads, monitoring systems, counter-drone, and selected integrated mission systems.

That mixed placement is a competitive advantage. It lets the group benefit from both volume-like instrumentation markets and smaller, stickier, qualification-heavy sovereign programs. It also means peers differ by business line rather than



collapsing into one clean comparable set.

8. Products, services, and core capabilities

This is one of the most important sections for Rohde & Schwarz because product depth is the heart of the franchise. In T&M, the company is one of the world's largest manufacturers of electronic test-and-measurement equipment and covers RF and microwave applications up to 500 GHz, with strong positions in wireless, EMC, automotive, aerospace and defense testing, and research use cases. Public product pages emphasize precision, innovation, and breadth rather than price competition.

In defense and sovereign systems, flagship capabilities include SOVERON software-defined radios, NAVICS naval integrated communications, airborne and military ATC communications, SIGINT/EW systems, counter-drone solutions, and KORA-family electronic support systems. SOVERON is described as mission proven with thousands of deliveries; NAVICS is marketed as a fully IP-based, multilevel-secure naval communications ecosystem; and the air-defense pages describe the company as a leading provider of airborne and ATC communications, SIGINT payloads, drone defense solutions, and ground-based air defense systems.

In security and dual-use infrastructure, the R&S QPS201 security scanner provides an unusually visible public quality marker because it achieved TSA Qualified Products List status in 2022. The air-traffic portfolio is also notable: the history page states CERTIUM supports air traffic control at more than 300 airports and flight operations centers. Those kinds of installed-base indicators are valuable because they show operational validation rather than just brochure breadth.

The product portfolio also continues to move forward. Publicly announced growth items include the Zurich Instruments quantum expansion, the ZES ZIMMER acquisition for power-electronics testing, Munich Innovation Labs for AI-based OSINT, Orbint for space-based SIGINT, and the TRUMPF counter-drone partnership. For FY2025/26, the annual report also highlights 6G, FR3, XR, holographic communication, AI-RAN, NTN, Wi-Fi 8, electrification, software architecture, SIGINT, secure radios, drone defense, FCAS, and digital ATC as ongoing focal areas.

At product level, the clearest strengths are technical breadth, sovereign-grade trust, and long-cycle reliability. The key risks are more subtle: some high-value programs are project-heavy and execution-sensitive, some markets are cyclical, and the company's economic dependence by product line is not publicly disclosed.

9. Customers and routes to market

Rohde & Schwarz sells to a broad mix of industrial customers, regulators, public authorities, governments, armed forces, research institutions, and infrastructure operators. In T&M, buyers include industry, network operators, regulators, and labs. In defense and security, the customer base includes all branches of the armed forces, government authorities, and aerospace-and-defense companies. In cyber and networking, the target is governmental and commercial customers with elevated security and certification needs.

Routes to market vary by business line. T&M appears to sell largely through direct global sales and service, supported by local subsidiaries and service infrastructure. Defense sales appear more program- and government-driven, often with integration content and long life-cycle support. Specific named public references include German Navy F124 work, Polish Navy Miecznik-related EW collaboration,



TSA-approved security scanners for airport screening, and ATC systems at more than 300 airports or flight operations centers.

Revenue quality likely differs across the group. Instrument sales are partly transactional but supported by service, calibration, repair, and long installed bases. Defense, cyber, and communications systems are more likely to involve multi-year programs, qualification cycles, and aftermarket support. Public sources do not quantify recurring revenue or customer concentration, so any deeper split would be inference.

10. Key financials and economic profile

The latest public audited figures are solid. For FY2024/25, group revenue was €3,160.1m, net income was €364.2m, cash at banks was €591.9m, total equity was €2,245.5m, total assets were €5,424.6m, and liabilities to banks were zero. Pension provisions were €1,221.4m, which is an important balance-sheet item and should be tracked alongside the otherwise conservative financing profile.

The recent top-line trajectory has been steady. Official company disclosures show revenue of €2.28bn in FY2020/21, €2.53bn in FY2021/22, €2.78bn in FY2022/23, €2.93bn in FY2023/24, and €3.16bn in FY2024/25. That implies an externally calculated CAGR of about 8.5% from FY2020/21 to FY2024/25. FY2023/24 order intake reached €3.87bn, which signaled strong demand momentum going into the following year.

The economic profile appears healthy, but not in the way a listed peer would present it. Public sources point to positive operating results, significant retained earnings, strong equity backing, and ongoing high R&D intensity of roughly 15% to 20% of revenue. What is not publicly visible is the divisional margin bridge. Our assumption is that T&M and selected software/certification-heavy niches likely earn structurally better economics than larger integration-heavy or project-heavy businesses, but that is not disclosed as fact.

11. Footprint and industrial setup

Rohde & Schwarz has a broad global footprint but remains strongly anchored in Germany. Official sources point to operations in around 70 countries, with the parent in Munich and key German manufacturing bases in Memmingen and Teisnach. The group also highlighted plants in Germany, the Czech Republic, Singapore, and Malaysia as part of its vertically integrated production setup.

The FY2025 annual report lists 22 domestic and 77 foreign consolidated subsidiaries. These include specialist operating companies across Europe, the US, Israel, South Africa, Singapore, and elsewhere, as well as production, sales, and solution entities such as Zurich Instruments, LANCOM, Federal Systems, and R&S Vimperk in the Czech Republic.

Recent investment signals suggest ongoing capacity expansion. The annual report states that a highly modern technology and manufacturing center is being built in Memmingen and that the Malaysia site is being expanded. That is consistent with management's broader messaging around resilience, value creation depth, and supply capability.

12. Supply chain and key dependencies

The most visible structural feature of Rohde & Schwarz's operating model is high vertical integration. Management repeatedly presents it as a source of flexibility,



resilience, and quality control, especially in volatile supply-chain conditions. That matters because many of the group's products depend on high-spec RF electronics, secure components, precision manufacturing, software integration, and certification discipline.

Hard evidence on supplier concentration is limited in public sources. Still, the likely dependencies are clear by business type: semiconductors and RF components for T&M, secure electronics and cryptographic components for sovereign communications, software and compute for analytics/cyber, export approvals for sensitive defense products, and skilled engineering talent across RF, embedded software, and systems integration. Those dependency points are partly based on assumptions, but they fit the portfolio and the company's own emphasis on supply-chain, export-control, and technology leadership roles.

Critical facilities likely include Munich, Memmingen, Teisnach, Vimperk, and selected international sites. That creates both resilience through geographical spread and concentration risk around specialized German engineering and production nodes.

13. Management, board, ownership, and governance

Ownership is a central governance fact. Rohde & Schwarz states that it is 100% owned by the founding families and that third-generation family members remain on the executive board. Christian Leicher, a family representative, is the current chairman and CEO. That structure provides strong continuity and long-term alignment, but also means governance is owner-led rather than market-disciplined through listed-company norms.

The current executive board consists of Christian Leicher, Andreas Pauly, and Markus Fischer. Mr. Pauly joined the executive board as CTO effective October 1, 2023. Following Peter Riedel's resignation effective end-July 2024, Markus Fischer became COO effective November 1, 2024. The public leadership record therefore shows continuity plus controlled internal succession rather than disruptive turnover.

Below the executive board, Rohde & Schwarz has also refreshed wider corporate management. Christina Gessner took over the Test & Measurement division in 2023, and Mihaela Seidl became EVP Human Resources in February 2025, succeeding Holger Schötz, who moved within management into operations. Headcount was more than 15,000 as of 30 June 2025.

Governance formalization appears solid for a private company. Public materials show a dedicated compliance framework, a whistleblower system, an LkSG complaints procedure, and an internal compliance organization with reporting lines into the group center. What is not publicly visible is the kind of supervisory-board detail, formal independence metrics, or remuneration granularity that one would expect from a listed issuer.

14. Competitive positioning

Competition depends on business line. In T&M, Rohde & Schwarz competes most directly with Keysight, Anritsu, Viavi, National Instruments-related platforms, and other instrumentation players. External public filings are useful here: both Teradyne and Keysight list Rohde & Schwarz as a competitor in wireless and aerospace/defense-related test markets, confirming that it is viewed as a top-tier global peer rather than a niche local player.



In defense and sovereign systems, the relevant peer set is more fragmented. Depending on the line item, competitors likely include Thales, Hensoldt, Leonardo, Indra, Saab, L3Harris, and ELT Group, plus national specialists. We label that as inference because Rohde & Schwarz does not publish a clean competitor matrix by defense business. What is clear is that competition is less about lowest cost and more about RF depth, sovereign trust, certification, integration, installed base, and long-standing government relationships.

The company's main strengths versus peers are likely technical credibility, vertical integration, breadth across adjacent RF and security domains, and private ownership that allows long-cycle investment. Its main vulnerabilities are lower public visibility than listed primes, less financial disclosure, and the challenge of scaling project businesses without diluting execution discipline.

15. Strategic relevance and market implications

Rohde & Schwarz matters beyond its own P&L because it occupies several constrained capability nodes in the European technology and defense stack. These include secure tactical communications, naval integrated communications, RF sensing and SIGINT, trusted networking, cyber, ATC communications, and test infrastructure for future wireless and aerospace systems. In several of these areas, the company is less a commoditized supplier than a capability anchor.

Its relevance to European sovereignty is explicit in public materials. The company frames its mission around technological and digital sovereignty, participates in the FCAS ecosystem, and is involved in the German FCMS construct that Bundeskartellamt reviewed in 2022. The EU-backed IPCEI microelectronics project in Memmingen also reinforces the view that Rohde & Schwarz is part of a broader European industrial-policy agenda, not just a commercial vendor.

For listed peers and suppliers, the implication is that Rohde & Schwarz is both competitor and partner depending on the node. In several markets, it can act as a bottleneck-quality subsystem or integration partner rather than just a rival. That dynamic is visible in collaborations with ELT Group and TRUMPF.

16. Dual-use and civilian exposure

Rohde & Schwarz should not be treated as a pure-play defense company. The group's roots and a large share of its visible portfolio remain civilian or dual-use, especially in test and measurement for wireless, automotive, EMC, industry, research, and broadcasting-adjacent applications. The company's own materials repeatedly highlight industrial, regulatory, and government customers together rather than presenting defense as the sole end market.

That said, defense and sovereign-security exposure is clearly increasing in strategic importance. The air, SIGINT/EW, tactical communications, naval, cyber, and counter-drone portfolios all align with military and national-security demand. Recent acquisitions and partnerships also skew toward security, analytics, and defense-oriented sovereignty themes rather than toward consumer or low-end commercial markets.

On balance, civilian and dual-use exposure diversifies rather than dilutes the defense case. It gives Rohde & Schwarz broader technology flow-through, greater resilience to defense budget timing, and better access to adjacent innovation cycles. But it also means the company's headline identity can be misunderstood if one looks only through a defense-lead contractor lens.



17. Legal, regulatory, and compliance risks

Observed legal and regulatory risk appears moderate rather than acute based on reviewed public sources, but the risk concentration is clear: export controls, procurement law, cyber and encryption regulation, sanctions screening, supply-chain due diligence, and antitrust or national-security review around sensitive transactions. The company publicly emphasizes compliance governance, whistleblowing, and an LkSG complaints procedure, and the FY2025 annual report states that no individual or aggregated risks are currently identifiable that would endanger the group's continued existence.

Public evidence of major active litigation or large regulatory penalties is limited in the sources reviewed. That is a positive signal, but it also reflects private-company disclosure limits. One notable 2025 external issue was not a confirmed wrongdoing finding, but a procurement-process controversy: the Financial Times reported that German Bundestag budget lawmakers objected to a planned non-competitive award for a Rohde & Schwarz mobile reconnaissance system. The article underscores procurement-law sensitivity around sovereign-defense contracts, even if it did not allege proven misconduct by the company itself.

Antitrust and industrial-policy exposure look manageable and in some cases favorable. Bundeskartellamt cleared the FCMS-related joint-venture structure in 2022, while the EU-backed IPCEI microelectronics project shows official support for selected capability development. Export-control exposure is clearly real: multiple public job listings and compliance materials show the group maintains dedicated export-control and sanctions functions, which is exactly what one would expect for a defense- and encryption-adjacent portfolio.

IP is part of the entry barrier and part of the compliance landscape. The annual report says the value of the patent portfolio increased in FY2024/25, but public sources reviewed do not provide a patent count. The R&S brand is publicly identified as a registered trademark in company materials.

18. Funding, financing, and strategic optionality

Rohde & Schwarz appears conservatively financed. The company says it finances growth with its own resources, and the FY2025 accounts show substantial cash, high equity, and no bank debt. That profile gives it unusual strategic flexibility for a private company of its size. It can fund R&D, acquisitions, and capacity investments without obvious dependence on public equity or stressed credit markets.

Strategic optionality is broad. The most obvious path remains continued private compounding, supported by selective M&A and capacity expansion. The group has actively used acquisitions in 2021 and 2025 to expand into quantum, power electronics testing, OSINT, and space-related intelligence adjacencies. A future IPO is theoretically possible, but public materials do not point in that direction, and the current ownership model seems structurally comfortable.

A more realistic optionality set is continued tuck-in M&A, deeper JV or partner structures in defense electronics, and selective expansion in high-security software and integrated systems. The ELT and TRUMPF partnerships fit that logic.

19. Moat analysis

The primary moat is a combination of technical depth, trust, certification, and long qualification cycles in RF-, communications-, and sovereign-security markets. This is not a network-effects business. It is an engineering, installed-base, and trust



moat. Customers do not switch easily in mission-critical measurement, secure communications, or naval and ATC system environments once qualification, integration, and life-cycle support are embedded.

The moat appears stable to slightly widening. It is stable because the underlying barriers have existed for years: RF know-how, long customer relationships, installed base, and deep product stacks. It may be widening because sovereign demand and certification intensity are increasing, while the company is adding software, analytics, OSINT, quantum, and counter-drone layers that deepen solution content. That widening conclusion is partly inference.

Cost advantage is not the main story, though vertical integration likely helps margins and resilience. The stronger moat elements are switching costs, intangible assets, qualification cycles, and process know-how. NAVICS stresses robustness, redundancy, and no single point of failure. SOVERON emphasizes secure architecture and certification. The annual report also states that the value of the patent portfolio increased, reinforcing the intangible-asset case.

20. SWOT analysis

Strengths

- Strong RF and test-and-measurement franchise with global credibility and broad technology reach.
- High vertical integration, conservative balance sheet, and owner-led long-term capital allocation.
- Growing strategic relevance in sovereign communications, SIGINT/EW, cyber, ATC, and counter-drone.

Weaknesses

- Public disclosure is materially lighter than for listed defense peers.
- Business mix is complex, which makes peer comparison and outside valuation framing harder.
- Some businesses are project-heavy and execution-sensitive, especially in sovereign systems.

Opportunities

- European rearmament and sovereign-tech policy support.
- Expansion into OSINT, space-based SIGINT, counter-drone, and software-rich defense architectures.
- Continued 6G, AI-RAN, NTN, electrification, and quantum-related test demand.

Threats

- Export-control, sanctions, and procurement-law friction in sensitive programs.
- Competitive pressure from larger public peers in T&M and defense electronics.
- Cyclicality in parts of wireless and industrial end markets.

21. Key risks and open questions

- How large is Technology Systems economically relative to T&M? Public sources do not disclose divisional revenue, so the defense mix is still partly inferential.
- How much of future profit growth will come from software- and analytics-heavy businesses versus hardware-heavy instrumentation and systems?
- Can the group scale project-heavy defense programs without pressuring margins or execution quality?
- How durable is the recent defense/security acceleration if civilian wireless and industrial markets remain mixed?
- Will procurement frictions in Germany slow program conversion even when strategic demand is strong?
- How large is the hidden value of the patent, installed-base, and service ecosystem given the limited public disclosure?
- Does management eventually formalize governance and disclosure further, or is the current private-company model the enduring equilibrium?

22. What to watch next

- Evidence that FY2025/26 stable-growth guidance is being met.
- Progress on the new Memmingen technology and manufacturing center and the Malaysia expansion.
- Further disclosure or contract news around air, naval, SIGINT/EW, and counter-drone programs.
- Integration progress at ZES ZIMMER and Munich Innovation Labs.
- Whether the Orbint stake evolves into a deeper space-intelligence platform play.
- Further cyber and secure-workplace wins around LANCOM and Rohde & Schwarz Cybersecurity.
- New 6G, AI-RAN, NTN, radar, and electrification-related T&M launches or adoption signals.

Any further governance or leadership changes below executive-board level.

23. News flow and recent developments

The recent-news picture is active and strategically coherent. In November 2025, the company reported FY2024/25 revenue of €3.16bn and emphasized resilience in both security/defense and diversified T&M. Over the same period, it divested Pixel Power to sharpen focus on core operations, acquired ZES ZIMMER and Munich Innovation Labs, took a stake in Orbint, and announced strategic partnerships with TRUMPF and ELT Group.

Leadership news also mattered. Peter Riedel exited the executive board in 2024, Markus Fischer entered as COO in November 2024, and Mihaela Seidl joined corporate management for HR in February 2025. These changes do not look destabilizing, but they are important because the group is large, private, and execution-heavy.

From a market and policy perspective, Rohde & Schwarz also remained visible in defense procurement and industrial-policy conversations. The FT's September 2025 reporting on Bundestag objections to a planned non-tendered contract

showed that even strategically favored domestic suppliers can face process friction. At the same time, the company continues to benefit from European and German sovereignty narratives in defense and microelectronics.

24. Source notes and transparency limits

Company sources used

Primary company sources used here include the official company overview, product and solutions pages, history pages, legal notice, compliance materials, and the company's own press releases for FY results, executive changes, acquisitions, and partnerships.

Government and regulatory sources used

The most important government and quasi-official sources were the Bundestag lobby register filing that hosts the FY2024/25 audited HGB report, Bundeskartellamt's FCMS-related release, and the EU industrial-policy page for the Memmingen IPCEI microelectronics project.

Independent external sources used

Independent external sources were used selectively. The most relevant was the Financial Times report on Bundestag objections to a planned non-tendered Rohde & Schwarz procurement. Peer competitive context was also cross-checked through SEC-filed disclosures by listed companies such as Keysight and Teradyne.

Transparency limits

The main transparency limit is that Rohde & Schwarz is private and reports under German HGB standards rather than public-equity reporting norms. That means public sources do not give a full divisional revenue split, backlog by business line, customer concentration tables, detailed program margins, or the governance detail typical of a listed large-cap. Some analytical conclusions in this note therefore rely on careful analysis from product breadth, public contracts, legal structure, and audited group-level figures.

Explicitly labeled estimates and inferences

The following points were explicitly treated as estimates or inferences rather than disclosed facts: the approximate 41.4% equity ratio was externally calculated from the FY2025 balance sheet; the 8.5% FY2020/21-FY2024/25 revenue CAGR was externally calculated from company-reported revenue points; the view that T&M is still likely the largest economic pillar is a deduction from product breadth and visibility, not a disclosed segment split; and the assessment of defense competitors outside the T&M peer set uses business-line logic rather than a company-issued peer list.

Figure 26: Helsing - overview

Helsing GmbH			
Ownership:	Privately held, venture- and strategic-backed company; publicly named investors include General Catalyst, Saab, Accel, Lightspeed, Plural, Greenoaks, and Prima Materia, but exact stakes are not disclosed		
HQ:	Munich, 81671, Germany	**Founded:**	2021
Domain:	Multi-domain – defense AI, autonomy and software-defined strike/reconnaissance systems		
Value chain:	Software-first prime / integrator hybrid		
Why it matters	Helsing is Europe's flagship defense-AI player, linking existing platforms into AI-enabled networks while also fielding its own autonomous air, land and underwater systems		
Key products	Altra – recce-strike software; HX-2 – AI strike drone; SG-1 + Lura – autonomous underwater surveillance; Cirra – AI for electronic warfare; CA-1 Europa – autonomous combat-air platform under development		
Core capabilities	AI-driven target acquisition and autonomy, software integration into legacy hardware, EW-resilient strike systems and rapid movement from software into producible hardware		
Customers	Helsing says it works with governments and industry partners and has multiple significant government contracts; the clearest named public end-user is Ukraine for HX-2, while other public programs include the FCAS AI backbone and Eurofighter EW software integration		

Key financials (31 Dec)	2023	2024
P&L (EURm)		
Sales	10	27
Gross profit	9	23
EBITDA	-9	-7
EBIT	-10	-8
EBT	-8	4
Net income	-8	4
Order intake	-	-
Balance sheet (EURm)		
Cash & cash equivalents	229	587
Tangible fixed assets	2	2
Goodwill & intangible assets	-	-
Inventories	0	0
Accounts receivable	0	16
Other assets		
Total assets	293	752
Equity	289	743
Minorities	-	-
Borrowings	-	-
Accounts payable	2	4
Other ST liabilities	-	-
Net debt (+) / net cash (-)	-	-

Source : Deutsche Bank research, company data, Bloomberg Finance LP, Gain.pro

Helsing

Germany's Most Valuable Startup

What type of company it is

Helsing's core mission is to deliver advanced AI and software-defined autonomy to the defense sector. The company develops AI-powered systems that enable unmanned platforms to process data and operate independently in complex environments, with its technology having been battle-tested in Ukraine. This focus on creating a combat-proven 'autonomy stack' for military hardware is what established Helsing as Germany's most valuable startup, with an $18bn valuation following its $1.8bn Series E in July 2026, positioning it as a critical enabler of next-generation defense capabilities.

Figure 27: Helsing's Underwater Drone SG-1



Source : Helsing

Investment-style or strategy-style summary

From a strategy standpoint, Helsing looks like one of the most serious European attempts to build a defense platform around software-defined capability, not just around hardware replacement. Its operational anchor remains software and AI, but the company is clearly pushing into a broader industrial model that includes autonomous mass, sovereign supply chains, and domain-specific manufacturing nodes. That makes the company strategically important in areas where Europe sees bottlenecks: electronic warfare, AI-enabled strike systems, underwater surveillance, and autonomous air systems.

From a business-mix standpoint, the company now spans land, air, sea, and adjacent space capabilities. In land, Altra and HX-2 look like the clearest near-term operational stack. In air, Cirra, Centaur, Eurofighter EW work, and CA-1 form the core narrative. In sea, SG-1 Fathom and Lura anchor the maritime push, supported by the Plymouth factory and the Blue Ocean transaction. In space, the Kongsberg



teaming agreement extends Helsing's software role into sovereign ISR architecture. This is a wider and more platform-oriented business than the "defense AI software company" label used in early funding rounds would suggest.

Financially, the visible picture is still incomplete because Helsing is private. What is visible is that the company has raised at least c.€1.36bn across the disclosed Series A, B, C, and D rounds, has a three-digit million-euro Eurofighter AI contract disclosed in November 2025, and is investing in physical capacity in Germany and the UK. That combination implies a company that is still in heavy build-out mode, where industrial scaling and strategic positioning matter as much as current-period profitability.

Governance and legal structure also matter here more than they would in a typical startup profile. The June 2025 SE conversion, the official legal notice showing Helsing SE alongside multiple operating subsidiaries, and the formal administrative board with Daniel Ek and Tom Enders as co-chairmen suggest a company that is institutionalizing quickly. That does not remove execution risk, but it does suggest Helsing is being built as a pan-European defense group rather than as a narrow venture-capital-backed product company.

Key takeaways

- Helsing is now best viewed as a private European defense-technology platform with software roots and a growing all-domain systems footprint, rather than as a narrow AI-software vendor.
- Its strongest strategic claim is sovereign European relevance in software, autonomy, EW, and decision systems.
- Publicly disclosed funding is very large by European standards; including the $1.8bn Series E in July 2026, cumulative disclosed funding now exceeds c.€3bn across visible rounds.
- Helsing's valuation rose to $18bn with its $1.8bn Series E in July 2026, led by a syndicate including Dragoneer, Lightspeed, Goldman Sachs Alternatives, JPMorgan, and CPP Investments; the company remains predominantly European-owned.
- Near-term visible traction is strongest where software is tied to real programs or production, especially Eurofighter EW, Ukraine-related strike-drone production, and UK maritime manufacturing.
- The largest external analytical gap is financial transparency: no public annual report, no public consolidated P&L, and no public segment margins were identified in the reviewed materials.
- The main open question is whether Helsing can convert high strategic relevance and strong capital backing into repeatable industrial execution at scale.

1. Why this company is relevant now

Helsing is relevant now because it is building exactly in the areas in which Europe sees both operational urgency and industrial scarcity. The company's current public narrative is built around AI-enabled defense innovation across domains, and the surrounding fact pattern supports that this is more than branding. In roughly the last 18 months, Helsing has raised major growth capital, converted into an SE holding structure, acquired Grob Aircraft, announced or expanded maritime production in the UK, formalized a large Eurofighter EW contract, and widened its ecosystem through Kongsberg and Hensoldt. That combination makes it relevant not just as a company, but as a marker of how European defense-tech

industrialization may evolve.

2. What type of company it is

Helsing is a software defense company that is becoming more industrial. Its early official descriptions stressed AI software, software-only approaches, and government-plus-industry partnerships. Its current official descriptions now speak in broader all-domain terms and explicitly refer to deploying "our own software-defined hardware." That progression matters. In practical terms, Helsing sits between a defense software house, an autonomy-platform provider, and an emerging systems company with selective in-house manufacturing and hardware ownership. It is therefore more accurate to describe Helsing as a hybrid European defense-tech platform than as a single-product or single-domain startup.

Figure 28: Helsing's Centaur, Integrating AI Pilots into Existing Fighter Jets



Source : Helsing

3. Three things that matter most

First, Helsing's real significance comes from its place in the European sovereignty ecosystem, not just from its venture valuation. The company has positioned itself where governments want more local control: AI, EW, autonomy, target processing, and integrated mission systems. That makes the company strategically relevant even before its full economics are visible.

Second, the company is moving from 'AI for defense' into 'AI plus producible systems for defense'. The HX-2 production program, the resilience-factory concept, the Grob deal, the Plymouth site, and the Blue Ocean deal all point in the same direction. Helsing wants to own more industrial choke points and more of the physical layer around its software.

Third, Helsing is one of the clearest examples of a private defense company where strategy and public visibility are outrunning audited disclosure. That does not invalidate the story, but it means any serious assessment must keep two truths in mind at the same time: the company has real programs, real products, and real strategic partners, but external readers still have limited visibility into revenue quality, margins, backlog, and cash generation.



4. At a glance

Helsing is headquartered in Munich and now organized around Helsing SE, with official company materials also showing operating entities including Helsing GmbH, Helsing Germany GmbH, Helsing SAS, and Helsing Limited. The founders are Gundbert Scherf, Niklas Köhler, and Torsten Reil, who remain central to leadership. The company's own pages show a formal administrative board including Daniel Ek and Tom Enders as co-chairmen, with the founders also represented at board level. This is already a more institutional governance setup than many venture-capital-backed European defense start-ups of similar age.

At a high level, Helsing's business today spans AI mission software, land strike software and drones, air autonomy and EW software, maritime autonomy, and emerging space-enabled intelligence systems. The industrial role is therefore wider than a normal middleware or analytics provider. It is becoming a company that wants to pair AI and autonomy with sovereign European production capability. The strongest public signals of that ambition are the June 2025 Grob acquisition, the November 2025 Plymouth maritime factory, and the February 2025 drone-production announcement for Ukraine.

5. Corporate history and legal structure

Helsing was founded in 2021 in Munich by Gundbert Scherf, Niklas Köhler, and Torsten Reil. The early company was presented as a defense-AI company focused on software-based capabilities to protect democracies, with operations across the UK, Germany, and France. The early funding arc was unusually strong: the Series B press release states that the 2023 round followed a €102.5m Series A led by Prima Materia, and in July 2024, Helsing announced a €450m Series C. In June 2025, the company announced a €600m Series D, also led by Prima Materia.

The legal structure has also evolved materially. On 2 June 2025, Helsing announced that it had concluded its transformation into a Societas Europaea, with Helsing SE entered in the commercial register on that date. The company said the new SE holding structure was intended to underscore its European identity and facilitate expansion across Europe. The official legal notice shows Helsing SE represented by Dr. Gundbert Scherf, Torsten Reil, and Niklas Darius Köhler, and it also lists subsidiary entities in Germany, France, and the UK. That combination strongly suggests a holding-company model over a multi-jurisdiction operating footprint.

The corporate timeline is fast-moving. In 2023, Helsing publicized its Series B and linked that round to Eurofighter EW selection and FCAS AI infrastructure work. In 2024, it announced Project Centaur and then the HX-2 strike drone and a €450m Series C. In 2025, it announced 6,000 additional strike drones for Ukraine, unveiled Lura and SG-1 Fathom, converted into an SE, signed the Grob acquisition, unveiled CA-1 Europa, opened the Plymouth resilience factory, and announced the signed Eurofighter AI contract. In late 2025 and early 2026, it expanded further into Blue Ocean, Kongsberg space cooperation, and the Hensoldt CA-1 partnership.

6. Business overview

The simplest way to describe the business is that Helsing uses AI and software to make defense systems faster, more autonomous, and more useful in contested environments. That original logic still sits at the heart of the company. What has changed is the breadth of the product stack. Helsing now presents itself as working across governments and industry to transform existing hardware assets and deploy its own software-defined hardware. In other words, it no longer looks like a pure software layer sitting on top of others' platforms.



The current business appears to have four practical pillars. The first is software for mission systems and AI-enabled decision-making. The second is autonomy stacks for land and air operations. The third is proprietary or quasi-proprietary hardware platforms such as strike drones, underwater gliders, and the CA-1 development effort. The fourth is industrial scaling through acquisitions, partner ecosystems, and resilience factories. The broad pattern is that Helsing is taking a software-led core and surrounding it with more physical and industrial control.

7. Position in the value chain

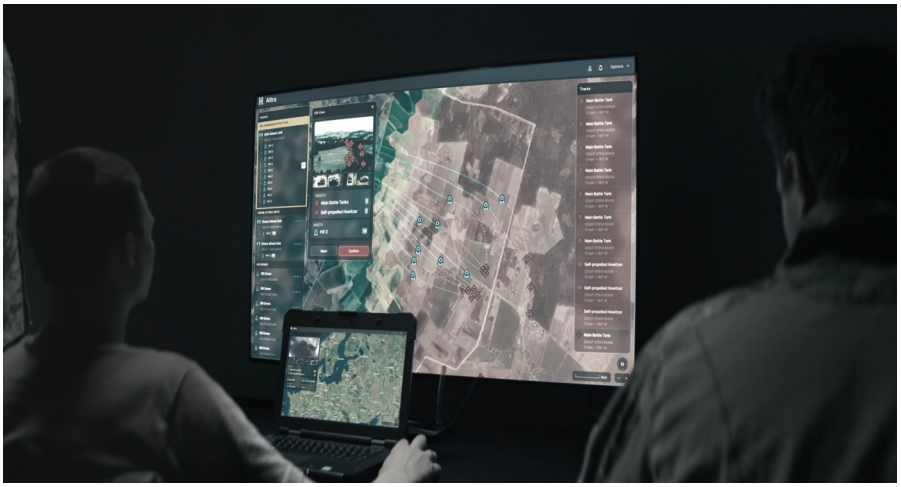
Helsing sits in multiple places in the value chain at once. In some programs, it acts as a subsystem or mission-software provider, as seen in Eurofighter EW where its Cirra AI software is integrated into Saab's Arexis sensor suite. In others, it looks more like a software-and-autonomy system house, as with Altra and HX-2, where software, targeting logic, network integration, and strike-drone functionality are bundled together. In still other areas, particularly CA-1 and maritime autonomy, it is moving closer to an OEM or systems-integrator position by combining proprietary software with owned or controlled manufacturing capability.

The important analytical point is that Helsing is not trying to occupy only the 'brain' layer of defense systems. It is trying to own more of the value chain where owning the stack improves speed, autonomy, and sovereign control. That rests on the analysis of the product portfolio and transactions, but it is a strong one. The Grob acquisition, the Blue Ocean transaction, the UK factory, and the resilience-factory concept all point to a company that wants more manufacturing and production leverage around its software core.

8. Products, services, and core capabilities

The most economically and strategically important part of Helsing is still likely the software and AI layer. The clearest publicly visible examples are Altra in land combat, Cirra in electronic warfare, and Centaur in air autonomy. Altra is positioned as a reconnaissance-and-strike software platform that connects battlefield elements, fuses ISR inputs, and supports target assignment, fires correction, and swarm control. Cirra is Helsing's AI-based EW software now contracted for integration into the German Eurofighter EK effort with Saab. Centaur is Helsing's autonomous air-combat AI, first disclosed in 2024 and then demonstrated in a Saab Gripen E in June 2025. These software-centric capabilities are likely the highest strategic-value part of the portfolio because they are harder to substitute, more reusable across platforms, and closer to the company's original technical identity. The margin point is an assumption, because Helsing does not disclose segment margins publicly.

Figure 29: Altra - An AI Battlefield Intelligence Platform



Source : Helsing

The second major pillar is strike and autonomy hardware. HX-2 is Helsing's software-defined, EW-resistant strike drone with a stated range of up to 100 km, swarm capability via Altra, and a human operator staying in or on the loop for critical decisions. In February 2025, the company announced production of 6,000 additional HX-2 strike drones for Ukraine, following an earlier order of 4,000 HF-1 drones. Those numbers matter because they make the drone business more than a demonstrator effort. They suggest that, at least in one category, Helsing has moved into meaningful production and operational deployment.

The third pillar is maritime autonomy. In May 2025, Helsing unveiled Lura, an AI software platform for underwater surveillance, and SG-1 Fathom, a mass-producible autonomous underwater glider powered by Lura. In November 2025, it opened its first UK Resilience Factory in Plymouth, where SG-1 is produced and where the site serves as the Maritime Centre of Excellence. The company later announced its October 2025 acquisition of Blue Ocean to add autonomous underwater vehicle hardware and manufacturing capability in the UK and Australia; the deal was completed in May 2026. This sequence suggests the maritime business is becoming a real vertical rather than a side project.

Figure 30: CA-1, Using Centaur to create a combat UAV



Source : Helsing

The fourth pillar is advanced air and adjacent space. CA-1 Europa, unveiled in September 2025, is Helsing's autonomous UCAV concept built around its Centaur autonomy stack and Grob's aircraft-design capability. In February 2026, Helsing and Hensoldt announced a partnership centered first on CA-1, with Hensoldt to supply sensors. In December 2025, Helsing and Kongsberg also announced a teaming agreement for a sovereign European space-based IST constellation with Hensoldt contributing sensors. These programs are not yet proof of scaled commercial success, but they materially deepen Helsing's strategic role in air and space-enabled defense systems.

9. Customers and routes to market

Helsing sells primarily to governments, armed forces, and government-backed defense programs, usually either directly or through industrial teaming. The public record supports three main routes to market. One is direct or near-direct government procurement, as in the Ukraine-related drone production and UK maritime investment. A second is co-selling or subsystem insertion through defense system providers and industrial partners, as seen with Saab on Eurofighter and with the FCAS-related consortium work mentioned in the Series B announcement. A third is newer, ecosystem-based program formation where Helsing's software is one layer in a broader sovereign stack, as with Kongsberg and Hensoldt in space and CA-1-related work.

Figure 31: AI-Enabled Eurofighter Jet, - Helsing & Saab Cooperation



Source : Helsing

From a revenue-quality perspective, the business likely mixes development contracts, production lots, and platform/software integration work. That means revenue quality is probably better than one-off prototype revenue, but still less annuity-like than a classic SaaS model. Publicly, the clearest evidence of monetizable traction is the signed three-digit million-euro Eurofighter AI contract, the large Ukraine drone-production announcement, and the UK maritime manufacturing commitment. What remains unclear is how much of Helsing's revenue base is recurring software-like support versus hardware deliveries and milestone-based program income. That ambiguity is one of the central limits of public analysis.

10. Key financials and economic profile

Helsing is private and does not publish the level of financial detail that a listed issuer would. We did not identify any public consolidated annual report with full P&L, balance sheet, cash-flow statement, segment reporting, or margin disclosure in the reviewed public materials. What is clearly public is the funding history: the Series B release states the company had previously raised €102.5m in Series A; Helsing then announced €209m Series B in September 2023, €450m Series C in July 2024, and €600m Series D in June 2025. On that basis, the publicly disclosed major rounds add up to at least c.€1.36bn.

On valuation, the company's own June 2025 announcement did not specify a valuation, but Reuters reported, citing the Financial Times, that the round boosted Helsing's valuation to about $12bn. That reported valuation places Helsing in the top tier of European private tech by headline value, but it should still be treated as externally reported rather than officially disclosed by the company.

The best public read-throughs on economic profile are therefore indirect. Helsing is spending like a company in scale-up mode: it is funding its own R&D, buying or building industrial capacity, producing drones at volume, opening a UK factory, and pushing into more capital-intensive air and maritime programs. Publicly visible commercial traction includes the three-digit million-euro Eurofighter AI contract



and the Ukraine-related drone production program. What remains unknown is whether the current mix is predominantly R&D-heavy and investment-led, or already generating substantial gross profit from software and systems delivery.

11. Footprint and industrial setup

The company's footprint is explicitly pan-European. Official company materials refer to offices in Munich, London, Paris, and Berlin, while legal-notice materials show German, French, and UK entities under the group. The headquarters and corporate center of gravity are clearly in Munich, but the operating footprint is broader and designed to meet sovereign requirements in multiple jurisdictions.

Industrial capacity is becoming more visible. Grob Aircraft in Tussenhausen gives Helsing composite-aircraft design and manufacturing capability, with Grob described as employing around 275 people. The Plymouth site gives Helsing a maritime center of excellence, production capacity for SG-1 Fathom, and a local UK R&D/manufacturing base. The Blue Ocean transaction extends maritime industrial capability into Australia and the UK. The emerging footprint is therefore not just geographical. It is increasingly domain-specific: Bavaria for air, Plymouth for maritime, and a multi-country software and program-management base around those physical nodes.

12. Supply chain and key dependencies

There is unusually clear publicly available information that Helsing sees supply chain and industrial control as strategic issues rather than back-office issues. The Grob transaction brings composite-aircraft know-how and airframe production in-house. The Blue Ocean deal adds underwater hardware and manufacturing capability. The Kongsberg-Hensoldt-Isar ecosystem around space shows Helsing relying on specialist partners where it does not own the full stack. And the Reuters report on Helsing's December 2025 cooperation with Schaeffler shows the company actively pursuing component supply and resilient drone production scaling with German industrial partners.

Beyond those hard signals, the likely dependencies are what one would expect in this category: sensors, propulsion, batteries and power systems, secure communications, EW resilience, advanced manufacturing, export permissions, and scarce software and AI talent. That second layer is an inference, but it is strongly supported by Helsing's product design and partner map. A company building EW software, autonomous aircraft, strike drones, underwater gliders, and space-enabled fusion systems inevitably depends on a mix of high-end electronics, certified manufacturing, and regulated cross-border technology flows.

13. Management, board, ownership, and governance

Leadership appears founder-led and relatively stable. The company page identifies Gundbert Scherf and Torsten Reil as co-CEOs and Niklas Köhler as President and Chief Product Officer. It also identifies a broader management team that includes Antoine Bordes as Chief Scientist, Mark Griffiths as VP Finance, Johannes Boie as Chief Marketing Officer and Communications, Sam Rogerson as COO, and Robert Fink as CTO. The official legal notice separately shows Helsing SE represented by Scherf, Reil, and Köhler, which indicates that founder control remains very direct at the legal-representative level.

Governance is more formal than at many private defense-tech peers. Helsing's company page shows an administrative board with Daniel Ek and Tom Enders as co-chairmen, with Jeannette zu Fürstenberg and Denis Mercier as members and



the founders also represented. That board composition matters because it combines capital, industrial credibility, and military experience. The June 2025 SE conversion further suggests an intentional step toward a more scalable European governance architecture. Ownership remains private. Public sources identify investors and strategic backers such as Prima Materia, General Catalyst, Saab, Lightspeed, Accel, Plural, and others, but we did not identify a public cap table with precise current percentages.

14. Competitive positioning

Helsing does not compete against one neat peer set. In land-strike and battlefield software, it competes against other reconnaissance-and-strike software providers, autonomy vendors, and drone-stack suppliers. In air, it overlaps with established defense groups and subsystem vendors in EW, autonomy, and future-combat-air architectures. In maritime autonomy, it faces specialist underwater and ISR players. And at a broader category level, it is part of the same global conversation as venture-capital-backed defense-tech platforms in the US that combine software, autonomy, and manufacturing. That means Helsing's competitive position should be understood by business line, not through a single simplistic peer group. The 'real' competition is fragmented by domain. This is a deduction from the company's disclosed activities and partner set.

Its strongest differentiator is probably the combination of European sovereign positioning with software depth and rising industrial ambition. Its vulnerabilities are the mirror image of that strength. It is still younger than the established defense groups, less transparent than a listed contractor, and now taking on the harder task of industrializing hardware-heavy programs at a pace that could expose execution weakness. In that sense, Helsing's competitive positioning is strongest where speed, AI capability, and sovereign narrative matter most, and weakest where long qualification histories, legacy installed bases, or mature production scale dominate procurement decisions.

15. Strategic relevance and market implications

Helsing matters beyond its own private-company profile because it sits in categories that have become strategically scarce in Europe. The company is active in air combat AI, EW software, strike-drone production, underwater surveillance, and adjacent space-based IST. Those are not random product lines. They are areas where European governments increasingly want sovereign or at least regionally controlled capability, faster iteration, and less dependence on foreign software and sensing stacks. Helsing therefore matters as a possible capability anchor in parts of the value chain that are becoming politically and militarily sensitive.

For the wider market, the company's rise signals that European defense value creation may move further toward software-led system design and vertically integrated autonomy rather than staying confined to legacy hardware cycles. The likely implications are strongest for established defense groups, sensor suppliers, propulsion/component vendors, and adjacent listed defense names that may increasingly need software-native partners or face pressure from software-led challengers. That second point is inference, but it is consistent with Helsing's partnership map and current industrial direction.

16. Dual-use and civilian exposure

Helsing should be thought of primarily as a defense company, not as a balanced dual-use group. Its official company language, product portfolio, funding narrative, and recent industrial announcements are overwhelmingly defense-oriented. The



one obvious nuance is that Grob Aircraft brings a heritage in both military and general aviation, and Blue Ocean came from broader ocean-technology roots. But Helsing is acquiring or integrating those capabilities explicitly to accelerate defense programs, not to create a diversified civil-industrial portfolio.

That means civilian exposure probably diversifies manufacturing know-how more than it diversifies earnings. Publicly, the economic anchor is clearly defense: democracies-only positioning, Ukraine-linked drone production, Eurofighter EW, maritime surveillance, sovereign space-based IST, and autonomous combat aircraft. On current evidence, Helsing is much closer to a defense pure-play with some dual-use industrial inputs than to a mixed civil-defense industrial group.

17. Legal, regulatory, and compliance risks

Because Helsing is private, the public litigation and regulatory picture is not as easy to read as it would be for a listed issuer. We did not identify a public annual-report-style legal proceedings schedule or a detailed disclosed regulatory-risk matrix. What is visible is the regulatory perimeter around the business model. Helsing operates in defense procurement, export-sensitive technologies, AI-enabled weapons-related systems, M&A subject to approvals, and cross-border sovereign-industrial partnerships. The company's own releases note that the Series C and Series D financings were subject to government or other approvals, and the Grob acquisition was announced as subject to customary regulatory approvals (obtained in the meantime). Those are normal for the sector, but they still matter because they show how dependent the company is on regulated pathways.

Compliance structure appears to be formalizing alongside scale. The official legal notice shows Helsing SE and multiple subsidiaries, legal representatives, and a modern-slavery statement that applies across the group including Helsing SE and subsidiaries. The company also repeatedly stresses ethics, transparency, explainability, and human oversight in its AI systems, including explicit human-in-or-on-the-loop language for HX-2. Those features are not proof against future controversy, but they show that Helsing understands the political and regulatory sensitivity of its category.

The most visible recent reputational risk in reviewed public sources is the April 2025 Bloomberg report alleging overpriced drones and glitchy software, with the company standing by its products. We did not identify a corresponding public major fine or formal enforcement action in the reviewed official materials, so the current public risk picture is better described as one of procurement, export-control, execution, and reputational sensitivity than of any disclosed regulatory sanctions. Still, the combination of weapons relevance, AI, and rapid scale-up means legal and compliance risk should be treated as structurally important even where the public enforcement record remains limited.

18. Funding, financing, and strategic optionality

Helsing has financed growth through private equity-style capital raising rather than through public markets. The disclosed path from Series A to Series D shows repeated ability to attract both financial and strategic backers, with Prima Materia, General Catalyst, Saab, Lightspeed, Accel, Plural, and others participating across rounds. That is strategically useful capital, not just passive valuation support, because parts of the investor base also bring industrial or geopolitical signaling value.



The strategic optionality is broad but mostly inferential. An eventual IPO is conceivable given the SE structure, the board formalization, and the sheer scale of capital already raised, but there is no official public indication of a listing plan (as of mid-2025, the company stated there were no plans for an IPO). Continued private compounding is equally plausible, especially if the company keeps converting funding into assets, programs, and sovereign partnerships. M&A is already part of the playbook through Grob and Blue Ocean. In other words, Helsing's optionality lies less in near-term financial engineering and more in whether it wants to become a large stand-alone private European defense platform, a future listed defense-tech champion, or the nucleus of a wider sovereign-defense ecosystem.

19. Moat analysis

Helsing's primary moat is best described as a combination of sovereign trust, software depth, and increasing control over the physical systems that matter most to its software. The moat is not a classic consumer brand moat. It is a defense moat based on government access, political alignment, integration into sensitive programs, autonomy algorithms, and the ability to pair those algorithms with producible systems. Eurofighter EW, Centaur integration into Gripen E, Ukraine drone production, and the build-out of maritime and air manufacturing all support that framing.

We would describe the moat as strategically widening but economically still unproven. It is widening strategically because the company is adding partners, manufacturing assets, programs, and domain breadth. It is still unproven economically because the public record does not yet show the durable installed-base economics, support revenues, or multi-year disclosed financial returns that would make the moat easy to quantify from outside. The strongest specific moat sources appear to be software know-how, trust and access in democratic defense markets, integration into regulated and qualification-heavy domains, and increasingly the ownership of capacity in constrained nodes such as aircraft composites and maritime autonomy manufacturing. That is partly inference, but it is the most evidence-consistent reading of the current public facts.

20. SWOT analysis

Strengths

- Strong sovereign-European positioning in AI, autonomy, and mission software.
- Rapid product expansion across land, air, sea, and adjacent space.
- High-quality partner and investor ecosystem including Saab, Kongsberg, Hensoldt, Prima Materia, and General Catalyst.
- Increasing industrial control through Grob, Plymouth, resilience factories, and Blue Ocean.

Weaknesses

- Limited public financial transparency for an externally rigorous valuation framework.
- Hardware expansion raises capital intensity and execution risk.
- Public economics are easier to see in fund-raising than in disclosed profitability or backlog.



Opportunities

- European demand for sovereign AI, EW, drones, maritime surveillance, and autonomous air systems.
- Potential to become a cross-domain integrator rather than a point-solution vendor.
- Space and sensor partnerships could widen the addressable system role.

Threats

- Procurement and industrial execution risk as the company scales physical systems.
- Competition from established defense contractors and other autonomy-led defense platforms. This is an inference from the market structure.
- Political, export-control, or regulatory friction around AI-enabled defense systems.

21. Key risks and open questions

The first open question is whether Helsing can convert strategic relevance into a financial profile that justifies the reported valuation. The second is whether the company can scale hardware-heavy programs without eroding the speed and capital efficiency that made its software-first story attractive in the first place. The third is how much recurring or sticky software-driven economics sit underneath public program announcements. The fourth is whether the current partner-heavy model will remain a strength or create future dependency in key domains such as sensors, satellites, propulsion, and certified manufacturing. The fifth is how much of Helsing's moat derives from true product performance versus timing, policy alignment, and capital abundance. The sixth is whether the company can institutionalize governance, compliance, and industrial discipline as fast as it is institutionalizing ambition.

22. What to watch next

The most important things to watch over the next 6 to 18 months are the pace of Eurofighter EW integration, additional evidence of HX-2 and related production scale, the ramp of the Plymouth maritime site, completion and operational integration of the Blue Ocean transaction, further technical and industrial milestones on CA-1 Europa, the depth of the Hensoldt relationship, and whether the Kongsberg-Helsing-Hensoldt space initiative turns into a more formal program with clearer budget backing. Also worth watching is whether Helsing adds more industrial partners on the German supply-chain side, as suggested by the Schaeffler cooperation.

23. News flow and recent developments

In February 2026, it added Hensoldt as a CA-1 partner, and Germany's budget committee approved an initial ~€269m HX-2 order (alongside STARK Defence), with options reported up to €1.46bn, to equip the brigade in Lithuania. In May 2026, Helsing conducted the first HX-2 launch from a coastal vessel off Plymouth and, with OHB, formed the space joint venture KIRK - taking joint lead of the Kongsberg-Hensoldt consortium bidding for the Bundeswehr's Spock 2 program - followed in June by MoUs with seven space start-ups. At ILA Berlin in June 2026 it unveiled the CA-1EA electronic-attack variant and confirmed a kinetic attack variant of CA-1 Europa, with first flight targeted for early 2027. In July 2026, it raised a $1.8bn Series E at an $18bn valuation - the largest defense-technology round in European history



- and selected Martinsburg, West Virginia for its first US Resilience Factory, a $50m HX-2 site targeting more than 2,000 drones per month. In August 2026, it enlisted Rakuten to broker an HX-2 sale to Japan's Ground Self-Defense Forces. Leadership was also deepened with a new CFO, CCO, and a dedicated Managing Director for Ukraine.

24. Source notes and transparency limits

Company sources used

The company-source base used here consisted mainly of Helsing's official company pages, management and board pages, product pages, legal-notice pages, and press releases covering funding rounds, SE conversion, product launches, acquisitions, factories, and program announcements.

Government and regulatory sources used

The reviewed public record shows Helsing describing approvals, procurement selection outcomes, and registry entry points, but the most directly visible public documentary layer in this pass was still company disclosure rather than a rich body of official regulator publications. Saab's own release on Arexis orders for the German Eurofighter program was used as an important quasi-primary partner source for program value and delivery timing.

Independent external sources used

Reuters was used primarily for the externally reported June 2025 valuation and for selected broader market or transaction context. Bloomberg was used only in the risk section to note publicly reported allegations and the company's stance. Those external sources help where the company does not disclose valuation or where reputational questions arise, but they do not replace the underlying limitation that Helsing is private.

Transparency limits

The main transparency limit is financial. We did not identify public audited consolidated financial statements, public segment margins, public backlog, public customer concentration disclosures, or a public cap table with current ownership percentages. That means the business can be described with reasonable confidence at the level of strategy, programs, products, structure, and funding, but not with the same precision one would expect for a listed defense contractor.

Explicitly labeled estimates and inferences

A few conclusions above are explicitly inferential rather than directly disclosed. These include the view that software likely carries the highest strategic and margin value in the portfolio, the view that the moat is strategically widening but still economically unproven, the interpretation that Helsing is moving up the value chain toward a broader systems role, and the arithmetic calculation that the disclosed major rounds total at least c.€1.36bn. The reported c.$12bn valuation was treated as externally reported by Reuters citing the Financial Times, not as an official company disclosure.

STARK Defence

Fusing AI-Enabled UAVs / USVs with a Unified Integration Platform

Figure 32: STARK Defense - overview

SKD SE			
Ownership:	Privately held, VC-backed defense-tech company; publicly named investors include Sequoia Capital, NATO Innovation Fund, Project A, and In-Q-Tel, while exact ownership percentages are not disclosed		
HQ:	Berlin, 10117, Germany	**Founded:**	2024
Domain:	Multi-domain – loitering munitions, unmanned surface vessels and C2 software		
Value chain:	OEM plus software-enabled defense-tech hybrid		
Why it matters	STARK moved from founding to frontline deployment in 13 months and already won a Bundeswehr framework contract for Virtus, making it one of the faster-scaling European autonomous-strike startups		
Key products	Virtus – loitering munition / one-way effector; Vanta-4 and Vanta-6 – unmanned surface vessels; Minerva – AI command-and-control software for multidomain operations		
Core capabilities	Rapid development-to-fielding, scalable production, software-linked multidomain operations and sovereign Europe/NATO positioning		
Customers	The named public customer is the Bundeswehr / BAAINBw for Virtus; Vanta has also been trialed in NATO exercises in Portugal, and STARK says it operates across Germany, Ukraine, the UK, Greece and Sweden		

Source : Deutsche Bank research, company data

What type of company it is

STARK is a venture-capital-backed European defense-technology scale-up that has crossed from startup into sovereign-supplier status. It is not a traditional prime contractor, but neither is it a niche component maker. In practical terms, it looks like a hybrid system house and integrator, vertically scaling around three pillars: a loitering-munitions family built on the Virtus system, an unmanned surface-vessel family, and a command-and-control software layer designed to connect unmanned systems across air, land, and sea. Rather than supplying subsystems, it develops its own effectors, software, and production capacity for volume output.

Its customer base is defined by state procurement. A February 2026 framework agreement to supply Virtus to the German armed forces contemplates deliveries from 2026 and several thousand systems within two years following qualification, alongside a separate loitering-munitions contract with an unnamed NATO member state. Geographically, it is a multi-country operator, with a footprint spanning Germany, Sweden, Greece, and Ukraine. In short, it is a young company with the contract validation, production ambition, and policy tailwind of a far more established defense prime. That characterization is partly factual and partly based on assumptions, but it fits the reviewed public materials closely.



Investment-style or strategy-style summary

STARK appears to be building a focused European unmanned-systems platform around strike, maritime autonomy, and mission software. The flagship product today is Virtus, which has become the company's visible commercial spearhead. Minerva matters because it suggests STARK wants to own the software and mission layer, not just the vehicle. Vanta broadens the company into maritime deterrence and helps support a multi-domain identity.

Strategically, the company sits in the right place in the current market. Europe is trying to rebuild domestic capacity in loitering munitions, autonomy, and scalable low-cost mass. STARK's product set maps directly onto those needs, and its recent footprint expansion into the UK, Sweden, Greece, and Ukraine reinforces the view that management is building a pan-European industrial network rather than a Germany-only supplier.

Financially, public visibility remains thin. The company discloses that more than 45% of expenditure is invested into R&D, and external reporting points to a $62m round at a $500m valuation in August 2025, followed by a €500m round in June 2026, co-led by Sequoia Capital and Founders Fund, that lifted the valuation above €3.5bn and brought total funding since founding to about €640m. Those figures imply capital-market support and rapid ambition, but they are not substitutes for revenue or profitability disclosure.

From a governance and regulatory perspective, the company is more formalized than many very young start-ups because it operates as an SE with a supervisory board. At the same time, ownership percentages, full supervisory-board composition, and detailed compliance infrastructure are not yet visible in the reviewed public materials. Political scrutiny around Peter Thiel-linked capital and the battlefield-performance debate in Ukraine show that reputation and procurement optics will matter almost as much as engineering.

1. Why this company is relevant now

STARK matters now because it is showing up where European defense demand is most acute: affordable, scalable, software-enabled unmanned strike systems. The February 2026 Bundeswehr Virtus framework (see 23. News flow) moved the company into the center of a strategically important capability build for the Bundeswehr and Brigade 45 in Lithuania. That is a stronger signal than general investor enthusiasm because it reflects buyer validation under a sovereign procurement process.

The timing is also favorable at the industrial level. STARK has paired procurement momentum with a visible pan-European manufacturing and R&D rollout: a Swindon factory in the UK, a Nordic office in Stockholm focused on EW-related capabilities, a new office in Greece, and an R&D hub in Ukraine. Publicly, the company is trying to position itself not as a niche German drone startup but as a distributed European defense manufacturer.

2. What type of company it is

STARK is best understood as a private defense-technology platform focused on unmanned systems rather than as a classic large weapons manufacturer. The reviewed product set spans one-way effectors, unmanned maritime systems, and a mission-software layer. That implies a mixed model: part hardware producer, part systems integrator, and part software company. The company's public materials consistently emphasize rapid iteration, modularity, and interoperability rather than

long-cycle platform primeship.

3. Three things that matter most

The first thing that matters is procurement credibility. STARK is no longer just selling a concept. Public sources show a German framework agreement, another NATO-country contract, battlefield use in Ukraine, and NATO maritime trials. That is enough to move the company into a higher strategic category than most 2024-vintage start-ups.

The second thing that matters is an integrated operating concept. Virtus is the product that wins attention, but Minerva may be the deeper strategic asset because it sits at the command, control, and integration layer. A company that controls both the effector and the mission stack can be harder to displace than a company that sells only airframes.

The third thing that matters is transparency risk. STARK is private. Public sources give strong evidence of momentum, footprint, and investor support, but weak evidence on revenue, gross margin, cash generation, and backlog. For an analyst or banker, that means the company's strategic relevance is easier to establish than its normalized economics.

4. At a glance

STARK is a Berlin-based defense technology company operating through SKD SE. It was founded in 2024 and has built a product portfolio around the Virtus loitering munitions, the Vanta unmanned surface-vessel family, and the Minerva command-and-control system. Publicly disclosed operations now span five countries: Germany, Ukraine, the UK, Greece, and Sweden.

The company's importance comes less from disclosed financial scale and more from where it sits in the current defense cycle. It addresses one of Europe's most urgent industrial gaps: scalable unmanned strike and autonomy capabilities. Its February 2026 Bundeswehr framework agreement, February 2026 NATO-country order, and UK production expansion give it more practical relevance than many private peers whose traction remains largely venture-led.

Public financial disclosure is limited. STARK's website discloses R&D intensity and footprint, but not revenue or earnings. External reporting suggests major funding support and rising valuation, which helps explain how the company has been able to scale staff, sites, and product development quickly.

Key takeaways

- Headquarters and legal seat are in Berlin.
- The company is private and venture-capital-backed, not publicly listed.
- Core identity is multi-domain unmanned systems, not just one drone program.
- Public disclosure supports strategic relevance, but not a full financial model.

5. Corporate history and legal structure

The current legal entity visible in public materials is SKD SE, registered in Berlin, with registered office at Französische Str. 12, 10117 Berlin. It is a Societas Europaea and is represented by an executive board that currently includes Uwe Horstmann,



Dr. André Schneider, Johannes Schaback, Jan-Patrick Helmsen, and Martin Rost. The chairman of the supervisory board is Robert Maier. Registry-based sources indicate an incorporation date of 14 February 2024. No listed securities were identified, and no parent holding structure or major subsidiaries are clearly laid out in the reviewed company materials. On current evidence, the group appears relatively centralized around the Berlin SE, though that last point is an inference.

STARK was founded in 2024 by Uwe Horstmann, Florian Seibel (who also founded Quantum Systems), and Johannes Schaback. Mr. Seibel has since stepped back to a founding-investor role, while Mr. Horstmann has served as CEO since October 2025.

Concise timeline of key inflection points

- 2024 - Incorporation of SKD SE: Registry sources show incorporation on 14 February 2024. Public company messaging later framed 2024 as the founding year.
- September 2025 - Launch of Vanta after NATO trials: STARK introduced the Vanta-4 and Vanta-6 unmanned surface vessels following NATO REP(MUS)/Dynamic Messenger activity in Portugal. This marked an explicit extension into maritime autonomy.
- October 2025 - Uwe Horstmann becomes CEO: This formalized leadership around scaling, production expansion, and entry into longer-range categories.
- November 2025 - Swindon factory opening: STARK opened its first production site outside Germany, with UK government support and more than 100 jobs announced.
- December 2025 - Governance and warhead milestone: STARK added Jan-Patrick Helmsen and Martin Rost to the management board and, with TDW, completed the first live German hot strike of the LS110 warhead integrated into Virtus.
- January to March 2026 - European expansion and contracts: STARK opened Stockholm, then added Ukraine and Greece, won an additional NATO-state order, signed the Bundeswehr Virtus framework agreement, and announced a Six Robotics reconnaissance-and-strike partnership (reconnaissance-strike).

6. Business overview

The business today is organized around three visible product pillars. Virtus is the flagship and likely the most commercially important current program. Vanta broadens the company into maritime autonomy. Minerva provides the software layer that links sensors, effectors, and multi-domain mission management. Taken together, the portfolio suggests a business model aimed at selling systems and mission outcomes rather than stand-alone components.

The company's public narrative also shows evolution. Early public attention centered on loitering munitions and battlefield deployment. Over time, STARK broadened into maritime systems, cross-domain software, and reconnaissance-and-strike integration partnerships. That widening of scope matters because it implies the company is trying to become a broader unmanned-systems platform, though Virtus still appears central and the rest of the portfolio remains earlier in commercial maturity.

7. Position in the value chain

STARK sits in the value chain as an emerging system developer and integrator in unmanned warfare. It is not just a component vendor, because it builds mission systems, software, and complete unmanned platforms. It is also not yet a full-spectrum defense manufacturer. In the air domain, it looks like a loitering-munitions system house. In maritime autonomy, it appears to be an unmanned-platform developer. In software, it positions Minerva as middleware and mission operating software that can integrate STARK and third-party systems.

The company also participates in constrained nodes rather than only in outer-shell assembly. The clearest example is its work with TDW on German warheads and fuzing for Virtus. That matters because energetics and qualified warhead supply are a real choke point in European loitering-munitions scaling. On current evidence, STARK's strongest position is not as a complete-platform manufacturer, but as a high-speed integrator that combines airframe, mission software, autonomy, and selected sovereign payload partnerships.

8. Products, services, and core capabilities

Virtus is STARK's flagship product and the clearest near-term economic driver. Official materials describe it as a software-defined one-way effector developed with frontline users, with over 130 km range, up to 90 minutes endurance, modular payloads up to 5kg, 250 km/h attack speed, and resilience features for low-connectivity and GNSS-denied environments. The system is integrated with Minerva, and the company says it can be reused for training thanks to VTOL capability. Margin data is not disclosed, but this is plainly the company's lead program today.

Figure 33: STARK Virtus one-way effector



Source : STARK Defence

Minerva is strategically important because it gives STARK a software layer rather than a hardware-only identity. Public materials describe it as interoperable middleware with standardized interfaces, human-in-the-loop control, full logging for certification, open modular framework, and support for swarming and

reconnaissance-and-strike missions. That implies value beyond STARK-owned vehicles and may be the foundation for recurring support, integration, or software-driven economics over time, though the company has not disclosed commercial terms.

Figure 34: Stark Minerva software



Source : STARK Defence

Vanta is STARK's maritime line. The public portfolio highlights Vanta-4 and Vanta-6 as unmanned surface vessels designed for scalable patrol, awareness, and deterrence, with up to 910 nautical miles of range, 45+ knots top speed, 200kg payload, and integration into naval and multi-domain command structures. This is strategically useful because it broadens STARK beyond one-way strike and positions it in a second fast-growing autonomous-defense niche. Economic relevance, however, likely still trails Virtus in the near term.

Figure 35: STARK Vanta unmanned surface vessels



Source : STARK Defence

The main product-level strengths are speed of iteration, software integration, and sovereign-European positioning around warheads and production. The main risks



are concentration in Virtus, qualification and scale-up risk, and the fact that public battlefield-performance debates can become reputation risk quickly in this market. Near-term catalysts include Bundeswehr qualification progress, delivery execution for the unnamed NATO-state order, the TDW warhead ramp, and any evidence that Minerva becomes a broader integration layer rather than a captive internal tool.

9. Customers and routes to market

The visible customer set is military and government led. Public disclosures point to Germany's armed forces, an unnamed NATO-country military, and Ukrainian military users. The company also refers to work with the armed forces of Germany, Ukraine, and the UK, though contract detail outside the disclosed orders is sparse. Route to market appears to be direct government procurement, live operational testing, and closely managed industrial-military collaboration rather than broad channel distribution.

Revenue quality is likely mixed. On one hand, defense procurement can be sticky once a system is qualified and integrated into doctrine. On the other hand, award timing is lumpy, contracts can be politically contingent, and public buyers can impose stepwise approvals. Because STARK does not disclose revenue concentration or backlog, customer concentration remains a major unknown. Our base assumption is that customer concentration is currently high simply because the company is young and a few public contracts dominate the visible narrative.

10. Key financials and economic profile

Public financial visibility is very limited. We did not identify disclosed revenue, EBIT, EBITDA, net income, cash, free cash flow, capex, or leverage figures in the reviewed company materials. What is public is narrower: STARK says more than 45% of expenditure is invested in R&D, its footprint covers 18,500 sqm of production facilities, and external reporting indicates a $62m August 2025 round at a $500m valuation, followed by a €500m round in June 2026 - co-led by Sequoia Capital and Founders Fund - that pushed the valuation above €3.5bn, with more than 80% of proceeds earmarked for research and manufacturing. Those valuation figures are external reporting, not company-confirmed financial guidance.

Economically, the company appears to be in a capital-intensive build phase rather than a harvest phase. The likely model is venture-capital-backed funding feeding R&D, warhead and production investment, geographical expansion, and certification work ahead of broader procurement conversion. That can create substantial upside if procurement scales, but it also means the market has more evidence of momentum than of normalized profitability.

11. Footprint and industrial setup

The company's industrial footprint is already pan-European by startup standards. STARK discloses operations in Germany, Ukraine, the UK, Greece, and Sweden, with total production facilities of 18,500 sqm. Berlin is the legal and symbolic center, but the public operating map suggests a deliberately distributed setup.

The UK site in Swindon is especially important because it is described as the first production site outside Germany, with 40,000 sq ft of space and more than 100 jobs in its first year. Stockholm is presented as an EW-focused capability node. Ukraine is explicitly an R&D hub tied to local battlefield learning. Greece appears to be a regional southern-Europe office rather than a disclosed manufacturing center. Public materials also point to production scaling across Germany, the UK, and Ukraine.

12. Supply chain and key dependencies

The clearest hard-evidence dependency is on sovereign payload and warhead supply. STARK's public partnership with TDW for the LS110 warhead and the emphasis on German-manufactured warheads and fuzing make clear that energetics and payload integration are critical dependencies. The company has also said it invested in thousands of warheads to support fully certified European loitering munitions system before 2027.

Beyond that, several dependencies are inferences but strongly grounded. Virtus and Minerva imply reliance on communications links, optical and navigation components, EW-resilient software, and fast update cycles. Vanta implies reliance on maritime navigation, sensors, and ruggedized hull production. The most important non-material dependency may be skilled labor across software, electronics, autonomy, and scaled manufacturing. A second key dependency is regulatory and procurement timing, because qualification and export permissions can matter as much as assembly throughput in this category.

13. Management, board, ownership, and governance

Current public governance looks relatively formal for such a young private company. The executive board listed in the company imprint comprises Uwe Horstmann, Dr. André Schneider, Johannes Schaback, Jan-Patrick Helmsen, and Martin Rost, with Robert Maier listed as chairman of the supervisory board. Registry-based sources also show board start dates in late 2025 and early 2026, reflecting a rapid formalization of leadership.

Management additions in December 2025 were meaningful. Helmsen came from Rheinmetall Waffe Munition GmbH (RWM) and took over global sales and market development. Rost came from Zalando and was tasked with scaling operations and production across Germany, the UK, and Ukraine. That pairing suggests STARK is intentionally combining defense-market access with large-scale operating discipline. Headcount growth is also notable: the company said it expanded from 120 to 400 employees during 2025.

Ownership remains the least transparent major governance area. Company-confirmed backers include Sequoia, NATO Innovation Fund, Project A, and In-Q-Tel. External reporting adds Peter Thiel-linked capital and a further 2026 funding round, but public ownership percentages and governance rights are not disclosed. That opacity is not unusual for a private company, but it matters because procurement politics in Germany have already touched on the investor base.

14. Competitive positioning

Competition is business-line specific. In German medium-range loitering munitions, the clearest direct peer is Helsing, which was selected alongside STARK in the February 2026 Bundeswehr procurement. In reconnaissance-and-strike operations and ISR-linked operational frameworks, the peer set broadens to include adjacent unmanned and autonomy players rather than only loitering-munitions vendors. STARK's market position is strongest where speed, software integration, and sovereign-European industrial posture matter more than legacy installed base.

STARK's likely strengths versus peers are rapid iteration, integrated mission software, and pan-European footprint. Its likely weaknesses are age, limited disclosed production history, limited disclosed installed base, and the absence of public long-run economic data. Larger incumbents can still challenge it on scale,



procurement relationships, and balance-sheet durability, while rival start-ups can compete on speed and autonomy. In short, STARK's competitive position looks promising but still formative.

15. Strategic relevance and market implications

STARK matters beyond its own economics because it sits inside a broader European effort to restore sovereign capacity in autonomy, loitering munitions, and distributed mass. Germany's willingness to contract with STARK, even under tighter parliamentary control, shows that fast-moving private defense technology suppliers are now part of mainstream force-design discussions. That has implications for listed incumbents, traditional procurement pathways, and suppliers to constrained nodes such as warheads, sensors, and software.

There is also a wider industrial point. STARK is one of the few visible private companies trying to stitch together a defense-manufacturing network across Germany, the UK, Ukraine, Nordics, and southern Europe. If that model works, it could become a template for how Europe scales certain classes of unmanned capability faster than legacy defense contractor-led structures normally allow.

16. Dual-use and civilian exposure

Publicly, STARK should be thought of as overwhelmingly defense-facing. The brand, website, news flow, and product pages all center on NATO, allied armed forces, loitering munitions, USVs, and military command-and-control. We did not identify a meaningful civilian revenue line in the reviewed public materials.

There is some theoretical dual-use adjacency. Autonomy software, maritime awareness tools, and parts of the communications and navigation stack could have civilian analogues. But that is not how the company currently presents itself, and there is no visible evidence that civilian diversification is a material part of the case today. On current evidence, STARK is better viewed as a defense pure-play in spirit, even if some underlying technologies are dual-use in principle.

17. Legal, regulatory, and compliance risks

Our overall assessment is medium legal and regulatory risk. The main concentration of risk is not visible litigation. It is export control, military procurement oversight, qualification and certification, and political sensitivity around ownership and battlefield performance claims. Germany's BAFA export-control system, EU dual-use controls, Bundeswehr procurement processes through BAAINBw, and UK export controls all matter to a company building military drones and software across borders.

The public chronology is relatively contained but worth noting. In October 2025, external reporting raised questions about the separation between STARK and Quantum Systems because of Florian Seibel's involvement, and management responded that the two were entirely separate. In February 2026, STARK issued a statement rebutting a critical video from a Ukrainian unit and emphasizing operational-security limits on what it could disclose. Also in February 2026, the Bundestag budget process capped future procurement exposure per supplier and intensified parliamentary control. These are not formal sanctions or fines, but they are real indicators of scrutiny.

18. Funding, financing, and strategic optionality

STARK is financed as a private venture-capital-backed growth company. Company-confirmed and reported backers include Sequoia, Founders Fund, the NATO



Innovation Fund, Project A, In-Q-Tel, Döpfner Capital, Air Street Capital, 201 Ventures and Advent. External reporting says the August 2025 round raised $62m at a $500m valuation, while the June 2026 round of €500m pushed the valuation above €3.5bn; ahead of the Bundeswehr contract vote, the defense ministry told lawmakers that Thiel Capital holds less than 10% of the company.

Strategic optionality looks broad. The most obvious path is continued private compounding through procurement-led scale-up. A later IPO is possible in principle but not publicly signaled. M&A or JV expansion also looks plausible, especially in payloads, autonomy, EW, or regional manufacturing, though we found no announced acquisition by STARK in the reviewed period. The company's current behavior favors capacity build and ecosystem partnering over near-term exit positioning. That is an inference, but it fits the evidence.

19. Moat analysis

STARK's primary moat is a combination of speed, integrated systems design, and sovereign positioning. The company is trying to couple rapid product iteration with multi-domain software integration and a pan-European industrial footprint. In a market where procurement speed and battlefield feedback loops matter, that can be more valuable than legacy brand alone.

We would describe the moat as forming and potentially widening, but still early. It is not yet a classic durable moat in the sense of a decades-old installed base. What could make it durable are qualification cycles, software integration into workflows, sovereign trust, and embedded supply relationships such as the warhead partnership with TDW. What limits durability today are youth, modest public financial transparency, and still-limited evidence of scaled installed base.

20. SWOT analysis

Strengths

- Visible sovereign procurement validation from Germany and another NATO member state.
- Integrated product stack spanning effectors, maritime systems, and C2 software.
- Pan-European footprint with disclosed manufacturing and R&D presence in five countries.

Weaknesses

- Limited public financial disclosure and no visible audited operating metrics.
- Product concentration still appears high around Virtus.
- Governance and cap-table transparency remain thinner than strategic relevance might warrant.

Opportunities

- Large European demand pull in loitering munitions, reconnaissance-and-strike, and autonomous maritime systems.
- Potential to turn Minerva into a broader integration layer across allied systems.



- Geographical expansion could support export access and localized sovereign production narratives.

Threats

- Procurement controls, qualification delays, and export-control friction could slow scaling.
- Fast-moving competitor field in European autonomy and loitering munitions.
- Battlefield-performance disputes or investor-politics controversies could damage trust.

21. Key risks and open questions

- Can Virtus move from early procurement wins to repeatable large-volume deliveries without cost or quality slippage?
- How large is the real revenue base today, and what portion is recurring versus program-based? Public disclosure does not yet answer that.
- How dependent is STARK on a small number of government customers in Germany and nearby allied markets?
- Will Minerva become a genuine platform layer across mixed fleets, or remain mainly attached to STARK hardware?
- How durable is the current investor coalition, and what governance rights attach to external capital?
- Can the company maintain separation from adjacent ecosystem players where overlap or founder relationships draw scrutiny?
- Does Vanta become a real second profit pool, or remain a strategically useful but commercially smaller adjunct to Virtus?

22. What to watch next

- Bundeswehr qualification progress and first 2026 Virtus deliveries.
- Whether Germany expands procurement beyond the initial approved tranche, despite tighter parliamentary controls.
- Evidence of repeat orders from the unnamed NATO customer or additional export wins.
- Ramp-up at the Swindon site and proof that the UK plant becomes a meaningful production node.
- Progress toward large-scale TDW-integrated warhead availability by end-2026 and a fully certified European loitering munitions system before 2027.
- Whether Stockholm, Greece, and Ukraine translate into actual capability and revenue, not just footprint.
- Signs that Minerva is being integrated into third-party systems or broader allied architectures.
- Any fresh disclosure on revenue, backlog, or ownership percentages.

23. News flow and recent developments

The recent-news pattern is unusually dense for a company this young. The major sequence began with Uwe Horstmann's appointment as CEO in October 2025, followed by the December 2025 board additions, the acquisition of AeroMass Technologies to build out defense-manufacturing supply chains, and the first live

German hot strike of the TDW-integrated LS110 warhead.

January and February 2026 then brought Sweden, Ukraine, and Greece expansion, a NATO-country order, and the Bundeswehr Virtus framework agreement, under which STARK and Helsing each secured an initial ~€269m contract - with options reported up to ~€2.86bn for STARK - to equip the 45th Armored Brigade in Lithuania. In March 2026, STARK added a Six Robotics partnership to reinforce its reconnaissance-and-strike positioning, and in April 2026 it partnered with Inleap Photonics on laser technology.

At ILA Berlin in June 2026 it unveiled two new products, Cascade and Gambit, broadening the portfolio beyond Virtus. Later in June 2026, STARK raised roughly €500m at a valuation above €3.5bn, led by Sequoia and Founders Fund, taking total funding since 2024 to about €640m; around the same time, founder Florian Seibel stepped back to a founding-investor role. The other major thread has been scrutiny. In October 2025, the Financial Times reported that STARK's Virtus drones failed to hit a single target across four attempts in two exercises - one with the British Army in Kenya, one with the German Army near Munster - with one drone crashing into woodland and another's battery catching fire on impact; STARK responded that repeated crashes are integral to its "test often, learn quickly" development philosophy. Procurement oversight also tightened: when the Bundestag budget committee approved the initial €540m tranche on 25 February 2026, it capped the longer-term framework at €2bn (down from €4.3bn) and required renewed parliamentary approval for each drawdown, while questions were raised over Peter Thiel's minority stake - which Defense Minister Pistorius characterized as "a single-digit percentage stake without access to or insight into operational matters." STARK's own February 2026 statements show that operational-credibility debates can spill into the open quickly in this segment. Overall, the news flow has been decisively expansionary, but not entirely frictionless.

Potential Quantum Systems combination
In early July 2026, alongside Quantum Systems' $1.2bn Series D, co-founder Florian Seibel - who founded both companies - publicly raised the idea of eventually folding STARK into Quantum. In an interview with the Financial Times, he said he "was never really happy that I was forced to build STARK outside of Quantum," adding that STARK "did great ... but if it was part of Quantum, there also would be benefits," while stressing there was "no roadmap" for a deal. STARK, for its part, was more categorical: a spokesperson stated "there are no plans to merge STARK with Quantum Systems," arguing that "a strong European defense ecosystem depends on independent companies that compete, innovate and collaborate where it makes sense." For now, the two remain separate, though the shared founder, overlapping investor bases, and their existing Hunter-Killer reconnaissance-and-strike integration (pairing Quantum's VECTOR with STARK's Virtus) mean the option is likely to resurface.

24. Source notes and transparency limits

Company Sources Used
STARK corporate website, including product pages, newsroom, careers, and imprint.

Government and Regulatory Sources Used
UK government press release regarding the Swindon factory. German/ Bundeswehr procurement-agency materials for BAAINBw context. BAFA export-control guidance. EU dual-use export-control framework. UK Export Control Joint



Unit (ECJU) export-control guidance. UK Companies House for context on UK entity searches.

Independent External Sources Used
Reuters and Financial Times reporting for context on funding, valuation, procurement, and ownership scrutiny.

Transparency Limits
STARK is a private company and, as such, does not disclose the full financial package typically desired by an analyst. Consequently, details such as revenue, EBITDA, cash, leverage, backlog, divisional sales, and customer concentration are not visible in the reviewed company materials. Furthermore, ownership percentages and the complete composition of the supervisory board are not clearly public within the sources reviewed. The company's founding history is also only partially clear across both primary and secondary sources.

Explicitly Labeled Estimates and Inferences
The description of STARK as a hybrid system house and integrator is an inference derived from its product architecture and public positioning. The judgment that customer concentration is likely high is an inference based on the company's age and visible contracts, rather than a disclosed figure. The view that Minerva could develop into a more substantial 'moat' (a sustainable competitive advantage) is an analytical assumption, not a company disclosure. All valuation comments rely on external media reports and not on audited company filings.

Figure 36: Quantum Systems - overview

Quantum-Systems Gmbh			
Ownership:	Privately held, founder-led and heavily VC/strategic-backed; named investors include Balderton, Hensoldt, Airbus Defence and Space, HV Capital, Project A, Porsche SE, DTCP, Omnes, Notion, and Peter Thiel		
HQ:	Gilching, 82205, Germany	**Founded:**	2015
Domain:	Air – eVTOL ISR drones and autonomy software, now expanding into multi-domain unmanned systems		
Value chain:	OEM plus software-enabled systems provider		
Why it matters	Quantum is one of Europe's strongest small-UAS defense names, pairing proven ISR drones with interoperable mission software and rapidly expanding production		
Key products	Vector – mid-range eVTOL ISR drone; Twister – short-range tactical ISR drone; Trinity – fixed-wing aerial data-collection platform; MOSAIC UXS – command, control and mission software across air/land/sea; emerging ground-autonomy		
Core capabilities	eVTOL endurance, multi-sensor ISR, AI-enabled autonomy, interoperable software and scaled industrialization		
Customers	Quantum serves government agencies and commercial customers; named public defense customers/programs include BAAINBw / German Armed Forces, the Ukrainian Ministry of Defense, the Romanian Ministry of Defence and U.S. government users via Blue UAS		

Key financials (31 Dec)	2021	2022	2023	2024	2025
P&L (EURm)					
Sales	11	23	43	111	~300
Gross profit	4	10	18	49	-
EBITDA	-4	-1	-3	12	-
EBIT	-4	-1	-3	10	-
EBT	-4	-1	1	10	-
Net income	-4	-1	1	10	-
Order intake	-	-	-	150	-
Balance sheet (EURm)					
Cash & cash equivalents	5	3	47	87	-
Tangible fixed assets	0	1	2	4	-
Goodwill & intangible assets	0	3	8	11	-
Inventories	3	8	21	45	-
Accounts receivable	1	2	3	8	-
Other assets	0	0	0	9	
Total assets	10	21	95	189	-
Equity	3	12	70	95	-
Minorities	0	0	0	0	-
Borrowings	5	5	0	2	-
Accounts payable	0	1	3	6	-
Other ST liabilities	0	0	0	0	-
Net debt (+) / net cash (-)	0	3	-47	-85	-
Key company metrics					
Book/bill (x)	-	-	-	1.4	-
Sales growth (%)	-	111	87	159	171
Net income growth (%)	-	-81	-278	682	-
EBITDA margin (%)	-33	-5	-6	11	-
EBIT margin (%)	-35	-6	-7	9	-
ROE (%)	-145	-6	2	11	-
Capex/sales (%)	-	16	17	6	-
Capex/depreciation (x)	-	16.4	11.8	3.0	-
Net debt/equity (%)	9	21	-68	-89	-
Net interest cover (x)	-33	2	1	-15	-

Source : Deutsche Bank research, company data

Quantum Systems

Fusing UAS with AI Software for Superior Aerial Intelligence

What type of company it is

In practical terms, Quantum Systems is a private German dual-use unmanned-systems company that started in fixed-wing eVTOL aerial intelligence and is now moving toward a broader unmanned-systems platform model. It still sells air systems, payloads, and mission software, but the 2025-2026 product and M&A record shows a deliberate expansion into multi-domain command software, propulsion, AI perception, ground autonomy, and localized production partnerships. The launch of MOSAIC UXS in June 2025, followed by MANDRILL modular high-mobility UGV, the MOSAIC Ground Autonomy Kit, and the FERNRIDE and Spleenlab acquisitions, marks a clear shift from ISR hardware supplier toward software-led, multi-domain orchestration and autonomy. It is therefore better described as an emerging system integrator and autonomy-stack provider than as a pure drone airframe vendor.

Investment-style or strategy-style summary

Quantum Systems appears to be transitioning from a dual-use drone specialist into a broader unmanned-systems platform company. The current business still centers on aerial intelligence, especially tactical ISR UAS such as Vector AI, Twister, Reliant, and Trinity variants, but management's recent product and M&A choices suggest the future business mix should include more software, AI, autonomy, and cross-domain systems integration. That matters strategically because software-led orchestration and open-platform control layers tend to sit closer to customer workflow and procurement stickiness than airframes alone.

The company's defense relevance is rising faster than its civilian narrative. Quantum still presents itself as an "AND company" serving commercial and defense markets, and Trinity Pro remains a meaningful commercial/GIS product family. But the recent disclosures overwhelmingly emphasize armed-forces use cases, sovereign European production, Blue UAS compliance, Bundeswehr-related tenders, and allied defense programs. The civilian business now looks more like a diversification layer than the core strategic driver.

Financially, the public signal is one of very rapid scale-up rather than full transparency. Quantum said it achieved triple-digit million turnover in 2024, while Bloomberg and Reuters later reported valuation above €3bn and media-cited revenue expectations of roughly €300m for 2025. The company describes itself as profitable, with more than 19,000 missions flown in Ukraine in 2025. Those figures imply substantial growth, but detailed audited line items remain outside normal public view. The confirmed funding base nevertheless gives the company unusually strong capacity to add sites, talent, acquisitions, and production.

Competitive positioning is strongest where battlefield-proven fixed-wing eVTOL ISR, open architecture, and European sovereign sourcing matter at the same time. Quantum does not need to beat every incumbent in every drone category. It needs to become difficult to replace in a subset of tactical ISR, modular autonomy, and multi-domain control missions where Europe wants trusted, non-Chinese, increasingly sovereign supply. The recent Airbus cooperation, Hacker Motor acquisition, and Blue UAS milestone all support that thesis.



Quantum Systems deserves attention because it is becoming one of the more important private European unmanned-systems companies at a time when defense customers are prioritizing speed, autonomy, resilient supply chains, and sovereign control layers. The opportunity is substantial, but so is the execution burden: the company is scaling rapidly across products, geographies, and domains while still offering only partial financial transparency to outside observers.

1. Why this company is relevant now

Quantum Systems is relevant now because the market is moving toward affordable, scalable, software-connected unmanned systems, and because European governments are increasingly willing to fund and procure those capabilities at speed. Reuters reported in June 2025 that Germany's procurement leadership was prioritizing rapid delivery over supplier origin as rearmament accelerated, which is a favorable backdrop for private, production-focused drone suppliers that can scale fast.

At the company level, Quantum has paired demand signals with capital formation and industrial expansion. It raised €160m in May 2025, extended that round with another €180m in November 2025 at a valuation above €3bn, and then added a €150m European financing package in February 2026. The July 2026 $1.2bn Series D (see 23. News Flow) brought valuation to ~$8bn. That is an unusually dense sequence of capital events for a private defense-tech business and points to both investor confidence and heavy capital needs.

The strategic relevance also comes from the company's widening role in the European and allied unmanned ecosystem. Quantum has expanded in Ukraine, the US, and the UK, won visible tenders, and launched software and ground-autonomy layers that suggest it wants to shape the operating model of unmanned operations, not just sell aircraft.

2. What type of company it is

Quantum Systems is best understood as a private dual-use unmanned-systems company with roots in eVTOL fixed-wing aerial intelligence. The company develops air vehicles, payload integrations, mission-planning software, and now a broader software and autonomy stack. Its own group description emphasizes in-house hardware, software, and AI, and recent launches show a transition from platform maker toward integrated unmanned-systems provider.

It is not a classic lead contractor. It is also not just a niche component supplier. In current practical terms, it sits between those categories: an emerging system house for tactical unmanned reconnaissance and autonomy, with increasingly visible ambitions in multi-domain command software, propulsion sovereignty, and localized industrial partnerships. That description is partly disclosed fact and partly derived from the product roadmap, acquisitions, and partnership pattern.

3. Three things that matter most

The first core takeaway is that Quantum's strategic focus is shifting upward in the stack. MOSAIC UXS, MOSAIC Ground Autonomy Kit, Spleenlab, and FERNRIDE all point toward software-defined coordination, autonomy, and control of mixed unmanned fleets. That can be strategically more valuable than selling stand-alone airframes.

The second is that the company's sovereignty narrative has substance. The AirRobot and Hacker Motor deals strengthen local manufacturing and propulsion



depth, while Airbus cooperation and EIB-backed financing reinforce Quantum's role inside a broader European defense-industrial architecture. That does not make the company unassailable, but it does make it more strategically relevant to European customers.

The third is that scale is arriving faster than transparency. Public sources show funding, sites, contracts, and headcount growth, but not the full financial anatomy that outside analysts would usually want. The company could become significantly more valuable as a strategic platform, but outside observers still lack a clean view on margins, recurring revenue, working-capital needs, and customer concentration.

4. At a glance

Quantum-Systems GmbH is a private German company headquartered in Gilching that develops unmanned systems for defense, security, public safety, and commercial geospatial users. It was founded in 2015 and has built a portfolio around fixed-wing eVTOL UAVs, mission software, payload integration, and increasingly a multi-domain autonomy stack. Publicly visible group entities now span Germany, the US, Australia, the UK, Romania, Ukraine, and Lithuania.

The company's business mix today appears defense-heavy in strategic significance, even though it still maintains commercial products and a dual-use identity. Vector AI, Twister, and Reliant address defense ISR missions, while Trinity Pro and Trinity Tactical bridge mapping, surveying, and tactical geospatial workflows. The launch of MOSAIC UXS and the move into ground systems suggest Quantum wants to own more of the command, autonomy, and orchestration layer around those platforms.

Financially, outside visibility is selective rather than comprehensive. The company disclosed triple-digit million turnover for 2024, confirmed total funding of €310m by May 2025, later added €180m in November 2025, and announced €150m of financing in February 2026. That creates a strong external signal of scale and balance-sheet support, but not a public-company-style earnings model.

Key takeaways

- Private German GmbH headquartered in Gilching near Munich, founded in 2015 by Florian Seibel, Dr. Michael Kriegel, Tobias Kloss, and Armin Busse.
- Core business is tactical unmanned ISR and autonomy systems, with a strategic expansion from aerial platforms into software, AI, ground autonomy, and broader multi-domain platform integration.
- Flagship products include Vector AI, Twister, Reliant, Trinity Pro, and Trinity Tactical.
- Major 2025-2026 strategic moves included the acquisitions/investments in AirRobot, Spleenlab, FERNRIDE, and Hacker Motor.
- Ownership remains private and venture-capital-backed; detailed cap-table percentages are not publicly disclosed.
- Funding visibility is unusually high for a private company, including a €160m Series C (May 2025), €180m extension (November 2025), €150m debt financing (February 2026), and a $1.2bn Series D (July 2026) at an implied valuation of roughly $8bn.
- Public disclosure on financing and strategic developments is relatively



strong, but detailed audited financials, margin data, and ownership information remain limited.

- The key analytical question is whether Quantum can translate its expanding product portfolio and substantial capital base into durable software-led economics without diluting execution focus.

5. Corporate history and legal structure

Quantum traces its origin to research work started around 2011 by Florian Seibel and colleagues at the University of the German Armed Forces in Munich. The founding team of Florian Seibel, Dr. Michael Kriegel, Tobias Kloss, and Armin Busse formally established Quantum-Systems GmbH in January 2015. The company's public origin story centers on a patented vertically launchable transition-aircraft concept that shaped the group's eVTOL fixed-wing identity.

Figure 37: Ownership-Quantum Systems

Legal entity	Entity number	Ownership %
Quantum-Systems	HRB 217004	
Quantum Founder Holding UG (haftungsbeschränkt)	HRB 286405	19.2%
Quantum-Systems GmbH	-	17.2%
HV Capital Fund IX Growth GmbH & Co. geschlossene Investment KG	HRA 117189	6.5%
Bee Holding GmbH	HRB 233325	6.1%
DTCP Growth Equity III SCSp SICAV-RAIV	201730514	6.0%
Project A Ventures IV GmbH & Co. KG	HRA 58954	5.8%
PXR 15. EIP Vorrats GmbH & Co. KG	HRA 63268	4.9%
Wachstumsfonds Bayern 2 GmbH & Co. KG	HRA 11924	4.2%
PORSCHE DRITTE BETEILIGUNG GMBH	HRB 751843	3.9%
QUA Investments LLC	201379594	3.6%
Rest		22.6%

Source : Gain.Pro

The current top-level legal entity visible in public company materials is Quantum-Systems GmbH, headquartered in Gilching, Germany. The legal notice lists Munich as the commercial-register court, HRB 217004 as the register number, and Florian Seibel and Sven Kruck as managing directors. Publicly visible country entities include Quantum-Systems Inc. in the US, Quantum-Systems Pty Ltd in Australia, Quantum-Systems Ltd in the UK, and Quantum-Systems Srl in Romania.

The structure appears partly centralized and partly federated. Strategy, leadership, and core product identity remain anchored in Gilching, but the company has built national operating entities and kept some acquisitions distinct. AirRobot, for example, was explicitly described as being integrated as a stand-alone entity within the Quantum Systems Group, while other moves such as Hacker Motor and Spleenlab deepen capability inside the broader group architecture. That implies a hub-and-spoke model rather than a fully centralized single-site structure.

A concise timeline of key inflection points is as follows. 2015: company founded in Gilching. 2021: EIB signed up to €10m of venture debt for RDI and growth. 2022: US operation opened in Moorpark. 2023-2024: Series B expanded to more than €100m and Bundeswehr-related programs became more visible. 2025: co-CEO structure introduced, Series C raised, MOSAIC UXS launched, AirRobot acquired, US facility opened, and UK expansion accelerated. 2025-2026: Series C extension, FERNRIDE, Spleenlab, Hacker Motor, and MANDRILL signaled expansion from aerial systems into broader unmanned-system domains, followed in mid-2026 by the WIY Drones and Tencore joint ventures, a $1.2bn Series D at roughly $8bn, and international manufacturing expansion for the NATO Drone Edge.

6. Business overview

Quantum's business is built around collecting and exploiting operational data from unmanned systems. The group description emphasizes advanced multi-sensor UAS, in-house software, and AI designed to convert aerial data into decision-useful intelligence across defense, public safety, commercial, industrial, and GIS use cases. That is the plain-language core of the model.

The current structure has at least four visible layers. First, tactical and professional aerial platforms such as Vector AI, Twister, Reliant, Trinity Pro, and Trinity Tactical. Second, mission software, especially QBase Tactical, QBase 3D, and MOSAIC UXS. Third, services and localization through subsidiaries, sustainment, training, and operational partnerships. Fourth, an emerging autonomy and systems-integration layer through ground robotics and acquired capabilities in propulsion, AI perception, and ground automation.

Historically, Quantum appears to have moved from commercial and dual-use mapping/survey roots toward a more defense-centered identity after 2022, while keeping commercial products alive. The company still openly presents itself as serving both commercial and defense customers, but the visible 2025-2026 cadence of contracts, sites, and product launches suggests defense and security now drive the strategic agenda. That is an assumption, but it is strongly supported by the public news flow.

7. Position in the value chain

Quantum sits primarily at the OEM and system-house level for tactical unmanned reconnaissance systems. It develops the aircraft, integrates payloads, supplies ground-control and mission software, and increasingly provides the autonomy and orchestration layer that connects assets across domains. That means it is above pure component supply and below the scale of traditional full-spectrum defense groups.

In aerial ISR, the company acts as a designer-manufacturer and direct supplier into government and defense channels. In software, it is moving toward a control-layer role through MOSAIC UXS. In ground autonomy, it is becoming an integrator by combining in-house systems like MANDRILL with partner platforms such as Daimler Truck and acquired software from FERNRIDE.

The most important inference is that Quantum is trying to move from a platform vendor to a systems integrator. The AirRobot, Hacker Motor, and Spleenlab acquisitions support that inference because they add adjacent capabilities in copter systems, propulsion, and AI perception rather than simply more of the same airframe capacity.

8. Products, services, and core capabilities

Quantum's aerial portfolio covers distinct operational tiers. Vector AI is a mid-range eVTOL ISR platform that the company describes as combat-proven, with thousands of mission hours and proven use in Ukraine. Twister targets short-range frontline ISR with a backpackable form factor and a small launch footprint. Reliant moves up the ladder into Group 2, long-range VTOL ISR with more than 10 hours endurance, heavy-fuel compatibility, and dual swappable payload bays for ISR, SIGINT, or EW missions.

Figure 38: Reliant VTOL

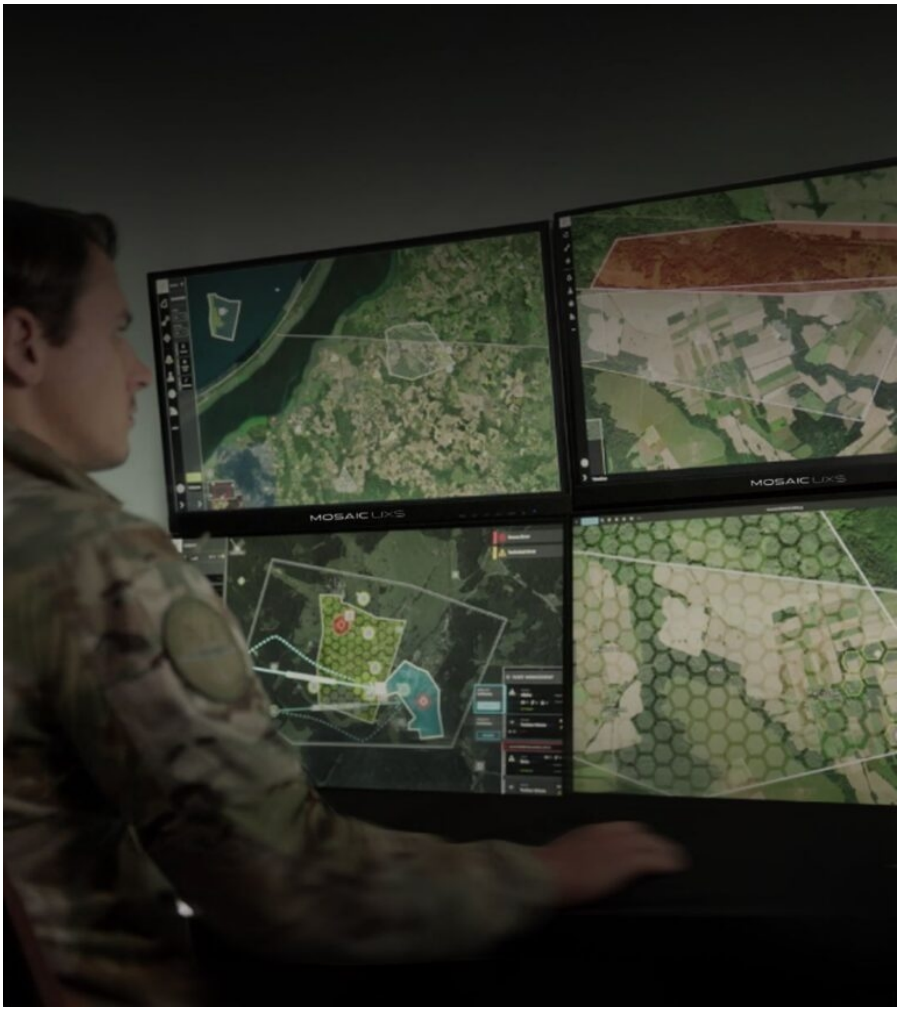


Source : Quantum Systems

The commercial and tactical geospatial side remains important because it preserves a dual-use technology base. Trinity Pro is a professional mapping platform with 90-minute flight time, broad sensor choice, IP55 protection, terrain awareness, and integrated QBase 3D mission planning. Trinity Tactical adapts that platform for tactical mapping, terrain visualization, line-of-sight analysis, and other intelligence collection missions. This side of the portfolio likely matters strategically because it keeps Quantum close to survey, inspection, and public-safety workflows rather than making it purely dependent on defense budgets.

Software is now central rather than accessory. QBase Tactical and QBase 3D remain mission-planning and control tools for individual systems, but MOSAIC UXS is the larger strategic step because it is designed to unify mission planning and mission execution across air, land, and sea systems from different vendors. That suggests Quantum is seeking switching costs and ecosystem relevance at the software layer.

Figure 39: MOSAIC UXS mission control software



Source : Quantum Systems

The company's pipeline and capability stack also broadened materially in 2025-2026. Public disclosures show radar integration work with Fraunhofer FHR on Trinity Pro, a drone-port or 'nest' concept in the company vision, MANDRILL as a modular UGV, and the MOSAIC Ground Autonomy Kit with Daimler Truck. Counter-UAS also appears on the horizon through the Jaeger counter-UAS drone, which Reuters said helped drive late-2025 valuation expansion.

The main product strengths are modularity, endurance in fixed-wing eVTOL ISR, and increasingly the combination of hardware with open-architecture software. The main product risks are portfolio sprawl, the difficulty of scaling across air and ground at the same time, and the possibility that airframes become more commoditized than the company expects. The near-term catalysts are software adoption, ground-domain rollout, and conversion of visible tenders into repeatable program positions.

9. Customers and routes to market

Quantum sells to a mix of defense, government, public-safety, and commercial customers. The public record names or identifies use by Germany, Ukraine, Australia, Romania, New Zealand, Spain, and US-linked procurement channels. It



also maintains commercial offerings for GIS, inspection, surveying, and related enterprise uses.

Routes to market vary by geography and mission. In defense, the company sells through direct tenders and framework contracts, local subsidiaries, and program partnerships. Examples include the Bundeswehr's FALKE and ALADIN-successor tenders, the Australian DEF129-SUAS contracts, the Romanian MoD order, and the US Department of State-linked subcontract through RRDS. In the UK, Quantum paired the Nordic Unmanned UK acquisition with a planned €50m investment and local leadership, which suggests a deliberate local-presence model rather than pure export sales from Germany.

Revenue quality is likely mixed. Hardware contracts and tenders can be large but lumpy. Software, sustainment, training, mission operations, and products such as Overwatch may create more recurring or service-like revenue over time, but the public sources do not quantify that split. Our assumption is that current economics still lean hardware-led, while management is trying to increase the software and lifecycle-service mix.

10. Key financials and economic profile

Annual accounts for Quantum-Systems GmbH are publicly available. Quantum stated in February 2025 that it achieved triple-digit million turnover in 2024. In May 2025, the company said total capital raised had reached €310m after the €160m Series C. In November 2025, it said it added €180m more, taking total capital raised in 2025 to €340m. In February 2026, it announced a further €150m financing package backed by several large European financial institutions, and in July 2026, it closed a $1.2bn Series D at a post-money valuation of roughly $8bn.

Secondary reporting adds some directional color, but it should be treated carefully. The company has since confirmed roughly €300m of revenue for 2025, up more than 160% year on year, alongside double-digit profitability and more than 19,000 missions flown in Ukraine during the year. Management indicated in July 2026 that revenue was on track to grow at a similar rate in 2026, implying a trajectory toward €600m or more. These remain company statements rather than audited public accounts, but they are now actuals rather than the earlier media-cited estimates of roughly €300m for 2025 and more than €500m for 2026.

The economic profile that emerges is one of high growth, heavy investment, and expanding industrial ambition. Publicly confirmed data support the view that the company is scaling aggressively in production, software, acquisitions, and international sites. What is still missing for a fuller economic assessment is line-item visibility on gross margin, EBITDA, cash burn or generation, working capital, R&D intensity, and leverage. That is the central limitation of any outside valuation discussion today.

11. Footprint and industrial setup

Quantum's visible industrial footprint has widened meaningfully. The company's locations page lists Gilching, Berlin, and Rheinbach in Germany, Moorpark in California, Redbank in Australia, Cranfield in the UK, Bucharest in Romania, plus undisclosed addresses in Ukraine and Lithuania. That suggests a distributed footprint oriented around both customer proximity and specialized technical nodes.

Several recent site investments look strategically significant. In April 2024, the

company opened a production facility and development hub in Ukraine with plans to scale to about 100 employees there and capacity of up to 1,000 drones annually including spare parts. In June 2025, it opened a 135,000-square-foot production and integration facility in Moorpark to scale US deliveries. In September 2025, it announced a long-term UK commitment and up to €50m of planned investment over five years.

The acquisition pattern also deepens the industrial map. AirRobot adds Arnsberg, Spleenlab adds Saalburg-Ebersdorf and Jena, and Hacker Motor adds retained existing sites in propulsion. Rheinbach matters because Quantum said it planned an R&D office near Fraunhofer FHR to build RF and radar expertise.

12. Supply chain and key dependencies

The clearest hard-evidence dependency is propulsion. Quantum's February 2026 acquisition of Hacker Motor was explicitly framed as a move to strengthen a sovereign European supply chain for unmanned systems, and the company said Hacker's motors had long been integrated into its platforms. That signals that propulsion was already strategically important and sufficiently concentrated to justify vertical integration.

Sensors, EW resilience, and AI perception are the next obvious dependency cluster. Fraunhofer FHR radar cooperation shows Quantum is investing in advanced sensing and countermeasures. Blue UAS inclusion shows that supply-chain integrity and cybersecurity are commercially important in the US market. The Spleenlab acquisition adds AI perception, GPS-denied navigation, and edge autonomy, which again looks like a response to dependencies the company would rather internalize.

Export permissions and regulatory compliance are structural dependencies rather than supplier dependencies. Germany's BAFA administers export control under national law, the EU Dual-Use Regulation, and sanctions frameworks, and the EU specifically highlights drones, drone software, and drone motors among sensitive categories in Russia-related sanctions. For a company selling dual-use and defense systems across multiple geographies, export licensing is not a side issue. It is a core gating mechanism for revenue realization.

Skilled labor is also a likely constraint. The company's visible growth in sites, product scope, and acquisitions implies large needs in engineering, software, autonomy, manufacturing, and legal/compliance. That is partly inference, but it is supported by the pace of footprint expansion and the broad role descriptions on the company's careers pages.

13. Management, board, ownership, and governance

Leadership has become more formalized as the company scaled. On 19 February 2025, Quantum announced that founder Florian Seibel moved from sole CEO to co-CEO alongside Sven Kruck, with Mr. Seibel focusing on product, technology, and strategy and Mr. Kruck on operations, sales, and finance. At the same time, Frank Thieser stepped back from active management, Jonas Jarosch became CFO, and Martin Karkour became CSO.

Founder continuity remains strong. The company states that Florian Seibel, Dr. Michael Kriegel, Tobias Kloss, and Armin Busse founded Quantum in 2015 and that three of the four founders still hold active positions. That continuity is a positive sign for technical coherence, but it also increases founder concentration in key



decisions.

Ownership is private and venture-capital-backed. The official investors page names Balderton, Porsche SE, Hensoldt, Airbus and Airbus Ventures, HV Capital, DTCP, Project A, Omnes, Notion, Thiel Capital, In-Q-Tel, Bullhound Capital, LP&E AG, Bayern Kapital, and others.

Public governance visibility is narrower than at a listed company. The legal notice clearly identifies managing directors, and the management pages show executive leadership, but we did not locate a detailed public supervisory-board or board-of-directors disclosure equivalent to a listed-company governance package. Headcount disclosures also vary by date: Quantum said it had over 400 employees by February 2025, while later secondary reporting pointed to 550 in May 2025 and roughly 1,000 by November 2025. That supports the broader conclusion that the company is scaling very rapidly, but public governance and reporting structures are still catching up.

14. Competitive positioning

Competition happens in different layers, not one single peer set. At the short-range tactical ISR end, Quantum's Twister overlaps more with systems like Skydio X10D and Parrot ANAFI USA in the sense that all target trusted, tactical, high-availability organic ISR for government users. Those platforms differ in aircraft design and endurance, but they compete for the same procurement logic around secure, fast-fielded ISR at the edge.

At the larger tactical and Group 2 end, Reliant and parts of Quantum's ISR offering compete more directly with systems such as AeroVironment's JUMP 20 and TEKEVER's AR3/AR5 families. Those competitors have strong credibility in tactical or longer-endurance ISR, and in some cases longer track records in formal programs of record. Quantum's differentiation is more likely to come from eVTOL flexibility, open architecture, and European sovereignty positioning than from simply being the longest-range or biggest-endurance provider in every class.

Where Quantum seems strongest is in the blend of modular eVTOL aircraft, in-house software, multi-domain integration, and a Europe-first industrial story. Where it remains vulnerable is in scale against established tactical UAS suppliers, in the challenge of supporting a widening portfolio, and in the fact that open-architecture software claims are increasingly becoming industry-standard rhetoric rather than a unique message. The company's recent Blue UAS milestone, Airbus cooperation, and supply-chain verticalization strengthen its hand, but they do not eliminate competition.

15. Strategic relevance and market implications

Quantum matters beyond its own P&L because it touches several current European defense priorities at once: tactical ISR, sovereign supply chains, local industrial capacity, software-enabled coordination, and rapid fielding. The company's links to Ukraine, German procurement, Airbus, Hensoldt, and the EIB-backed financing package make it relevant to how Europe is trying to rebuild autonomous-systems capacity without relying on non-aligned supply chains.

There are also broader market implications. For listed defense names, Quantum is both a partner and a pressure point. Airbus and Hensoldt have chosen to work with or invest in the company rather than dismiss it. That suggests that larger incumbents increasingly see private autonomy players as complementary



capability nodes or future ecosystem anchors rather than as marginal start-ups.

Strategically, Quantum looks more like a capability anchor than a pure consolidator today. But the 2025-2026 acquisition pace shows it already has a consolidation instinct in software, propulsion, and adjacent unmanned domains. If that continues, the company could become one of the more consequential private aggregation platforms in European unmanned systems.

16. Dual-use and civilian exposure

Quantum still has real civilian and dual-use exposure. The company openly describes itself as an "AND company" serving both commercial and defense customers, and Trinity Pro remains positioned for mapping and industrial data capture. Its public materials also reference use cases in public safety, infrastructure, industry, and GIS.

That said, the direction of travel is clearly defense-heavy. The visible capital raises, site openings, program announcements, AI/software messaging, Blue UAS milestone, and acquisitions all lean toward defense, security, and autonomy for contested environments. Civilian businesses likely still diversify the technology base and may help with product iteration speed, but they no longer appear to set the strategic narrative.

We think Quantum should no longer be thought of primarily as a civilian drone company with defense adjacency. It is better understood as a defense-led dual-use company whose civilian business still matters technologically and commercially, but is now secondary to the defense scaling thesis. That is an assumption, but it is strongly supported by the company's public activity mix.

17. Legal, regulatory, and compliance risks

The overall legal and regulatory risk profile looks medium to high, not because public sources show major disclosed litigation, but because the business is deeply exposed to export control, procurement law, cybersecurity, data protection, and sovereign-security regulation. BAFA is the relevant German export-control authority, the EU maintains dual-use rules and sanctions that explicitly cover drone-related categories, and US government procurement relevance depends in part on frameworks such as Blue UAS.

The clearest publicly visible recent compliance event was the 16 July 2025 security incident at a subcontractor. Quantum said the affected account was at a subcontractor, that its own systems were not compromised, and that the company reported the matter to the competent data-protection authority while coordinating with Bavarian cybercrime and police authorities. Even if operational impact was limited, the episode highlights the cyber and third-party control burden that comes with defense-adjacent software and hardware ecosystems.

Procurement law and national-security review are also meaningful. Quantum's visible growth increasingly depends on government tenders, framework agreements, and allied procurement pathways. In Germany, BAAINBw-related wins are important opportunities but also create exposure to delivery timing, qualification requirements, and public-procurement scrutiny. In the US, Blue UAS status lowers a barrier, but it also embeds the company in a compliance-heavy ecosystem where supply-chain assurance matters.



IP is important but only partially transparent in reviewed sources. Quantum's public narrative repeatedly references its patented transition-aircraft concept, and DPMAregister is the official German source to verify patents, utility models, trademarks, and their legal status. We also found secondary registry mirrors reflecting 2025-2026 trademark registrations tied to Quantum-Systems GmbH, but we would treat the official DPMAregister as the correct diligence source for any hard IP mapping.

18. Funding, financing, and strategic optionality

Quantum has funded itself through a blend of venture equity, strategic capital, and debt. Earlier support included up to €10m of EIB venture debt in 2021. Public company announcements then show more than €100m of Series B financing by September 2024, a €160m Series C in May 2025, a €180m Series C extension in November 2025, a €150m financing package in February 2026, and a $1.2bn Series D in July 2026 at roughly $8bn. That funding history gives the company unusual optionality for a private European defense-tech business.

The strategic options visible today are straightforward. One path is continued private compounding through acquisitions, site build-outs, and software expansion. A second is selective JV and localization, as shown in Ukraine, Taiwan, and the UK. A third is eventual public listing. Reuters reported in February 2026 that Manager Magazin said Quantum was preparing for a possible IPO by early 2027, though the company declined to comment. The $1.2bn Series D in July 2026 gives the company ample private capital and reduces any near-term funding pressure to list, so an IPO now looks more like optionality than necessity.

The practical implication is that Quantum does not appear strategically boxed in. It can keep compounding privately if demand remains strong. It can sell more capability into European sovereign ecosystems. Or it can eventually use an IPO as a capital and credibility event. The main constraint is likely execution bandwidth, not lack of financing access.

19. Moat analysis

Quantum's primary moat is the combination of field-relevant fixed-wing eVTOL ISR hardware, in-house software and AI, and growing sovereign credibility in Europe. None of those elements alone is a perfect moat. Together, they create a stronger position because the customer is buying not just a drone, but an integrated operating concept with mission software, supply-chain assurances, and an increasingly broad unmanned ecosystem.

The moat appears to be widening, though from a relatively early base. It is widening because Quantum keeps adding adjacent control points: AirRobot for copters and programs, Hacker Motor for propulsion, Spleenlab for AI perception, FERNRIDE for ground autonomy, and MOSAIC UXS for system orchestration. That pattern is more consistent with moat-building than with opportunistic product collection.

There are real barriers to entry, but they are mixed rather than absolute. Certifications, trust, Blue UAS acceptance, procurement relationships, integration know-how, and battlefield feedback all matter. Switching costs also rise when customers adopt software workflows and multi-domain integrations rather than stand-alone aircraft. The risk is that unmanned airframes themselves become less differentiated over time, which is why the software and systems layer matters so much to Quantum's long-term defense.

20. SWOT analysis

Strengths

- Strong alignment with current defense demand for tactical ISR, autonomy, and rapid fielding.
- Expanding software and autonomy layer through MOSAIC UXS, Spleenlab, and FERNRIDE.
- Deepening sovereign industrial position through AirRobot, Hacker Motor, Airbus cooperation, and EIB-backed financing.

Weaknesses

- Limited public financial transparency.
- Very broad expansion agenda across air, software, and ground domains raises execution complexity.
- Governance visibility remains lighter than for listed peers.

Opportunities

- European rearmament and faster procurement cycles.
- Growth in allied procurement channels such as Blue UAS, Australia, Romania, and UK localization.
- Potential IPO or further strategic financing once scale and reporting maturity improve.

Threats

- Export-control and sanctions exposure across dual-use and defense sales.
- Intense competition from US and European unmanned-system providers.
- Cybersecurity and third-party security risks in a distributed supply chain.

21. Key risks and open questions

The most important open questions are these:

- Can Quantum convert rapid product expansion into a coherent software-led operating model rather than a wide but uneven portfolio?
- How much of future revenue will come from repeat software, sustainment, and service streams versus one-time hardware programs?
- How concentrated is the customer base by ministry, program, or country? Public sources do not answer this cleanly.
- Can the company maintain quality and delivery performance while scaling across Germany, Ukraine, the US, the UK, and additional localized production models?
- Does the co-CEO structure improve operating leverage, or does it add decision complexity as the group broadens?
- Will Quantum become primarily a defense systems company, or preserve a meaningful dual-use commercial hedge?
- How durable is the current valuation if defense-tech funding cools before recurring economics become more visible?

22. What to watch next

The most important forward markers over the next 6-18 months are:

- Production ramp and fulfillment against Bundeswehr-related tenders, especially Twister and broader German demand signals.
- Commercial adoption and customer stickiness of MOSAIC UXS and MOSAIC Ground Autonomy Kit.
- Evidence that MANDRILL becomes a real product family, not just a launch demo.
- Integration progress on Spleenlab, Hacker Motor, AirRobot, and FERNRIDE.
- New wins in the UK, Taiwan, and other localization-heavy markets.
- Whether counter-UAS products such as Jaeger become a meaningful adjacent category.
- Any movement from IPO press reports to confirmed financing or governance preparation.
- Whether the company begins to disclose more standardized financial information as it matures.

23. News flow and recent developments

Recent news flow has been dense and strategically coherent rather than random. February 2025 brought a co-CEO structure and CFO transition. May 2025 brought the €160m Series C. June 2025 brought MOSAIC UXS, a major US facility opening, and cooperation announcements with Airbus and Fraunhofer FHR. July 2025 brought the Frontline investment agreement and a disclosed subcontractor security incident. September 2025 brought UK expansion and Taiwan localization. October through December 2025 added Spleenlab, FERNRIDE, the Frontline JV (QFI) for European production, and the ALADIN-successor win for Twister.

February to March 2026 added Hacker Motor, the €150m financing package, MANDRILL, the Daimler Truck partnership for ground autonomy, and a contract for 15,000 STRILA interceptor drones for Ukraine alongside an investment in Ukrainian maker WIY Drones. From April 2026 the pace accelerated further: the US Army selected Vector AI under a company-level sUAS requirement (a roughly $15m award); in June, Quantum formed the Quantum Tencore Industries joint venture to build 2,000 TerMIT unmanned ground vehicles on German lines and won a roughly $36m Romanian contract financed through the EU's SAFE facility. The defining event came on July 2, 2026, when Quantum closed a $1.2bn Series D at roughly $8bn, co-led by Blackstone, Noteus, Airbus, and Advent, with Airbus Defence and Space simultaneously deepening a strategic partnership.

The rest of July brought a rapid sequence of expansion: added international manufacturing capacity for the NATO Drone Edge, a further Ukraine contract for MANDRILL UGVs and unmanned Daimler Zetros trucks, entry into Estonia, selection for Japan's counter-UAS proof-of-concept program, founding membership of both the EU-Ukraine Drone Alliance and the Project Conexus consortium for NATO targeting interoperability, and a major upgrade to the long-range RELIANT system. Financially, the company reported roughly €300m of 2025 revenue, double-digit profitability, and over 19,000 missions flown in Ukraine, anchored by a €246m German Army contract signed in December 2025 - its largest to date.



One strategic option concerns STARK Defence. Alongside the Series D, Quantum co-founder Florian Seibel - who founded both companies - publicly floated eventually folding STARK into Quantum, telling the Financial Times he "was never really happy that I was forced to build STARK outside of Quantum," while stressing there was "no roadmap" for a deal. STARK was more categorical, stating "there are no plans to merge STARK with Quantum Systems." For now, the two remain separate, but the shared founder, overlapping investor bases, and their existing VECTOR-plus-Virtus reconnaissance-strike integration keep the option in view.

24. Source notes and transparency limits

Sources used Primary company sources included Quantum Systems' official company overview, product pages (Vector AI, Twister, Reliant, Trinity Pro, Trinity Tactical), the MOSAIC UXS and ground-autonomy pages, the locations and careers pages, the investors page, the legal notice, and the company's own press releases covering funding rounds, acquisitions, site openings, contracts, and management changes.

Government and regulatory sources used Official and quasi-official public sources included the German commercial register for entity and shareholder-structure detail, BAFA and the EU Dual-Use Regulation for export-control context, BAAINBw materials for German procurement context, and DPMAregister as the correct source for verifying patents, utility models, and trademarks tied to Quantum-Systems GmbH.

Independent external sources used Independent external support included Reuters and CNBC for the Series D and broader market context, Bloomberg and Reuters for earlier valuation and Bundeswehr order reporting, the Financial Times for the STARK-merger commentary, and selected trade outlets for site, contract, and headcount color. These were used only as support, not as substitutes for company or official filings.

Transparency limits Quantum Systems is a private company and does not disclose the full financial package an analyst would normally expect. Revenue, EBITDA, gross margin, cash flow, working capital, leverage, backlog, divisional splits, and customer concentration are not visible in the reviewed company materials. Cap-table percentages beyond the Figure 37 register extract, control-rights hierarchy, and shareholder-agreement terms are also not public, and headcount figures vary by source and date. Reported 2025 revenue (~€300m) and profitability are company statements rather than audited public accounts.

Explicitly labeled estimates and inferences The following were treated as inference rather than disclosed fact: that Quantum is best understood as an emerging system integrator and autonomy-stack provider rather than a pure airframe vendor; that the business is now defense-led despite its dual-use "AND company" positioning; that current economics still lean hardware-led while the software and lifecycle-service mix grows; that the moat is widening from a relatively early base; and that the $1.2bn Series D reduces near-term IPO pressure to optionality rather than necessity. These judgments are grounded in the disclosed product set, transactions, funding, and partner map, but are not themselves company disclosures.

Alpine Eagle

Coordinated Air-To-Air Defence

Figure 40: Alpine Eagle - overview

Alpine Eagle GmbH			
Ownership:	Privately held VC-backed startup; public investors include IQ Capital, HTGF, Expeditions Fund, Sentris Capital, General Catalyst, and HCVC		
HQ:	Muenchen, 80538, Germany	**Founded:**	2023
Domain:	Air – airborne counter-UAS for low-altitude defense		
Value chain:	OEM / software-defined counter-drone provider		
Why it matters	Alpine Eagle is one of the few European startups pursuing air-to-air counter-drone defense as a scalable, lower-cost answer to mass drone threats		
Key products	Sentinel – airborne counter-UAS system combining radar, EO/IR, fire control and interceptors in a networked architecture		
Core capabilities	Machine learning, computer vision, multi-sensor airborne detection, hardware-agnostic integration and rapid scalable deployment		
Customers	The clearest public customer is the Bundeswehr as launch customer; public/secondary materials also reference other government agencies, three additional European customers, participation in a Dutch defense innovation program, tests in Ukraine and Project Vanaheim with U.S. and UK armed forces		

Source : Deutsche Bank research, company data

What type of company it is

Alpine Eagle is a Munich-based defense technology start-up focused on airborne counter-UAS systems. Its flagship Sentinel platform combines sensors, autonomy software, and interceptor capabilities to address one of the fastest-growing requirements in modern warfare: protection against low-cost drones. Since its founding in 2023, the company has gained early customer traction, secured venture-capital backing, and begun expanding beyond Germany.

Why this company is relevant now

Alpine Eagle is best described as a private, venture-capital-backed European defense technology company focused on airborne counter-UAS systems. In practical terms, it is not a prime contractor in the classic sense. It looks more like a niche system house and subsystem integrator that combines software, sensor fusion, onboard sensing, interceptor logic, and selected third-party hardware into an operational counter-drone architecture. The company's flagship offering, Sentinel, is positioned as a networked airborne counter-drone system that can integrate with existing radar, C2, and short-range air-defense layers. That places Alpine Eagle in a strategically important layer of the defense stack: not at the level of large platform defense contractors, but in a constrained node where battlefield urgency, modularity, and rapid deployment can matter more than size.

Investment-style or strategy-style summary

Alpine Eagle looks like an early-stage but increasingly relevant European defense company focused on a narrow and urgent problem set. Its business mix appears concentrated around Sentinel and related counter-UAS architecture, with



software, sensor fusion, airborne sensing, and interceptor integration as the core competencies. Public materials suggest the company is deliberately building around modularity and interoperability rather than a closed monolithic stack.

Financially, public disclosure remains thin, as expected for a private company. The most useful hard data point is that Alpine Eagle raised a €10.25m seed round in March 2025 and that one investor, HTGF, stated the company had already reached 7-digit revenue in its first 12 months of operation. That indicates early commercial traction, but it is not enough to judge margin durability, capital intensity, or cash conversion with confidence.

From a defense relevance perspective, Alpine Eagle appears more important than its current size implies. It is operating in a choke point category, has publicly available information of Bundeswehr engagement, has referenced Ukrainian operational deployment, and has expanded into the UK and the Netherlands. If execution continues, the company could become a strategically interesting subsystem specialist, acquisition target, or European niche champion. If execution slips, the same concentration around one product family and one urgent market can become a source of fragility.

Alpine Eagle deserves attention because it appears to be an early but credible European entrant in a defense segment that has moved from peripheral to central. The company is still too opaque for a fully conviction-based financial view, but publicly available information supports a balanced conclusion that it has meaningful strategic relevance, unusually fast early traction, and real optionality if it can scale production, sustain technical performance, and navigate defense procurement and export-control complexity.

1. Why this company is relevant now

Alpine Eagle is relevant now because counter-UAS has become a front-line operational requirement rather than a niche capability. European defense demand is expanding in drones, loitering munitions, and layered air defense, which also raises demand for systems that can detect and neutralize small autonomous threats quickly and at useful range. Alpine Eagle is directly exposed to that shift.

The company also matters because it is addressing a specific weakness in many existing C-UAS concepts. Alpine Eagle argues that ground-based solutions often detect small drones only at short range, while drones themselves can operate at much longer distances. Its answer is to move sensing and interception into the air. Whether or not one accepts every marketing formulation, the underlying operational logic is clear and timely.

Lastly, Alpine Eagle is relevant because it seems to have advanced unusually quickly from founding to field use. Public disclosures point to founding in 2023, Bundeswehr trials in 2024, a seed round in 2025, Ukrainian combat-environment deployment in 2025, and production scaling in 2026. That sequence makes the company strategically noteworthy despite its short track record.

2. What type of company it is

Alpine Eagle is a private defense technology company rather than a classic defense group. It is best understood as a specialized counter-UAS system developer that combines software, sensor fusion, airborne sensing, interceptor logic, and selected third-party components into a mission system. The legal entity publicly visible in Germany is Alpine Eagle GmbH, based in Munich.



In value-chain terms, the company appears closest to a niche system house or subsystem integrator. It does not look like a broad mixed industrial group and there is no publicly available information that it operates large diversified civilian divisions. Instead, the public record suggests a focused defense player whose product logic is modular and integration-heavy.

3. Three things that matter most

The first thing that matters most is speed. Alpine Eagle's public timeline suggests rapid movement from founding to military trials, then to international expansion and manufacturing scale-up. In defense, especially in counter-UAS, speed of iteration can be as strategically important as scale.

The second thing that matters most is architecture. The company's core value proposition is not merely a single interceptor drone. It is a broader airborne C-UAS system that combines radar, EO/IR, sensor fusion, operating software, and modular effectors, while also integrating with existing radar, C2, and SHORAD environments. That gives it a potentially wider role than a pure hardware vendor.

The third thing that matters most is transparency risk. Because Alpine Eagle is private, many standard diligence anchors remain unavailable in public sources. Investors or bankers can verify the legal entity, seed round, some management information, and certain public milestones, but not a full audited financial picture, detailed ownership percentages, or long-form customer concentration data.

4. At a glance

Alpine Eagle is headquartered in Munich and operates as Alpine Eagle GmbH. Public company materials describe it as building scalable defense capabilities against autonomous threats, with Sentinel as its flagship airborne counter-UAS system. The company's operating posture is clearly defense-led, even if some marketed use cases extend to border, airport, shipping, and infrastructure security.

The company was founded in 2023 and raised a €10.25m seed round in March 2025 led by IQ Capital, with participation from HTGF, Expeditions Fund, Sentris Capital, General Catalyst, and HCVC. HTGF stated that Alpine Eagle had already generated 7-digit revenue in its first 12 months. That is a useful traction signal, but detailed financial accounts remain limited in public view.

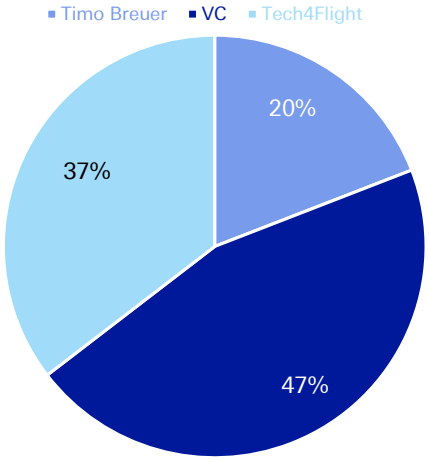
Operationally, Alpine Eagle appears to be moving from prototype-stage credibility to industrialization. Public reports indicate that first systems were delivered to the Bundeswehr in 2024, that Sentinel was deployed with Ukrainian forces in Q2 2025, and that by March 2026 the company was opening a 2,000 sqm facility near Munich to scale interceptor production.

Figure 41: Alpine Eagle Ownership

Legal entity	Entity number	Ownership %
Alpine Eagle GmbH	HRB 286074	
Tech4Flight GmbH	HRB 747662	36.7%
Timo Breuer	-	19.8%
Elephants & Ventures SAS	201462925	16.8%
IQ Capital Fund IV A LP	-	8.9%
GC XII IGNITION - GERMANY HOLDINGS, LLC	201715435	5.6%
HIGH-TECH GRÜNDERFONDS IV GMBH & CO. KG	HRA 9754	4.6%
IQ Capital Partners Nominee Limited	-	4.0%
Rest		3.6%

Source : Gain.Pro

Figure 42: Alpine Eagle Ownership Structure



Source : Gain.Pro

Key takeaways

- Alpine Eagle GmbH is a private defense startup founded in Munich in 2023, focused on airborne counter-UAS capabilities.
- Its core product is Sentinel, an airborne counter-drone system built around a modular, software-defined, and interoperable C-UAS architecture.
- Early public customer references include the Bundeswehr and other government agencies, while public reporting also points to operational use in Ukraine.
- The company raised a €10.25m seed round in March 2025 and has since expanded into the UK and the Netherlands while scaling interceptor-production capacity near Munich.
- Publicly available information suggests strong early commercial traction, with HTGF reporting seven-digit revenue during the company's first 12 months of operation.
- Public disclosure is still limited versus listed defense peers.



5. Corporate history and legal structure

Alpine Eagle's German operating entity is publicly identified as Alpine Eagle GmbH, registered in Munich under HRB 286074. The company imprint lists the address as Prinzregentenstraße 54, 80538 Munich, and identifies Jan Hendrik Boelens as the legal representative and managing director. Open register data shows the company was founded on 17 July 2023 and remains active.

Public sources suggest a founder-led structure centered on Jan Hendrik Boelens, who is presented by the company as co-founder and CEO. Alpine Eagle's public team materials also identify Timo Breuer as co-founder and Director Sensor Fusion. The company was incorporated as a German GmbH, which is typical for venture-capital-backed German start-ups before any later holding-company optimization or larger financing event. No listed securities were identified.

The publicly visible capital structure is only partially transparent. Official register information shows registered share capital of €44,283 as of March 2025 after successive capital increases. That figure should not be confused with enterprise funding or valuation. It is legal share capital, not a measure of balance-sheet scale. Public sources do confirm a €10.25m seed round in March 2025, but not a full cap table or ownership breakdown.

The group structure appears to be relatively centralized, but with signs of international extension. In the UK, ALPINE EAGLE DEFENCE LTD was incorporated on 28 March 2025, with Jan Hendrik Boelens listed as director and person with significant control. That suggests early cross-border structuring to support commercial, government, or partnership activity outside Germany. Publicly accessible sources did not reveal a complex multi-tier holding structure.

A concise timeline of visible inflection points follows. In 2023, Alpine Eagle GmbH was founded in Munich. In 2024, the company says first systems were delivered to the Bundeswehr for operational trials. In March 2025, it announced a €10.25m seed round. In Q2 2025, it said Sentinel was deployed with the Ukrainian Armed Forces in combat environments. In July 2025, it announced UK expansion and public association with UK Ministry of Defence Project Vanaheim. In March 2026, public reporting pointed to a new production site near Munich and scaling to multiple new European customers.

6. Business overview

Alpine Eagle's business is currently best understood as concentrated around a single mission area: airborne counter-UAS. The company markets Sentinel as a networked system combining airborne sensors, operating software, and interceptors to detect, track, and defeat autonomous aerial threats. This is a focused business model, not a broad portfolio conglomerate.

The current structure appears to have been built around an explicit problem statement. Alpine Eagle argues that ground-based detection is often too short-ranged against modern drones and that effective defense therefore requires a mobile, airborne layer. That concept shapes the product architecture and likely also the company's commercial identity.

There is some evidence of adjacent civilian or homeland-security applicability. Alpine Eagle's use-case materials reference defense, border security, airports, shipping, and high-value infrastructure. Still, disclosed traction is overwhelmingly defense and government-oriented. The current center of gravity is therefore military

and public-sector security rather than commercial enterprise.

Because the company is private, there is no fully disclosed segment reporting. No publicly available information reviewed here supports a formal divisional structure akin to listed defense groups. The business should therefore be viewed as product-led and capability-led rather than segment-led at this stage.

7. Position in the value chain

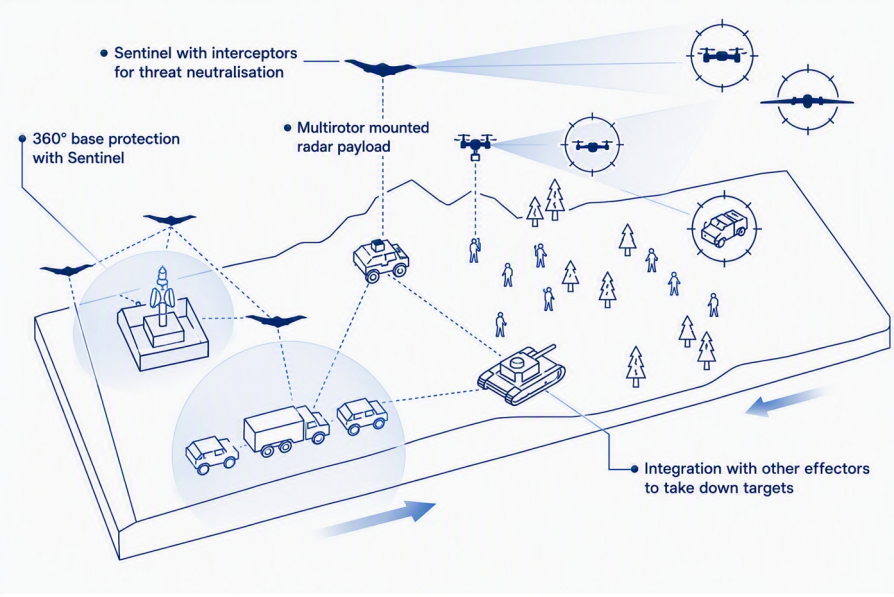
Alpine Eagle appears to sit in the defense value chain as a specialist C-UAS system developer and integrator. It is not a large complete-platform manufacturer for multi-domain platforms, but it is more than a simple component supplier. The company seems to assemble a mission architecture spanning sensing, software, mission control, and airborne interception.

Its role is best described as a mixed model of subsystem specialist and system integrator. The software layer, sensor fusion, and interoperability logic appear central to its differentiation, while parts of the airframe and radio architecture can be sourced or integrated from partners. Public product materials and manuals point to use of third-party elements such as Auterion Mission Control, Silvus radios, and situational-awareness components.

That position in the value chain is strategically attractive because it targets a constrained node in modern defense architecture. If the inference is correct, Alpine Eagle is trying to own the mission logic and airborne counter-drone workflow while remaining modular at the hardware layer. That can be capital-efficient and deployment-friendly, but it also creates dependency on integration execution and partner availability.

8. Products, services, and core capabilities

Figure 43: How Sentinel works

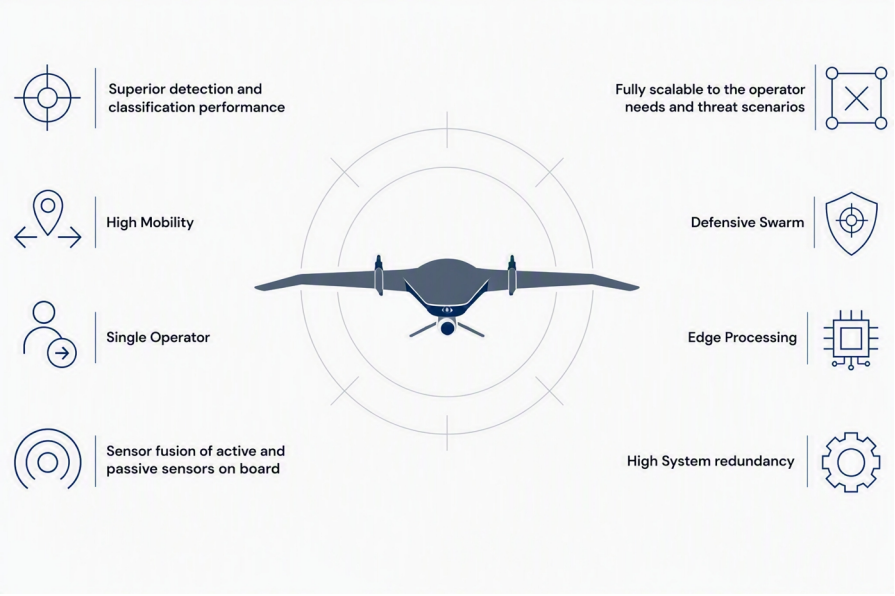


Source : Deutsche Bank Research, company data

Alpine Eagle's flagship offering is Sentinel, which the company describes as an airborne counter-UAS system and, in some materials, as the world's first air-to-air

counter-drone system. That 'first' claim should be treated as company positioning rather than an independently verified market fact. What is more important is the disclosed product concept: Sentinel is a networked architecture combining detection, tracking, classification, and interception in the air rather than relying only on fixed ground-based layers.

Figure 44: Sentinel key features



Source : Deutsche Bank Research, company data

The public product architecture is more substantive than a simple drone interceptor label suggests. Company materials reference integrated airborne radar, EO/IR, edge processing, onboard sensor fusion, high mobility, redundancy, single-operator use, and modular payload options ranging from ISR to kinetic defeat. The product page also stresses the ability to integrate with existing radar, C2, SHORAD, and other effector systems. That interoperability claim is strategically important because it lowers the barrier to insertion into existing defense architectures.

Manual excerpts provide additional technical color. They indicate use of Auterion Mission Control as the main ground-control software, a Silvus StreamCaster radio, and situational awareness systems. Those details support the broader view that Alpine Eagle is building an open, software-heavy operational stack rather than a closed, proprietary ecosystem. This is consistent with management's public emphasis on modularity and integration.

Public disclosure does not provide a segment-level margin profile or product gross margins. That said, the likely economic logic is that software, mission systems, and integration carry structurally better margin potential than pure airframe hardware, while interceptor manufacturing and rapid scaling can be margin-dilutive in the near term. This is an inference, not a disclosed fact.

In terms of portfolio quality, Alpine Eagle's strongest public indicators are operational rather than financial. The company cites Bundeswehr trials and



Ukrainian use, which matter more in this niche than consumer-style reviews or broad installed-base metrics. The main product-level strengths appear to be mobility, fast deployment, modularity, and software-defined integration. The main risks are product concentration, execution complexity, and the challenge of proving sustained effectiveness across varied threat environments.

The Sentinel product family is now presented as a family - including the Sentry sensor node, the Sparrowhawk airborne element, and the hardware-agnostic Sentinel-OS software backbone - reinforcing the company's positioning around a networked, modular C-UxS operating system rather than a single interceptor.

External trends are favorable to the product roadmap. European rearmament, anti-drone urgency, procurement adaptation, and the need for layered air defense all support demand. The main offsetting risks come from export controls, regulatory compliance, and the possibility that incumbent ground-based and EW-heavy vendors expand faster than Alpine Eagle can industrialize.

9. Customers and routes to market

Publicly disclosed customer data is limited, but enough exists to sketch the commercial shape of the business. The company and investors have pointed to the Bundeswehr as an early customer or launch customer, have referenced other government agencies, and have publicly stated deployment with Ukrainian armed forces. That implies a predominantly government and defense customer base at present.

The route to market appears to be direct and programmatic rather than channel-heavy. This is typical for defense start-ups selling emerging capability into military users, innovation programs, and government agencies. Public materials also point to participation in UK Ministry of Defence Project Vanaheim and a Dutch defense innovation program, suggesting that trials, pilot programs, and challenge-led procurement mechanisms may be a meaningful part of commercial entry.

Geographically, the most visible customer and engagement footprint includes Germany, Ukraine, the UK, and the Netherlands. The company has also said it expanded into the UK and the Netherlands in 2025. That cross-border pattern is strategically useful because it reduces dependence on a single national procurement cycle, although publicly available information is still too limited to estimate country revenue mix.

Revenue quality likely has both attractive and risky elements. Defense and security customers can support sticky repeat business once systems are qualified and adopted, especially if software, upgrades, training, sustainment, and interceptor replenishment develop into recurring streams. But near-term revenue is likely still lumpy and program-driven. No public source reviewed here disclosed top-customer concentration or backlog, so concentration risk should be assumed meaningful until proven otherwise.

10. Key financials and economic profile

Public financial disclosure for Alpine Eagle is limited. The most important verified funding fact is the €10.25m seed round announced on 6 March 2025. HTGF, one of the participating investors, also stated that the company had generated 7-digit revenue in its first 12 months of operation. Those are useful anchors, but they are not substitutes for a full income statement or balance sheet.



The public record does not provide a complete set of current sales, EBIT, EBITDA, net income, cash, free cash flow, capex, or R&D figures. There is evidence that annual accounts for 2024 were published in February 2026, but the accessible registry summaries reviewed here do not expose detailed line items. As a result, any detailed financial model would require direct company information or paid registry access beyond the visible public snippets.

What can still be said is that Alpine Eagle's economic profile looks typical of an early-stage defense technology company moving from development into production. The company appears to have enough external funding to support scaling, public traction strong enough to support early revenues, and enough hiring and facility expansion to imply rising operating expense. That combination usually produces a business that is strategically promising but still early on financial maturity. The statement is an inference, not a disclosed margin conclusion.

The registered share capital of €44,283 should be read only as legal capitalization, not as enterprise scale. The economically meaningful capital data point is the seed round. Beyond that, public financial visibility remains one of the main diligence gaps.

11. Footprint and industrial setup

Alpine Eagle is headquartered in Munich, with public references also to teams in London and Karlsruhe. That suggests a still-concentrated but increasingly multi-site setup centered on Germany with selective international expansion.

The industrial footprint appears to be evolving rapidly. In March 2026, public reporting stated that the company was opening a 2,000 sqm production facility near Munich to scale production of its self-developed interceptor. The same reporting said the company expected output to scale from several hundred units per month to more than a thousand per month. That is a major industrial step-change if executed.

There is also evidence of partnership-enabled manufacturing. Reporting tied Alpine Eagle's scale-up to cooperation with DeltaQuad, a Dutch drone producer whose Evo platform and associated industrial processes are already in use. That reduces time-to-scale but also means Alpine Eagle's industrial model likely blends in-house mission integration and selected partner-enabled platform production.

From a strategic standpoint, the footprint suggests a company moving from R&D-heavy startup mode toward a more industrial defense posture. The main question is whether the physical expansion is sufficient to support repeatable delivery at acceptable cost and quality under military demand conditions.

12. Supply chain and key dependencies

Several dependencies are visible even though the company does not disclose a formal supplier map. First, Alpine Eagle depends on radios, sensing hardware, software infrastructure, and likely specialized components that are common choke points in defense electronics. Manual references to Silvus radios and Auterion mission-control software illustrate at least some third-party technology reliance.

Second, the company likely depends on partner-enabled platform availability. Public reporting around DeltaQuad suggests Alpine Eagle is not building every hardware element from scratch, which helps speed but also introduces external manufacturing dependency. This should be read as a mixed positive and risk: it

supports fast scaling, but reduces vertical control.

Third, export control is an operational dependency as much as a legal one. For German defense companies, export permissions and product classification can shape delivery timing and addressable markets. BAFA, the Federal Office for Economic Affairs and Export Control, is central in the German licensing system, alongside war-weapons, foreign-trade, and EU dual-use frameworks. For Alpine Eagle, this is likely material because drones, counter-drone effectors, sensors, and software can fall into sensitive categories.

Fourth, talent is clearly a key dependency. Job postings point to demand in robotics, computer vision, software, simulation, and UAV systems. In a market where technical defense talent is scarce, Alpine Eagle's ability to recruit and retain high-quality engineering teams is likely one of its most important operating constraints.

13. Management, board, ownership, and governance

The most visible senior executive is Jan Hendrik Boelens, identified by the company as co-founder and CEO and by the legal imprint and registry as managing director. His background includes prior senior engineering roles at Volocopter, Quantum Systems, Airbus Helicopters, and the German Aerospace Center, which is directly relevant to Alpine Eagle's product domain. Public sources also identify Timo Breuer as co-founder and Director Sensor Fusion, with prior research experience at Fraunhofer and Microsoft Research.

Public sources also identify several additional senior leaders, including Klaus Kittmann as Engineering Director, Jamie Bennett as Commercial Director (a former Royal Air Force officer with NATO HQ experience), Theresa Prietz as Head of People & Talent Acquisition, and Claudia Steinhoff as Head of Communications & Marketing. That suggests the company is formalizing functional leadership as it scales, though the public record does not yet show the depth of management disclosure one would expect from a listed company.

Governance transparency remains limited. No public supervisory board composition or full board-of-directors architecture was clearly disclosed in the reviewed sources. In German legal terms, the visible formal representative is the managing director, while UK filings show Jan Hendrik Boelens as director and person with significant control for the UK entity. That supports the view that governance remains founder-centered.

Ownership disclosure is also limited. Public sources confirm the investor syndicate in the March 2025 seed round, but not post-money percentages. The founder appears to retain significant control at least in the UK entity, but the exact ownership structure at the German parent level cannot be established from the sources reviewed.

Headcount data has improved over time. HTGF stated in March 2025 that the team was expected to grow to more than 40 people. By March 2026, press reporting said Alpine Eagle had grown from 12 employees in 2024 to about 50, with plans to reach 100 in 2026. That points to rapid scaling and therefore rising management and process complexity.

14. Competitive positioning

Competition in counter-UAS is fragmented and should be viewed by business line rather than through one forced peer set. Alpine Eagle's closest direct comparison



appears to be companies pursuing drone-on-drone or airborne interception concepts, such as Fortem Technologies and its DroneHunter family. A second competitive bucket includes broader C-UAS ecosystems built around sensors, RF detection, electronic attack, and command-and-control, such as DroneShield, Dedrone by Axon, and elements of Anduril's force-protection stack.

Alpine Eagle's main differentiator appears to be its emphasis on an airborne interception layer within a modular, software-defined architecture. That can be attractive where ground-based systems lack range, mobility, or reaction time. Fortem has some similar logic with autonomous interceptors, while DroneShield and Dedrone are more strongly associated with detection, EW, and C2 layers. Anduril is broader, larger, and more integrated, but also more complex and less directly comparable as a private European niche specialist.

Relative strengths likely include speed, European sovereign positioning, modular integration, and real-world operational learning. Relative vulnerabilities likely include scale, procurement reach, installed base, and limited public financial resources compared with better-capitalized US and Australian peers. In practical terms, Alpine Eagle does not need to outscale every competitor globally to matter. It needs to become one of the credible European specialists in its chosen layer of the stack.

15. Strategic relevance and market implications

Alpine Eagle matters beyond its current size because it operates in a capability area that has become strategically visible for Europe. Counter-UAS is no longer an ancillary market. It sits close to the center of force protection, infrastructure security, border defense, and battlefield survivability. A successful European specialist in this category has significance that exceeds its current revenue base.

The company also has relevance to broader sovereign industrial policy. Europe is trying to add capacity not just in traditional munitions and large platforms, but also in agile defense technologies that can iterate on modern conflict lessons. Alpine Eagle's UK and Dutch activity, and its operational links to Germany and Ukraine, fit that pattern.

For listed peers, suppliers, and potential partners, Alpine Eagle is relevant in three ways. It may become a partner to larger system houses that need airborne C-UAS modules. It may become an acquisition candidate if its technology proves sticky. Or it may become a pressure point for incumbents if it accelerates faster than traditional procurement models. At present, the partner-or-target scenario appears more visible than the stand-alone defense-group scenario. That final point is an inference.

16. Dual-use and civilian exposure

Alpine Eagle's marketed use cases extend beyond military customers. Public materials mention defense, commercial shipping, high-value assets, border security, and airport security. That indicates genuine dual-use adjacency.

Even so, the company should currently be thought of as defense-led rather than a balanced dual-use platform. The strongest disclosed customer evidence relates to the Bundeswehr, government agencies, UK defense programs, and Ukrainian deployment. There is no comparable publicly available information of major civilian revenue today.



Civilian exposure could diversify the story over time, especially in critical infrastructure and transport security. But in the current phase, civilian adjacency looks more like optionality than the main economic engine. That means Alpine Eagle is closer to an emerging defense pure-play with dual-use extension potential than to a diversified commercial drone company.

17. Legal, regulatory, and compliance risks

Alpine Eagle's overall legal and regulatory risk should be rated Medium, with the main concentration in export control, defense procurement rules, product compliance, and national-security sensitivity rather than in publicly visible litigation. The company operates in a domain where licensing, classification, export permissions, radio compliance, and defense customer requirements can materially affect operations.

German export-control exposure is likely the most important structural risk. BAFA plays a central role in licensing military and dual-use exports, while German and EU frameworks govern war weapons, other military goods, and dual-use items. For a company selling airborne C-UAS systems, especially across borders and into conflict-adjacent settings, the regulatory path can be strategically important. Future growth could be slowed or redirected by licensing decisions, destination restrictions, or evolving interpretation of controlled technologies.

Procurement-law exposure is also relevant. Early-stage defense suppliers often win through trials, pilots, innovation competitions, or urgent procurement pathways before facing more formal scaled procurement. The UK Project Vanaheim connection is a good example of how opportunity can come through challenge-led or capability-driven processes rather than classic long-cycle procurement. That can accelerate adoption, but it can also make revenue timing less predictable.

On intellectual property, public registry-derived sources indicate trademark protection for 'Alpine Eagle' and 'Sentinel'. One trademark source notes that the 'Sentinel' mark is registered but subject to opposition proceedings. A patent family has since become verifiable: 'Distributed sensor network implemented by swarm of unmanned autonomous vehicles' (filed February 2024) is now granted in Europe (EP-4610765-B1) and Australia (AU-2025201334-B2), with additional filings pending. This gives the moat a modest formal-IP anchor beyond trademarks and know-how, though the portfolio remains small and early.

18. Funding, financing, and strategic optionality

The most important visible financing event is the €10.25m seed round announced in March 2025. The investor base is notable because it mixes deep-tech and defense-relevant capital, including IQ Capital, HTGF, Expeditions Fund, Sentris Capital, General Catalyst, and HCVC. That syndicate gives Alpine Eagle both capital and signaling value.

There is no publicly available information of listed debt, public bonds, or a broad leveraged capital structure. This looks like a venture-capital-backed private company financing model rather than a debt-heavy industrial one. The company's funding posture therefore likely remains equity-led. That fits an early-stage defense technology business that is still scaling product and production.

Strategic optionality appears relatively broad. If Alpine Eagle continues to prove product effectiveness and scale production, it could remain independent and compound privately. It could also become an acquisition candidate for a larger



defense electronics, air-defense, or autonomous-systems player. A later IPO is conceivable in theory, but based on current public disclosure and company maturity, M&A or continued private scaling looks more plausible in the medium term. The last sentence is an inference.

The March 2026 production expansion also suggests that the next major strategic issue is not merely raising money, but deciding how aggressively to build capacity versus preserving flexibility. That balance often determines whether an early defense startup becomes a durable supplier or remains a promising niche innovator.

19. Moat analysis

Alpine Eagle's primary moat appears to be a combination of mission-specific software integration, operational relevance, and early qualification in a fast-moving defense niche. In simple terms, the company is trying to build a system that is hard to replicate not because the airframe alone is unique, but because the full counter-UAS workflow, airborne sensor fusion, and system integration become operationally credible only through repeated iteration and field use.

The moat appears potentially widening, but still early. It is widening to the extent that Alpine Eagle adds real deployments, customer references, manufacturing scale, and interoperability into existing defense architectures. It remains early because the company is young, private, and not yet visibly entrenched in the way mature defense suppliers are entrenched.

Switching costs may become meaningful over time if customers rely on Alpine Eagle's system integration, mission software, training, and replenishment workflow. Today, those switching costs are probably moderate rather than high, because the category itself is still evolving and many buyers are still experimenting. Certifications, export permissions, and qualification cycles may ultimately create stronger barriers than brand alone.

Intangible-asset support now includes a granted patent family covering distributed swarm sensing, alongside trademarks and know-how, though it is still too small to constitute a strong patent wall on its own; the more durable barrier remains integration know-how, qualification, and field-proven performance.

20. SWOT analysis

Strengths

- Clear positioning in a high-priority defense niche.
- Evidence of early military traction and operational use.
- Modular, software-defined architecture with interoperability emphasis.
- Founder and engineering team with relevant aerospace and autonomy backgrounds.

Weaknesses

- Limited public financial transparency as a private company.
- High current product concentration around Sentinel and related architecture.
- Small scale relative to established global C-UAS peers.

- Likely founder and key-engineering person dependence.

Opportunities

- European rearmament and anti-drone demand growth.
- Expansion through UK, Dutch, and other European government channels.
- Potential recurring revenue from software, upgrades, sustainment, and interceptors. This is an inference.
- Strategic optionality as partner or acquisition target. This is an inference.

Threats

- Export-control and licensing friction.
- Rapid competitive response from better-capitalized peers.
- Production scaling risk and supplier dependency.
- Procurement timing and budget volatility.

21. Key risks and open questions

1. Can Alpine Eagle convert early deployments and trials into repeatable scaled procurement, or will it remain pilot-heavy?

2. How differentiated is Sentinel's real-world performance versus Fortem-style airborne interceptors and broader C-UAS stacks from DroneShield, Dedrone, and Anduril?

3. How much of the company's value lies in software and integration versus hardware and interceptor manufacturing? Public sources do not fully resolve this.

4. What does the true revenue mix look like between system sales, trials, software, sustainment, and replenishment? Public disclosure is limited.

5. How concentrated is customer exposure by country and by agency? Public sources do not disclose this.

6. How exportable is the current product set under German and EU control regimes?

7. Can the company scale production without sacrificing quality, speed, or capital discipline?

8. Does the current governance structure remain fit for purpose as headcount, sites, and sovereign customer exposure increase?

22. What to watch next

1. Evidence of repeat Bundeswehr or wider German government orders beyond trials.

2. Additional named European defense customers or framework agreements.

3. Progress of UK Ministry of Defence Project Vanaheim and any associated procurement outcomes.

4. Scaling of the near-Munich interceptor facility and proof of delivery cadence.



5. Whether Alpine Eagle discloses more detail on software, operating system, and recurring revenue components.

6. Further hiring in engineering, simulation, autonomy, and operations, which would signal continued ramp.

7. Any follow-on funding round or strategic investor with industrial defense relevance.

8. Any disclosed export approvals, international deployments, or restrictions that clarify commercial addressable market.

9. Evidence of product family expansion beyond the current visible Sentinel-centered architecture.

10. Any formalization of board, compliance, or governance structures as the company matures.

23. News flow and recent developments

Recent news flow has been dense for a company of Alpine Eagle's age. The most important financing event was the 6 March 2025 seed round of €10.25m, disclosed by HTGF and others. That round mattered not just because of capital raised, but because it publicly validated the company's early traction and investor syndicate quality.

The second major theme has been international and operational expansion. Alpine Eagle says it delivered first systems to the Bundeswehr in 2024, deployed with the Ukrainian Armed Forces in Q2 2025, expanded into the UK and the Netherlands in 2025, and engaged in UK Project Vanaheim. That pattern suggests the company is building relevance through operational credibility and allied-market penetration, not only through fundraising.

The third major theme has been industrialization. March 2026 reporting pointed to a new 2,000 sqm facility near Munich, a production ramp from hundreds to over one thousand units per month, a team size of about 50, and three new European customers. If accurate and sustained, that marks the transition from startup novelty to emerging defense manufacturing relevance.

A final theme is management build-out. The executive team has been strengthened around Engineering Director Klaus Kittmann and Commercial Director Jamie Bennett, alongside dedicated people/talent and communications leadership, consistent with a company moving from founder-led early execution toward broader organizational scaling.

In June 2026, at Eurosatory in Paris, Alpine Eagle signed a partnership with Latvian firm Origin Robotics to integrate Origin's BLAZE interceptor into Sentinel, creating a layered sensor-to-effector architecture. This deepens Alpine Eagle's move toward an open, best-of-breed effector layer rather than a single proprietary interceptor.

24. Source notes and transparency limits

Company sources used

Alpine Eagle's homepage, About page, product page for Sentinel, imprint, contact page, careers pages, and accessible product-manual excerpts were used as the core company-source base. These sources support the company's self-

description, product architecture, visible management team, legal identity, and some timeline claims.

Government and regulatory sources used
German registry data, UK Companies House, BAFA export-control materials, German government export-policy descriptions, EU dual-use regulation sources, and UK procurement material for Project Vanaheim were used to support legal-structure and regulatory analysis.

Independent external sources used
HTGF and IQ Capital were the most useful independent-public sources on funding, traction, and investor narrative. Reuters and selected secondary outlets were used for market context and recent developments. Official competitor websites were used for business-line comparison.

Figure 45: Dynamit Nobel Defence - overview

Dynamit Nobel Defence GmbH			
Ownership:	German private subsidiary within the Rafael group; public filings state Rafael Ltd., Haifa is the highest parent		
HQ:	Burbach, 57299, Germany	**Founded:**	2004
Domain:	Land – shoulder-launched infantry weapons and force-protection systems		
Value chain:	OEM / system house		
Why it matters	DND is one of Germany's key shoulder-fired weapons specialists and says it controls critical technologies from R&D and design through production, qualification and verification, while expanding capacity		
Key products	RGW 60 family – lightweight multipurpose launchers; RGW 90 family – anti-tank and multi-purpose systems; RGW 110 – larger shoulder weapon; Panzerfaust 3 – heavy anti-tank weapon; DND Digital / vehicle and fire protection –		
Core capabilities	Disposable/recoilless shoulder-weapon design, warhead technology, qualification-heavy manufacturing and broad land-force protection know-how		
Customers	Public customers include the Bundeswehr, Belgian Armed Forces via NSPA, and Latvia, Denmark & Sweden for Skorpion; DND also says RGW systems are used by numerous European and international armed forces		

Key financials (31 Dec)	2020	2021	2022	2023	2024
P&L (EURm)					
Sales	94	58	126	132	146
Gross profit	50	14	62	54	51
EBITDA	8	-4	15	9	12
EBIT	5	-7	12	5	8
EBT	4	-8	12	4	8
Net income	2	-8	11	3	7
Order intake	-	-	-	-	-
Balance sheet (EURm)					
Cash & cash equivalents	13	8	42	105	53
Tangible fixed assets	21	24	28	30	34
Goodwill & intangible assets	1	1	1	1	1
Inventories	51	68	57	87	120
Accounts receivable	8	3	6	20	25
Other assets					
Total assets	109	108	142	257	253
Equity	31	23	34	37	43
Minorities	-	-	-	-	-
Borrowings	7	8	9	10	10
Accounts payable	8	5	14	10	11
Other ST liabilities	-	-	-	-	-
Net debt (+) / net cash (-)	-7	-1	-33	-95	-43
Key company metrics					
Book/bill (x)	-	-	-	-	-
Sales growth (%)	-41	-38	116	5	10
Net income growth (%)	-77	-	-	-70	105
EBITDA margin (%)	8	-7	12	7	9
EBIT margin (%)	5	-11	10	4	6
ROE (%)	8	-35	32	9	15
Capex/sales (%)	-	10	5	5	5
Capex/depreciation (x)	-	2.3	2.1	1.7	1.8
Net debt/equity (%)	-22	-3	-99	-259	-99
Net interest cover (x)	4	-5	22	5	24

Source : Deutsche Bank research, company data, Bloomberg Finance LP, Gain.pro

Dynamit Nobel Defence

From the Creator of Dynamite to Modern Anti-Tank Weaponry

Investment-style or strategy-style summary

DND is a private, medium-sized German defense manufacturer centered on land-combat systems. Its disclosed 2023 revenue mix shows a core serial-products business supported by newer digital, counter-mobility, development, and service activities, spanning RGW and Panzerfaust shoulder-fired weapons, explosive reactive armor and other vehicle protection, SKORPION^2 and counter-mobility offerings, tactical communications, and civilian and military fire-protection systems. The company serves product areas where demand has grown under the post-2022 European defense cycle and where sovereign industrial capacity matters, and it has moved beyond a purely domestic posture through its Hungarian manufacturing expansion and Latvian mine-production initiative - both fitting current European security-of-supply priorities. As a German operating platform inside Rafael Ltd.'s consolidation perimeter, DND combines a domestic industrial base and Bundeswehr-facing credibility with the backing of a larger defense parent.

1. Why this company is relevant now

Dynamit Nobel Defence GmbH (DND) matters because it sits in several of the land-warfare categories that have moved from peacetime niche demand to rearmament priority. The company is moving from a Germany-centered producer toward a distributed European defense industrial footprint. DND secured a January 2025 framework agreement with BAAINBw for shoulder-fired weapons and Panzerfaust 3 cartridges and won a roughly €50m Latvian SKORPION^2 contract in October 2025. This was followed by a Danish order for eight SKORPION^2 systems in December 2025 and a Swedish FMV framework agreement worth approximately SEK 4bn (€360m) through 2033 in June 2026. Additionally, DND signed a February 2026 MoU for anti-tank mine production in Latvia and is expanding production in Hungary with a site intended to be fully operational by summer 2026.

2. What type of company it is

DND is best described as a niche land-warfare systems and survivability company. Factually, it develops and manufactures shoulder-fired weapons, vehicle protection systems, counter-mobility solutions, fire-protection systems, digital products, and certain test and environmental services. Inference: the company sits between a specialist OEM and a focused systems house, rather than fitting neatly into one category such as 'complete manufacturer' or 'component supplier'.

Its ownership context matters. The legal entity is German, but the annual report says its statements are included in Rafael Ltd.'s consolidation perimeter, and the disclosed supervisory board is heavily populated by Rafael-linked executives. That suggests DND should be viewed as a German operating platform with clear strategic linkage to a larger Israeli defense parent.

3. Three things that matter most

The first core takeaway is product concentration with improving breadth. DND still looks primarily powered by mature serial defense products, but the 2023 accounts also show a step change in digital revenue and an initial contribution from counter-mobility, indicating that the company is not just a single-product launcher story anymore.

The second core takeaway is that cash generation and liquidity are real, but they are contract-model liquidity. Customer prepayments, public-sector order timing, and program execution matter more here than recurring-revenue concepts that would apply to software or service businesses.

The third core takeaway is that DND appears to be turning strategic relevance into industrial geography. Hungary and Latvia are not just export destinations. They point to localization, security-of-supply positioning, and potentially deeper integration into European defense production networks.

4. At a glance

DND is headquartered in Burbach, Germany, and operates as Dynamit Nobel Defence GmbH, a private company registered in Siegen under HRB 6701. The company website currently has more than 500 professionals.

Scale is meaningful but still modest compared with major listed European primes. DND reported €132m of 2023 sales, with 56.4% generated outside the EU and only 28.1% in Germany, underscoring an export-oriented model. The disclosed mix shows a defense-centered business in which serial products dominate, while digital is emerging as a material second leg.

Key takeaways

- Private German defense company headquartered in Burbach, ultimately controlled by the Rafael group.
- 2024 revenue was €146m, with net income of €6.5m; the business remains centered on serial defense products, while digital systems have become an increasingly relevant growth area.
- Core activities include shoulder-fired weapon systems, vehicle protection, counter-mobility solutions, tactical digital systems, and fire protection.
- The company occupies several strategically important land-warfare niches that are benefiting from European rearmament and defense modernization.
- Liquidity appeared robust at year-end 2024, although the balance sheet was materially influenced by customer prepayments.
- Public disclosure is above average for a private defense company but remains well below listed-peer standards.
- The key analytical question is whether newer growth areas such as digital systems, counter-mobility, and localized production can scale fast enough to reduce dependence on the mature launcher business.

5. Corporate history and legal structure

The current legal entity is Dynamit Nobel Defence GmbH, seated in Burbach. OpenRegister shows that the company was first entered on 26 January 2004 as SPETEK GmbH and was renamed Dynamit Nobel Defence GmbH on 7 June 2004. The company's own history materials link it to the older Dynamit Nobel industrial lineage and to the Würgendorf site, whose industrial roots go back to land



acquisition in 1902 and plant approval in 1903.

Historically, the business appears to have evolved out of German energetics and launcher know-how into a broader land-systems portfolio. Company history materials identify Panzerfaust 3 development beginning in 1986 under Dynamit Nobel AG, a 1998 move into civil fire protection via Dynameco, and the 2004 emergence of DND as an independent company in Burbach. Later milestones include the 2017 handover of RGW90 LRMP to the Bundeswehr, the 2023 creation of Dynamit Nobel Defence Zrt. in Hungary, the 2025 BAAINBw framework agreement, and the 2026 Latvia mine-production MoU.

The public legal-structure picture is clear at the operating-company level and less clear above it. The annual report states that DND is included in Rafael Ltd.'s consolidated financial statements, which identify the ultimate parent. The full immediate shareholder chain, however, is not visible in the accessible source set used here. Inference: the group appears relatively centralized around the German operating company, with Hungarian expansion as the main visible structural extension. No listed securities were identified in the registry, company, or annual-report materials reviewed.

Concise timeline of key inflection points

- 1902-1903 - Würgendorf industrial site established: Land was acquired in 1902 and the factory was approved in 1903, creating the industrial base that later became central to the company's physical footprint.
- 1986 - Panzerfaust 3 development begins: Dynamit Nobel AG started development of Panzerfaust 3, establishing a core lineage in shoulder-fired anti-armor systems.
- 1998 - Dynameco fire protection introduced: The business broadened into fire-protection systems, creating one of the clearest civilian and dual-use strands in today's portfolio.
- 2004 - DND founded as independent company: The legal entity emerged from SPETEK GmbH and was renamed Dynamit Nobel Defence GmbH, marking the start of the current operating structure.
- 2017 - RGW90 LRMP handover to Bundeswehr: This reinforced DND's relevance in modern shoulder-launched systems and Bundeswehr-facing programs.
- 2023 - Hungarian production platform established: The annual report discloses the foundation of Dynamit Nobel Defence Zrt. in Hungary to build additional European production capacity.
- 2025-2026 - New procurement and localization phase: BAAINBw framework business, the Latvian SKORPION^2 contract, follow-on SKORPION^2 orders from Denmark (December 2025) and Sweden (June 2026), the Latvia anti-tank mine MoU, and the DOK-ING integration partnership show a rapidly widening operational footprint.

6. Business overview

The clearest picture of DND's current business comes from the 2023 revenue bridge. Serial products represented €100.0m of revenue, communications and reconnaissance €18.0m, counter-mobility €3.5m, development €1.3m, small-series and fire extinguishers and services €6.1m, and site services €3.5m. That mix shows a company whose economic base is still hardware-heavy and defense-led, but no longer limited to a single launcher franchise.



In plain terms, DND's current structure combines mature launcher programs, survivability products, newer tactical digital offerings, and smaller peripheral service and civilian activities. Public materials also indicate that vehicle protection, counter-mobility, environmental services, and qualification support have been added over time rather than existing as original core businesses. Inference: the central businesses today are serial defense products, vehicle protection, and digital, while fire protection, site services, and some testing functions look economically peripheral.

7. Position in the value chain

DND occupies more than one place in the defense value chain. In shoulder-fired weapons, it appears to act primarily as an OEM and systems house, designing and manufacturing finished end-user products for armed forces. In vehicle protection, it can be thought of more as a survivability specialist and subsystem supplier, while in counter-mobility it appears to operate at the level of systems and enabling technology. In tactical digital solutions, the company markets complete communication products, but public disclosure does not fully separate in-house IP from group-sourced technology.

The route to the customer also shows a mixed value-chain role. DND sells into direct government procurement, framework agreements, and localized industrial partnerships. That means it is not just a component maker. It can also be a program-facing industrial partner with local manufacturing and technology-transfer relevance, as seen in Hungary and Latvia.

8. Products, services, and core capabilities

DND's flagship product family is its shoulder-fired weapons portfolio. Public company materials emphasize Panzerfaust 3 and the RGW family, including anti-tank, anti-structure, and multipurpose variants, with RGW110 described as the next-generation successor extending the portfolio into 110mm. These products likely matter most economically because the annual report's serial-products category dominates revenue and because the company has more than 60 years of experience in shoulder-launched weapons.

The second major capability area is vehicle protection. DND describes explosive reactive armor and broader protection solutions as core competencies and says its protection kits are fielded and combat-proven. The company also claims that certain reactive solutions can reduce vehicle weight by up to 80% compared with purely passive concepts, which, if program-specific, is a relevant selling point for mobility and integration. Product-level margin disclosure is not public.

The third important area is counter-mobility, where DND points to roughly six decades of experience in terrain shaping and obstacle concepts. This area has recently become more visible through the Latvian Skorpion^2 contract and the later anti-tank mine-production MoU. The fourth is tactical digital communications, where DND markets B-NET radio and networking products with public specifications including high data throughput, AES-256 encryption, and MANET capabilities. Inference: digital is strategically important because it has grown much faster than legacy lines and may carry better long-term differentiation if it becomes a scaled business.

DND also retains a smaller but notable civilian and dual-use capability in Dynameco fire protection, plus environmental and qualification services. Public quality indicators are stronger at the site and system level than at the individual product

level. The 2023 annual report confirms recertification to DIN EN 9100:2018 and surveillance audits for ISO 14001 and ISO 50001. The company also states that its products are used by armed forces in more than 20 nations. Product-level return rates or warranty metrics are not publicly disclosed, although the 2023 balance sheet did contain €9.4m of warranty provisions.

The main product-level strengths are installed base and qualification depth in launchers, breadth across adjacent land survivability categories, and rising relevance of digital. The main risks are dependence on mature launcher demand, procurement lumpiness, and limited public visibility on the economics of newer product lines. The key near-term catalysts are RGW110 industrialization, the Hungary ramp, digital scaling, and Latvia-related counter-mobility production.

9. Customers and routes to market

DND's primary customers appear to be defense ministries, procurement agencies, and allied armed forces. The company's own materials say its products are used by armed forces from more than 20 nations. Public customers include the Bundeswehr, the Belgian Armed Forces via NSPA, and Latvia, Denmark and Sweden for SKORPION[2]; DND also says RGW systems are used by numerous European and international armed forces. Vehicle protection and digital products also imply relationships with vehicle programs, military integrators, or operational users beyond a pure launcher customer base.

Routes to market appear to combine direct procurement, framework agreements, and local-industrial partnership structures. The January 2025 BAAINBw framework agreement is a direct domestic procurement channel. The Hungarian and Latvian activities show another route: in-country manufacturing, technology transfer, and local partnership. That matters because it can strengthen position in sovereign or politically sensitive markets where supply security matters as much as product performance.

DND's business is international. In 2023, Germany accounted for 28.1% of revenue, other EU countries 15.5%, and non-EU countries 56.4%. Revenue quality looks better than a pure one-off export merchant model because public-sector contracts can produce advance payments and follow-on orders, but it is still not recurring in the software sense. Public customer-concentration disclosure is limited, yet the 2022 annual report states that roughly 75% of annual sales could be generated by only 5 to 10 orders. That implies meaningful concentration and contract lumpiness even if receivables risk is relatively low because customers are mainly public-sector counterparties.

10. Key financials and economic profile

Publicly available financial data indicate that DND expanded further in 2024, with continued top-line growth and a recovery in earnings after a softer 2023. Statutory net revenue increased from €132.4m in 2023 to €145.6m in 2024. A broader operating-performance measure, which also includes inventory changes, capitalized own work, and other operating income, rose from €131.2m in 2022 to €142.2m in 2023 and €161.9m in 2024. The narrower statutory sales figure is the better anchor for revenue analysis, while the broader measure is useful for understanding total operating output.

Profitability shows a clear dip-and-recovery pattern. EBITDA amounted to €15.3m in 2022, fell to €8.8m in 2023, and recovered to €12.4m in 2024. EBIT moved from €12.3m in 2022 to €5.2m in 2023 and €8.2m in 2024. Net income followed the same



trajectory, declining from €10.6m in 2022 to €3.2m in 2023 before improving to €6.5m in 2024. On that basis, 2024 appears to represent a year of renewed earnings progress, although profitability remained below the stronger 2022 level. Gross profit also increased in absolute terms to €67.7m in 2024, from €64.3m in 2023.

The balance sheet remains sizable relative to the income statement and continues to reflect the project-based nature of the business. Total assets stood at €256.7m at year-end 2023 and €252.9m at year-end 2024. Equity increased from €36.7m to €43.2m, lifting the equity ratio from about 14.3% to about 17.1%. Cash declined from €105.3m in 2023 to €52.9m in 2024, but the company still appears to have remained in a net cash position, with reported net cash of €95.0m in 2023 and €42.9m in 2024. A central balance-sheet feature remains the high level of advance payments received on orders, which amounted to €152.7m in 2023 and €142.4m in 2024. This supports liquidity, but it also underlines that part of the cash position is operationally tied to customer-funded program execution rather than representing fully surplus cash.

Working capital increased materially in 2024, suggesting that growth and production ramp-up absorbed additional balance-sheet resources. Inventories rose from €86.8m in 2023 to €120.1m in 2024, while receivables increased from €19.5m to €24.8m. On the reported definition, trade working capital expanded from €84.6m in 2023 to €123.8m in 2024, after €47.6m in 2022. Capex also moved higher, reaching €7.7m in 2024 versus €6.3m in 2023. Overall, the financial profile suggests a company with solid recent growth, improving earnings, and continued customer-funded liquidity support, but also with a more working-capital-intensive operating model as activity scales.

11. Footprint and industrial setup

DND's center of gravity is Burbach in North Rhine-Westphalia. The company says its site in Burbach covers roughly 1,000,000 sqm and 260 buildings. The 2023 annual report places research and development in Würgendorf, with 40 R&D employees there, and describes ongoing investment in production facilities. The company also maintains a Berlin representation office disclosed in the German Lobbyregister.

The most important non-German industrial footprint is Hungary. Official Hungarian material points to a December 2022 joint-venture agreement, and DND's 2023 report discloses a financial investment in the newly founded Dynamit Nobel Defence Zrt. The company later said construction of the Vecsés production site began in January 2025 and that the facility should be fully operational by summer 2026. An earlier secondary report indicated phased series production could start earlier, suggesting the ramp may occur in stages.

Inference: this industrial setup looks more like controlled expansion around a core German site than a fully decentralized multinational network. That matters because capacity additions are visible, but the business still seems dependent on a relatively small number of core facilities and specialized teams.

12. Supply chain and key dependencies

Hard evidence from the annual reports points to several dependencies. Management flagged inflationary procurement costs, raw-material and energy bottlenecks, and longer lead times, especially in the period following 2021 and the start of the war in Ukraine. The same filings also highlight the importance of export approvals, skilled labor, and a relatively small number of very large orders.



Beyond that, the publicly available information becomes thinner and assumptions becomes necessary. Because DND builds shoulder-fired weapons, reactive armor, counter-mobility systems, and digital radios, likely dependencies include energetics and explosives, metals and fabricated barrels, electronic components, encrypted communications hardware, and specialized testing and safety infrastructure. The EU's own ASAP defense-production program is useful context because it identifies bottlenecks in explosives, powders, missiles, and testing capacity across Europe. Public sources reviewed here do not identify a single dominant DND supplier, so supplier-concentration mapping remains a research gap.

The company also appears dependent on key facilities. The Burbach and Würgendorf complex is not just an office site. It is the hub for production, testing, R&D, and compliance-heavy operations. The existence of Würgendorf Genehmigungshaltergesellschaft mbH, which coordinates public-law obligations on the site, reinforces the importance of site-level regulatory and safety infrastructure.

13. Management, board, ownership, and governance

The executive picture is relatively clear. The company's imprint identifies Michael Humbek as managing director, and OpenRegister shows him in that role since 7 August 2017. OpenRegister also lists procuration holders including Sarah Huf, Sven Alexander Volk, Miriam Daum, and, since 25 July 2024, Jens Beimforde, with Markus Fitterling no longer holding procuration from that date. That suggests top-level management stability with some refresh below the CEO level.

Governance is more formalized than at many privately held Mittelstand companies because the 2023 annual report discloses a supervisory board. Avishay Regev served as chair in 2023, with several additional members tied to Rafael functions in finance, legal, and precision tactical weapon systems. That supports the view that parent oversight is strong and that DND is strategically embedded in the Rafael group rather than operating as a loosely affiliated portfolio company.

Ownership transparency is partial. The annual report clearly identifies Rafael Ltd. as the highest parent for consolidation purposes. However, the exact immediate shareholder percentages are not visible in the accessible public material reviewed here.

14. Competitive positioning

Competition needs to be analyzed by business line, not by forcing one peer set across the whole company. In shoulder-fired disposable anti-armor weapons, obvious comparison points are Saab's AT4 and Nammo's M72 families, while MBDA's Enforcer is a more advanced adjacent benchmark in the lightweight precision-guided category. DND appears differentiated by its long launcher lineage, the breadth of its anti-structure and anti-armor portfolio, and its emphasis on confined-space use and upgraded next-generation systems such as RGW110.

In vehicle protection, the competitive benchmark is closer to companies like Rheinmetall with modular armor and survivability systems. Here DND's public differentiation rests on reactive armor know-how, combat-proven protection kits, and the weight-performance argument around add-on protection. In tactical digital systems, the competitive field is broader and public ranking is harder, but DND's B-NET offer indicates a bid to compete on high-throughput, secure tactical networking rather than simple voice radios.

Overall, DND appears to compete more on qualification, combat utility, local industrial fit, and trust than on absolute scale. That is a credible positioning advantage in sovereign defense procurement. Its vulnerability is that larger established defense groups and bigger listed peers have more financial breadth, broader customer access, and often deeper integration into large vehicle and munitions programs. Public sources do not support a precise market-share estimate, so any share claim would be speculative.

15. Strategic relevance and market implications

DND matters beyond its own size because it sits in several land-warfare categories that have moved from peacetime niche demand to rearmament priority: shoulder-fired anti-armor weapons, vehicle survivability, and counter-mobility. These are exactly the areas where Europe is trying to rebuild depth, qualified capacity, and security of supply, giving a mid-sized producer relevance that exceeds its revenue base. Its clearest strategic signal is that it is turning that relevance into industrial geography: the Hungarian production platform and the Latvian mine-production initiative point to localization, technology transfer, and closer integration into European defense networks, fitting current 'buy European' themes and strengthening DND's position in sovereign markets where supply security matters as much as unit performance.

DND's ownership context adds a second dimension. As a German operating platform inside Rafael Ltd.'s consolidation perimeter, the company links a domestic industrial base and Bundeswehr-facing credibility to a larger Israeli defense parent. That connection is visible in the supervisory board and, more recently, in the jointly issued May 2026 Firefly loitering-munitions demonstration, which suggests the relationship could widen DND's addressable portfolio beyond its legacy launcher and protection lines over time - though this is an inference from governance and product-adjacency signals rather than a stated strategy.

For the wider market, DND is best read as a qualified capability anchor in supply-constrained niches rather than a broad-spectrum prime. Its most relevant connections run to larger lead contractors and integrators - Rheinmetall in vehicle protection and munitions adjacencies, and vehicle programs such as Puma where its protection kits are fielded - and to allied procurement bodies through frameworks like BAAINBw and the SKORPION^2 customer group of Latvia, Denmark, and Sweden. In a more consolidated European land-systems market, a trusted, qualification-heavy specialist with expanding sovereign-production credentials is more likely to matter as a partner, localization vehicle, or bolt-on target than as a stand-alone challenger to the largest established defense groups.

16. Dual-use and civilian exposure

DND is not a pure one-line military company in the narrowest sense, but it appears strongly defense-dominant. The clearest civilian and dual-use strand is Dynameco fire protection, which the company markets for both military and civilian use. The company also mentions environmental and qualification services.

That said, the 2023 disclosed revenue mix suggests that civilian activities are not the economic center of the company. Serial defense products, digital, and counter-mobility dominate the strategic narrative, while small-series, fire extinguishers, and services are only a minor share of revenue. Inference: civilian exposure probably offers some diversification benefit, but it does not dilute the basic conclusion that DND should be thought of primarily as a defense company.



17. Legal, regulatory, and compliance risks

Overall legal and regulatory risk appears Medium. The main concentration of risk is not visible litigation. It is the intersection of export control, public procurement, explosives and site regulation, and program concentration. The 2022 annual report explicitly said that more than 50% of budgeted 2023 sales were expected to come from exports, which creates political and regulatory approval risk, and that roughly 75% of annual sales could depend on only 5 to 10 orders.

The public-source review did not identify major active public antitrust proceedings, sanctions actions, or material court cases against DND itself. That absence should not be overread. It means only that no such major matters were visible in the sources reviewed here. What is clearly visible is a compliance-heavy operating environment: quality recertification to EN 9100, environmental and energy audits, site-level public-law obligation management, defense procurement exposure, and export-control policy engagement via the German Lobbyregister.

Chronological regulatory and compliance timeline

- 2022: The annual report flagged export-approval sensitivity and high dependence on a limited number of very large orders.
- 2023: The company reported quality and environmental certifications and carried €9.4m in warranty provisions, which is not in itself unusual for an industrial defense producer but is relevant for product-liability and execution monitoring.
- 2024-2026: The Lobbyregister shows active policy engagement on defense and export matters, while Hungarian and Latvian projects increase cross-border compliance, technology-transfer, and end-use oversight intensity.

Procurement-law exposure is structurally high because the company serves state buyers such as BAAINBw and Latvia's defense ministry. National-security review exposure also exists by inference because DND is a German defense manufacturer with an Israeli ultimate parent and expanding European footprint. Antitrust exposure is presently low in the visible record. IP exposure exists but does not appear to be the central risk driver. Public patent records show DND-linked filings in launchers and fire-protection technology, and registry-derived sources show trademarks such as Dynameco and RGW.

18. Funding, financing, and strategic optionality

Publicly visible financing is conservative and private. DND is not equity-market funded. The 2023 accounts show strong cash, bank lines of €97.6m, and parent-backed guarantees. Customer advances appear to be a major working-capital funding source, which is common in defense manufacturing when large public contracts are involved.

Capital deployment has recently focused on capacity and geographic optionality rather than financial engineering. The company invested in production facilities and in its Hungarian subsidiary or JV structure. Latvia now points to a second localization vector through technology transfer and future mine production. Inference: the most realistic strategic options are continued private compounding inside the Rafael orbit, further joint ventures, selective European capacity partnerships, and deeper localization in customer countries. An IPO or carve-out is theoretically possible but not supported by any public indication in the sources reviewed.



19. Moat analysis

DND's primary moat appears to be qualification-heavy defense know-how in shoulder-launched weapons and survivability products combined with sovereign customer trust and a long installed-base history. This is not a network-effects business. Its advantage is more industrial and regulatory than platform-like.

The moat looks stable to modestly widening. It is stable because product qualification cycles, training familiarity, safety certification, installed base, and trusted supplier status are real entry barriers. It may be widening at the margin because current European rearmament has increased the value of existing qualified capacity and because DND is adding footprint in Hungary and Latvia. The counterpoint is that moat widening is not guaranteed. Larger peers can also invest, and some product categories may face substitution pressure from more advanced guided systems over time.

Cost advantages are likely limited relative to larger primes, although Hungarian manufacturing could improve selected cost positions. Switching costs are meaningful at the program level because militaries do not change launcher families, protection kits, or tactical networking solutions casually. Intangible assets include patents, trademarks, certifications, and the reputational weight of being a qualified defense supplier. Efficient scale also matters because some of DND's niches are specialized enough that too many entrants would struggle to earn acceptable returns.

20. SWOT analysis

Strengths

- Established German defense industrial base with a long lineage in shoulder-launched weapons.
- Portfolio breadth across anti-armor, vehicle protection, counter-mobility, digital, and fire protection.
- Strong liquidity support from customer advances and parent-backed banking lines.

Weaknesses

- Limited private-company transparency on backlog, exact ownership, segment margins, and current-year figures.
- High dependence on a relatively small number of large orders.
- Still modest scale versus major listed European peers.

Opportunities

- European rearmament and higher land-force spending.
- Hungary and Latvia localization could deepen market access and capacity.
- Digital and counter-mobility businesses could diversify the earnings base.

Threats

- Export-control and procurement-policy risk.
- Input-cost inflation and lead-time disruptions.

- Competitive pressure from larger and better-capitalized peers.

21. Key risks and open questions

- How much did sales, profitability, and backlog progress in 2025? The registry shows accounts were filed, but detailed public figures were not visible in the accessible source set.
- How durable is the digital revenue step-up seen in 2023, and what margin profile does it carry?
- How concentrated is the customer base beyond the disclosed concentration in a handful of very large orders?
- What are the economics and ownership details of the Hungarian and Latvian expansion programs?
- How much of DND-Digital is proprietary versus sourced from Rafael group technology?
- Can DND keep broadening beyond legacy launcher dependence without diluting returns?
- Does the low equity ratio remain acceptable as the balance sheet expands with advances and capacity investment?

22. What to watch next

- Ramp-up of the Vecsés, Hungary production facility toward the company's stated summer 2026 fully operational target.
- Whether Latvia's anti-tank mine facility moves from MoU to binding industrial implementation and whether local-capacity details become clearer.
- Additional drawdowns or follow-on orders under the January 2025 BAAINBw framework agreement.
- Evidence of RGW110 customer adoption and industrial scaling.
- Whether digital stays a material second growth leg after recent strong growth.
- Further Puma or other vehicle-protection follow-on activity.
- Public release of readable 2024 financial details or newer audited figures.
- Any move from marketing or demonstration to actual program activity around the announced May 2026 Firefly loitering-munitions demonstration with Rafael.
- Conversion of the new Danish and Swedish SKORPION^2 frameworks into follow-on call-offs, and whether the DOK-ING KOMODO integration attracts an unmanned-platform customer.

23. News flow and recent developments

The last 6 to 18 months have been active. On 9 January 2025, DND announced a framework agreement with BAAINBw for RGW90 family launchers and Panzerfaust 3 cartridges. On 22 October 2025, Latvia announced a roughly €50m SKORPION^2 contract, and on 10 December 2025, Denmark's DALO ordered eight SKORPION^2 systems. On 5 February 2026, Latvia and DND signed an MoU for an anti-tank mine production facility targeted to be operational in 2028. On 27 March 2026, DND broke ground on its roughly €3.5m 'Anbau Ost - Rohrfertigung' expansion in Burbach (completion summer 2027); on 30 April 2026, DND and Croatia's DOK-ING unveiled an unmanned mine-scattering system integrating SKORPION^2 onto the MV-8 KOMODO platform; and on 19 June 2026, Sweden's



FMV signed a SKORPION[2] framework worth approximately SEK 4bn (€360m) through 2033, with an initial call-off of around SEK 900m (€82m).

Meanwhile, DND has continued public positioning around capacity and strategic relevance, including its Hungarian production expansion, a 26 February 2026 visit from North Rhine-Westphalia's economics minister to Burbach, and a May 2026 Firefly demonstration invitation issued jointly with Rafael.

The pattern in this news flow is more important than any single headline. It suggests DND is moving simultaneously on procurement traction, localized manufacturing, political visibility, and product adjacency. What remains missing is equally important: public recent financial detail is still sparse, so the commercial consequences of these announcements cannot yet be fully measured from the visible source base.

24. Source notes and transparency limits

Company sources used

Primary company sources included the DND website's company overview, history page, product pages, news pages, upcoming exhibitions page, and imprint. These were used for legal identity, portfolio mapping, milestones, current public messaging, and recent announcements.

Government and regulatory sources used

Official and quasi-official public sources included OpenRegister for entity history and officers, the German Lobbyregister for filings and public-affairs disclosure, Latvia's Ministry of Defence for procurement and localization developments, and Hungarian official defense material for the JV context.

Independent external sources used

Independent external support included SIPRI for defense-spending context, Doing Business in Hungary and Euro-SD for color on the Hungary ramp, official competitor pages for product benchmarking, and public patent databases for IP references. These sources were used only as support, not as substitutes for company or official filings.

Transparency limits

DND is a private company. That creates real disclosure limits. The accessible source set did not provide a readable 2024 annual report, current backlog figures, product-level margins, exact current ownership percentages, detailed customer concentration, or contract-level profitability. Some business-line interpretation therefore relies on disclosed revenue categories plus product pages rather than on full segment reporting.

Explicitly labeled estimates and inferences

The following judgments were treated as inference rather than disclosed fact: that DND is best understood as a niche land-warfare systems house rather than a prime; that the group appears relatively centralized around Burbach with Hungarian extension; that digital may offer a structurally more attractive long-term business mix; that the company is defense-dominant despite some dual-use activity; and that its moat is mainly driven by qualification, trust, and survivability know-how rather than cost leadership. Those inferences are grounded in the cited sources but are not themselves directly disclosed company statements.

ARX Robotics

Dual-Use Autonomous Ground Systems

Figure 46: ARX Robotics - overview

ARX Robotics GmbH			
Ownership:	Privately held, VC-backed defense-tech company; named investors include HV Capital, Omnes, NATO Innovation Fund, Project A, and Speedinvest		
HQ:	Neubiberg, 85579, Germany	**Founded:**	2022
Domain:	Land – unmanned ground systems and legacy-fleet digitalization		
Value chain:	OEM plus software-platform hybrid		
Why it matters	ARX is a fast-scaling European UGV player focused not only on new robots but on upgrading legacy fleets with autonomy and AI		
Key products	Gereon RCS – modular UGV family; Mithra OS – AI operating system for autonomous and legacy platforms; Autonomy Kit – retrofit autonomy package; Hector – optionally manned medium-class UGV		
Core capabilities	Software-defined land systems, retrofit of existing fleets, modular autonomy and sensor integration, plus production scaling for European and Ukrainian demand		
Customers	ARX says its systems are deployed by several European armed forces and are in use in Ukraine; named public deliveries include Ukrainian Armed Forces units, with additional public programs involving the European Defence Agency		

Source : Deutsche Bank research, company data, Bloomberg Finance LP, Gain.pro

What type of company it is

In practical terms, ARX is a private German defense-technology company focused on unmanned ground systems and autonomy software rather than a classic defense prime. Its model combines deployable robotic platforms, especially the Gereon family and the newer Hector platform, with Mithra OS, which is positioned as an autonomy and fleet-digitalization layer for both ARX vehicles and legacy platforms. That makes ARX best understood as a software-enabled land-systems specialist with aspirations to become a broader land-domain defense system supplier over time.

Investment-style or strategy-style summary

ARX's business mix is centered on unmanned ground vehicles and autonomy software. The flagship Gereon RCS family is already fielded, while Hector extends the portfolio upward into a wheeled, medium-class, optionally manned platform designed for higher speed and longer reach. Mithra OS is strategically important because it expands the addressable market from new-build robotic systems into retrofits for legacy vehicles, which could be far larger if procurement budgets favor modernization over all-new fleets.

Strategically, ARX is positioning itself as a European land-autonomy champion rather than a niche robot maker. Its messaging emphasizes European supply chains, interoperability, and sovereign control. The company's partner set supports that ambition: Daimler Truck for defense mobility, RENK for drivetrain and international scaling, Helsing for networked reconnaissance-and-strike systems, and another German company for drive systems, energy infrastructure, and



production network access.

The financial profile is harder to pin down. Public sources clearly show venture-capital backing from Project A, NATO Innovation Fund, HV Capital, Omnes, Speedinvest, Lakestar, Discovery Ventures, Interface Capital, and Heliad, plus a strategic funding round involving a German industrial investor. But no reliable open-source P&L, cash flow, or margin history was identified in the reviewed materials. That means the company's operational and strategic momentum is visible, while its underlying economics remain largely private.

The most important governance and regulatory considerations are not visible controversies or litigation, but rather the structural realities of being a defense company in Germany and Europe. Export control, war-weapons licensing, procurement rules, and the pace of adoption by ministries and procurement agencies will likely matter more than conventional commercial-governance issues in the near term. At the same time, ARX's public materials do show a more formalized leadership structure than in its early phase, including a strengthened management team and a board adviser appointment in 2025.

1. Why this company is relevant now

ARX Robotics operates at the intersection of three key European defense trends: the automation of land forces, the software-led modernization of existing vehicle fleets, and the strategic objective to strengthen Europe's sovereign industrial capacity. Since late 2024, the company has progressed from initial product development to scaling its operations, a phase demonstrated by the opening of a new production facility near Munich, product deliveries to Ukraine, participation in an EDA-backed experimentation program, and expansion into the UK. ARX has also established partnerships with companies including Daimler Truck, RENK, and Helsing, and these combined factors contribute to the company's current strategic position within the defense sector.

2. What type of company it is

ARX does not fit neatly into any single legacy defense category. It is not yet a full-spectrum defense group with a broad land-systems portfolio, nor is it merely a component supplier, and its practical identity sits between the two. On new-build systems, it behaves like an OEM and system house for light and medium autonomous ground vehicles through the Gereon and Hector families. On legacy platforms, it looks more like a subsystem and autonomy-stack partner, because Mithra OS is designed to digitize and autonomize existing fleets rather than only ARX's own hardware. That dual role - vehicle maker and software modernization layer - is the clearest way to understand the company.

Two further features sharpen the picture. First, ARX is deliberately software-led: Mithra OS is arguably its most strategic asset, since it can expand the addressable market from greenfield robots to the far larger installed base of vehicles armies already operate. Second, ARX is partner-enabled rather than fully vertically integrated, leaning on Daimler Truck, RENK, and others for mobility, drivetrain, energy, and production scale, and on Helsing for networked reconnaissance-and-strike integration. Taken together, ARX is best described as a software-enabled land-autonomy specialist that is trying to grow into a broader full-line land-systems manufacturer - an ambition consistent with its own messaging, though still an inference rather than a settled fact.

3. Three things that matter most



ARX's most important strategic asset may be Mithra OS rather than any single robot. The software expands ARX from a vehicle maker into a modernization layer for legacy fleets, which could matter materially if defense customers prioritize retrofits and mixed fleets over pure greenfield procurement. That is especially relevant given Daimler Truck and RENK partnerships that explicitly frame Mithra as a way to digitize and autonomize existing and future vehicle platforms.

The second major point is that battlefield relevance is not theoretical. ARX has publicly available information of real deployment in Ukraine, British Army testing, German military use, EDA experimentation, and follow-on Ukrainian scale-up. That gives the company a stronger field-validation narrative than many earlier-stage defense start-ups.

The third point is that ARX's industrial model is still forming. The company clearly has momentum, but it remains young, venture-capital-backed, and partly dependent on partner ecosystems for powertrain, mobility, scale manufacturing, and possibly route-to-market leverage. That is not a weakness by itself, but it means ARX's long-term economics and strategic position will depend heavily on execution over the next several years.

4. At a glance

ARX Robotics GmbH is a privately held German defense-technology company headquartered in the Munich area. The company builds unmanned ground vehicles and autonomy software, with a product stack centered on the Gereon RCS family, Mithra OS, and the newer Hector platform. Its public footprint now spans Germany, the UK, and Ukraine, and company materials indicate more than 100 employees, rising to about 140 by winter 2025 and roughly 150 by 2026.

Ownership is private and VC-backed. The best-documented investors are Project A, NATO Innovation Fund, HV Capital, Omnes Capital, and Speedinvest. The company also disclosed in October 2025 that a German industrial company intended to act as lead investor in a further strategic funding round, though completion details were not visible in the reviewed open sources.

Industrial relevance comes from ARX's role in a constrained and newly important niche: software-defined land autonomy. ARX is not yet a full-spectrum full-line defense group, but it is increasingly relevant as a robotics and autonomy node inside the broader European land-systems ecosystem. The company matters because it is trying to connect modular robots, legacy vehicles, and networked battle-management concepts in a European-controlled industrial stack.

Key takeaways

- German private defense-tech company focused on unmanned ground vehicles (UGVs) and autonomy software.
- Core products include the Gereon RCS vehicle family, Mithra OS autonomy software, and the newer Hector platform.
- The company has gained visible traction in Ukraine, Germany, the UK, and European defense programs, supported by a strategically important partner network including Daimler Truck, RENK, and Helsing.
- Funding support appears strong, although public disclosure on revenue, profitability, backlog, and total capital raised remains limited.
- The key analytical question is whether ARX evolves into a scaled

autonomous land-systems platform company or remains a specialist technology supplier within larger defense ecosystems.

5. Corporate history and legal structure

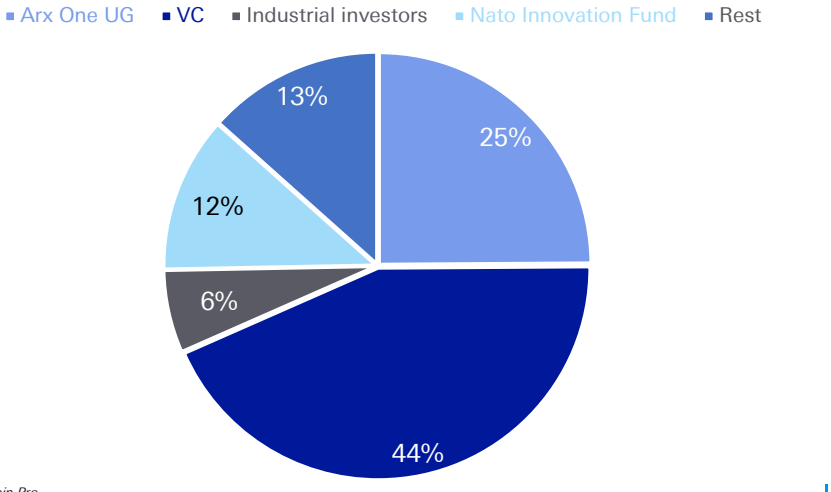
There are two different 'founding' dates in the public record, and both are relevant. ARX's own materials date the company's origin to 2021 for the initial idea and to 2022 for the founding of ARX Landsysteme, later renamed ARX Robotics. Commercial-register summaries, however, show the legal entity ARX Landsysteme GmbH entered the Munich commercial register on 7 March 2023. The cleanest interpretation is that the operating project began in 2021-2022 and the current GmbH vehicle was incorporated in early 2023.

Figure 47: ARX Robotics Ownership

Legal entity	Entity number	Ownership %
ARX Robotics GmbH	HRB 282951	
ARX One UG (haftungsbeschränkt)	HRB 278118	24.9%
Project A Ventures IV GmbH & Co. KG	HRA 58954	15.6%
HV Capital Fund IX Growth GmbH & Co. geschlossene Investment KG	HRA 117189	13.1%
NATO Innovation Fund SCSp SICAV-SIF - NATO Sub-Fund I	201679866	11.9%
Industrial investor		3.8%
Katana Venture Capital UG (haftungsbeschränkt)	HRB 40243	3.5%
Discovery Ventures III GmbH & Co. KG	HRA 59004	3.5%
Lakestar Early IV LP	201290183	3.2%
Interface Capital SPV ARX Robotics GmbH & Co. KG	HRA 64067	2.5%
Helsing SE	HRB 135822	2.5%
Speedinvest Industry & Climate Opportunity EuVECA GmbH & Co KG	201877082	2.1%
Rest		13.4%

Source : Gain.Pro

Figure 48: Ownership Structure ARX Robotics



Source : Gain.Pro

The current legal entity is ARX Robotics GmbH. The company's legal notice and terms identify the registered office as Munich, the operating address as Möslstraße 19B in Oberding, and the commercial-register number as HRB 282951 at Amtsgericht München. Public register aggregators also show the company's earlier address at the Bundeswehr University in Neubiberg and confirm that the earlier name ARX Landsysteme GmbH was later changed to ARX Robotics GmbH.

The legal form is a German GmbH. Publicly accessible registry summaries show a



nominal registered capital that increased over time, reaching €69,941 by early 2026. That number should not be confused with enterprise value or external funding raised. It is simply nominal share capital as reflected in register publications. The same register summaries indicate repeated shareholder-list and articles-of-association updates through 2025 and early 2026, which is consistent with a venture-capital-backed company that has raised multiple rounds.

A concise chronology of inflection points is as follows.

- 2021 - idea stage: Marc Wietfeld began work on the first battlefield-assistance robot prototype.
- 2022 - operating founding: company materials say ARX Landsysteme was founded in 2022.
- 7 March 2023 - legal incorporation: commercial-register summaries show initial entry of ARX Landsysteme GmbH.
- 2023 - pre-seed: €1.15m pre-seed led by Project A.
- 2024 - seed and rebrand period: €9m seed led by NATO Innovation Fund with Discovery Ventures and Project A; company renamed from ARX Landsysteme to ARX Robotics in register summaries.
- 4 December 2024 - Mithra OS launch: ARX publicly launched its autonomy operating system.
- 7 February 2025 - production facility: ARX announced a Munich-region production facility.
- 15 February 2025 - Ukraine delivery: ARX announced delivery of 30 systems to Ukraine.
- 3 March 2025 - EDA contract: ARX disclosed its EDA experimentation award.
- 9 April 2025 - UK expansion: ARX announced UK headquarters and planned facility.
- 28 April 2025 - Series A: ARX announced a €31m Series A.
- Mid-2025 - management build-out: CTO, legal, and program leadership were added.
- 9 July 2025 - RENK partnership: mobility and scaling partnership announced.
- 9 September 2025 - Helsing partnership: reconnaissance-and-strike ecosystem partnership announced.
- 16 October 2025 - Industrial partnership: drivetrain, energy, production-network, and strategic-funding intent disclosed.
- Late 2025 - Board adviser and Ukraine scale-up: Detlef Kayser joined as board adviser and ARX announced a major Ukrainian order for several hundred systems.
- Winter 2025 - 140 employees: company timeline says headcount reached 140.
- May 2026 - fivefold Ukraine fleet expansion: a new GEREON contract increases the deployed fleet to roughly five times its initial size.

On group structure, public visibility is limited. The top-level operating entity is clear, but no full free public organizational chart was identified. The company's contact



pages show Germany, the UK, and Ukraine locations, and public articles reference both a Ukrainian plant and a UK subsidiary footprint. That suggests a central German parent with a still-evolving international operating structure, but the exact subsidiary architecture is not fully visible in the reviewed open sources.

6. Business overview

ARX's business is organized around a simple but increasingly broad proposition: build unmanned ground systems for modern land operations, and pair them with software that can also upgrade existing vehicle fleets. The current public product stack points to two main business engines. One is platform sales and associated integration around the Gereon and Hector families. The other is software, autonomy, and retrofit value via Mithra OS.

The structure appears to have evolved from a narrower robotics platform concept into a more comprehensive land-digitization model. Earlier company materials emphasize modular robotic systems for both military and civil uses. Newer materials are more explicit about software-defined defense, connected fleets, autonomy layers, and legacy-fleet modernization. The strategic shift is not away from robots, but toward robots plus digital infrastructure.

Defense is now clearly the central business. Public evidence of customer use, media focus, partnerships, and capital raising all revolve around defense programs and defense-adjacent land mobility. Civilian and humanitarian use cases are still present in product marketing, but no public segmentation suggests that non-defense revenue is currently the main economic driver. That is an inference from the evidence, not a disclosed segment split.

7. Position in the value chain

ARX does not fit neatly into one legacy defense bucket. It is not yet a classic lead contractor with a full-spectrum land-systems portfolio, nor is it merely a component supplier. Based on publicly available information, ARX currently sits in the value chain as a mixed platform-and-software specialist. It supplies deployable unmanned systems, but it also provides the software layer that can be integrated into third-party or legacy vehicles.

On new-build systems, ARX looks like an OEM or system house for light and medium autonomous ground platforms. On retrofit programs, it behaves more like a subsystem supplier or autonomy-stack partner. On partnered programs with companies such as Daimler Truck and RENK, ARX appears to move one step deeper into systems integration, where its software and autonomy modules enhance much larger incumbent vehicle platforms.

The most likely strategic destination, if execution holds, is a higher-value role as a land-domain autonomy integrator with its own platforms plus partner-enabled access to heavier vehicle categories. That is an inference, not a disclosed corporate target, but it is consistent with the company's language about building a new-generation full-line land-systems manufacturer and with the partner set it has assembled.

8. Products, services, and core capabilities

The core ARX product family today consists of the **Gereon** RCS series, Mithra OS, and Hector. Gereon is the battle-proven tracked UGV family. Public specifications on the current product page show a 500kg payload, up to 4 km control range, thermal night vision, about 40 km range, 72 hours operating time, and modular use



cases spanning logistics, medevac, surveillance, and other mission roles. Janes reporting adds evidence from real-world use, including logistics, ISR, casualty evacuation, electronic warfare adaptation, and UK Army testing. Public reporting (Welt via Defender) puts a single GEREON at roughly €30,000 to €150,000 depending on configuration.

Mithra OS is strategically central because it is the software layer that turns ARX from a robotics company into a fleet-modernization company. Company materials describe it as an AI-based operating system that can connect autonomous and legacy platforms, enable manned-unmanned teaming, and support retrofits. Daimler Truck and RENK both framed their partnerships around Mithra integration, which is important external validation that the software is not just internal middleware for ARX's own vehicles.

Hector is the portfolio expansion that matters most for the next phase. ARX presents it as an optionally manned, wheeled, medium-class UGV built for speed, range, and software-defined operation. Public specs show up to 120 km/h speed, up to 350 km range for the combustion version, 500kg payload, and common control/autonomy architecture across combustion and electric variants. In practical terms, Hector is ARX's attempt to move from small tracked logistics and utility robots into a more operationally mobile land node that can keep pace with maneuver units.

ARX's product strengths appear to be modularity, open interfaces, and rapid mission adaptation. Gereon and Hector are both presented as modular payload carriers rather than single-purpose machines, and the company repeatedly emphasizes that software updates and integrations can be rolled out across fleets. That software-centric modularity is important because it supports multiple routes to value: new vehicle sales, add-on modules, software features, and possibly long-tail upgrade work. The weakness is that public margin data does not exist, so it is impossible to say from open sources whether the economic engine is mostly hardware, software, or services.

Public quality indicators are better on use-case credibility than on classical manufacturing KPIs. ARX does not publish return rates, warranty metrics, or formal SLA data in reviewed sources. What it does disclose is battlefield use in Ukraine, customization through frontline feedback, UK Army testing, and participation in NATO and EDA-linked experimentation. Those are useful proxies for operational credibility, though not substitutes for normal industrial quality disclosure.

The most important product-level catalysts are threefold. First, Hector broadens ARX into faster and higher-endurance missions. Second, Mithra OS could unlock retrofit demand across non-ARX fleets. Third, the Ukrainian connected-fleet program could deepen software-and-network effects if it becomes a recurring architecture rather than a one-off hardware order.

9. Customers and routes to market

The visible customer base is overwhelmingly governmental and defense-related. Company materials state that ARX systems are deployed or procured by six European armed forces, and publicly available information specifically points to Ukraine, British Army testing, German special-forces or training-related use, EDA experimentation, and activity on NATO's eastern flank. ARX therefore appears to sell primarily to military end-users or defense agencies rather than to commercial buyers at present.



The route to market seems mixed. Some sales appear direct, such as Ukrainian deliveries and the announced Ukrainian follow-on order. Other activity is clearly partnership-led or demonstration-led, including British Army exercises, EDA campaigns, and industrial tie-ups with larger defense-mobility players. That suggests ARX is using three channels at once: direct ministry or unit sales, operational trials that can convert into procurement, and partner-enabled access into larger programs. This channel description is partly inference, but it is strongly supported by the public pattern of announcements.

Revenue quality is harder to judge because ARX does not publish a mix of recurring versus non-recurring revenue. Our best evidence-based view is that current revenue is likely hybrid but skewed toward non-recurring hardware, development, and integration work, with software and retrofit layers offering the possibility of more repeatable revenue later. That is an inference rather than a disclosed fact. Customer concentration is also likely high given the company's stage and the visibility of Ukraine, but no public concentration schedule or named top-customer revenue contribution was identified.

10. Key financials and economic profile

Public financial disclosure is extremely limited. We did not identify open, free public sources that reliably provide current sales, EBIT, EBITDA, net income, free cash flow, capex, or R&D spend for ARX. Registry aggregators confirm the existence of filed annual accounts for 2024 and earlier years, but the underlying detailed figures were not visible in the reviewed open-access materials. The same sources explicitly note that revenue information is not available to them in the public-facing summary.

What is visible is the financing history. Company and legal-adviser releases document a €1.15m pre-seed, a €9m seed, a €31m Series A, and an €11m Series A extension. However, current company pages also state "€42m capital raised to date" and describe the €11m extension as bringing total funding to €42m. The apparent conflict resolves once the €42m is read as the total Series A (the €31m initial round plus the €11m extension) rather than lifetime funding. Including the earlier €1.15m pre-seed and €9m seed, cumulative disclosed equity is roughly €52m.

Economically, the visible profile is that of a capital-consuming scale-up company rather than a mature cash compounding business. Public markers that support this view include rapid headcount growth, new production capacity, a planned £45m UK investment, and repeated equity raises. At the same time, the company is already far enough along operationally that the valuation story can no longer rest only on concept-stage technology. The unknown is whether gross margins and software attach rates will become strong enough to support prime-like economics.

11. Footprint and industrial setup

ARX's visible footprint today spans Germany, the UK, and Ukraine. The current contact page lists a German headquarters at Oberding near Munich, a Berlin office, a UK headquarters in London, and a Ukraine headquarters in Kyiv. That suggests a company that is both commercializing and internationalizing quickly, while keeping its center of gravity in southern Germany.

Industrial capacity has become a major part of the story. In February 2025, ARX announced what it describes as Europe's largest production facility for autonomous defense systems in the Munich metropolitan region. In April 2025, the company then announced UK expansion with a planned production and R&D

facility in south-west England and a stated £45m investment plan creating 90 jobs over roughly 30 months. It has also disclosed plant activity in Ukraine as part of the major connected-fleet order.

The footprint matters strategically because it suggests ARX is not building a single-site prototype shop. It is trying to become a multi-country industrial platform with German engineering, UK market presence, and Ukrainian field-proximate manufacturing. That architecture may improve customer intimacy and resilience, although it also adds execution complexity. ARX has publicly stated a goal of producing up to 1,800 GEREON units per year.

12. Supply chain and key dependencies

Hard evidence on supply chain points to a deliberate Europeanization strategy. ARX consistently emphasizes European technological sovereignty and a European-controlled supply chain. The most concrete visible dependencies are now partner-based: an established German industrial company for drive systems, energy infrastructure, and manufacturing network; RENK for mobility solutions and industrial scaling; and Daimler Truck for integration into Mercedes-Benz Special Trucks platforms. Those partnerships reduce some technical and industrial dependencies, but they also create reliance on partner execution.

On product inputs, public sources do not disclose a detailed bill of materials. That means there is no clear open-source view on sensor bottlenecks, compute sourcing, communications modules, energetics, batteries, or other critical subsystems. Still, given the product set, likely dependencies include autonomous-navigation sensors, thermal and night-vision systems, communications links, powertrain components, batteries, software talent, and export-licensable subsystems. This list is inference, not disclosed company fact.

Two dependencies stand out most. First, industrial-scale delivery likely depends on how successfully ARX translates partner announcements into stable throughput. Second, skilled labor is likely a real choke point because the company is scaling across robotics, embedded software, AI, integration, and regulated defense production all at once. Public evidence of rapid hiring and leadership expansion supports that view.

13. Management, board, ownership, and governance

The core management team is still founder led. The current public leadership page lists Marc Wietfeld as CEO, Stefan Röbel as COO, Maximilian Wied as CFO, Ciaran Murphy as CTO, and Dr. Volker Hartmann in a legal and regulatory role. The legal notice also names Mr. Wietfeld, R. Röbel, and Mr. Wied as the managing directors legally representing the GmbH. That points to a governance structure in which operational authority remains concentrated in the founding trio, but with an increasingly formalized second layer around technology and legal functions.

Management formalization clearly increased in 2025. The company announced the addition of Ciaran Murphy as CTO, Dr. Volker Hartmann as Chief Legal Officer and CHRO, and Peter Schulze as Vice President of Programs and Product Strategy. Later in 2025, ARX added Dr. Detlef Kayser as Board Adviser. Registry aggregators also show added signatory authority for several senior executives in August 2025. All of that is consistent with a business moving from founder intensity toward more structured scaling governance.

Ownership visibility is limited. Public materials clearly identify the investor base,



and one registry aggregator says the company has 35 owners, but exact stakes were not visible in the reviewed open sources. There is no evidence in the reviewed materials of public listing, state ownership, or a disclosed controlling industrial shareholder at the top level. Governance-wise, we did not identify a public supervisory board in the sense common to larger German corporates. Instead, the governance signals are managing directors, a management team, investors, and an advisory-board layer.

Headcount is one of the few operational scale markers the company discloses. ARX's timeline says the team reached 140 in winter 2025, while other company pages still refer more broadly to 100-plus employees across more than three countries. That is enough to indicate meaningful organizational scaling, but not enough to judge productivity or labor efficiency.

14. Competitive positioning

ARX competes in a crowded but still fast-forming UGV and land-autonomy field. The most relevant direct comparators are Milrem's THeMIS family, Rheinmetall's Mission Master family, and the QinetiQ-Milrem TITAN offering. All three address overlapping military roles such as logistics, reconnaissance, fire support, communications relay, and force protection. Milrem's THeMIS has a broader installed-base narrative, with acquisitions by 19 nations, while Rheinmetall benefits from a much larger defense-industrial system and integrated battle-management stack.

ARX's differentiation is centered on three ideas. One is software-defined platform via Mithra OS. The second is modularity and fleet interoperability across both new and legacy vehicles. The third is European sovereign positioning backed by industrial partners rather than by a single incumbent parent. Compared with Milrem, ARX seems earlier in installed-base maturity. Compared with Rheinmetall, it is likely faster-moving and more software-first, but far smaller and less embedded in legacy procurement channels. That last comparative judgment is inference.

Competition will likely be based less on pure sticker price and more on a bundle of operational factors: speed of fielding, ease of integration, autonomy quality, fleet networking, and trusted industrial support. ARX's product story is strongest where armies want modular, rapidly fieldable systems and where legacy fleets need software upgrades. It is weaker wherever customer preference leans heavily toward very large incumbents, installed command systems, or heavier armored platforms.

15. Strategic relevance and market implications

ARX matters beyond its own size because land autonomy is becoming strategically important in Europe. The battlefield in Ukraine has accelerated acceptance of unmanned and semi-autonomous systems, and German and European policymakers are trying to compress procurement cycles and deepen sovereign technology capacity. In that environment, ARX is relevant as one of the visible startup nodes in European land autonomy, especially in Germany's rearming ecosystem.

The company's broader market implication is not that it will displace large established defense groups immediately. It is that it could become an important enabling layer in the land-domain ecosystem, especially for autonomy, retrofits, and connected robotics. That matters for listed suppliers and partners such as RENK and Daimler Truck, because ARX gives them a pathway into software-



defined defense mobility without having to build all of that capability internally.

ARX also matters as a signal. If it succeeds, it would reinforce the case that European defense start-ups can move from prototype to fielding and then into industrial partnerships fast enough to matter. If it stalls, that would underline how hard it remains for Europe to turn operational urgency into scaled procurement and repeatable defense-company economics. The second sentence is inference.

16. Dual-use and civilian exposure

ARX's product language remains explicitly dual-use in parts of the website. Gereon is still marketed for defense, commercial, and governmental sectors, and company materials refer to logistics, inspection, surveillance, and humanitarian applications. The original business purpose in commercial-register summaries also covered civil applications such as surveillance, agriculture, civil protection, and construction.

That said, the visible company reality is now defense dominated. The investor base is defense aligned, the announced contracts and demonstrations are military, and the partner set is centered on defense mobility and battlefield networking. We therefore would not frame ARX as a balanced dual-use company in today's market context. It is better described as a defense company that still retains some dual-use design logic in the underlying platforms.

Civilian exposure may still matter strategically in three ways. It may broaden the technical addressable market for modular robots, it may support testing and non-defense revenue over time, and it may give ARX a slightly wider narrative than a pure-play weapons supplier. But publicly available information at present does not justify assuming a meaningful civilian revenue contribution.

17. Legal, regulatory, and compliance risks

Our overall assessment of ARX's public legal and regulatory exposure is Medium. The rating is not driven by known public litigation or enforcement. In the reviewed open sources, we did not identify a material public lawsuit, sanctions action, antitrust case, or securities-style governance controversy. The risk concentration instead comes from the company's operating environment: German war-weapons control, EU and German export controls, procurement regulation, and defense-specific licensing.

In Germany, BAFA is the central administrative authority for export control, and the War Weapons Control Act requires prior licensing for activities involving war weapons. At EU level, Regulation (EU) 2021/821 governs dual-use exports, and the control list is updated periodically. For a company like ARX, that means commercial timing, customer access, and international scaling can all be influenced by licensing and classification decisions, even absent any misconduct.

Procurement-law exposure also matters. Germany's BAAINBw is the central equipment and procurement authority for the Bundeswehr, and Reuters reported in 2025 that Berlin was moving to accelerate defense procurement and make it easier for innovative suppliers to participate. That should help start-ups like ARX at the margin, but it also reinforces that scaling depends on government process and budget timing, not just product quality.

On intellectual property, public open-web evidence is clearer for trademarks than patents. DPMA-derived search results show the GEREON trademark registered in June 2025 and ARX word-mark applications filed in February 2026. We did not

identify a clearly attributable patent family for ARX in the reviewed public search results. That does not mean patents do not exist. It means publicly available information is presently more visible for brand protection than for patents.

A concise timeline of public legal and regulatory visibility is therefore mostly about framework rather than disputes.

- 2023-2026: no material public litigation or regulator enforcement identified in reviewed open sources.
- 3 March 2025: ARX disclosed an EDA experimentation contract, which is positive regulatory-program engagement rather than adverse scrutiny.
- 2025: Germany moved to accelerate defense procurement and ease startup participation, affecting the demand environment.
- 2025: EU updated the dual-use control list, underlining continuing export-control complexity for advanced technologies.

The tail risk is not a known legal case. It is that export-control classification, end-use review, or procurement-policy shifts slow deployment into priority markets or constrain international growth. That would be strategically meaningful even if the direct financial cost were hard to quantify. We would mark that impact as unquantifiable from public sources.

18. Funding, financing, and strategic optionality

ARX is financed primarily through equity capital rather than through public markets. The documented funding rounds are pre-seed, seed, Series A, and Series A extension, with backing from both venture-capital investors and defense-relevant capital such as the NATO Innovation Fund. That investor mix is important because it brings credibility, but it also signals that ARX is still in a private scaling phase where capital formation and strategic execution remain intertwined.

Strategic optionality is broad. A future IPO is conceivable in theory, but no public steps point that way now. The nearer-term strategic options look more like continued private compounding, deeper industrial partnerships, JV-style manufacturing expansion, or eventually a strategic stake by a larger defense or mobility player. The company's own partner map and capital history make all of those more plausible than a near-term public listing. The final sentence is inference.

19. Moat analysis

ARX's primary moat is its emerging combination of battlefield-relevant land-robotics hardware and an autonomy software stack that appears usable beyond its own vehicles. In other words, the moat is not just a robot. It is a modular platform model plus a software layer plus field feedback loops. That combination is more defensible than a chassis alone, but it is still early enough that we would call the moat forming rather than fully proven.

Switching costs are potentially meaningful, but not yet fully locked in. Once a customer adopts ARX hardware, payload interfaces, autonomy workflows, and possibly Mithra-based mixed-fleet concepts, switching should become harder. Yet the company is still too young and the procurement base still too early to assume high embeddedness across many armies. So switching costs likely exist at the program level, but not yet at the scale of entrenched land primes. This is partly inference.



Intangible assets are visible mainly in know-how, field feedback, brand, and regulatory positioning rather than in a large public patent estate. The strongest barriers are probably certification and trust barriers, procurement relationships, export-control competence, and the ability to integrate across European partners while staying sovereign-friendly. That is a real moat candidate in defense, where process, trust, and interoperability often matter as much as pure invention.

On moat trend, we would say widening modestly. The widening case is supported by funding, fielding, facilities, and partnerships. The caution is that larger incumbents and established UGV players are not standing still. So ARX's moat is improving, but it is still exposed to execution risk and ecosystem dependence.

20. SWOT analysis

Strengths

- Strong positioning in a fast-rising niche: land autonomy and legacy-fleet digitization.
- Real-world defense validation through Ukraine, UK exercises, and EDA experimentation.
- High-quality industrial and ecosystem partners.

Weaknesses

- Limited public financial transparency.
- Young company with still-forming governance and industrial processes.
- Dependence on partners for some scale, mobility, and manufacturing leverage.

Opportunities

- Retrofits of legacy fleets via Mithra OS.
- Faster European procurement and rising defense budgets.
- Expansion into UK and broader European programs through sovereign industrial positioning.

Threats

- Larger incumbents and established UGV peers can outspend and outscale ARX.
- Export controls and procurement timing can slow scaling.
- Rapid growth can expose execution and supply-chain strain.

21. Key risks and open questions

- Can Mithra OS become a true multi-platform standard, or will it remain mainly an ARX-adjacent software layer?
- How much of future revenue will be recurring software and upgrade revenue versus one-time platform sales? Public sources do not yet answer this.
- How concentrated is the customer base, especially around Ukraine and a small number of European defense users? Public disclosure is limited.



- Can ARX industrialize quickly enough to meet large orders without margin dilution or delivery slippage?
- Will ARX remain independent, or does its partner-heavy model eventually pull it toward strategic ownership or JV structures? This is an inference-based question.
- How durable is the dual-use narrative if actual commercial revenue remains limited?

22. What to watch next

- Any evidence that Mithra is being adopted on third-party fleets beyond pilots and partnership announcements.
- Conversion of UK expansion into visible production, contracts, or sustained British Army programs.
- Execution against the Ukrainian several-hundred-system order and further local manufacturing build-out.
- Whether Hector moves from portfolio addition into formal procurement traction.
- Signs of repeatable EDA or NATO-linked experimentation converting into procurement relevance.
- Further management and governance build-out as the company scales.
- Any disclosure of real financial metrics, backlog, or unit economics. None are visible today.

23. News flow and recent developments

The last 18 months have been dense. The clearest early scaling sequence was: Mithra OS launch on 4 December 2024, production-facility announcement on 7 February 2025, Ukraine delivery announcement on 15 February 2025, EDA contract disclosure on 3 March 2025, UK expansion on 9 April 2025, and the €31m Series A on 28 April 2025. That alone would have marked a significant year for a private defense startup.

The second half of 2025 deepened the industrial and ecosystem story. ARX strengthened leadership in June, partnered with RENK in July, partnered with Helsing in September, and announced an industrial strategic partnership in October. Around the same period it added Dr. Detlef Kayser as board adviser, expanded the Series A by €11m, and announced a major Ukraine order for several hundred systems in November 2025, and then in May 2026 secured a further contract for several hundred additional GEREON UGVs that expanded its deployed Ukrainian fleet roughly fivefold - the third expansion - cementing ARX as the largest Western UGV supplier to Ukraine, with part of production localized at its Ukrainian plant.

Portfolio-wise, the most notable product development has been Hector, which broadens ARX from the Gereon tracked family into a faster, optionally manned, medium-class wheeled platform. The company also demonstrated a British Armed Forces reconnaissance-and-strike chain with Helsing in Kenya and refreshed its public press materials in January 2026, suggesting continued outward-facing positioning as it scales.

24. Source notes and transparency limits

Company sources used

The report relied heavily on ARX's own website, especially the About, Investors, News, Press, Contact, Imprint, Terms, Gereon RCS, Hector, and Mithra OS pages, as well as individual company press releases on funding, production, Ukraine, UK expansion, EDA participation, leadership, and partnerships. These are the most authoritative sources for the company's own stated strategy, structure, and milestones.

Government and regulatory sources used

For legal and regulatory context, we relied on BAFA export-control materials, German ministry pages on war-weapons control, BAAINBw official descriptions, EU dual-use regulation summaries, and EDA / HEDI pages. These were used to frame the regulatory environment rather than to identify adverse findings against the company.

Independent external sources used

Independent external support came from Reuters for defense spending and procurement reform context, Janes for operating detail on Gereon and production ambition, SIPRI for broader market-research source guidance, and partner press releases from Daimler Truck, RENK, and Helsing, to cross-check the industrial ecosystem around ARX. We also used a limited number of registry-aggregator sources for commercial-register chronology, while treating them as supporting rather than primary evidence.

Transparency limits

ARX is a private company, and that sharply limits visibility into revenue, profitability, backlog, customer concentration, and exact ownership percentages. The public record is rich on strategic direction and visible milestones, but thin on core economic disclosure. There are also a few genuine inconsistencies across public materials, most notably around founding year versus legal incorporation date and around lifetime funding totals. Where that happened, we flagged the conflict rather than forcing a false precision.

Explicitly labeled estimates and inferences

Several judgments in this report are inferences rather than disclosed facts. These include: that ARX's current revenue mix is likely still hardware-heavy; that the company's partner-led model may evolve toward JV or strategic-ownership optionality; that civilian revenue is likely modest today despite dual-use marketing; and that the moat is widening but still early. Those conclusions are grounded in the visible public pattern of products, customers, partnerships, and financing, but they are not direct company disclosures.

Reflex Aerospace

Custom-Built, Dual-Use Satellite Technology

Figure 49: Reflex Aerospace - overview

Reflex Aerospace GmbH			
Ownership:	Privately held, founder-led New Space company; latest public funding was a €50m Series A led by Human Element, with Alpine Space Ventures, Bayern Kapital, HTGF, and Renovatio Financial Investments also named		
HQ:	Berlin, 10245, Germany	**Founded:**	2021
Domain:	Space – rapid-built LEO satellite platforms for ISR, SAR, Earth observation and SATCOM		
Value chain:	OEM / satellite-platform supplier		
Why it matters	Reflex is building European sovereign satellite manufacturing capacity with faster, more adaptable platforms for defense and dual-use missions		
Key products	Custom satellite platforms / bus for SAR, optical EO, ISR and SATCOM missions; mission-specific LEO spacecraft such as SIGI; OPTIMAS – secure optical communication for multi-domain defense		
Core capabilities	Rapid customized satellite design and assembly, adaptable high-performance LEO platforms, secure-by-design communications modules and end-to-end mission support		
Customers	Reflex says it targets government agencies, defence organizations and commercial satellite data providers; named public customers/programs include Media Broadcast Satellite, LiveEO and the EDF-backed OPTIMAS consortium, with Umbra as a public SAR partner		

Source : Deutsche Bank research, company data

What type of company it is

Reflex is best understood as a private German satellite-bus manufacturer and mission-specific system integrator for dual-use low-earth-orbit satellites. It does not present itself as a launch provider, payload builder, data seller, or satellite operator. Instead, it designs and manufactures the bus around the customer payload and integrates that payload into a customized satellite platform.

In practical value-chain terms, Reflex is a specialist upstream manufacturer with some prime-like characteristics on small, mission-defined spacecraft programs. On fully scoped customer missions such as SIGI or MIRI, it appears to function as the spacecraft prime for the bus and integration layer. On consortium programs such as OPTIMAS, it appears more like a space-segment subsystem lead within a broader defense program. That second point is an inference from the public description of Reflex's responsibilities.

Investment-style or strategy-style summary

Reflex Aerospace is a German company that builds the main body, or "bus," for satellites. It acts like a specialized manufacturer, creating custom platforms for customers who have their own instruments (payloads) they want to put into space. The company does not operate the satellites itself or sell the data they collect.

A key part of Reflex's strategy is its focus on being a sovereign European provider. It uses ITAR-free components, meaning it avoids reliance on US parts and regulations, which is very attractive for European government and defense customers. It also promises much faster delivery times than traditional space

companies, aiming to get a satellite ready in under 18 months.

The company has already proven it can deliver, with its first satellite launched in early 2025. It also recently raised €50 million to expand its manufacturing capabilities. While its financials are private, the investment story is based on its ability to meet Europe's growing demand for secure, independent space technology. The main challenge is scaling from building single, custom satellites to producing larger constellations efficiently.

1. Why this company is relevant now

Closing Europe's "Sovereignty Gap" in Space: The company's core mission directly addresses a major political and military priority. European leaders, including the German defense minister, have identified over-reliance on non-European satellite technology as a critical weakness. Reflex's focus on building ITAR-free, European-controlled satellite platforms for reconnaissance, secure communications, and early warning aligns perfectly with the stated goal of creating a "European Space Shield" and achieving strategic autonomy. ITAR stands for International Traffic in Arms Regulations; it is a set of US government rules that control where American defense and military technology can be sold or sent.

Speed and Responsiveness: Traditional satellite programs are notoriously slow and expensive. Reflex's business model is built around delivering complex, customized satellites in under 18 months. This speed is critical in the current security environment, where the ability to rapidly deploy new space assets in response to emerging threats is a significant strategic advantage.

Filling an Industrial Need: Beyond the political goals, Reflex is filling a tangible gap in the European industrial base. It provides new manufacturing capacity for high-performance small satellites, acting as a crucial link between payload innovators and orbit. This makes it a key enabler for the entire European space ecosystem, from government agencies to commercial data companies that need a reliable, sovereign path to space.

2. What type of company it is

Reflex is a private commercial aerospace manufacturer focused on adaptable satellite platforms. Its own reference page describes it as a commercial aerospace manufacturer and space systems company with primary customers including government agencies, defense organizations, commercial satellite data providers, and commercial companies seeking space-data access.

Operationally, Reflex is a payload-centric bus designer and integrator. It repeatedly states that it does not build payloads, does not provide launch services, does not operate satellites, and does not sell Earth observation data. That makes it closer to a highly customized satellite platform supplier and integrator than to a traditional vertically integrated space-services company.

The closest practical label is dual-use smallsat system integrator with manufacturing capability. For some customers it likely behaves like a small-scale lead contractor for spacecraft. For others it is a platform supplier embedded inside a wider mission architecture.

3. Three things that matter most

The first thing that matters is execution speed. Reflex's public materials center on getting from contract to deployment faster than traditional satellite programs. The precise claim has tightened over time, from a marketed 12-18 months down to a stated 'under nine months' to build a unique, complex satellite from scratch (per CEO comments at the 2025 Series A), with SIGI's 13-month order-to-flight-readiness as the concrete proof point.

The second thing that matters is sovereign positioning. Reflex is not just selling fast satellites. It is selling a European, ITAR-free, dual-use manufacturing proposition tailored to customers that care about control over supply chain, security architecture, and European command-and-production ownership. That matters more now than it would have a few years ago.

The third thing that matters is scale-up risk. Reflex now has funding, flight heritage, and visible programs, but the central question is whether it can industrialize beyond early bespoke missions. Public evidence points to a current manufacturing base in Munich and an ambition to reach materially higher annual capacity, but the large-step scale transition still lies ahead.

4. At a glance

Reflex Aerospace GmbH is a German private satellite manufacturer founded in 2021 by Walter Ballheimer and Alexander Genzel. It operates from Berlin and Munich and is structured as a GmbH registered with the District Court of Munich under HRB 267113. The company presents itself as a B2B manufacturer of adaptable satellite platforms for low-earth-orbit missions.

Public scale indicators are directional rather than complete. The company has one flown satellite, several publicly named follow-on missions, and a Munich MicroFactory that management expects to scale to roughly 60 to 70 highly complex satellites per year following the 2025 Series A. Public headcount indicators are inconsistent across sources, ranging from 70+ on one company page to 90+ on a third-party Bavarian innovation profile, so workforce scale should be treated as approximate.

Key takeaways

- Private German satellite manufacturer and integrator founded in 2021, focused on customizable 75-500kg LEO satellites for dual-use, commercial, and defense missions.
- The company has moved from founding to flight heritage unusually quickly, with its first satellite, SIGI, reaching orbit in January 2025.
- Strategic relevance is high due to its focus on sovereign European ISR and communications capabilities, with programs including SIGI, MIRI, OPTIMAS, and Twinspector.
- A €50m Series A round in 2025 supports scaling of satellite manufacturing and constellation capabilities.
- Public financial, backlog, and ownership disclosure remains limited because the company is private.
- The key analytical question is whether Reflex can convert early mission success into repeatable, higher-volume constellation production.



5. Corporate history and legal structure

Reflex was founded in Munich in May 2021. The founder names that are consistently public are Walter Ballheimer and Alexander Genzel. The company's origin story, as told on its own history page, is that the founders saw falling launch costs, more accessible space technology, and a gap between NewSpace speed and high-performance spacecraft requirements.

The current legal entity is Reflex Aerospace GmbH. The imprint lists Grünberger Str. 48b, 10245 Berlin, as the address, with registration at District Court Munich under HRB 267113 and Walter Ballheimer as CEO. The Bundestag Lobby Register also describes Reflex as a GmbH and lists Walter Ballheimer and Alexander Johannes Genzel among the authorized representatives.

No parent company or detailed holding structure is clearly disclosed in the primary public materials reviewed. Likewise, a detailed subsidiary list is not publicly visible. One historically relevant structural item is the HTGF portfolio description that says Reflex is a shareholder in the JV UNIO together with Mynaric, Isar Aerospace, and SES, aimed at a European broadband constellation. Current strategic relevance of UNIO is not clear from Reflex's own current core materials, so it should be treated as a historical or adjacent item rather than a central disclosed operating pillar.

Key milestones are straightforward. In 2021, the company was founded and expanded into Berlin. In late 2021, it says it was selected by the European Commission to lead the foundational study that became IRIS^2. In December 2022, it announced a first seed closing of about €7m, and by April 2023 the history page says total seed funding had reached nearly €9m. It signed MBS in May 2023, Marble Imaging in March 2024, and OPTIMAS in late 2024. SIGI launched in January 2025, and the company announced its €50m Series A in November 2025. The company history page dates the Series A closure to October 2025, but the press release, HTGF note, Reuters coverage, and Munich Startup coverage all place the public announcement on 4 November 2025. We treat the November 2025 date as the better public anchor.

6. Business overview

Reflex's business is built around mission-specific satellite platforms for low Earth orbit (LEO). The company describes its core focus as designing and manufacturing adaptable satellite platforms for complex missions in the 75-500kg range, typically around customer payloads, with delivery targeted in less than 18 months and sometimes marketed at 12 months.

The business does not appear to be segmented publicly in the way a listed industrial company would be. Instead, the visible structure is product-configuration based: OmniFlex baseline, HiPr, TSAR, UltraSharp, DataSphere, and the Praetora platform, which the company now uses in production to develop new satellite types rapidly using generative design and additive manufacturing. Those are better understood as mission and configuration families than as separately reported business units. Public disclosure does not provide revenue by configuration or any formal segment contribution.

The strategic drift of the business is increasingly defense-oriented, though not defense-exclusive. Early public narrative mixed commercial and dual-use demand. By late 2025, the Series A announcement was explicitly framed around sovereign optical, SAR, SDA, and SIGINT constellations, while the public mission set still included commercial EO and infrastructure-monitoring customers. The company is



therefore more defense-relevant than a pure commercial EO supplier, but still more mixed than a pure-play defense prime.

7. Position in the value chain

Reflex sits in the defense-space value chain at the satellite-platform and integration layer. It is not upstream in raw components and not downstream in data analytics or mission operations. Its economic role is to design and build the spacecraft bus, integrate payloads, and deliver a flight-ready platform around a customer mission need.

In narrower aerospace terms, Reflex is closest to an OEM and system integrator for small, customized satellites. It also has some characteristics of a "system house" because it markets full mission configurations, handles design trade-offs around payload, power, storage, communications, and pointing, and emphasizes end-to-end tailoring rather than catalog parts.

Where Reflex is not yet visible is in full platform control at national scale. Its current position appears to be below the very largest European satellite primes but above many narrow subsystem suppliers. That is an inference from the public product and mission record.

8. Products, services, and core capabilities

Reflex's core offering is the OmniFlex satellite bus, a customizable platform for satellites weighing 75-500kg. Designed to be built around customer-specific payloads, it features in-house avionics, high-capacity data storage, agile stabilization, and a wide range of communication options. The company's true value lies not in specific brand names but in this underlying capability stack, which combines payload-centric design, precision pointing, and flexible power and communication systems into an integrated solution.

The company offers several mission-specific configurations based on this platform. HiPr and TSAR are high-value versions for demanding intelligence, surveillance, and reconnaissance (ISR) missions, with TSAR adding advanced synthetic-aperture radar (SAR) for all-weather monitoring. For civilian and dual-use markets, UltraSharp is designed for very-high-resolution Earth observation, while DataSphere is optimized for large constellations requiring advanced data fusion. A next-generation architecture, Praetora, is also in development for future critical ISR missions, though details remain limited.

Reflex's key strengths are its ability to deliver customized, high-performance platforms, its flight-proven technology, and its focus on defense-relevant applications. However, significant risks remain. The company has a very small installed base, limited publicly available information on long-term field reliability, and has not yet proven it can scale from building individual bespoke satellites to manufacturing larger constellations efficiently.

9. Customers and routes to market

Reflex's customer base appears to fall into four buckets: government agencies, defense organizations, commercial satellite data providers, and commercial companies that need access to space-based data but do not operate spacecraft themselves. That customer map comes directly from the company's own reference page.

Named customers and programs in the public record are MBS for SIGI, Marble



Imaging for MIRI, LiveEO for Twinspector, and the EDF/Navantia-led OPTIMAS consortium for secure optical communications. The MBS partnership was Reflex's first commercial customer. Marble Imaging is especially important because its mission is described as the first step toward a planned 200-satellite EO constellation. Twinspector matters because it places Reflex in infrastructure monitoring and sovereign 3D EO. OPTIMAS matters because it embeds Reflex in a multi-domain defense communications program.

Routes to market appear to be direct enterprise-style sales and consortium participation rather than channel-based distribution. Reflex also seems to use partnership-led market extension. The Umbra agreement is an example for SAR, while the Sidus MOU is an example for international go-to-market and possible JV development.

Revenue quality is probably mixed but potentially attractive if repeat orders emerge. The current visible business looks milestone and contract driven rather than recurring. However, constellation follow-ons, production ramps, and long mission families could create repeat manufacturing revenue once initial qualification is complete. That conclusion is an inference because no contract terms, backlog, or customer concentration metrics are publicly disclosed.

10. Key financials and economic profile

The public financial picture is partial. The preliminary 2023 HGB statement shows total assets of about €4.10m, cash of about €1.82m, inventories and advances of about €560k, other receivables of about €754k, and equity of about €3.91m. It also shows capital reserves of about €9.73m and a net loss of about €3.71m for 2023. On a simple external calculation, the 2023 equity ratio was about 95.4%.

The same filing does not present a clean public revenue number in the extracted summary, but it does show "Rohergebnis" of about €0.55m in 2023 versus about €1.41m in 2022, and personnel expense of about €2.68m versus about €1.27m. That suggests a company still in aggressive build-out mode in 2023, with cost base growth outpacing visible gross result.

The 2021 statement shows how early stage the company still was at founding: revenue of €5k, personnel expense of about €219k, and a net loss of about €398k for the May-December 2021 period. That makes the later funding raises far more informative than the old P&L in understanding current scale.

The major financial event since those filings is the €50m Series A announced in November 2025. That round should have transformed liquidity relative to the 2023 balance sheet, but no updated post-raise balance sheet, revenue run rate, EBITDA, or cash-burn detail is publicly visible. Strategically, this means the company looks far better funded than the last published HGB figures alone would imply.

11. Footprint and industrial setup

Reflex's footprint is concentrated in Germany. Public materials consistently point to Berlin and Munich as the two main locations, with Berlin functioning as the design center and Munich as the MicroFactory for manufacturing, assembly, integration, and testing.

The Berlin site is described as containing an electronics lab, mechanics workshop, and test equipment, with BSI Grundschutz and ISO27001 certification "in process." The Munich site is described as having an ISO 8 clean room, cell manufacturing

concept, and current capacity of 12 satellites per year.

The future industrial story is expansion. The company history page says the Munich MicroFactory reached initial completion in December 2024, and the Series A announcement says part of the new capital will be used to expand manufacturing capacity in Bavaria. Reflex's public site also speaks of future capacity of 60-70 satellites per year, though the exact site description is not perfectly consistent across pages and snippets.

12. Supply chain and key dependencies

The clearest visible supply-chain theme is sovereignty. Reflex emphasizes ITAR-free components, purposeful verticalization, and deployable MicroFactory manufacturing. That implies the company sees export-control friction, lead times, and sovereign control over production as commercially important differentiators rather than just technical details.

Visible dependencies still remain. Reflex does not build payloads, so it depends on customer or partner payload providers. It also does not provide launch, which makes it dependent on rideshare or dedicated launch providers. MBS/SIGI used SpaceX, MIRI was framed around a European carrier in Marble's announcement, and future defense missions may involve additional launch and ground-segment dependencies that are not yet publicly detailed.

There are also likely dependencies on specialized electronics, secure communications modules, propulsion, optical terminals, and skilled engineering labor. The company's own materials strongly support the labor and facility sensitivity point, while the rest is inference from the type of spacecraft it is building. Public disclosure does not provide a supplier list or single-source exposure table.

13. Management, board, ownership, and governance

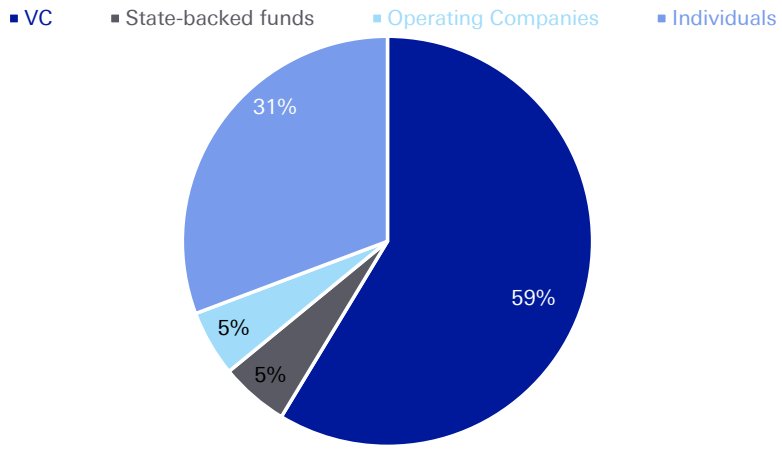
Publicly visible leadership is founder-led. Walter Ballheimer is CEO and Alexander Genzel is COO and co-founder. The leadership page also lists Felix Motzki as CTO, Dennis Moore as CCO, Dr. Sascha Weiß as Head of Missions, Cecile Schirmer as Head of Finance, and Henrik Rutzky as Head of IT.

Figure 50: Reflex Aerospace Ownership

Legal entity	Entity number	Ownership %
REFLEX aerospace GmbH	HRB 267113	
Alpine Space Ventures Fund I GmbH & Co. KG	HRA 113881	47.4%
Walter Ballheimer	-	30.7%
Schwarz Holding GmbH	HRB 171585	5.2%
TY Invest KG	HRA 102431	2.4%
Valeto Invest GmbH	HRB 290920	2.4%
SCALEUP-FONDS BAYERN GMBH & CO. KG	HRA 12153	2.3%
HIGH-TECH GRÜNDERFONDS IV GMBH & CO. KG	HRA 9754	1.8%
REFLEX aerospace GmbH	-	1.7%
Bayern Kapital Innovationsfonds EFRE GmbH & Co. KG	-	1.3%
Human Element Venture Fund I LP		1.2%
Others		3.6%

Source : Gain.Pro

Figure 51: Ownership Structure Reflex Aerospace



Source : Gain.Pro

Recent management formalization is visible. Dennis Moore was promoted to CCO in August 2024, and Felix Motzki was promoted to CTO in September 2024. That suggests the company is institutionalizing its commercial and technical leadership as it scales beyond the founder-only phase.

Ownership data are accessible via third-party data providers. The company's own grounding page says it is privately held, and the publicly disclosed investor base includes Human Element, Alpine Space Ventures, Bayern Kapital, and HTGF, with HTGF and Reuters also naming Renovatio Financial Investments in the 2025 round coverage. No cap table, founder ownership, or employee option pool disclosure is publicly visible.

Governance visibility is modest. As a German GmbH, the company does not publicly present a listed-company style board structure. We did not identify a public supervisory-board composition in the reviewed materials. Compliance signals are limited but not absent: the site references a whistleblowing channel, the Berlin site notes BSI Grundschutz and ISO27001 certification in process, and the company operates in explicitly European and international export-control frameworks according to its own reference page.

Human-capital disclosure is inconsistent. The current accessible "Why Reflex" page shows 70+ team members, while the company history page marked the 50th employee in July 2024, and a Bavarian innovation profile says the team is over 90 employees. The correct current number is therefore not fully clear from public sources, but the direction of travel is clearly upward.

14. Competitive positioning

Reflex competes in the European and broader smallsat platform market against companies such as Aerospacelab, EnduroSat, NanoAvionics, Open Cosmos, OHB LuxSpace, and SSTL, depending on mission category. NASA's 2024 small spacecraft technology report includes Reflex in its ESPA-class provider table alongside many of these peers, which is a useful external anchor for the competitive set.

Reflex's strongest competitive message is not scale. It is a combination of

customization, speed, and sovereign European positioning. The company is explicitly arguing that customers do not need to choose between "classic space" performance and "NewSpace" speed. That is a powerful commercial pitch if delivered, especially for defense and dual-use missions where schedule and sovereignty matter.

Its main competitive weakness is relative maturity. Reflex has one flown mission and strong early traction, but some peers have broader flight heritage, larger installed bases, or parent-company support. NASA's table also labels Reflex as "under development" in that benchmarking context, which underscores that it is still earlier on the maturity curve than the most established spacecraft providers.

Competition likely happens on a business-line basis, not across one monolithic peer set. Reflex may compete against nimble smallsat specialists on commercial EO programs, against more defense-oriented sovereign suppliers on ISR missions, and against larger European incumbents where procurement bodies want industrial resilience and national control. That segmentation is an inference, but it is the most sensible reading of the public mission mix.

15. Strategic relevance and market implications

Reflex matters beyond its own P&L because it represents new European manufacturing capacity in a part of the defense-space stack where speed, sovereignty, and mission customization are becoming more valuable. Germany's stated space-defense buildout and the EU's Space Shield agenda both point toward more demand for responsive domestic spacecraft production.

Its industrial relevance is strongest in bottleneck-sensitive areas. Europe does not only need launchers and payloads. It also needs more bus builders, integrators, secure communications infrastructure, and scalable production capacity for smaller constellations. Reflex is trying to fill exactly that gap.

If Reflex succeeds, the market implication is not just another startup win. It would reinforce the idea that Europe can build a middle layer of sovereign space manufacturers below the very largest satellite primes and above pure subsystem vendors. That could matter for procurement optionality, industrial policy, and supplier ecosystems across communications, EO, onboard computing, and launch. The last point is an inference.

16. Dual-use and civilian exposure

Reflex is explicitly dual-use. Its own materials say its custom satellites are designed for dual-use applications, and the mission mix confirms that. Commercial EO and infrastructure monitoring sit alongside defense optical communications, SAR, and ISR-linked capabilities.

The civilian side still matters. MIRI is framed around sustainability and high-resolution EO data, while Twinspector is framed around critical infrastructure monitoring. Those programs give Reflex non-defense reference cases and broaden the addressable market beyond pure military procurement.

That said, the company's strategic narrative has clearly tilted more defense-relevant over time. By late 2025, the flagship funding announcement emphasized optical, SAR, SDA, and SIGINT sovereign constellations. The right label today is therefore dual-use with increasing defense weight, not pure-play defense and not simply commercial EO.



17. Legal, regulatory, and compliance risks

The main legal and regulatory exposure is structural rather than event-driven. Reflex operates in space and defense-adjacent markets, so its risk set likely centers on export controls, procurement-law compliance, cyber and information security, debris mitigation, possible national-security review on partnerships, and IP ownership around payload integration and platform architectures. The company's own reference page explicitly says it operates within European and international space and export-control frameworks.

Public evidence of live legal controversy is thin. We did not identify any material public litigation, enforcement action, or sanctions issue in the reviewed sources. That should not be read as proof of zero exposure. It simply means nothing material surfaced in the public company, registry, and official-policy record reviewed for this pass.

Procurement and public-funding compliance matter because Reflex already touches public programs and grants. OPTIMAS is linked to the European Defence Fund, and the Lobby Register discloses public grants from EU and German authorities for additive-manufacturing and rapid-response related projects. That creates normal audit, program-performance, and eligibility obligations even if no public dispute is visible.

IP visibility is limited. Public company materials highlight internal technologies such as Reflex ARC and branded product names such as Praetora, but we did not establish a reliable official patent portfolio from the sources reviewed. A formal EUIPO and DPMA follow-up would therefore be part of next-step diligence rather than a concluded finding from this note.

18. Funding, financing, and strategic optionality

Reflex has followed a classic venture-capital-backed deep-tech funding path so far. The public seed story began with about €7m announced in December 2022, with a target of up to €12m after second closing. By spring 2023, public company materials pointed to total seed funding of nearly €9m, and a May 2023 announcement referenced a further €1.75m expansion.

The financing step-change was the €50m Series A in November 2025. Investors publicly named across company, HTGF, Reuters, and Munich Startup sources include Human Element, Alpine Space Ventures, Bayern Kapital, HTGF, and Renovatio Financial Investments. The strategic use of proceeds is explicit: scale manufacturing and field sovereign ISR and communications capabilities by 2027.

Reflex also has non-dilutive support in the public record. The Bundestag Lobby Register discloses public grants from the EU, the government of Swabia (Bavarian region, *Schwaben*), and Land Berlin, with amounts shown in ranges rather than exact numbers.

Strategic optionality exists in several directions. The most likely near-term path is continued private compounding through manufacturing expansion and program execution. Beyond that, possible options include additional strategic JVs, deeper payload partnerships, selective M&A, or eventually a public-market event if the company achieves repeatable constellation-scale production. At present, no IPO or transaction plan is publicly disclosed.

19. Moat analysis

Reflex's sovereign positioning provides a strong defense against non-European firms, but it does not eliminate competition within Europe. Reflex faces a number of European rivals, including Aerospacelab, EnduroSat, OHB LuxSpace, and SSTL.

Against these peers, its competitive advantage shifts away from simply being European toward a specific combination of deep customization around the customer's payload, significantly faster delivery, and a sharp focus on high-value defense and ISR missions. The key risk is Reflex's relative immaturity, as some of these competitors have more flight experience and a larger installed base.

20. SWOT analysis

Strengths

- Real flight heritage from SIGI.
- Strong fit with Europe's sovereign ISR and communications agenda.
- Clear product differentiation around speed, customization, and ITAR-free European positioning.

Weaknesses

- Limited public financial transparency.
- Small installed base relative to mature peers.
- Scale-up execution remains to be proven at constellation volume.

Opportunities

- German and EU defense-space investment tailwinds.
- Repeat orders from named constellation customers such as Marble Imaging and LiveEO.
- Expansion into sovereign SAR and secure optical communications through Umbra and OPTIMAS.

Threats

- Competition from more established European and international smallsat manufacturers.
- Export-control, procurement, and cybersecurity burdens as defense exposure rises.
- Program delays or qualification issues on next missions.

21. Key risks and open questions

- Can Reflex convert one flown mission into repeat constellation-scale manufacturing economics?
- What is the current revenue run rate after the 2025 Series A and pre-2026 launches? Public sources do not say.
- How concentrated is the customer base, and how much of future demand depends on a small number of flagship programs? Public disclosure is limited.
- How much of the future order book is defense-funded versus commercial

dual-use?

- How durable is the ITAR-free and sovereign-positioning advantage if larger European incumbents respond more aggressively?
- What is the real post-Series-A capacity roadmap, and where exactly will the jump toward 60-70 satellites per year occur? Public wording is not fully consistent.
- Does the company own enough protected IP, or is the moat more about speed, process, and relationships than formal patent protection?

22. What to watch next

- MIRI production progress and planned Q4 2026 launch.
- Any update on Twinspector launch timing and whether the program expands beyond the first two platforms.
- OPTIMAS execution milestones and whether Reflex's role expands within European defense optical communications.
- Evidence of manufacturing expansion funded by the 2025 Series A.
- Whether Praetora moves from teaser status into a fully specified platform offering.
- Additional sovereign SAR developments with Umbra and whether the first 2027 launch target holds.
- Any public disclosure on repeat orders, backlog, or commercial conversion beyond the currently named missions.
- Any new government or European program alignment under the Space Shield and broader readiness agenda.

23. News flow and recent developments

Recent news flow has been dense and constructive. In December 2024, Reflex announced shipment of its first commercial satellite, SIGI, and in January 2025 SIGI launched on SpaceX Transporter-12. In February 2025, Reflex announced OPTIMAS, linking the company more clearly to European defense communications. In April 2025, it announced partnerships with Umbra and Sidus. In November 2025, it announced the €50m Series A. In February 2026, it was selected to build two satellite platforms for LiveEO's Twinspector Earth-observation constellation. Reflex is also building MIRI for Marble Imaging, which will host ThrustMe's JPT150 iodine Hall thruster in a 2026 in-orbit demonstration.

The tone of that news flow is important. It shows a company that is shifting from technical validation to industrial scaling and partnership-led market expansion. It also shows growing defense adjacency, even while maintaining commercial and civilian dual-use references.

Independent media coverage broadly supports that reading. Reuters framed the Series A as a response to Europe's growing focus on autonomous defense capabilities in space. Munich Startup and HTGF both emphasized manufacturing expansion and a target of operational readiness by 2027.



24. Source notes and transparency limits

Company sources used

Primary company sources included Reflex's grounding page, company history, leadership page, imprint, FAQs, solutions overview, mission pages for SIGI, MIRI, OPTIMAS, and Twinspector, product pages for OmniFlex configurations, and company press releases on seed funding, shipping, partnerships, and Series A.

Government and regulatory sources used

Government and regulatory sources used included the Bundestag Lobby Register and linked annual-statement filings, the German Federal Ministry of Defence speech at the BDI Space Congress 2025, and European Commission defense-readiness materials.

Independent external sources used

Independent external sources used included Reuters, HTGF, Munich Startup, NASA's Small Spacecraft Technology report, ESA's space-economy publication, and Novaspace market material.

Transparency limits

Reflex is private, so several items that would normally matter for a bank-grade underwriting view are not publicly disclosed. These include a cap table, current revenue, backlog, customer concentration, detailed contract values, margin by program, and a current audited post-Series-A balance sheet. Headcount disclosure is also inconsistent across public materials.

Explicitly labeled estimates and inferences

The following points in this note are inferences rather than disclosed facts: likely revenue quality, the exact value-chain role on certain programs, the degree of moat widening, the competitive segmentation by business line, and the interpretation that Reflex's strategic importance currently exceeds its public financial visibility. Those inferences are grounded in the disclosed mission set, product architecture, funding, and policy backdrop, but they remain analytical judgments rather than management disclosures.

Figure 52: AUTOFLUG - overview

Autoflug GmbH

Ownership:	Privately held family-owned company, with public materials identifying Martin Kroell as managing partner and Andreas Sedlmayr as partner/family shareholder		
HQ:	Rellingen, 25462, Germany	**Founded:**	1919
Domain:	Air – aviation safety, rescue, seating, cabin and fuel systems for defense and aerospace		
Value chain:	Tier-1 supplier plus MRO/service partner		
Why it matters	Autoflug occupies safety-critical niches in military aviation and support, with deep certification know-how and long embedded support roles for German defense aviation		
Key products	airSEAT / FLYWEIGHT seats; airFLEX / MULTIBASE cabin solutions; safeFLIGHT and safeCHUTE rescue and protection equipment; fuelSENS, fuelDATA and fuelPLUS fuel-management products		
Core capabilities	Certified textile, mechanical and electronic safety systems, civil/military quality management, lifecycle support and cabin/safety integration		
Customers	Public customers/programs include the Bundeswehr, Airbus Defence & Space for Eurodrone fuel equipment, Airbus Helicopters, the A400M program and the Franco-German air transport squadron in Évreux		

Key financials (31 Mar)	2021	2022	2023	2024	2025
P&L (EURm)					
Sales	47	53	46	48	53
Gross profit	31	34	32	31	31
EBITDA	4	5	13	2	1
EBIT	3	5	12	1	0
EBT	3	4	1	-1	0
Net income	3	4	0	-1	1
Order intake	45	38	47	82	37
Balance sheet (EURm)					
Cash & cash equivalents	0	0	0	0	0
Tangible fixed assets	2	3	3	2	2
Goodwill & intangible assets	0	0	0	1	1
Inventories	14	16	15	18	22
Accounts receivable	13	12	12	11	11
Other assets					
Total assets	29	31	31	33	36
Equity	9	9	9	9	13
Minorities	0	0	0	0	0
Borrowings	1	3	1	3	4
Accounts payable	4	3	2	4	4
Other ST liabilities	-	-	-	-	-
Net debt (+) / net cash (-)	1	3	1	3	4
Key company metrics					
Book/bill (x)	1.0	0.7	1.0	1.7	0.7
Sales growth (%)	5	13	-14	3	11
Net income growth (%)	-	-	-	-	-
EBITDA margin (%)	9	10	28	5	2
EBIT margin (%)	7	9	26	3	0
ROE (%)	36	46	5	-6	4
Capex/sales (%)	3	2	2	1	1
Capex/depreciation (x)	1.9	1.4	1.0	0.5	0.4
Net debt/equity (%)	15	29	6	37	33
Net interest cover (x)	21	13	1	1	-1

Source : Company data, Bloomberg Finance LP, Gain.pro

AUTOFLUG GmbH

Custom Seating & Cabin Solutions

What type of company it is

In practical terms, AUTOFLUG is a privately held German aviation and defense subsystem supplier and service partner, rather than a builder of complete aircraft or vehicles. Its portfolio spans rescue-and-safety systems, aircraft and vehicle seating, modular cabin solutions, fuel sensing and control, and maintenance, repair, overhaul, and engineering support, with clear exposure to both military and adjacent civil aviation applications.

Investment-style or strategy-style summary

As a privately owned German company, AUTOFLUG GmbH is a specialist supplier of safety-critical subsystems for the aviation and defense sectors. Its core products include crashworthy seating for helicopters and military vehicles, advanced rescue equipment, and modular cabin systems for rapid aircraft reconfiguration. The company's competitive advantage is rooted in its deep engineering know-how for survivability and mission-specific customization, creating high switching costs for customers on qualified platforms.

Financially, AUTOFLUG is a mid-sized industrial player with approximately €53m in revenue (FY2024/25), driven primarily by manufacturing and recurring maintenance (MRO) contracts. This mixed business model provides a stable economic foundation, further secured by critical German military and civil aviation certifications that create significant barriers to entry for competitors.

Strategically, AUTOFLUG is deeply embedded in the European defense ecosystem. An investment would provide targeted exposure to German and European defense modernization through its roles on core programs like Eurodrone and with primes like Airbus. This offers a stake in enabling technologies for mission flexibility and the stable aviation aftermarket, plus a call option on the emerging urban air mobility market through its dual-use capabilities.

Despite its strategic importance, transparency is limited due to its private status. Ownership is linked to the founding family and managing partner, but detailed financials, ownership percentages, and profit margins are not publicly disclosed.

1. Why this company is relevant now

AUTOFLUG is relevant now because current public disclosures show it participating in programs and capability areas that align closely with today's defense-aviation priorities: survivability, rapid mission reconfiguration, helicopter seating, unmanned-system support, and lifecycle readiness. The 2024 Eurodrone contract, the Airbus Helicopters seat award, NH90 survival-pack delivery, and public funding for modular crashworthy seats all reinforce the view that AUTOFLUG is not simply a legacy safety-equipment vendor, but an active participant in current military and dual-use aviation demand.

The company is also relevant because its own lobbying register states that it is materially dependent on decisions of Parliament and the Federal Government, especially in defense-related areas. That makes AUTOFLUG not just an industrial supplier, but a company whose outlook is tied directly to procurement, budget, and



policy direction in Germany.

2. What type of company it is

AUTOFLUG is best described as a niche aviation and defense systems supplier with a mixed model spanning subsystem manufacturing, systems integration, and aftermarket support. It is not a lead contractor and not a pure component shop either. Its practical identity is closer to a specialist system house for rescue, safety, seating, cabin reconfiguration, fuel measurement, and sustainment.

The company also has a meaningful dual-use character. Its products serve military aircraft and vehicles, but it also addresses civil helicopters, VIP/passenger transport, medevac, and urban air mobility. That broadens its addressable market, though it also makes the investment case less clean than that of a pure-play defense electronics or munitions business.

3. Three things that matter most

AUTOFLUG appears to make money where defense customers often place the highest value: survivability, safety certification, and mission-ready availability. Those capabilities usually have longer qualification cycles and higher switching friction than generic fabricated parts.

The company's business mix looks more resilient than a single-program supplier's mix. Public sources show seats, cabin modules, rescue gear, fuel systems, engineering, and MRO. That should provide some internal diversification, even if customer and geographic concentration are still likely meaningful.

The main caveat is transparency. AUTOFLUG's public profile is strong on products and programs, but weak on ownership percentages, margins by business line, backlog, customer concentration, and medium-term capital allocation. Any banking or coverage decision would need a management conversation to close those gaps.

4. At a glance

AUTOFLUG is headquartered in Rellingen and operates as an owner-managed German Mittelstand defense-and-aviation supplier. Public company pages present it as a globally recognized supplier and service partner for international aviation and defense, with competencies in rescue and safety, ergonomics, textiles, mechanics, fine mechanics, electronics, software, and measurement and control. The legal entity is AUTOFLUG GmbH, registered at Amtsgericht Pinneberg under HRB 5365.

The strongest readable financial disclosure indicates a group revenue base of €53m in FY2024/25, with the reporting perimeter including Autoflug GmbH, Autoflug Gruppe GmbH, and Autoflug USA. Activity mix is led by manufacturing and maintenance, and geography is heavily concentrated in Germany.

Industrial relevance stems less from size than from position. AUTOFLUG appears to sit inside several narrow but valuable capability zones: military and civil helicopter seating, aircrew survival equipment, modular transport-aircraft cabin systems, fuel quantity indication, and specialist MRO for safety-critical aviation systems.

Key takeaways



- Private German aviation and defense supplier headquartered in Rellingen, focused on safety-critical subsystems, seating solutions, and support services for military and civil aerospace applications.
- FY2024/25 revenue was approximately €53m with around 219 employees currently (277 on a FY23/24 basis), with manufacturing and maintenance appearing to be the economic core.
- Key programs include Eurodrone, Airbus Helicopters seating systems, A400M-related cabin solutions, and Bundeswehr support activities.
- The company also has meaningful dual-use exposure through civil helicopters and emerging eVTOL/urban-air-mobility platforms.
- Public disclosure is limited due to its private ownership structure, although industrial relevance and program participation are well established.
- The key analytical question is how much future growth will come from newer defense-and-aviation programs versus the more established sustainment and maintenance activities.

5. Corporate history and legal structure

AUTOFLUG traces its origins to 1 October 1919, when Gerhard Sedlmayr founded the business in Berlin-Johannisthal as "Spezialhaus für das AUTOmobil und FLUGwesen, AUTOFLUG." The official company history highlights early parachute work, a long Martin-Baker relationship, post-war reconstruction in Hamburg in 1955, relocation to Rellingen in 1958, later expansion into helicopter safety seats, fuel measurement and control, protective equipment, military land-vehicle seating, and modular cabin systems.

Today's legal operating entity is AUTOFLUG GmbH, registered in Rellingen, Schleswig-Holstein, under HRB 5365 PI, with Martin Kroell listed in the imprint as managing director. Public registry aggregators also show Autoflug Gruppe GmbH in Rellingen, and the strongest readable consolidated filing is at SECO GmbH & Co. KG level, which explicitly includes Autoflug GmbH, Autoflug Gruppe GmbH, and Autoflug USA. That indicates AUTOFLUG sits inside a broader privately held group structure, though the exact full ownership chain and ownership percentages are not transparently disclosed in the reviewed public sources.

A reasonable inference is that the group is operationally centered on AUTOFLUG GmbH, with holding entities above or alongside it. AUTOFLUG is privately held by the founding Sedlmayr family. Public commercial-register records show Martin Kroell as managing partner of AUTOFLUG GmbH and CEO of the family holding, Autoflug Gruppe GmbH, having entered the shareholder circle in 2019 as the first non-Sedlmayr co-shareholder; Andreas Sedlmayr remains an owner/shareholder.

Concise timeline of key inflection points:

- 1919 - Founding in Berlin-Johannisthal by Gerhard Sedlmayr. The initial business emerged from early aviation and mobility needs.
- 1955 - Reconstruction in Hamburg. The company re-established itself after the war.
- 1958 - Move to Rellingen and beginning of German Martin-Baker ejection-seat activities. This linked AUTOFLUG to long-cycle military aviation safety work.
- 1989 to 2002 - Expansion into helicopter safety seats, fuel measurement/control, protective equipment, and military land-vehicle seating. This

broadened the company beyond legacy rescue equipment.

- 2006 to 2009 - Entry into cabin solutions and A400M seat-system production. This moved the company further into modular aircraft interiors and transport-platform relevance.
- 2017 to 2019 - Management/ownership transition toward Martin Kroell. This marked a governance shift away from purely family-only ownership.
- 2020 - Volocopter cooperation. This expanded the narrative into advanced air mobility and lightweight seating.
- 2024 - Eurodrone contract, Airbus Helicopters seat award, NH90 PSP deliveries, and first MULTIBASE handover. This is the clearest recent cluster of evidence for current strategic relevance.

6. Business overview

AUTOFLUG's current business is organized around a fairly coherent safety-and-mission-systems logic. Public product pages group the company around Cabin Solutions, Vehicle Solutions, Rescue & Safety, Measurement & Control, and Services & Engineering. The resulting mix spans physical products, engineered subsystems, software-enabled sensing, and recurring support work.

The structure appears to have evolved from classic aviation rescue-and-safety equipment toward a broader defense-aviation specialist profile. The historical additions of helicopter seating, fuel systems, land-vehicle seats, modular cabin pallets, and MRO support show a stepwise extension from personal safety products into mission equipment and lifecycle services.

The business lines that appear central today are crashworthy seating, safety/rescue systems, fuel measurement and management, and service support. Peripheral or smaller activities likely include more experimental or emerging dual-use efforts such as urban-air-mobility applications and certain research projects, though these may still matter strategically. That ranking is an inference drawn from program visibility and the group activity mix, not from disclosed segment margins.

7. Position in the value chain

AUTOFLUG sits mainly in the subsystem and specialist-system layer of the defense-and-aviation value chain. It is not a complete-platform manufacturer. Instead, it supplies and supports specific safety-critical modules and integrated subassemblies such as seats, cabin pallets, rescue equipment, fuel indication systems, and sustainment services.

In several product areas, AUTOFLUG also behaves like a mini system integrator. MULTIBASE is not just a seat but a modular cabin system. Fuel Quantity Indication equipment includes hardware, software, sensors, and harnessing. Rescue-and-safety offerings are described as system solutions that extend across the product lifecycle.

8. Products, services, and core capabilities

AUTOFLUG's major product families include MULTIBASE cabin pallets, airSEAT and FLYWEIGHT aircraft seating, landSEAT and landFLEX for armored vehicles, rescue-and-safety equipment under lines such as safeFLIGHT and PROTÆCT, fuel systems under fuelSENS/fuelPLUS/fuelDATA, and lifecycle support under fullSERVICE and engineeringSERVICE. The portfolio breadth is notable for a company of this scale.



MULTIBASE is one of the clearest strategic products. It enables rapid reconfiguration of aircraft such as the A400M or C-130 into passenger, VIP, medevac, or mixed-mission layouts, with 16g crashworthy seating and cabin conversion in less than two hours according to the company. That directly links AUTOFLUG to flexibility, medical logistics, and transport-platform mission adaptation.

FLYWEIGHT and airSEAT matter because they speak to both military and civil rotorcraft demand. The FLYWEIGHT family is positioned around lightweight composite construction, crashworthiness, and configurable variants including foldable, rotating, sliding, and wall-mounted options. airSEAT covers troop seats and helicopter safety seats, including ballistic protection options for military use.

In its Vehicle Solutions unit, AUTOFLUG unveiled the Suspended Dynamic Seat (SDS) 2.0 for armored-vehicle crews at DSEI London in September 2025, featuring a new backward-compatible single-lever adjustment system; the company states the seat has already been ordered by a NATO customer for a current program.

The rescue-and-safety and service businesses add depth. AUTOFLUG describes itself as a system provider for rescue-and-safety equipment, and its fullSERVICE offering includes maintenance and overhaul of Tornado and Eurofighter ejector-seat-related equipment and other specialist aircraft gear for the Bundeswehr. That combination of product and support work is important because it usually improves customer stickiness.

Measurement and control is another strategically useful capability. The Eurodrone award specifically covers development, manufacture, support, and delivery of Fuel Quantity Indication equipment, including fuel data conditioners, software, sensors, and cable harnesses. fuelPLUS and related products also show that AUTOFLUG is not only a seat supplier, but an aviation-systems supplier with sensing and integration know-how.

Public quality indicators are better than average for a private company. AUTOFLUG states it is approved by the Luftfahrtamt der Bundeswehr and the Luftfahrt-Bundesamt, holds EASA Part 21/G and Part 145 approvals, and is certified to ISO 9001 and EN 9100. FLYWEIGHT marketing also references ETSO approval for certain variants. No public margin split by product line was found in the reviewed set.

Product-level strengths, risks, and catalysts:

Strength - safety-critical specialization backed by civil and military approvals.

Risk - some products are likely niche and program-linked, which can make volume lumpy. That is an deduction from the program-specific public wins.

Catalyst - new program traction in Eurodrone, Airbus Helicopters, and advanced-air-mobility seating.

9. Customers and routes to market

AUTOFLUG's visible customer base spans military end-users, prime contractors, and civil/dual-use aviation partners. Publicly named relationships include Bundeswehr/Luftwaffe support work, Airbus Helicopters, Airbus Defence & Space on Eurodrone, and Volocopter in urban air mobility. The mix implies both direct

government-linked activity and indirect sales through OEM supply chains.

Figure 53: MULTIBASE program



Source : Autoflug

The route to market appears to vary by business line. Some work is clearly direct sustainment and support for German military users. Other work is OEM-facing, with AUTOFLUG supplying qualified systems into Airbus or other integrators' platforms. The company also uses alliances and cooperation agreements to widen market access, including US and European partnerships.

Revenue quality is probably mixed but relatively solid. Manufacturing of original systems can be cyclical, but maintenance, overhaul, spare parts, and ongoing in-service support are usually stickier and can smooth program volatility. The fact that maintenance represented €14.8m of FY2023/24 group revenue supports the view that recurring or semi-recurring support income is meaningful. Customer concentration, however, is not publicly quantified in the reviewed sources.

10. Key financials and economic profile

AUTOFLUG is a mid-sized, privately held supplier with a stable but specialist financial profile. Revenue has moved in a €45-56m range, from €45.0m in 2019 to €56.2m in 2025, with a cyclical dip to €47.9m in 2023 before recovering - a pattern consistent with program-driven rather than recurring demand. Gross margin is healthy and fairly steady, typically in the low-to-mid 60s as a percentage of revenue, though it eased to 55.6% in 2025.

Operating profitability is thin and uneven: excluding an anomalous 2023 (EBITDA €12.9m, likely a one-off), EBIT has generally run between roughly €1.4m and €4.6m. The balance sheet is small and working-capital-intensive - total assets of €35.5m at end-2025, net debt of €3.5m, effectively no cash, and inventory that has climbed from €10.2m (2019) to €23.2m (2025). Capex is light at €0.3m in 2025, underlining an asset-light, certification-led model. The main limitation is disclosure: no public net income, segment margins, backlog, or customer concentration - so these are inferences from group figures rather than disclosed detail.

11. Footprint and industrial setup

AUTOFLUG's operational center is Rellingen, where the company's legal seat, public contact address, and core corporate identity are located. The company has described itself as having more than 6,000 sqm of production area there. That points to a concentrated, Mittelstand-style industrial setup rather than a broad multi-site manufacturing network.

The Luftfahrtamt der Bundeswehr's published list of approved aviation organizations also identifies AUTOFLUG locations in Rellingen and Ummendorf. In



addition, the SECO consolidated filing includes Autoflug USA, suggesting at least a small US corporate or commercial footprint inside the reporting perimeter. Still, the visible industrial gravity remains in Germany.

Recent public cooperation agreements suggest that AUTOFLUG is willing to extend engineering reach through partners rather than only through wholly owned sites. Examples include the ALTHOM cooperation around seat-system design and documentation, and prior US market-access partnerships.

12. Supply chain and key dependencies

Hard evidence on named suppliers is limited in public sources, so this section necessarily mixes fact and inference. Factually, AUTOFLUG publishes supplier quality documentation and export-control classification information, and emphasizes regulated production and maintenance. That indicates a structured supply chain and compliance environment.

The visible hard dependencies are regulatory and industrial rather than vendor-specific. AUTOFLUG depends on civil and military approvals, on government procurement and budget decisions, and on maintaining quality-assured manufacturing for safety-critical products. It also appears dependent on large customers' programs and schedules, especially where it supplies into Airbus or Bundeswehr-related activity.

Our inference is that the main hidden dependencies likely include specialist textiles and composites, metal fabrication, restraint systems, electromechanical components, sensors, harnessing, software validation, and skilled engineering labor. There is also likely a meaningful site-concentration risk because publicly available information points to a Germany-centered setup. None of those assumed dependencies are unusual for this type of business, but they do matter for ramp execution.

13. Management, board, ownership, and governance

Martin Kroell is the clearly visible current operating leader. The company imprint lists him as managing director, and company releases show he entered management in 2017 after already being with the company since 2013. In 2019, AUTOFLUG announced that he had also entered the shareholder circle of the Autoflug Gruppe, becoming the first non-Sedlmayr family co-shareholder in the company's then nearly 100-year history.

Ownership remains only partly transparent in public sources. The 2025 Bundestag lobby register lists Martin Kroell as managing partner and Andreas Sedlmayr as owner/shareholder, which suggests continuing family influence alongside management ownership. The company also describes itself as an owner-managed SME. Exact ownership percentages were not found in the reviewed public material.

On wider leadership, AUTOFLUG announced Sylvia Schmenk as Vice President of Sales and Marketing in November 2024. Public disclosures do not provide a listed-company style board pack, and we did not identify a publicly disclosed supervisory board composition in the reviewed set. For a private GmbH, that is not surprising, but it limits governance analysis. Headcount of 277 average employees in FY2023/24 (around 219 currently) suggests a sufficiently large organization for professionalization, and the existence of a whistleblower portal and formal export-control/quality materials points to at least a basic compliance infrastructure.

Management stability looks reasonable rather than obviously fragile. The main governance questions are succession depth beyond Kroell, the exact current role and influence of Andreas Sedlmayr, and the visibility of second-line leadership in operations, engineering, and finance. Those are open questions rather than disclosed weaknesses.

14. Competitive positioning

AUTOFLUG's real competitors differ by business line. In crashworthy aircraft seating, Martin-Baker is a relevant benchmark because it markets a broad crashworthy seat range for military and civil rotorcraft and fixed-wing aircraft. AUTOFLUG competes there with airSEAT and FLYWEIGHT, though the relationship is also partly cooperative in ejection-seat sustainment and historically in Germany. This is a classic case of line-by-line competition rather than whole-company overlap.

In blast-mitigating land-vehicle seating, Jankel is an obvious comparator. Jankel says it has sold more than 60,000 BLASTech seats globally, which suggests greater scale in that niche. AUTOFLUG's landSEAT and OVERMATCH offerings, by contrast, appear narrower and more specialized, likely stronger in specific German or European programs than in sheer global seat count. That relative-size judgment is an inference.

AUTOFLUG's advantages appear to be customization, survivability focus, mixed military/civil certification know-how, and a combination of product plus MRO/service capability. Its disadvantages are limited scale, lower public visibility, and likely less purchasing power and geographic breadth than larger global aerospace-seat or defense-subsystem suppliers.

Competition also depends on customer channel. Against primes, AUTOFLUG competes on specialist know-how and responsiveness. Against other seat houses, it competes on crashworthiness, modularity, and mission fit. Against generic suppliers, it competes on certification and installed-base familiarity.

15. Strategic relevance and market implications

AUTOFLUG matters strategically because it serves enabling nodes in aviation readiness and survivability rather than headline platforms. Seats, survival systems, cabin reconfiguration, fuel quantity indication, and sustainment do not attract as much attention as missiles or radars, but they are still required for deployability and mission availability.

The company's relevance is especially visible in German and European contexts. Eurodrone is a multinational European program. Airbus Helicopters is a major European rotorcraft prime. Bundeswehr/Luftwaffe work ties AUTOFLUG to domestic readiness. The implication is that AUTOFLUG looks more like a niche capability anchor inside the European defense-industrial base than like a generic industrial supplier.

For listed peers and partners, AUTOFLUG is most relevant as a supply-chain signal. Its wins and capacity trajectory can matter to complete-aircraft manufacturers, airframe suppliers, support contractors, and adjacent seating/survivability players. In a more consolidated European defense market, companies like AUTOFLUG can become attractive partners, bolt-on acquisitions, or protected national specialists depending on policy mood and ownership preferences. The final point is an inference.

16. Dual-use and civilian exposure

AUTOFLUG is not a strict pure-play defense name. Civil helicopter seating is clearly part of the portfolio, and the company has openly pursued urban-air-mobility and air-taxi opportunities through FLYWEIGHT and its Volocopter partnership. MULTIBASE also has VIP/passenger and medevac applications beyond purely military transport.

That said, current publicly available information still points to defense as the more strategically important side of the business. The lobbying register focuses on defense and government-linked policy fields. Revenue is strongly Germany-centered. Publicly named work includes Bundeswehr, Eurodrone, NH90, and Airbus Helicopters. Our judgment is that AUTOFLUG should be thought of as defense-led with meaningful dual-use and civil adjacency, not as a civilian aerospace company with incidental defense exposure.

Civilian diversification is still valuable. It broadens product applicability, helps engineering reuse, and may reduce dependence on a single procurement cycle. The risk is that mixed exposure can sometimes dilute narrative clarity and management focus if defense growth accelerates faster than civil opportunities. That is an inference, not a disclosed issue.

17. Legal, regulatory, and compliance risks

Our overall legal and regulatory risk assessment is Medium. The main concentration of risk is structural rather than event-driven: export control, civil and military airworthiness approvals, procurement-law exposure, and dependence on government decisions. In the reviewed public-source set, we did not identify a material disclosed company-specific pattern of fines, sanctions, or litigation. That should be read as a transparency limit, not as proof of absence.

The core regulatory environment is clear. AUTOFLUG says it is approved by the Luftfahrtamt der Bundeswehr as an aviation organization for Bundeswehr aircraft and aeronautical equipment, and by the Luftfahrt-Bundesamt as an EASA Part 21/G production organization and Part 145 maintenance organization. The BAFA is Germany's central export-control authority, and export control is explicitly relevant because AUTOFLUG publishes export-control classification documentation and operates in defense and dual-use areas.

- A concise visible timeline looks like this:
- Ongoing - civil and military approvals under LBA/LufABw frameworks.
- 2022 onward - formal Bundestag lobbying register presence, reflecting direct public-policy interaction.
- 2024/25 - public grants above €10,000 disclosed in the current register, including ModSeat and LH2 measurement-related projects.
- Current - public whistleblower portal, supplier quality manual, and export-control documentation available on the company site.

On IP, the company states that its logos and product names are protected. Secondary registry aggregation also points to patents, utility models, designs, and trademarks including FLYWEIGHT and helicopter seating-related filings. We treat the detailed IP list as secondary rather than primary evidence, but it supports the idea that AUTOFLUG does maintain a real IP estate around seating and protection systems.



18. Funding, financing, and strategic optionality

Publicly visible funding appears to be a combination of internal cash generation, bank financing, and targeted public R&D support. The FY2024/25 consolidated filing shows equity of €13.8m, borrowings of €3.5m, and positive but modest profit. The current lobbying register also discloses several public grants in FY2024/25, including support for modular crashworthy seats and LH2 tank-content measurement.

That profile is typical of a capable private industrial company rather than of a heavily financialized roll-up. AUTOFLUG does not look dependent on public equity markets, and there is no visible sign of IPO preparation in the reviewed sources. At the same time, the company is active enough in current programs and technology areas that strategic optionality is real. Plausible paths include continued private compounding, targeted acquisitions or alliances, deeper JV/cooperation structures with primes, or an eventual transaction if ownership priorities change. The last sentence is inference.

19. Moat analysis

AUTOFLUG's moat rests on three reinforcing barriers rather than scale. First, qualification and certification: LufABw, LBA/EASA Part 21G and Part 145, EN 9100 and ETSO approvals take years to obtain and are platform-specific, which deters new entrants. Second, switching costs: once a seat, restraint or fuel-measurement system is qualified into an airframe, replacing it triggers re-certification, re-testing and crew retraining, so incumbency tends to persist across a platform's decades-long life. Third, installed-base and sustainment lock-in: the MRO and spares relationship (e.g., Tornado/Eurofighter ejection-seat support) generates recurring, sticky revenue that a pure component vendor cannot easily contest. The double-edge is NATO standardization: alliance-wide adoption of a system like MULTIBASE would enlarge the addressable market, but a competitor's product becoming the standard could erode AUTOFLUG's home-field advantage and force price competition against larger players. On balance the moat looks stable to modestly widening, anchored in certification depth rather than cost leadership – an inference from the disclosed approvals and program roles rather than a disclosed metric.

Strategically, AUTOFLUG is deeply embedded in the European defense ecosystem, but faces a double-edged sword from the trend towards NATO standardization. On one hand, standardization presents a major opportunity; if a product like its MULTIBASE cabin system were adopted alliance-wide, it would unlock a massive new market and solidify its competitive position. On the other hand, it poses a significant threat, as a competitor's product becoming standard could erode AUTOFLUG's "home-field advantage" and force it to compete on price and scale against larger global players, even within Germany. This high-stakes dynamic overlays its current roles on programs like Eurodrone and with primes like Airbus, which offer exposure to European defense modernization and a call option on the emerging urban air mobility market.

20. SWOT analysis

Strengths

- Long operating history and strong specialization in rescue, safety, seating, and sustainment.
- Civil and military certifications create credibility and barriers to entry.
- Product mix spans hardware, systems, and support rather than a single

niche.

Weaknesses

- Private-company disclosure is limited.
- Revenue appears heavily concentrated in Germany.
- Scale is modest relative to larger global aerospace-seat and defense-subsystem peers.

Opportunities

- Eurodrone, Airbus Helicopters, and other European defense-aviation programs.
- Growth in modular cabin and medevac solutions.
- Dual-use expansion via civil helicopters and eVTOL seating.

Threats

- Procurement and budget dependence on the German public sector.
- Export-control and certification complexity.
- Competitive pressure from larger specialist seat and survivability suppliers.

21. Key risks and open questions

- How much of current revenue and profit comes from defense sustainment versus newer production programs? Public sources do not break this out.
- How concentrated is AUTOFLUG by customer, platform, and ministry budget line? That is not disclosed publicly in the reviewed set.
- What is the exact ownership and succession structure between Martin Kroell, Andreas Sedlmayr, and any holding entities above AUTOFLUG GmbH?
- Can the company scale output smoothly if current European rotorcraft and defense-aviation demand strengthens further?
- How durable is the civil/eVTOL option value versus near-term defense demand?
- Are current public R&D grants translating into commercially meaningful products or mostly into option value?
- Is the current Germany-heavy revenue mix a strength, because of embeddedness, or a risk, because of concentration?

22. What to watch next

- Eurodrone development milestones and whether AUTOFLUG's fuel-system role expands beyond current scope.
- Airbus Helicopters seat-delivery cadence and evidence of follow-on platform penetration.
- Further MULTIBASE deployments, especially in A400M, medevac, or follow-up heavy-lift applications.
- Whether FLYWEIGHT gains additional civil helicopter or eVTOL reference wins.
- Conversion of grant-funded R&D projects such as ModSeat or LH2



measurement into visible products or contracts.

- Any new disclosures on ownership, governance, or management-layer expansion.
- Hiring intensity and any sign of site/capacity expansion.

Availability of a fuller FY2024/25 financial filing in machine-readable form. A filing date is visible publicly, but the reviewed set did not yield a fully readable updated financial extract.

23. News flow and recent developments

The most important recent development is a cluster of program and market-access announcements rather than a single transformative transaction. In July 2024, AUTOFLUG signed the Eurodrone contract for Fuel Quantity Indication equipment. In November 2024, it disclosed an Airbus Helicopters seating contract covering development, production, and in-service support for multiple helicopter families. In November 2024, it also highlighted Sylvia Schmenk's appointment as Vice President of Sales and Marketing.

The 2025 public flow added technology and cooperation themes. AUTOFLUG publicized work on smart textiles with the German Air Force's aerospace medicine center, and the company remained active in trade-show and market-facing communication around FLYWEIGHT and fuel systems. That indicates continued product development and go-to-market effort rather than a pure harvest phase.

Stepping back, the last 18 months support a simple conclusion: AUTOFLUG's recent news flow has been program-led, partnership-led, and technology-led, not balance-sheet-led. That fits the profile of a private specialist supplier focused on capability relevance and position in the value chain.

In June 2026, AUTOFLUG was named among the eight founding members of the Airbus-led "Team Gen 6" consortium, a German-backed sixth-generation fighter initiative formed after the collapse of the Franco-German-Spanish FCAS program, alongside MBDA, Hensoldt, Diehl Defence, MTU Aero Engines, Liebherr and Rohde & Schwarz.

AUTOFLUG also received a BAAINBw order for 46 palletized qualified-patient-transport kits for the A400M, deepening its two-decade role on the platform.

24. Source notes and transparency limits

Company sources used

- AUTOFLUG website, including home, history, imprint, product, service, and quality-management pages.
- AUTOFLUG press releases on Eurodrone, Airbus Helicopters, management appointments, Volocopter, smart textiles, and historical management/ownership changes.
- Government and regulatory sources used
- Bundestag Lobbyregister entry for AUTOFLUG.
- BAFA export-control overview.
- Luftfahrtamt der Bundeswehr list of approved aviation organizations.
- Independent external sources used



- Secondary registry aggregators for legal-structure and IP checks, used cautiously.

Competitor official sites such as Martin-Baker and Jankel for business-line comparison.

Transparency limits
AUTOFLUG is private. There is no public investor-relations package, no disclosed ownership percentages, no public segment-margin table, no disclosed customer concentration table, and no clean listed-company style governance disclosure. The strongest readable financial view available in this review is the SECO consolidated filing rather than a stand-alone AUTOFLUG management presentation.

Explicitly labeled estimates and inferences
Where we referred to AUTOFLUG as a subsystem specialist, mini system integrator, niche capability anchor, or stable-to-widening moat business, those are inferences based on the disclosed product mix, approvals, activity mix, and program references rather than direct management guidance. Where we discussed likely supplier dependencies or strategic optionality, those were also inferences and not disclosed facts.

Figure 54: Anschütz - overview

Anschütz GmbH

Ownership:	Since 2023, Anschütz has been owned by DMB Dr. Dieter Murmann Beteiligungsgesellschaft mbH, the Murmann family's industrial holding company		
HQ:	Kiel, 24106, Germany	**Founded:**	1905
Domain:	Sea – naval navigation, bridge and mission systems		
Value chain:	Tier-1 supplier and systems integrator		
Why it matters	Anschütz is a long-established naval navigation specialist with a large installed base and a credible path into autonomous maritime operations		
Key products	SYNAPSIS Naval – warship integrated navigation and bridge system; SYNAPSIS WECDIS NX – naval chart/tactical display; gyrocompasses and repeaters; naval radars, autopilots and steering systems; Autonomics – autonomous		
Core capabilities	Certified mission-critical navigation, deep systems integration for warships and submarines, long-life support and software-defined upgrades for autonomy		
Customers	Anschütz says more than 50 navies rely on its naval technology; public programs include the German Navy plus Royal Navy Type 31 and Type 26-related bridge/navigation integrations		

Key financials (31 Dec)	2020	2021	2022	2023	2024
P&L (EURm)					
Sales	91	97	104	122	128
Gross profit	65	59	70	80	82
EBITDA	0	-4	-1	-2	11
EBIT	-2	-6	-3	-4	9
EBT	-3	-7	4	-4	10
Net income	-3	-7	4	-6	9
Order intake	108	104	156	-	-
Balance sheet (EURm)					
Cash & cash equivalents	18	18	16	2	10
Tangible fixed assets	9	9	9	8	9
Goodwill & intangible assets	2	2	1	1	1
Inventories	35	33	40	47	54
Accounts receivable	14	16	19	26	21
Other assets					
Total assets	83	82	94	106	121
Equity	19	12	16	9	18
Minorities	-	-	-	-	-
Borrowings	-	-	-	-	-
Accounts payable	4	2	4	5	3
Other ST liabilities	-	-	-	-	-
Net debt (+) / net cash (-)*	10	11	16	29	20
Key company metrics					
Book/bill (x)	1.2	1.1	1.5	-	-
Sales growth (%)	-5	6	8	17	5
Net income growth (%)	20	154	-148	-281	-234
EBITDA margin (%)	0	-4	-1	-2	9
EBIT margin (%)	-2	-6	-3	-3	7
ROE (%)	-15	-62	23	-71	49
Capex/sales (%)	3	2	1	-	-
Capex/depreciation (x)	1.5	1.0	0.5	-	-
Net debt/equity (%)	49	93	104	316	115

Source : Deutsche Bank research, company data, Bloomberg Finance LP, Gain.pro; FY22 is the last standalone reporting year

Anschütz GmbH

Navigation Systems Maritime Excellence

Anschütz is one of Germany's most established maritime technology providers, supplying navigation, automation, and command-support systems across commercial and naval fleets. Its position in surface vessels, submarines, and lifecycle support gives it exposure to both global shipping markets and growing naval modernization demand.

Investment-style or strategy-style summary

Anschütz is a German company that makes critical navigation and control systems for ships and submarines. It's not a massive shipbuilder, but a key supplier whose technology is trusted by over 50 navies and found on more than 35,000 vessels worldwide. The company's main strength is this huge, credible installed base, which creates a strong moat built on decades of trust and reliability in safety-critical environments. Strategically, Anschütz is becoming more important by moving beyond simple hardware into more valuable software-driven systems, including integrated warship bridges, submarine automation, and autonomous navigation.

This focus on higher-value systems aligns perfectly with current naval modernization programs, where software, cyber security, and integration are crucial. Since being acquired by a German owner in 2023, the company has increased its defense focus and is expanding in key markets like Australia and the US

The main challenge for an investor is the lack of public financial data, as the company is private. The last public balance sheet showed high inventory and thin equity, suggesting it is a program-heavy business where managing cash flow is vital. While its strategic position in the naval supply chain is strong and growing, the underlying financial health is not fully transparent. Therefore, the investment story relies on trusting its operational momentum and its entrenched position in a specialized, high-barrier market.

1. Why this company is relevant now

Anschütz is relevant because naval procurement is increasingly constrained by subsystem availability, software integration, certification, and support capacity rather than by hull fabrication alone. The company sits directly in those constrained layers. Its products affect heading, control, radar picture, charting, bridge integration, and increasingly autonomous or semi-autonomous operation. That makes it relevant to both warship newbuilds and lifecycle modernization. Recent company disclosures show tangible momentum in Australia, Brazil, and submarine automation, alongside new cyber approvals and continued investment in production and service infrastructure.

Timing also matters. The company is now three years into life under German private ownership after the 2023 DMB takeover. Since then, public disclosures have shown expansion into the US and Australia, more visible defense messaging, stronger autonomous-navigation positioning, and explicit participation in Germany's maritime and defense-policy dialogue through the Bundestag lobby register. In other words, Anschütz is no longer only a legacy navigation supplier. It is presenting itself as a more software-defined, defense-relevant maritime systems house.



2. What type of company it is

Anschütz is best described as a maritime systems specialist with a mixed model. In commercial shipping, it behaves like an OEM and integrator of navigation and bridge systems. In naval surface vessels, it behaves like a subsystem provider and bridge-system house. In submarines, it looks closer to a specialist automation and control supplier with meaningful systems-integration capability. In after-sales and refit, it operates as a lifecycle service provider with a global support footprint.

This is also a dual-use company. The official site still markets solutions to merchant vessels, passenger ships, offshore and special-purpose vessels, megayachts, coast guards, refit customers, naval ships, and submarines. Older filings confirm that civilian and military activities were both significant enough to be separately described. The right mental model is therefore not a pure defense group, but a technically deep maritime company whose defense relevance has increased within a still broader commercial maritime base.

3. Three things that matter most

Anschütz has real depth in the installed base and in qualification-heavy maritime niches. More than 25,000 Standard 22 installations, more than 600 Standard 30 MF installations, more than 50 navies, more than 100 submarines, and more than 35,000 vessels overall are not proof of high margins, but they are proof of trust and operational relevance.

The company appears to be moving toward higher-value software and systems content. Public disclosures on submarine platforms, Aegis integration, autonomous navigation, and cyber approvals suggest that its value increasingly depends on systems integration, software, compliance, and integration rather than only on standalone hardware boxes.

The main analytical handicap is transparency. Public sources make the company's products, locations, and strategic programs relatively visible, but they do not make the current P&L equally visible. That means the strategic case is easier to understand than the current earnings power.

4. At a glance

Anschütz GmbH is a Kiel-based German maritime systems company with a history dating back to 1905. The company's current headquarters remain in Kiel, where more than 500 employees work across administration, development, purchasing, production, quality, sales, and service. The broader group says it has more than 700 employees and 200-plus sales and service partners worldwide. Publicly visible group entities now include Germany, China, Singapore, the UK, Brazil, Panama, the US, and Australia.

Its business mix spans commercial navigation and bridge systems, naval bridge and navigation suites, submarine automation and control systems, radar, ECDIS and WECDIS, autopilots, gyrocompasses, and after-sales support. That is broader than a single-product company but narrower than a diversified defense group. Historically, the company expanded materially under Raytheon ownership into radar, ECDIS, sophisticated bridge integration, and mission systems. Since the 2023 transfer to DMB, the public narrative has shifted toward German ownership, maritime defense relevance, autonomous operation, and software-defined naval systems.

Strategically, Anschütz matters because it occupies a credible position in a



bottleneck layer between complete-warship manufacturers or shipyards and end-users. It is not the headline platform builder, but it can be embedded in the ship's safe-navigation and operator-control framework, which makes it hard to replace once selected. That positioning is visible in the Hunter program, the Tamandaré class, submarine automation disclosures, and the company's expanding local-presence strategy in Australia and the US.

Key takeaways:

- Private German maritime systems supplier headquartered in Kiel and owned by DMB since 2023.
- The company combines commercial navigation and bridge systems with naval and submarine automation, integration, and lifecycle-support activities.
- Its installed base spans more than 35,000 vessels, over 50 navies, and more than 100 submarines, underlining a strong position in global maritime markets.
- Strategic relevance is increasing through naval modernization, cyber resilience, automation, and submarine programs.
- Public disclosure is limited relative to listed defense suppliers, with only partial visibility on financial performance and business mix.
- The key analytical question is how much of current and future earnings are driven by naval and submarine programs versus the larger commercial maritime business.

5. Corporate history and legal structure

Anschütz traces its origin to Dr. Hermann Anschütz-Kaempfe's effort to solve reliable navigation near the poles. The official history says the first gyrocompass was patented in 1904, the company was founded in Kiel in 1905, and the first operational gyrocompass followed shortly afterward. The same official material links later milestones to the first ship autopilot, the "Iron Helmsman," the two-gyrosphere compass developed with Max Schuler and Albert Einstein, and later innovations such as chart plotting, track control, and integrated navigation systems. The company later came under Carl Zeiss and then Raytheon ownership before returning to German ownership under DMB in 2023.

The current legal entity is Anschütz GmbH. The official imprint lists the registered address as Zeyestraße 16-24, 24106 Kiel, Germany, with commercial registration at Amtsgericht Kiel under HRB 4086. The Bundestag lobby register also identifies the company as a GmbH and shows the register entry last updated on 16 February 2026. Public sources reviewed do not indicate any listed securities. The 2023 acquisition press release states that DMB took over the entire group, including subsidiaries in Panama City, Portsmouth, Rio de Janeiro, Shanghai, and Singapore. Current public pages additionally show group entities in the US and Australia. On balance, the structure appears strategically centralized around Kiel with outward-facing regional subsidiaries and service hubs. That last point is an inference based on the visible footprint and headquarters concentration.

Concise timeline of key inflection points:

- 1905 - Company founded in Kiel after the gyrocompass invention. This established the technological foundation of the business.



- 1920 - Launch of the "Iron Helmsman," described by the company as the first autopilot for ships. This widened the company from core heading to automated control.
- 1930 - Majority ownership transferred to Carl Zeiss. This marked the first major ownership transition.
- 1945-1952 - Wartime destruction followed by reconstruction in Kiel and renewed commercial-shipping production. This reset the industrial base and reoriented the company toward postwar shipping demand.
- 1995 - Entry into Raytheon ownership. Under Raytheon, Anschütz broadened its portfolio into radar, ECDIS, bridge systems, and mission systems.
- 2021 - Organizational rework from civil and military nautics into "Products" and "Programmes." This indicates a move toward a customer- and process-led structure.
- 2023 - DMB acquisition and return to the traditional Anschütz name. This restored German ownership and appears to have increased strategic emphasis on maritime-defense positioning.
- 2025-2026 - Expansion into autonomous navigation, US and Australian subsidiaries, and major program milestones in Australia and Brazil. This suggests a broadening of defense and regional-market relevance.

6. Business overview

The company's current business can be understood through five visible pillars. First, there is a core navigation and sensor business built around gyrocompasses, autopilots, and related bridge-control products. Second, there are integrated bridge and navigation systems for commercial vessels. Third, there is a naval business that includes integrated bridge, navigation, and warship-specific charting and control functions. Fourth, there is a submarine business focused on steering, automation, monitoring, and control. Fifth, there is a global after-sales and refit business that supports the installed base.

The current structure appears to be the result of two long-term evolutions. One was product broadening under Raytheon from core navigation hardware into radar, ECDIS, bridge systems, and mission systems. The other was the 2021 internal reorganization away from a simple civilian-versus-military split toward "Products" and "Programmes." In that 2021 structure, Products covered Products & Partner plus Customer Service & Refit, while Programmes covered Submarine, Naval Surface Combatants, Integrated Mission Solutions, and Integrated Bridge Systems. That older structure is useful because it clarifies where the company itself saw the strategic operation anchor.

Central businesses appear to be integrated navigation and bridge systems, naval solutions, submarine automation, and installed-base service. Peripheral businesses are harder to define because the company is private and does not disclose full current segment economics, but public project pages suggest that autonomous navigation and research-linked control architectures are moving from peripheral R&D toward commercially relevant future offerings. That is an inference, but it is grounded in the 2025 launch of Autonomics, the approval-ready autonomous system, and multiple public grants tied to CAPTN, OCCUMAR, and related projects.

7. Position in the value chain

Anschütz does not sit at the top of the naval value chain as a complete-warship manufacturer. It sits one layer down, where navigation, bridge integration, control logic, automation, and operator interface meet platform integration. In commercial shipping, that means OEM supply of compasses, autopilots, radar, ECDIS, and integrated bridge systems. In naval vessels, it means a systems-house role for bridge and navigation systems, including warship-specific solutions. In submarines, it means control, steering, automation, battery monitoring, and related subsystem integration.

That value-chain position matters because it gives Anschütz exposure to both newbuilds and retrofits. It can be specified by shipyards and primes at the build stage, but it also participates in refit and lifecycle support. The 24/7 support structure in Germany, Singapore, and Panama reinforces that aftermarket role. The company therefore combines elements of OEM, system integrator, and service provider. The precise weight of each role is not fully disclosed, but the route-to-market and product architecture strongly support that mixed model.

8. Products, services, and core capabilities

Anschütz offers a comprehensive portfolio of maritime navigation and control systems, ranging from individual hardware components to fully integrated solutions. The core product lineup includes its famous gyrocompasses, autopilots, steering controls, radar systems, and digital chart displays (ECDIS). However, the company's highest strategic value comes from its integrated systems, particularly its custom suites for naval warships and its advanced automation systems for submarines. These integrated solutions are extremely difficult and costly to replace once they are designed into a vessel, as they require extensive qualification, testing, and training, effectively "locking in" naval customers for the long term.

The company's market credibility is anchored in its vast and proven installed base. Its flagship gyrocompass, the Standard 22 NX, has over 25,000 installations, while the more advanced, maintenance-free Standard 30 MF has accumulated millions of operating hours with exceptional reliability. In total, Anschütz systems are found on over 35,000 vessels and more than 100 submarines globally. The flagship integrated product, the SYNAPSIS bridge system, comes in a commercial version and a military-grade "WECDIS" variant that meets NATO standards. A crucial recent development is that these core products have received modern DNV cybersecurity approvals, a key requirement for navies concerned with digital threats.

Beyond maintaining its legacy dominance, Anschütz is heavily invested in future technology. It has developed an approval-ready system for autonomous and unmanned maritime operations and formally launched "Anschütz Autonomics," a new division dedicated to this field. This is complemented by hardware innovation, such as a new solid-state radar that reduces maintenance. This strategic push demonstrates the company's evolution from a traditional hardware supplier into a modern, software-defined maritime systems house focused on automation, cyber resilience, and integrated control.

Product-level strengths, risks, and catalysts:

- Strength - deep installed base with strong quality and reliability signals, especially in compasses and naval navigation.
- Risk - no publicly available information yet that autonomy products have reached large-scale contracted deployment. The pipeline is visible, but commercial ramp remains an open question.



- Catalyst - Hunter-class, U212CD, Tamandaré follow-on ships, and further cyber approvals could all validate a higher-value systems role.

9. Customers and routes to market

Anschütz serves a wide variety of customers, from commercial cargo ships and megayachts to the most advanced naval warships and submarines. The key distinction lies in what these customers are buying. A commercial shipowner typically seeks efficient, safe, and reliable navigation with good service support. In contrast, a defense customer—like a navy or a major shipyard—prioritizes systems that are highly rugged, cyber-secure, and guaranteed to be interoperable with complex weapon and command systems.

The company uses a mixed strategy to reach these customers. It sells directly through its own international offices in key maritime hubs like Germany, the US, Singapore, and Australia. This direct presence is crucial for winning large naval contracts. Alongside this, Anschütz relies on a global network of over 200 sales and service partners, ensuring that its massive installed base of ships can get support anywhere in the world, a critical factor for both commercial and military clients.

Anschütz has significant, publicly named roles in major naval programs, which validates its credibility. These include providing the integrated navigation system for Australia's new Hunter-class frigates, Brazil's Tamandaré-class warships, and contributing to the highly advanced U212CD submarine project for Germany and Norway. This participation demonstrates real traction in the high-value defense market. Consequently, the company's revenue is a mix of one-time equipment sales, long-term program revenue tied to these large naval builds, income from refitting older ships, and recurring service fees. This business model involves long project timelines and significant upfront investment in inventory, which can lead to lumpy or uneven cash flow.

10. Key financials and economic profile

Public financial visibility is limited and somewhat dated, but the readable disclosures point to a program-heavy, working-capital-intensive business rather than a high-margin one. Reported sales stagnated around €100m before accelerating in 2023 and 2024. Profitability also improved markedly, with EBIT jumping to €11m (after a small EBITDA loss in 2023). The cash position also improved as a consequence, despite further inventory growth. That balance-sheet shape is not unusual for a naval-program business with long milestones and heavy pre-build inventory, but it does mean liquidity discipline, milestone execution, and parent-level financing support matter more than headline revenue growth.

11. Footprint and industrial setup

Kiel is clearly the core site. Official company materials say the headquarters have been in Kiel since the company was founded and that more than 500 employees there cover administration, development, purchasing, production, quality management, sales, and service. That implies that Kiel is not only corporate headquarters but also the main engineering and industrial center. The official imprint also lists a Flensburg site.

The international footprint is designed around sales, service, and regional market access. Publicly visible group locations include China, Singapore, the UK, Brazil, Panama, and, more recently, the US and Australia. The US subsidiary in Bethesda was launched in November 2025 to support commercial, shipyard, and government customers, including the Coast Guard, Military Sealift Command,



Navy, and Army. The Australian subsidiary was launched the same month to support sovereign naval shipbuilding and sustainment in Australia and New Zealand, with explicit links to the ANZAC and Hunter programs.

Recent public investments suggest some selective strategic-capacity expansion. The most concrete industrial example is the February 2025 completion of a major compass-production expansion in Kiel, lifting capacity to more than 5,000 compass sensors and gyrospheres per year. The service side has also been reinforced through 24/7 coordination hubs in Germany, Singapore, and Panama. Together, that suggests management is investing not only in new products but also in manufacturing resilience and lifecycle support capacity.

12. Supply chain and key dependencies

Hard evidence on supply chain and dependency risk is limited but still useful. The 2020 filing reproduction states that inventories were raised to mitigate interrupted supply chains during the COVID period. The 2021 reproduction mentions material quality problems within a program that led to higher program costs, claims risk, and rework-related provisions. The 2023 audited filing also shows heavy inventories and large provisions, including program-related reserves and warranty methodologies tied to prior years' experience. That collectively points to a business where execution quality, component availability, and warranty discipline matter materially.

Visible hard dependencies likely include specialized sensors and gyros, electronics, software, standards compliance, export licenses, and skilled engineering labor. The first part of that sentence is an inference, but it is grounded in the product architecture, the compass-production investments, the cyber-certification work, and the fact that the company supplies systems that must comply with IMO, IEC, STANAG, DNV, and military standards. The company also remains highly export exposed, which adds foreign-exchange and export-control complexity.

A further dependency is critical-facility concentration. Public sources strongly suggest that Kiel is the key design and manufacturing site. That gives the company a coherent center of competence, but it also creates concentration risk if that site is disrupted. The company's multi-region service network helps mitigate field-service risk, but not necessarily core manufacturing or engineering concentration. This is an assumption based on the footprint disclosed.

13. Management, board, ownership, and governance

Ownership is clear at the top level. Anschütz is privately owned within the DMB group after the February 2023 acquisition from Raytheon. The official imprint names four current managing directors: Martin Graßmann, Thomas Harlander, Niels Möller, and Andreas Weidner. The Bundestag lobby register repeats those four as legal representatives and also lists two Prokurists, Dr. Andreas Mues and Stefan Köhler. That indicates a more distributed leadership model than the one-man managing-director structure visible in the 2021 filing reproduction.

Management stability appears mixed. The company is not showing signs of disorder in public materials, but there has clearly been leadership evolution since the 2023 ownership change. The 2023 transaction press release quoted Michael Schulz as managing director of the future Anschütz GmbH, while current official sources show a different four-director structure. Because the company is a private German GmbH, public sources do not present a classic CEO/CFO governance map in the way a listed issuer would. That means analysts should not assume direct



comparability with public-company C-suite structures.

Board visibility is partial. The 2023 audit report explicitly references an Aufsichtsrat, which confirms that supervisory governance exists. However, the current public source set reviewed did not provide a clean, current list of supervisory-board members. A 2021 filing reproduction said the supervisory board had three members, two of them women. That historical point is useful but may no longer be current. Headcount disclosure is also uneven but directionally informative: 549 employees in 2020, 536 in 2021, more than 500 in Kiel today, and more than 700 at group level today. Direct comparability is imperfect because some figures refer to the legal entity and others to the broader group.

Governance formalization appears stronger than disclosure depth. The compliance page describes a comprehensive ethics and compliance program, a code of conduct, a supplier code, a SpeakUp channel, a declaration of principles on human rights, and a named human-rights officer. The supplier code also references anti-corruption, competition law, labor and environmental standards, and due-diligence frameworks. That does not eliminate governance risk, but it does suggest that the company has built more formal compliance infrastructure than a loosely run founder business.

14. Competitive positioning

Competition should be understood by business line, not by one peer set. In integrated bridge and navigation systems, the clearest direct comparators are Sperry Marine and Kongsberg Maritime. Both market integrated bridge systems, both emphasize lifecycle support, and both have stronger global scale. In maritime radar, Hensoldt UK is a meaningful adjacent competitor through Kelvin Hughes' SharpEye radar heritage. In naval and submarine programs, complete-warship manufacturers and combat-system houses such as BAE Systems, TKMS, Thales, and others are sometimes partners or customers rather than pure direct competitors.

Anschütz's likely strengths versus larger peers are installed-base trust, naval and submarine specialization, German engineering reputation, integrated lifecycle service, and willingness to tailor systems into specific program architectures. Its visible role in WECDIS, submarine automation, and Aegis-connected naval-navigation work suggests depth in operational integration rather than only in commodity equipment. Its vulnerabilities are more obvious on corporate scale, disclosure, and possibly financial firepower, especially when compared with larger publicly listed maritime and defense electronics groups.

Competition appears to be driven less by headline price and more by lifecycle cost, certification, interoperability, installed base, and service reach. There is no public pricing grid to prove that. That is an inference. But the product pages and program disclosures consistently emphasize reliability, long life, low maintenance, standards compliance, and integration. In a safety-critical and military environment, those factors often outweigh the lowest upfront price.

15. Strategic relevance and market implications

Anschütz matters beyond its own P&L because it sits in a capability layer that can become a bottleneck during naval recapitalization. A navy cannot field a modern frigate or submarine without trusted navigation, steering, situational awareness, and operator-control architecture. That makes companies like Anschütz important even if they are far smaller than the shipyard that builds the vessel. The Hunter-



class, Tamandaré-class, and U212CD references all reinforce that role.

The company is also relevant to German and broader European industrial policy. Its Bundestag lobby-register description explicitly frames its advocacy around the importance of the German security and defense industry, especially in the maritime domain, and references export control and ESG as policy themes. Public-grant disclosures show federal support for autonomy, 5G, and maritime digital-testfield projects. That matters because it connects the company not only to ship programs but also to future operating concepts around autonomy and digital maritime infrastructure.

For listed peers, suppliers, and partners, the implication is that Anschütz should be viewed as a niche maritime-electronics and systems-integration node, not as a broad-spectrum challenger. It is more likely to matter as a supplier, partner, or selective competitor within naval navigation and automation subdomains than as a full-line defense group. That is an inference, but it is the most grounded way to interpret the publicly available information.

16. Dual-use and civilian exposure

Anschütz should be considered a dual-use company. The current website still actively markets commercial-vessel solutions, while also foregrounding naval and submarine work. Historical financial disclosure reinforces that point. In 2020, the reproduced filing showed roughly balanced activity between civil nautics and marine equipment. In 2021, the company reorganized away from a civil-versus-military framing, which makes current public mix harder to observe directly, but it does not suggest that civilian business disappeared.

The defense case has strengthened. Recent news flow is more defense-heavy than civilian-heavy, with Hunter, Tamandaré, submarine automation, WECDIS, autonomous naval operations, and Australian sovereign shipbuilding all prominent. However, civilian business still matters economically and strategically because it supports installed base, scale, service density, and product learning. The right conclusion is that civilian exposure diversifies the business, but also dilutes any pure-play defense classification. Anschütz is not best thought of as a pure-play defense company today.

17. Legal, regulatory, and compliance risks

Overall legal and regulatory risk looks Medium based on the public record reviewed. The main concentration of risk is not visible enforcement history. It is the forward-looking regulatory burden that comes with export control, defense procurement, cyber certification, product safety and classification, and international program execution. We did not identify a material active public enforcement case or major litigation in the primary source set reviewed. That is not proof of absence. It is simply what the public record reviewed showed.

A concise chronology of relevant public regulatory touchpoints is as follows. In February 2023, the DMB acquisition became effective after “official approvals” referenced in the company press release. In late 2023, the company’s public compliance architecture included an updated Supplier Code of Conduct and human-rights materials. In 2025 and 2026, the Bundestag lobby register made visible the company’s engagement on export-control and defense-industry issues, and the company announced DNV UR E27 cybersecurity approvals for core navigation products. Those are not negative events, but they show where regulatory intensity is concentrated.



The most relevant regulatory frameworks are clear. In Germany, BAFA is the central licensing authority for export control. At EU level, Regulation (EU) 2021/821 governs exports, brokering, technical assistance, transit, and transfer of dual-use items. For maritime products, classification and operational approvals also matter, including IMO, IEC, DNV, and related standards. For defense sales into Germany, BAAINBw is the central Bundeswehr procurement authority. For human-rights due diligence, the company's compliance page explicitly references the German Supply Chain Due Diligence Act.

IP risk exists but is only partly visible. The official imprint says trademarks including "ANSCHÜTZ" are registered and owned by Anschütz GmbH in Germany and other countries. DPMAregister is the official German register for patents, utility models, trademarks, and designs and should be the first stop for deeper IP work. We did not establish a current patent count from the public source set reviewed, so we would not present one as fact.

The main realistic tail risk is regulatory delay rather than major challenges. A denied or delayed export authorization, a cyber-compliance shortfall, or a procurement-rule issue would be more likely to delay deliveries, defer revenue, or damage eligibility for future programs than to destroy the business outright. That judgment is an inference, but it is the most plausible one given the absence of a major public enforcement history and the presence of high regulatory intensity in the company's end markets.

18. Funding, financing, and strategic optionality

Anschütz is privately financed and strategically anchored within DMB rather than reliant on public equity markets. The 2020 reproduced filing said financing came from own funds at that point. 2024 accounts still show loans from related companies. Public sources reviewed did not show listed bonds, public equity raises, or an IPO process.

The balance sheet suggests that liquidity discipline matters. Cash was only €2.7m at year-end 2023, while inventories and receivables were high and provisions were large. That does not necessarily indicate distress, because industrial program businesses often carry heavy working capital and milestone timing. But it does imply that internal funding support, working-capital control, and milestone execution likely matter more than in a cash-light software model.

State-backed funding is visible on the innovation side. The Bundestag lobby register lists public grants tied to OCCUMAR, CAPTN Förde Areal, Förde 5G, CAPTN Förde Areal II, ENDURE, CAPTN X-FERRY, and PRS-AHRS. Those grants do not finance the whole company, but they do show that external public money supports parts of the autonomy, digital, and maritime-innovation roadmap.

Strategic optionality appears broad. The most likely path is continued private compounding under DMB with targeted capacity additions, software development, and regional expansion. Bolt-on M&A or joint ventures around autonomy, software, and naval integration would also be logical. An IPO is possible in theory, but there is no publicly available information that it is being prepared now, and the current public disclosure profile is not close to listed-company depth. That part is an inference.

19. Moat analysis

Anschütz possesses a durable moat founded on decades of accumulated trust in a



safety-critical market. Its primary defense is a massive installed base, with systems on over 35,000 vessels, creating a powerful credibility barrier that new competitors struggle to overcome. Switching costs are exceptionally high, as replacing its integrated navigation and automation systems requires significant re-engineering, recertification, and crew retraining.

The moat is further reinforced by stringent regulatory and military qualification standards, which act as a significant barrier to entry. This position is considered stable and is modestly widening as the company integrates higher-value software, cybersecurity, and autonomy features, which deepen customer dependency.

20. SWOT analysis

Strengths

- Deep installed base across commercial and naval fleets
- Strong heritage and engineering credibility in navigation
- Visible niche strength in naval bridge systems and submarine automation
- Global service and refit network with 24/7 support
- Growing capability in cyber-resilient and autonomous maritime systems

Weaknesses

- Private-company opacity limits confidence in current earnings quality
- Mid-scale player versus larger public peers in adjacent markets
- Civilian exposure remains meaningful, which can dilute a pure defense narrative

Opportunities

- Naval recapitalization and sovereign shipbuilding in Europe and Australia
- Higher software and autonomy content in maritime systems
- Cyber resilience, GNSS-jamming resilience, and refit demand
- US and Australian local-presence expansion for defense programs

Threats

- Export-control and procurement-compliance delays
- Program execution issues, warranty costs, or material-quality problems
- Component and skilled-labor constraints at a concentrated industrial base
- Competitive pressure from better-capitalized maritime-electronics peers

21. Key risks and open questions

- How much of current revenue and margin now comes from defense and naval programs versus commercial navigation after the 2023 ownership change? Public sources do not answer this clearly.
- Are the Hunter, U212CD, and Tamandaré references economically large enough to shift group profitability, or are they strategically important but still modest in revenue terms?
- How sustainable is cash conversion if inventories, receivables, provisions,

and milestone timing stay elevated?

- What is the current supervisory-board composition and how formalized is governance after the DMB transition? Public visibility is incomplete.
- Can Autonomics move from demonstrations and prototypes into scaled defense or commercial contracts?
- How concentrated is manufacturing in Kiel, and which components or suppliers represent the real bottlenecks? Public disclosure is limited.
- What is the actual IP depth beyond trademarks, especially in software, filtering, and autonomy layers? DPMAregister offers a route to answer this, but the public source set reviewed does not provide a patent count.

22. What to watch next

- Further Hunter-class production and integration milestones after critical design review.
- Evidence that Aegis-connected SYNAPSIS functionality converts into broader naval-system adoption.
- Additional disclosed workshare or milestones on U212CD and other submarine programs.
- Follow-on vessels and lifecycle scope in the Tamandaré-class program.
- Commercial proof points for Anschütz Autonomics and the approval-ready autonomous navigation system.
- Whether the company publishes more useful 2024 or 2025 financial detail than was accessible in this review.
- Further regional expansion, hiring, or localization in Australia and North America.
- More UR E27 or related cybersecurity approvals across the broader SYNAPSIS stack.
- Capacity additions or process modernization beyond the 2025 compass expansion.

23. News flow and recent developments

Recent news flow has been constructive and defense-tilted. In March 2026, the company said its Hunter-class WINBS passed critical design review, moving the program into production and integration. In February 2026, it announced DNV cyber approvals under UR E27 for multiple navigation and steering products. Also in February 2026, it disclosed delivery and commissioning of SYNAPSIS for the first Brazilian Tamandaré-class frigate. Those are the most material recent signals because they touch live naval programs and product qualification at the same time.

The 2025 news flow showed the company broadening both geography and technology. It founded subsidiaries in the US and Australia, launched Anschütz Autonomics, introduced a new solid-state radar sensor, disclosed an approval-ready autonomous navigation and command system, and highlighted a role in the U212CD submarine program. It also completed a major compass-production expansion in Kiel. Taken together, these announcements suggest a business investing in both installed-base support and future architecture.

A secondary recent theme is service and resilience. In January 2026, Anschütz emphasized upgraded 24/7/365 customer service across Germany, Singapore, and Panama. In other recent product communications, the company also stressed



resilience to jamming and spoofing and the operational benefits of reference architectures and modular software. That matters because market demand is increasingly shifting toward systems that are not just compliant on paper but resilient in degraded operating environments.

24. Source notes and transparency limits

Company sources used

The report relied heavily on the official Anschütz website, including the imprint, history, company pages, product pages, naval and submarine solution pages, compliance page, Supplier Code of Conduct, and company news archive. Audited annual filings were also used as a primary company-linked financial source.

Government and regulatory sources used

Government and regulatory sources used included the Bundestag lobby register, BAFA export-control materials, EUR-Lex for Regulation (EU) 2021/821, DPMAregister, BAAINBw procurement materials, and the Australian Department of Defence's Hunter-class program page. These sources were most useful for legal form, public grants, policy exposure, export-control context, procurement context, and program validation.

Independent external sources used

Independent external sources were used mainly for competitor benchmarking and market framing. These included Sperry Marine, Kongsberg Maritime, Hensoldt UK, SIPRI, and CAPTN. A Heuking press item was also available as external confirmation of the DMB transaction, although the company press release was used as the main transaction source.

Transparency limits

The company is private, and disclosure is materially thinner than for a listed issuer. The 2023 filing explicitly states that the company relied on Section 264(3) HGB relief, which limited stand-alone disclosure. A 2024 annual report is publicly linked via the Bundestag lobby register, but detailed line-item extraction was not available in the sources accessible in this session. Historical 2020 and 2021 financial details were available through secondary reproductions of official filings rather than through clean primary registry downloads within this session. No material public enforcement case was identified in the reviewed source set, but that should not be mistaken for proof that no issue exists anywhere outside the reviewed set.

Explicitly labeled estimates and inferences

The following points were treated as estimates or inferences rather than disclosed fact: the implied 2023 equity ratio of about 8.6%, derived from disclosed equity and total assets; the view that Kiel-centered group operations imply a strategically centralized structure; the judgment that defense weight has likely increased since 2023; the assessment that integrated bridge and submarine-system content likely carries higher strategic value than standalone hardware; and the interpretation that the moat is stable to modestly widening. These conclusions are grounded in publicly available information, but they remain analytical inferences rather than direct company disclosures.

Glossary

Abbreviations, technical terms, and German-language expressions used in this report. German terms are shown in italics with an English gloss.

A2/AD (Anti-Access, Area-Denial) — capabilities designed to prevent an adversary from entering or operating freely in a theater.

Amtsgericht — German local district court; maintains the Handelsregister (commercial register).

ASAP (Act in Support of Ammunition Production) — EU instrument to expand European munitions output.

Aufsichtsrat — supervisory board; the non-executive body overseeing management in German companies.

BAAINBw (*Bundesamt für Ausrüstung, Informationstechnik und Nutzung der Bundeswehr*) — Germany's federal office for defense procurement, IT and in-service support.

BAFA (*Bundesamt für Wirtschaft und Ausfuhrkontrolle*) — Germany's federal export-control and licensing authority.

Beteiligungsgesellschaft — German investment or holding company.

BMS (Battle Management System) — system supporting commanders in planning, coordination and operational oversight.

BMVg (*Bundesministerium der Verteidigung*) — Germany's Federal Ministry of Defense.

BSI (*Bundesamt für Sicherheit in der Informationstechnik*) — Germany's federal cyber-security office; sets the IT-Grundschutz baseline-protection standards.

Bundeskartellamt — Germany's federal competition authority, responsible for antitrust oversight and merger review.

Bundestag — German federal parliament; its budget committee (Haushaltsausschuss) must approve defense contracts above €25m.

Bundeswehr — the German federal armed forces.

BVLOS (Beyond Visual Line of Sight) — unmanned-aircraft operations beyond the operator's direct visual observation.

BWB (*Bundesamt für Wehrtechnik und Beschaffung*) — Germany's former defense procurement office; predecessor of BAAINBw.

BwBBG (*Bundeswehrbeschaffungsbeschleunigungsgesetz*) — the 2026 Procurement Acceleration Act, in force 14 February 2026.

C2 (Command and Control) — capability enabling commanders to direct, coordinate and manage forces and operations.

C4ISR — Command, Control, Communications, Computers, Intelligence, Surveillance and Reconnaissance.

CBRN (Chemical, Biological, Radiological and Nuclear) — threat category and associated protective equipment.

CIR (*Cyber- und Informationsraum*) — the Bundeswehr's Cyber and Information Domain Command.

COTS (Commercial-off-the-Shelf) — commercially available products acquired without significant modification.

C-UAS / C-UxS — Counter-Unmanned Aerial Systems / Counter-Unmanned Systems (air, ground and maritime).

DALO — Denmark's Defence Acquisition and Logistics Organisation.

DeepTech — ventures built on significant scientific or engineering innovation rather than incremental software development.

DIANA (Defence Innovation Accelerator for the North Atlantic) — NATO's dual-use innovation network.

DIB (Defense Industrial Base) — the industrial capacity, workforce and supply chain required to meet national and allied defense requirements.

D-LBO (*Digitalisierung Landbasierter Operationen*) — Bundeswehr program to digitize land operations.

DMB — Dr. Dieter Murmann Beteiligungsgesellschaft mbH; the family holding that has owned Anschütz since 2023.

DNV / DNV UR E27 — international classification society; UR E27 is its unified requirement for cyber security of shipboard computer-based systems.

DPMAregister — official German register for patents, utility models, trademarks and their legal status.

EASA (European Union Aviation Safety Agency) — the EU's aviation-safety regulator.

ECDIS / WECDIS — Electronic Chart Display and Information System / its warship variant.

EDA (European Defence Agency) — EU body for capability development and defense research.

EDF / EDIRPA / EDIP — European Defence Fund / European Defence Industry Reinforcement through common Procurement Act / European Defence Industry Programme; EU defense-industrial funding instruments.

EK (*Elektronischer Kampf*) — German for electronic warfare; also the designation of the Eurofighter EW configuration.

EN 9100 — aerospace-industry quality-management standard.

EO/IR (Electro-Optical / Infrared) — sensor category used for surveillance, targeting and reconnaissance.

ESSI (European Sky Shield Initiative) — multinational European air-and-missile-defense procurement group.

ETSO (European Technical Standard Order) — European aviation equipment approval standard.

eVTOL / VTOL — (electric) Vertical Take-Off and Landing.

EW (Electronic Warfare) — military use of the electromagnetic spectrum for sensing, attack, protection and countermeasures.

FCAS (Future Combat Air System) — Franco-German-Spanish next-generation air program; its fighter component was discontinued in 2026 (see Team Gen 6).

FDI screening — foreign direct investment review; national-security vetting of cross-border acquisitions.

Fraunhofer FHR (Fraunhofer-Institut für Hochfrequenzphysik und Radartechnik) — Fraunhofer Institute for High Frequency Physics and Radar Techniques.

Fuzing — the arming and detonation mechanism (the fuze) of a munition; standard US-defense spelling, distinct from an electrical fuse.

GBAD / SHORAD / MRAD — Ground-Based Air Defense / Short-Range Air Defense / Medium-Range Air Defense.

Geschäftsführer — managing director of a German GmbH.

GIS (Geographic Information System) — software and data framework for capturing, storing, analyzing and visualizing spatially referenced information

GmbH (*Gesellschaft mit beschränkter Haftung)* — German limited-liability company.

GNSS (Global Navigation Satellite System) — satellite navigation such as GPS or Galileo.

Handelsregister — Germany's commercial register, maintained by the local Amtsgericht.

Haushaltsausschuss — the Bundestag's budget committee; approves major defense procurement.

HGB (*Handelsgesetzbuch*) — Germany's Commercial Code; the accounting standard under which these private companies report.

HRA / HRB — German commercial-register identifiers for partnerships (HRA) and registered companies (HRB).

HTGF (*High-Tech Gründerfonds*) — Germany's public-private seed-stage venture fund.

IAMD (Integrated Air and Missile Defense) — integrated detection, tracking, command and engagement of air and missile threats.

Impressum — the legally required imprint (legal notice) on German websites; a source for management and legal-entity data.

IRIS-T (SLS / SLM / SLX) — Diehl's infrared-guided missile family and its surface-launched air-defense variants for short, medium and extended range.

ISR (Intelligence, Surveillance and Reconnaissance) — mission category covering intelligence collection, surveillance and reconnaissance.

IST (Intelligence, Surveillance and Targeting) — mission category combining intelligence collection and surveillance with target identification and designation.

ITAR (International Traffic in Arms Regulations) — US controls on the export of defense technology; "ITAR-free" denotes hardware free of such restrictions.

KG (*Kommanditgesellschaft*) — German limited partnership.

KWKG (*Kriegswaffenkontrollgesetz*) — Germany's War Weapons Control Act.

LBA (*Luftfahrt-Bundesamt*) — Germany's Federal Aviation Office.

LEO (Low Earth Orbit) — orbital region used for Earth-observation, communications and reconnaissance satellites.

LkSG (*Lieferkettensorgfaltspflichtengesetz*) — Germany's Supply Chain Due Diligence Act.

Loitering munition — a strike drone that loiters over a target area before self-detonating on impact; also "one-way effector."

LRMP (Long-Range Multi-Purpose) — capability designation, e.g. the RGW90 LRMP.

LufABw (*Luftfahrtamt der Bundeswehr*) — Germany's military aviation authority.

MANET (Mobile Ad-hoc Network) — self-forming tactical radio network.

MBDA — European multinational missile manufacturer; parent of TDW.

MGCS (Main Ground Combat System) — Franco-German next-generation land-combat program.

Mithra OS — ARX Robotics' autonomy operating system for unmanned and legacy ground vehicles.

Mittelstand — Germany's specialized medium-sized business sector, often export-oriented "hidden champions."

MOSAIC UXS — Quantum Systems' multi-domain command-and-autonomy software ecosystem (UXS = Unmanned Systems).

MOTS (Military-Off-The-Shelf) — military-grade products acquired substantially in existing form.

MRO (Maintenance, Repair and Overhaul) — sustainment activities for fielded equipment and platforms.

MULTIBASE — AUTOFLUG's modular cabin and pallet system.

NAVICS — Rohde & Schwarz's naval integrated communications system.

NATO IF / NIF (NATO Innovation Fund) — NATO-backed venture fund for defense and dual-use technology.

NSPA (NATO Support and Procurement Agency) — NATO's principal procurement and logistics organization.

OPTIMAS — European Defence Fund-backed secure optical-communications program.

OSINT (Open-Source Intelligence) — intelligence derived from publicly available information.

OWE-V (One-Way Effector – Vertical) — the VTOL designation of STARK's Virtus loitering munition.

Panzerfaust 3 / RGW (90 / 110) — Dynamit Nobel Defence shoulder-fired weapons (*RGW = Rückstoßfreie Granatwaffe*, recoilless grenade weapon).

PGM (Precision-Guided Munition) — guided weapon designed for high targeting accuracy.

Prokura / Prokurist — German statutory commercial power of attorney / an authorized signatory, subordinate to a managing director.

RAA (Reciprocal Access Agreement) — proposed Germany-Japan agreement easing joint military training and deployments.

Rafael Ltd — Rafael Advanced Defense Systems; the Israeli state-owned group that is the ultimate parent of Dynamit Nobel Defence.

RCS (Robotic Combat System) — e.g. ARX Robotics' Gereon RCS.

RF (Radio Frequency) — spectrum domain used for communications, radar, EW and test and measurement.

SAFE (Security Action for Europe) — the EU's defense-loan facility.

SAR (Synthetic Aperture Radar) — radar imaging technology producing high-



resolution imagery regardless of weather or daylight.

SATCOM (Satellite Communications) — communications via satellite across long distances and contested environments.

Schuldenbremse (debt brake) — Germany's constitutional cap on new federal borrowing; reformed in 2025 to exempt defense spending above 1% of GDP.

SDA (Space Domain Awareness) — monitoring and understanding of activity in the space domain.

SDR (Software-Defined Radio) — radio in which traditionally hardware-based functions are implemented in software.

SE (Societas Europaea) — European public-company legal form.

SHORAD — see GBAD.

SIGINT (Signals Intelligence) — collection and analysis of electronic signals and communications.

SIPRI (Stockholm International Peace Research Institute) — independent source of defense-spending and arms-trade data.

SKORPION^2 — Dynamit Nobel Defence's vehicle-mounted, scatterable anti-tank mine-laying system.

Sondervermögen — a German off-budget "special fund," e.g. the €100bn Bundeswehr fund and the €500bn infrastructure fund.

SOVERON — Rohde & Schwarz's software-defined tactical radio family.

STANAG (Standardization Agreement) — NATO standards governing interoperability among allied forces.

STEM (Science, Technology, Engineering and Mathematics) — technical disciplines underpinning defense capability, and an area of German workforce shortage.

Stiftung — German foundation; a common ownership vehicle for family-held companies.

sUAS (Small Unmanned Aircraft System) — small-class unmanned aircraft.

SYNAPSIS — Anschütz's integrated navigation and bridge system; WECDIS NX is its warship electronic-charting module.

SYNTACS — Anschütz's naval command-and-control system.

T&M (Test and Measurement) — instrumentation business segment, e.g. at Rohde & Schwarz.

TDW (TDW Gesellschaft für verteidigungstechnische Wirksysteme mbH) — MBDA's warhead and energetics subsidiary; supplies STARK's LS110 warhead.

Team Gen 6 — Airbus-led German consortium for a sixth-generation fighter, formed after the FCAS fighter component was discontinued in 2026.

TKMS — TKMS AG & Co. KGaA (formerly thyssenkrupp Marine Systems); listed on the Frankfurt Stock Exchange since October 2025, with thyssenkrupp AG retaining a majority stake.

U212CD — the German-Norwegian Type 212 Common Design submarine class.

UAV / UGV / USV / UAS / UxS — Unmanned Aerial / Ground / Surface Vehicle; Unmanned Aircraft System; Unmanned Systems (umbrella term).

UCAV (Unmanned Combat Aerial Vehicle) — armed unmanned aircraft designed for strike missions.

UG (*Unternehmergesellschaft*) — German limited-liability form, a "mini-GmbH."

Virtus / Minerva / Vanta / Cascade / Gambit — STARK Defence's product family: loitering munition, command-and-control software, unmanned surface vessel, tube-launched effector and reconnaissance quadcopter.

WINBS (Warship Integrated Navigation and Bridge System) — Anschütz naval navigation and bridge system.

XR (Extended Reality) — umbrella term for augmented, virtual and mixed reality.

Zeitenwende — literally "turning point"; the term for Germany's post-2022 shift in defense and security policy.



Appendix 1

Important Disclosures

*Other information available upon request

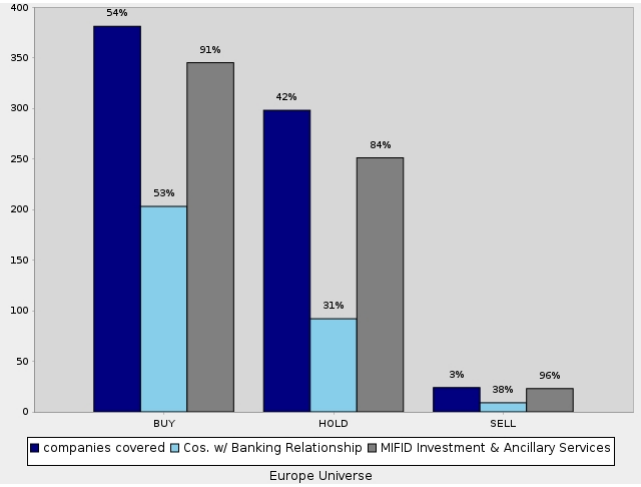
*Prices are current as of the end of the previous trading session unless otherwise indicated and are sourced from local exchanges via Reuters, Bloomberg and other vendors . Other information is sourced from Deutsche Bank, subject companies, and other sources. For disclosures pertaining to recommendations or estimates made on securities other than the primary subject of this research, please see the most recently published company report or visit our global disclosure look-up page on our website at https://research.db.com/Research/Disclosures/EquityResearchDisclosures. Aside from within this report, important risk and conflict disclosures can also be found at https://research.db.com/Research/Disclosures/Disclaimer. Investors are strongly encouraged to review this information before investing.



Analyst Certification

The views expressed in this report accurately reflect the personal views of the undersigned lead analyst about the subject issuers and the securities of those issuers. In addition, the undersigned lead analyst has not and will not receive any compensation for providing a specific recommendation or view in this report. Michael Kuhn, Christophe Menard.

Equity rating dispersion and banking relationships



Equity Rating and Dispersion Key

The Equity Rating Dispersion Chart depicts the following:

The proportion of recommendations that are rated "buy", "sell" and "hold" over the previous 12 months. This is shown for securities issued in the stated region e.g. "Europe Universe". See rating definitions below. This is represented by the "Companies Covered" bars in the chart. The percentage value displayed above the bar is the proportion as a percentage. E.g. 50% above the “buy” / “Companies Covered” bar means that 50% of DB’s equity research covered companies over the past 12 months have a “buy” rating.

Next to each of the three respective bars showing the proportion of "buy", "sell" and "hold" recommendations we provide two additional bars to show:

- The proportion of "buy", "sell" or "hold recommendations where Deutsche Bank and or/Affiliates provided MIFID Investment or Ancillary Services in the past 12 months. This is represented in the "MIFID Investment and Ancillary Services" bar. The percentage value displayed above the bar shows the proportion of Companies Covered with the given rating where DB has also provided MIFID Investment and Ancillary Services in the past 12 months. E.g. 50% above the “Cos. w/ MIFID Investment and Ancillary Services” bar means 50% of the Companies Covered with the rating stated have also received MIFID Investment and Ancillary Services from DB.

- The proportion of "buy" (or "sell" or "hold) recommendations where Deutsche Bank and or/Affiliates has provided Investment Banking services in the past 12 months for which it has received compensation. The percentage value displayed above the bar shows the proportion of Companies Covered with the stated rating where DB has also provided Investment Banking services in the past 12 months. E.g. 50% above the “Cos. w/ Investment Banking relationship” bar means 50% of the Companies Covered with the rating stated also have an Investment Banking Relationship with DB.

Buy: Based on a current 12- month view of TSR, we recommend that investors buy the stock.

Sell: Based on a current 12-month view of TSR, we recommend that investors sell the stock.

Hold: We take a neutral view on the stock 12-months out and, based on this time horizon, do not recommend either a Buy or Sell.

TSR = Total Shareholder Return. Percentage change in share price from current price to projected target price plus projected dividend yield

Newly issued research recommendations and target prices supersede previously published research.



Additional Information

The information and opinions in this report were prepared by Deutsche Bank AG or one of its affiliates (collectively 'Deutsche Bank'). Though the information herein is believed to be reliable and has been obtained from public sources believed to be reliable, Deutsche Bank makes no representation as to its accuracy or completeness. Hyperlinks to third-party websites in this report are provided for reader convenience only. Deutsche Bank neither endorses the content nor is responsible for the accuracy or security controls of those websites.

If you use the services of Deutsche Bank in connection with a purchase or sale of a security that is discussed in this report, or is included or discussed in another communication (oral or written) from a Deutsche Bank analyst, Deutsche Bank may act as principal for its own account or as agent for another person.

Deutsche Bank may consider this report in deciding to trade as principal. It may also engage in transactions, for its own account or with customers, in a manner inconsistent with the views taken in this research report. Others within Deutsche Bank, including strategists, sales staff and other analysts, may take views that are inconsistent with those taken in this research report. Deutsche Bank issues a variety of research products, including fundamental analysis, equity-linked analysis, quantitative analysis and trade ideas. Recommendations contained in one type of communication may differ from recommendations contained in others, whether as a result of differing time horizons, methodologies, perspectives or otherwise. Deutsche Bank and/or its affiliates may also be holding debt or equity securities of the issuers it writes on. Analysts are paid in part based on the profitability of Deutsche Bank AG and its affiliates, which includes investment banking, trading and principal trading revenues.

Opinions, estimates and projections constitute the current judgment of the author as of the date of this report. They do not necessarily reflect the opinions of Deutsche Bank and are subject to change without notice. Deutsche Bank provides liquidity for buyers and sellers of securities issued by the companies it covers. Deutsche Bank research analysts sometimes have shorter-term trade ideas that may be inconsistent with Deutsche Bank's existing longer-term ratings. Some trade ideas for equities are listed as Catalyst Calls on the Research Website (https://research.db.com/Research/) , and can be found on the general coverage list and also on the covered company's page. A Catalyst Call represents a high-conviction belief by an analyst that a stock will outperform or underperform the market and/or a specified sector over a time frame of no less than two weeks and no more than three months. In addition to Catalyst Calls, analysts may occasionally discuss with our clients, and with Deutsche Bank salespersons and traders, trading strategies or ideas that reference catalysts or events that may have a near-term or medium-term impact on the market price of the securities discussed in this report, which impact may be directionally counter to the analysts' current 12-month view of total return or investment return as described herein. Deutsche Bank has no obligation to update, modify or amend this report or to otherwise notify a recipient thereof if an opinion, forecast or estimate changes or becomes inaccurate. Coverage and the frequency of changes in market conditions and in both general and company-specific economic prospects make it difficult to update research at defined intervals. Updates are at the sole discretion of the coverage analyst or of the Research Department Management, and the majority of reports are published at irregular intervals. This report is provided for informational purposes only and does not take into account the particular investment objectives, financial situations, or needs of individual clients. It is not an offer or a solicitation of an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Target prices are inherently imprecise and a product of the analyst's judgment. The financial instruments discussed in this report may not be suitable for all investors, and investors must make their own informed investment decisions. Prices and availability of financial instruments are subject to change without notice, and investment transactions can lead to losses as a result of price fluctuations and other factors. If a financial instrument is denominated in a currency other than an investor's currency, a change in exchange rates may adversely affect the investment. Past performance is not necessarily indicative of future results. Performance calculations exclude transaction costs, unless otherwise indicated. Unless otherwise indicated, prices are current as of the end of the previous trading session and are sourced from local exchanges via Reuters, Bloomberg and other vendors. Data is also sourced from Deutsche Bank, subject companies, and other parties. Artificial intelligence tools may be used in the preparation of this material, including but not limited to assist in fact-finding, data analysis, pattern recognition, content drafting and editorial corrections pertaining to research material.

The Deutsche Bank Research Department is independent of other business divisions of the Bank. Details regarding our organizational arrangements and information barriers we have to prevent and avoid conflicts of interest with respect to our research are available on our website (https://research.db.com/Research/) under Disclaimer.

Macroeconomic fluctuations often account for most of the risks associated with exposures to instruments that promise to pay fixed or variable interest rates. For an investor who is long fixed-rate instruments (thus receiving these cash flows), increases in interest rates naturally lift the discount factors applied to the expected cash flows and thus cause a loss. The longer the maturity of a certain cash flow and the higher the move in the discount factor, the higher will be the loss. Upside surprises in inflation, fiscal funding needs, and FX depreciation rates are among the most common adverse macroeconomic shocks to receivers. But counterparty exposure, issuer creditworthiness, client segmentation, regulation (including changes in assets holding limits for different types of investors), changes in tax policies, currency convertibility (which may constrain currency conversion, repatriation of profits and/or liquidation of positions), and settlement issues related to local clearing houses are also important risk factors. The sensitivity of fixed-income instruments to macroeconomic shocks may be mitigated by indexing the contracted cash flows to inflation, to FX depreciation, or to specified interest rates - these are common in emerging markets. The index fixings may - by construction - lag or mis-measure the actual move in the underlying variables they are intended to track. The choice of the proper fixing (or metric) is particularly important in swaps markets, where floating coupon rates (i.e., coupons indexed to a typically short-dated interest rate reference index) are exchanged for fixed coupons. Funding in a currency that differs from the currency in which coupons are denominated carries FX risk. Options on swaps (swaptions) the risks typical to options in addition to the risks related to rates movements.



Derivative transactions involve numerous risks including market, counterparty default and illiquidity risk. The appropriateness of these products for use by investors depends on the investors' own circumstances, including their tax position, their regulatory environment and the nature of their other assets and liabilities; as such, investors should take expert legal and financial advice before entering into any transaction similar to or inspired by the contents of this publication. The risk of loss in futures trading and options, foreign or domestic, can be substantial. As a result of the high degree of leverage obtainable in futures and options trading, losses may be incurred that are greater than the amount of funds initially deposited - up to theoretically unlimited losses. Trading in options involves risk and is not suitable for all investors. Prior to buying or selling an option, investors must review the 'Characteristics and Risks of Standardized Options", at https://www.theocc.com/company-information/documents-and-archives/options-disclosure-document. If you are unable to access the website, please contact your Deutsche Bank representative for a copy of this important document.

Participants in foreign exchange transactions may incur risks arising from several factors, including the following: (i) exchange rates can be volatile and are subject to large fluctuations; (ii) the value of currencies may be affected by numerous market factors, including world and national economic, political and regulatory events, events in equity and debt markets and changes in interest rates; and (iii) currencies may be subject to devaluation or government-imposed exchange controls, which could affect the value of the currency. Investors in securities such as ADRs, whose values are affected by the currency of an underlying security, effectively assume currency risk.

Unless governing law provides otherwise, all transactions should be executed through the Deutsche Bank entity in the investor's home jurisdiction. Aside from within this report, important conflict disclosures can also be found at https://research.db.com/Research/ on each company's research page or under the 'Disclosures' tab. Investors are strongly encouraged to review this information before investing.

Deutsche Bank (which includes Deutsche Bank AG, its branches and affiliated companies) is not acting as a financial adviser, consultant or fiduciary to you or any of your agents (collectively, "You" or "Your") with respect to any information provided in this report. Deutsche Bank does not provide investment, legal, tax or accounting advice, Deutsche Bank is not acting as your impartial adviser, and does not express any opinion or recommendation whatsoever as to any strategies, products or any other information presented in the materials. Information contained herein is being provided solely on the basis that the recipient will make an independent assessment of the merits of any investment decision, and it does not constitute a recommendation of, or express an opinion on, any product or service or any trading strategy.

The information presented is general in nature and is not directed to retirement accounts or any specific person or account type, and is therefore provided to You on the express basis that it is not advice, and You may not rely upon it in making Your decision. The information we provide is being directed only to persons we believe to be financially sophisticated, who are capable of evaluating investment risks independently, both in general and with regard to particular transactions and investment strategies, and who understand that Deutsche Bank has financial interests in the offering of its products and services. If this is not the case, or if You are an IRA or other retail investor receiving this directly from us, we ask that you inform us immediately.

In July 2018, Deutsche Bank revised its rating system for short term ideas whereby the branding has been changed to Catalyst Calls ("CC") from SOLAR ideas; the rating categories for Catalyst Calls originated in the Americas region have been made consistent with the categories used by Analysts globally; and the effective time period for CCs has been reduced from a maximum of 180 days to 90 days.

United States: Approved and/or distributed by Deutsche Bank Securities Incorporated, a member of FINRA and SIPC. Analysts located outside of the United States are employed by non-US affiliates and are not registered/qualified as research analysts with FINRA.

European Economic Area (exc. United Kingdom): Approved and/or distributed by Deutsche Bank AG, a joint stock corporation with limited liability incorporated in the Federal Republic of Germany with its principal office in Frankfurt am Main. Deutsche Bank AG is authorized under German Banking Law and is subject to supervision by the European Central Bank and by BaFin, Germany's Federal Financial Supervisory Authority.

United Kingdom: Approved and/or distributed by Deutsche Bank AG acting through its London Branch at 21 Moorfields, London EC2Y 9DB. Deutsche Bank AG in the United Kingdom is authorised by the Prudential Regulation Authority and is subject to limited regulation by the Prudential Regulation Authority and Financial Conduct Authority. Details about the extent of our authorisation and regulation are available on request.

Hong Kong SAR: Distributed by Deutsche Bank AG, Hong Kong Branch except for any research content relating to futures contracts within the meaning of the Hong Kong Securities and Futures Ordinance Cap. 571. Research reports on such futures contracts are not intended for access by persons who are located, incorporated, constituted or resident in Hong Kong. The author(s) of a research report , and the entities for which they act and/or are affiliated with, may not be licensed to carry on regulated activities in Hong Kong and, if not licensed, do not hold themselves out as being able to do so. The provisions set out above in the 'Additional Information' section shall apply to the fullest extent permissible by local laws and regulations, including without limitation the Code of Conduct for Persons Licensed or Registered with the Securities and Futures Commission. This report is intended for distribution only to 'professional investors' as defined in Part 1 of Schedule of the SFO. This document must not be acted or relied on by persons who are not professional investors. Any investment or investment activity to which this document relates is only available to professional investors and will be engaged only with professional investors.



India: Prepared by Deutsche Equities India Private Limited (DEIPL) having CIN: U65990MH2002PTC137431 and registered office at 14th Floor, The Capital, C-70, G Block, Bandra Kurla Complex, Mumbai (India) 400051. Tel: + 91 22 7180 4444. It is registered by the Securities and Exchange Board of India (SEBI) as a Stock broker bearing registration no.: INZ000252437; Merchant Banker bearing SEBI Registration no.: INM000010833 and Research Analyst bearing SEBI Registration no.: INH000001741. DEIPL's Compliance / Grievance officer is Ms. Rashmi Poddar (Tel: +91 22 7180 4929 email ID: complaints.deipl@db.com). Registration granted by SEBI and certification from NISM in no way guarantee performance of DEIPL or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. DEIPL may have received administrative warnings from the SEBI for breaches of Indian regulations. Deutsche Bank and/or its affiliate(s) may have debt holdings or positions in the subject company. With regard to information on associates, please refer to the "Shareholdings" section in the Annual Report at: https://www.db.com/ir/en/annual-reports.htm. For latest India research audit report, refer https://country.db.com/india/deutsche-equities-india/index?language_id=1.

Japan: Approved and/or distributed by Deutsche Securities Inc.(DSI). Registration number - Registered as a financial instruments dealer by the Head of the Kanto Local Finance Bureau (Kinsho) No. 117. Member of associations: JSDA, Type II Financial Instruments Firms Association and The Financial Futures Association of Japan. Commissions and risks involved in stock transactions - for stock transactions, we charge stock commissions and consumption tax by multiplying the transaction amount by the commission rate agreed with each customer. Stock transactions can lead to losses as a result of share price fluctuations and other factors. Transactions in foreign stocks can lead to additional losses stemming from foreign exchange fluctuations. We may also charge commissions and fees for certain categories of investment advice, products and services. Recommended investment strategies, products and services carry the risk of losses to principal and other losses as a result of changes in market and/or economic trends, and/or fluctuations in market value. Before deciding on the purchase of financial products and/or services, customers should carefully read the relevant disclosures, prospectuses and other documentation. 'Moody's', 'Standard Poor's', and 'Fitch' mentioned in this report are not registered credit rating agencies in Japan unless Japan or 'Nippon' is specifically designated in the name of the entity. Reports on Japanese listed companies not written by analysts of DSI are written by Deutsche Bank Group's analysts with the coverage companies specified by DSI. Some of the foreign securities stated on this report are not disclosed according to the Financial Instruments and Exchange Law of Japan. Target prices set by Deutsche Bank's equity analysts are based on a 12-month forecast period.

Korea: Distributed by Deutsche Securities Korea Co.

South Africa: Deutsche Bank AG Johannesburg is incorporated in the Federal Republic of Germany (Branch Register Number in South Africa: 1998/003298/10).

Singapore: This report is issued by Deutsche Bank AG, Singapore Branch (One Raffles Quay #18-00 South Tower Singapore 048583, 65 6423 8001), which may be contacted in respect of any matters arising from, or in connection with, this report. Where this report is issued or promulgated by Deutsche Bank in Singapore to a person who is not an accredited investor, expert investor or institutional investor (as defined in the applicable Singapore laws and regulations), they accept legal responsibility to such person for its contents.

Taiwan: Information on securities/investments that trade in Taiwan is for your reference only. Readers should independently evaluate investment risks and are solely responsible for their investment decisions. Deutsche Bank research may not be distributed to the Taiwan public media or quoted or used by the Taiwan public media without written consent. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation to trade in such securities/instruments.

Qatar: Deutsche Bank AG in the Qatar Financial Centre (registered no. 00032) is regulated by the Qatar Financial Centre Regulatory Authority. Deutsche Bank AG - QFC Branch may undertake only the financial services activities that fall within the scope of its existing QFCRA license. Its principal place of business in the QFC: Qatar Financial Centre, Tower, West Bay, Level 5, PO Box 14928, Doha, Qatar. This information has been distributed by Deutsche Bank AG. Related financial products or services are only available only to Business Customers, as defined by the Qatar Financial Centre Regulatory Authority.

Russia: The information, interpretation and opinions submitted herein are not in the context of, and do not constitute, any appraisal or evaluation activity requiring a license in the Russian Federation.

Kingdom of Saudi Arabia: Deutsche Securities Saudi Arabia (DSSA) is a closed joint stock company authorized by the Capital Market Authority of the Kingdom of Saudi Arabia with a license number (No. 37-07073) to conduct the following business activities: Dealing, Arranging, Advising, and Custody activities. DSSA registered office is Faisaliah Tower, 17th Floor, King Fahad Road - Al Olaya District Riyadh, Kingdom of Saudi Arabia P.O. Box 301806.

United Arab Emirates: Deutsche Bank AG in the Dubai International Financial Centre (registered no. 00045) is regulated by the Dubai Financial Services Authority. Deutsche Bank AG - DIFC Branch may only undertake the financial services activities that fall within the scope of its existing DFSA license. Principal place of business in the DIFC: Dubai International Financial Centre, The Gate Village, Building 5, PO Box 504902, Dubai, U.A.E. This information has been distributed by Deutsche Bank AG. Related financial products or services are available only to Professional Clients, as defined by the Dubai Financial Services Authority.

Australia and New Zealand: This research is intended only for 'wholesale clients' within the meaning of the Australian Corporations Act and New Zealand Financial Advisors Act, respectively. Please refer to Australia-specific research disclosures



and related information at https://www.dbresearch.com/PROD/RPS_EN-PROD/PROD0000000000521304.xhtml . Where research refers to any particular financial product recipients of the research should consider any product disclosure statement, prospectus or other applicable disclosure document before making any decision about whether to acquire the product. In preparing this report, the primary analyst or an individual who assisted in the preparation of this report has likely been in contact with the company that is the subject of this research for confirmation/clarification of data, facts, statements, permission to use company-sourced material in the report, and/or site-visit attendance. Without prior approval from Research Management, analysts may not accept from current or potential Banking clients the costs of travel, accommodations, or other expenses incurred by analysts attending site visits, conferences, social events, and the like. Similarly, without prior approval from Research Management and Anti-Bribery and Corruption ("ABC") team, analysts may not accept perks or other items of value for their personal use from issuers they cover.

Additional information relative to securities, other financial products or issuers discussed in this report is available upon request. This report may not be reproduced, distributed or published without Deutsche Bank's prior written consent.

Backtested, hypothetical or simulated performance results have inherent limitations. Unlike an actual performance record based on trading actual client portfolios, simulated results are achieved by means of the retroactive application of a backtested model itself designed with the benefit of hindsight. Taking into account historical events the backtesting of performance also differs from actual account performance because an actual investment strategy may be adjusted any time, for any reason, including a response to material, economic or market factors. The backtested performance includes hypothetical results that do not reflect the reinvestment of dividends and other earnings or the deduction of advisory fees, brokerage or other commissions, and any other expenses that a client would have paid or actually paid. No representation is made that any trading strategy or account will or is likely to achieve profits or losses similar to those shown. Alternative modeling techniques or assumptions might produce significantly different results and prove to be more appropriate. Past hypothetical backtest results are neither an indicator nor guarantee of future returns. Actual results will vary, perhaps materially, from the analysis.

The method for computing individual E,S,G and composite ESG scores set forth herein is a novel method developed by the Research department within Deutsche Bank AG, computed using a systematic approach without human intervention. Different data providers, market sectors and geographies approach ESG analysis and incorporate the findings in a variety of ways. As such, the ESG scores referred to herein may differ from equivalent ratings developed and implemented by other ESG data providers in the market and may also differ from equivalent ratings developed and implemented by other divisions within the Deutsche Bank Group. Such ESG scores also differ from other ratings and rankings that have historically been applied in research reports published by Deutsche Bank AG. Further, such ESG scores do not represent a formal or official view of Deutsche Bank AG. It should be noted that the decision to incorporate ESG factors into any investment strategy may inhibit the ability to participate in certain investment opportunities that otherwise would be consistent with your investment objective and other principal investment strategies. The returns on a portfolio consisting primarily of sustainable investments may be lower or higher than portfolios where ESG factors, exclusions, or other sustainability issues are not considered, and the investment opportunities available to such portfolios may differ. Companies may not necessarily meet high performance standards on all aspects of ESG or sustainable investing issues; there is also no guarantee that any company will meet expectations in connection with corporate responsibility, sustainability, and/or impact performance.

Copyright © 2026 Deutsche Bank AG

David Folkerts-Landau
Group Chief Economist and Global Head of Research

Pam Finelli
COO and Head of Fixed Income Research

Steve Pollard
Global Head of Company Research and Sales

Jim Reid
Global Head of Macro and Thematic Research

Tim Rokossa
Head of European Company Research

Matthew Barnard
Head of Americas Company Research

Debbie Jones
Global Head of Sustainability and Data Innovation, Research

Robin Winkler
Head of German Macro Research

Sameer Goel
Global Head of EM & APAC Research

Francis Yared
Global Head of Rates Research

George Saravelos
Global Head of FX Research

Peter Hooper
Vice-Chair of Research

Nilendra de-Mel
Head of APAC & Middle East Product Development

International Production Locations

Deutsche Bank AG
Deutsche Bank Place
Level 16
Corner of Hunter & Phillip Streets
Sydney, NSW 2000
Australia
Tel: (61) 2 8258 1234

Deutsche Bank AG
Equity Research
Mainzer Landstrasse 11-17
60329 Frankfurt am Main
Germany
Tel: (49) 69 910 00

Deutsche Bank AG
Filiale Hongkong
International Commerce Centre,
1 Austin Road West,Kowloon,
Hong Kong
Tel: (852) 2203 8888

Deutsche Securities Inc.
1-3-1 Azabudai
Azabudai Hills Mori JP Tower
Minato-ku, Tokyo 106-0041
Japan
Tel: (81) 3 6730 1000

Deutsche Bank AG
21 Moorfields
London EC2Y 9DB
United Kingdom
Tel: (44) 20 7545 8000

Deutsche Bank Securities Inc.
The Deutsche Bank Center
1 Columbus Circle
New York, NY 10019
Tel: (1) 212 250 2500

Deutsche Bank AG
Filiale Singapur
One Raffles Quay, South Tower,
Singapore 048583
TeL: +65 6423 8001